Python for computational Problem Solving · Unit_03 · Unit 3: Functions, Functional Programming and Modules

DRAFT — UE26CS151A · Unit_03 — Unit 3: Functions, Functional Programming and Modules · Complete Unit Master

Source-complete searchable master · transcript, slide text, page visuals and archive references

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Search this single unit master by lesson, timestamp, slide text or source page. Read the transcript in timestamp order, then compare extracted slide text with its page-faithful source figure. Formula candidates and archive question mappings are explicitly flagged for review. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Unit_03 · Unit 3: Functions, Functional Programming and Modules

1 UE26CS151A · Unit_03: Unit 3: Functions, Functional Programming and Modules

This unit master organizes source material for Python for computational Problem Solving, Unit_03, officially titled Unit 3: Functions, Functional Programming and Modules. It includes 11 video transcripts and 18 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Official catalogue entry: UE26CS151A, Unit_03 (Unit 3: Functions, Functional Programming and Modules). Bucket folder present: yes. Videos: 11; slide decks: 18.

EV0000001

2 Recursion

This lesson is indexed from the source video titled Recursion. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000002 · EV0000003

Graphs and diagrams

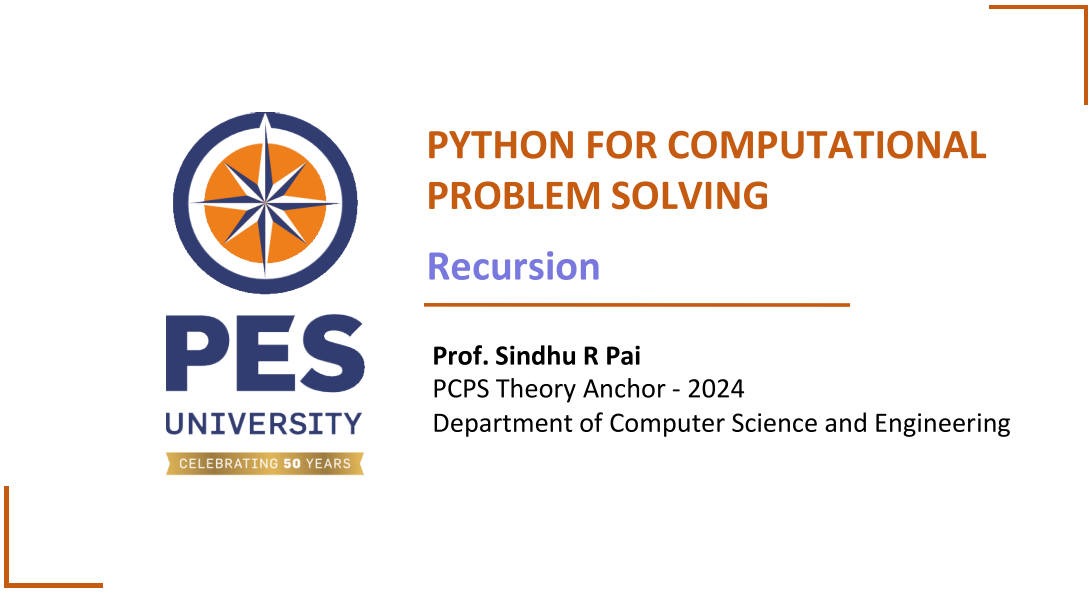

Figure G000001. UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion · source slide page 1.

EV0000119

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

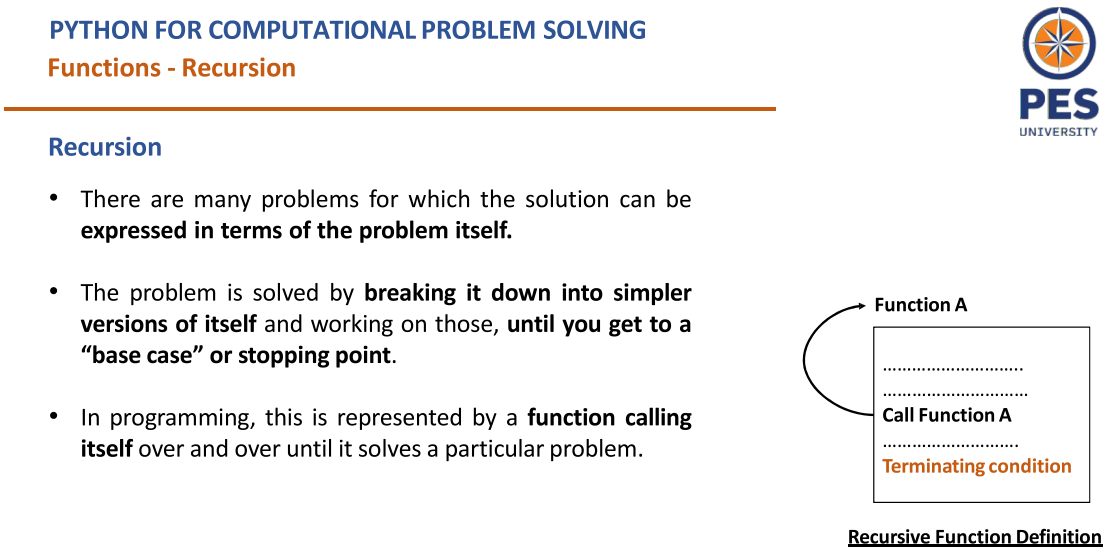

Figure G000002. UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion · source slide page 2.

EV0000122

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Functions - Recursion

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Recursion

Every recursive problem can be divided into a base case and a recursive case.

- **Base Case:** This is the simplest subproblem the problem can be reduced to. In a recursive solution, **the base case is the one case you know the direct solution for**.
- **Recursive Case:** Any case that isn't the base case is automatically a recursive case. We **don't know the solutions** for these cases; instead, we **split them further into simpler cases** until we reach the base case.

Figure G000003. UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion · source slide page 3.

EV0000125

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Functions - Recursion

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Recursion – Basic Example

Take the example of opening a set of Russian nesting dolls.

Base Case: The doll doesn't open. This means we've reached the smallest doll, and represents the stopping condition for our recursion.

Recursive Case: The doll opens. This means we still haven't finished opening the set, and we need to perform some action here called the recursive step.

Recursive Step: If you have a doll that can open (i.e., recursive case), then open it and see if the doll inside it can open. This is the action we perform on the recursive case to get it closer to the base case.

Figure G000004. UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion · source slide page 4.

EV0000128

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Functions - Recursion

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Recursion – Basic Example

We can illustrate this using pseudocode as follows:

```
function OpenDoll(doll):
    # Base Case
    if doll is the smallest doll:
        print("This is the smallest doll. Stop opening.")
    # Recursive Case
    else:
        print("Opening doll to find a smaller one inside.")
        OpenDoll(smaller_doll_inside)   # Recursive Step

# Start the recursive process by opening the largest doll
OpenDoll(largest_doll)
```

Figure G000005. UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion · source slide page 5.

EV0000131

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

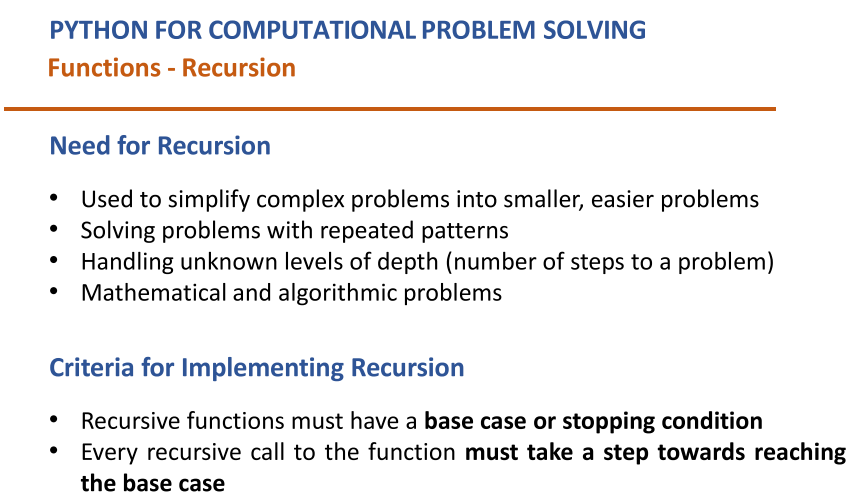

Figure G000006. UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion · source slide page 6.

EV0000134

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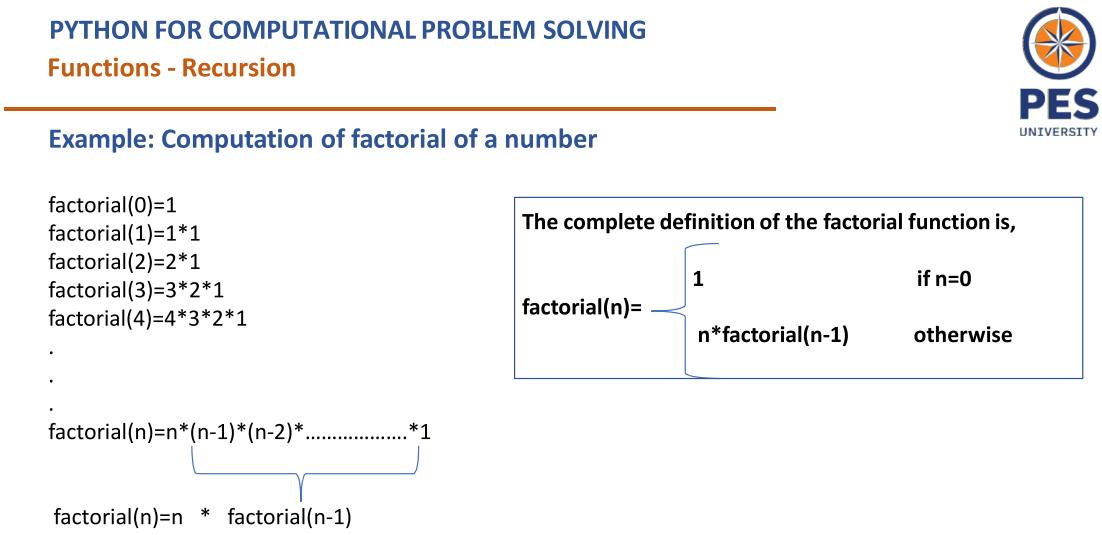

Figure G000007. UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion · source slide page 7.

EV0000137

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Functions - Recursion

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Implementation of Recursion

- Recursion is implemented using stack because activation records are to be stored in LIFO order (last in first out).
- An activation record of a function call contains arguments, return address and local variables of the function.
- Stack is a linear data structure in which elements are inserted to the top and deleted from the top.

Figure G000008. UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion · source slide page 8.

EV0000140

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Functions - Recursion

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Implementation of Recursion

- Consider the example of computing the factorial of a number.

```
def fact(n): #Recursive Function
        if n == 0 : #terminating condition
                res = 1
        else:
                res = n * fact(n - 1)
        return res
```

- Suppose we want to compute **fact(3)**. The above function gets executed using a stack.

Figure G000009. UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion · source slide page 9.

EV0000143

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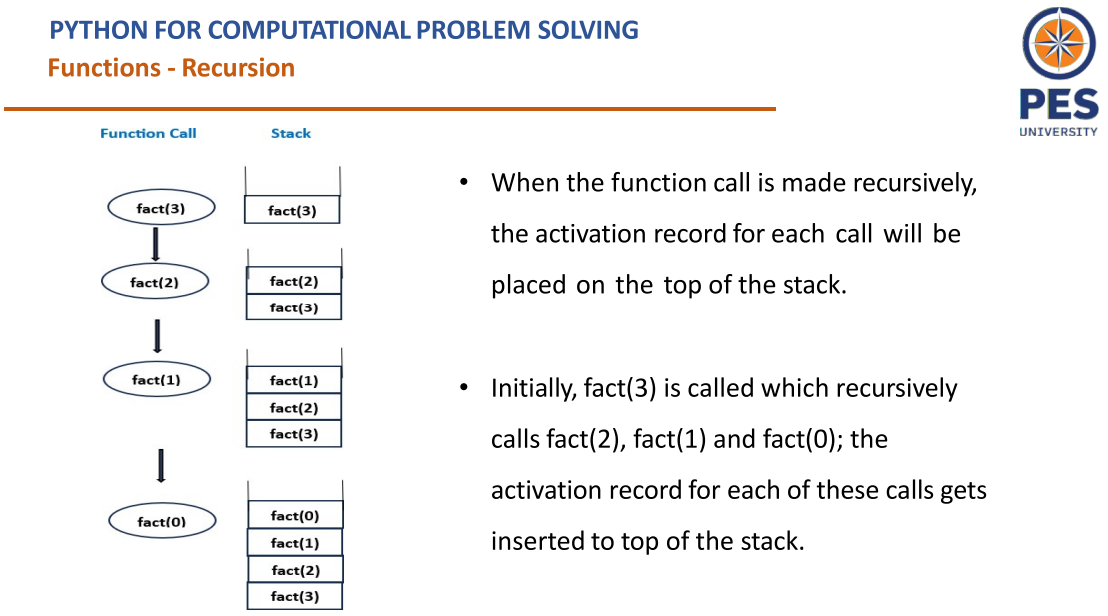

Figure G000010. UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion · source slide page 10.

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Functions - Recursion

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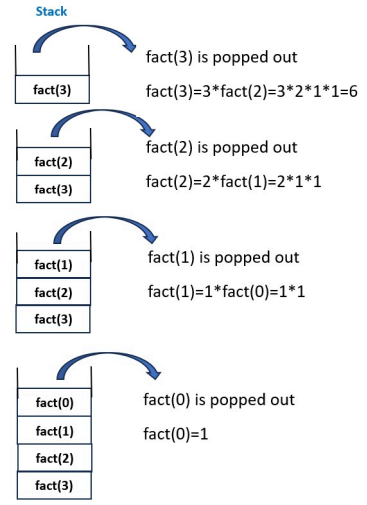

- For n=0 i.e. base case (Termination Condition), the function call stops and fact(0) is popped out from the top of the stack.
- It returns 1, then the recursion backtracks and solves the pending function calls. These are popped out of the stack as they get computed.

Figure G000011. UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion · source slide page 11.

EV0000149

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

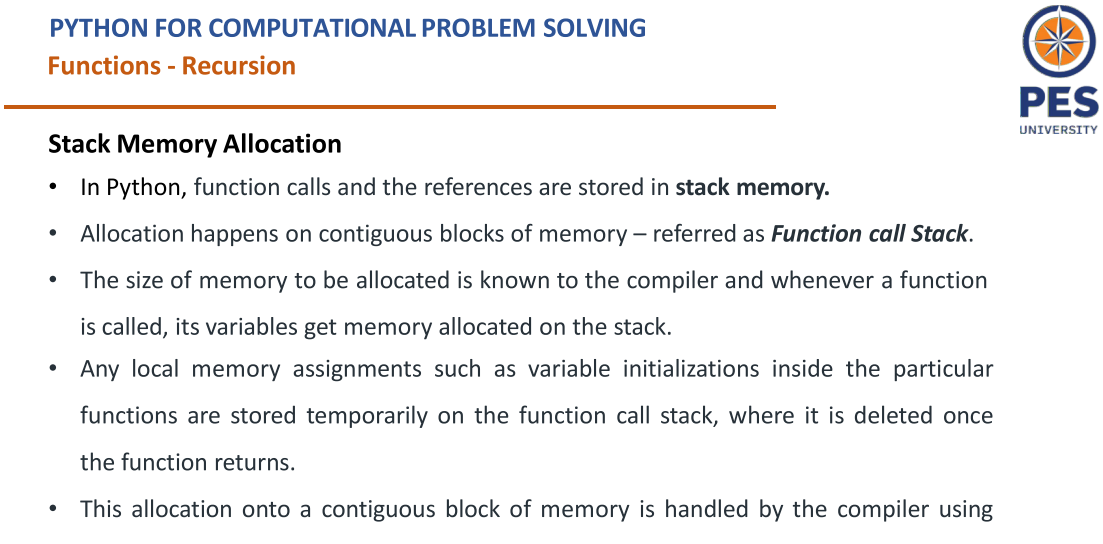

Figure G000012. UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion · source slide page 12.

EV0000152

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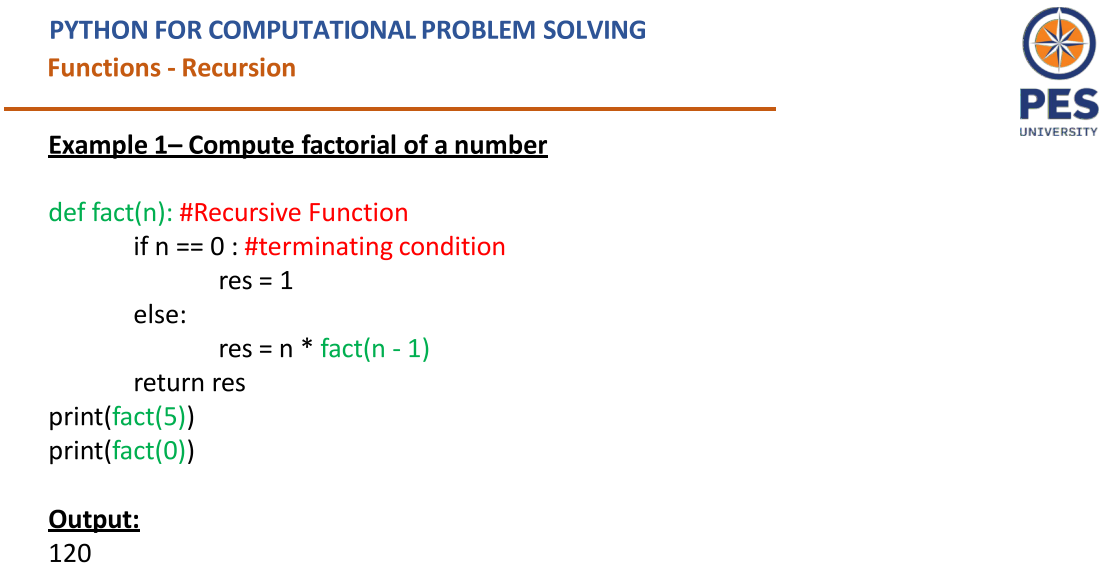

Figure G000013. UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion · source slide page 13.

EV0000155

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

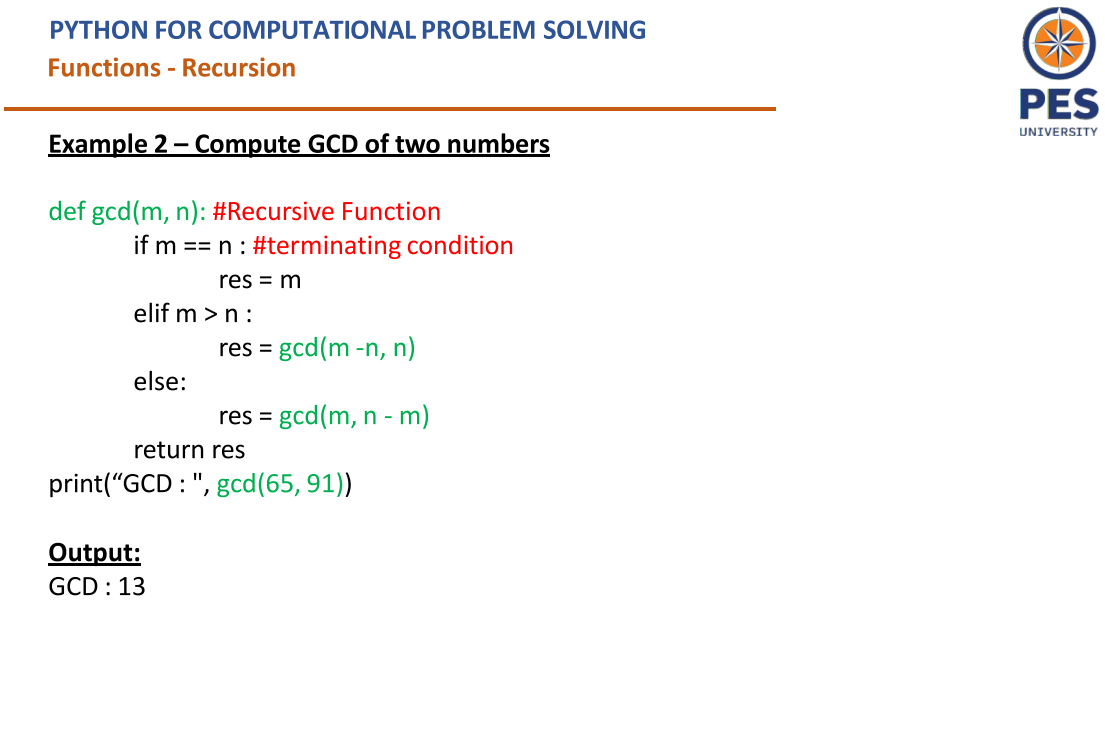

Figure G000014. UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion · source slide page 14.

EV0000158

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

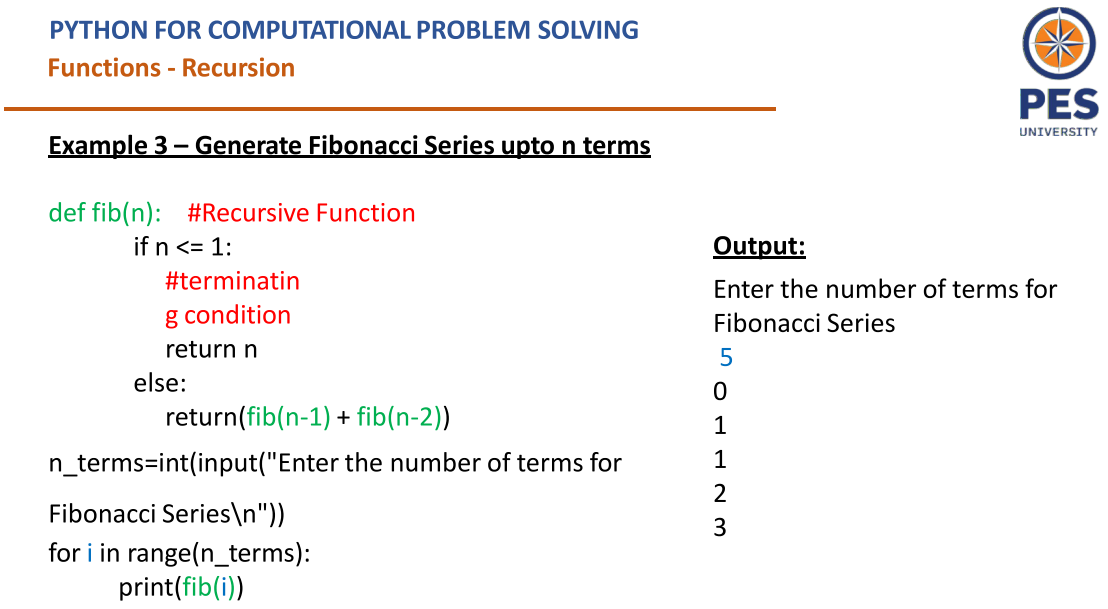

Figure G000015. UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion · source slide page 15.

EV0000161

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

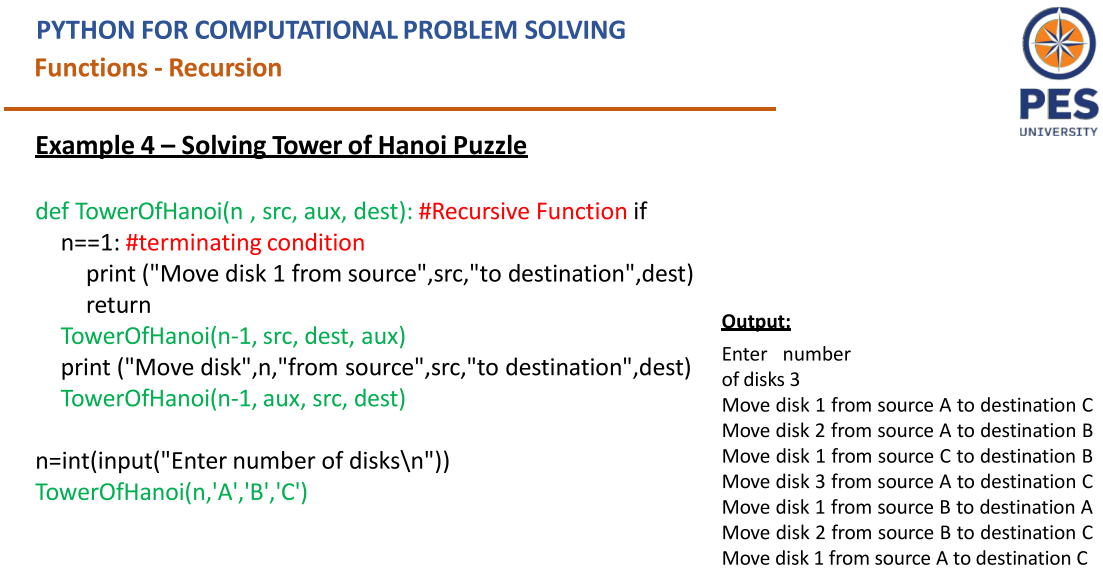

Figure G000016. UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion · source slide page 16.

EV0000164

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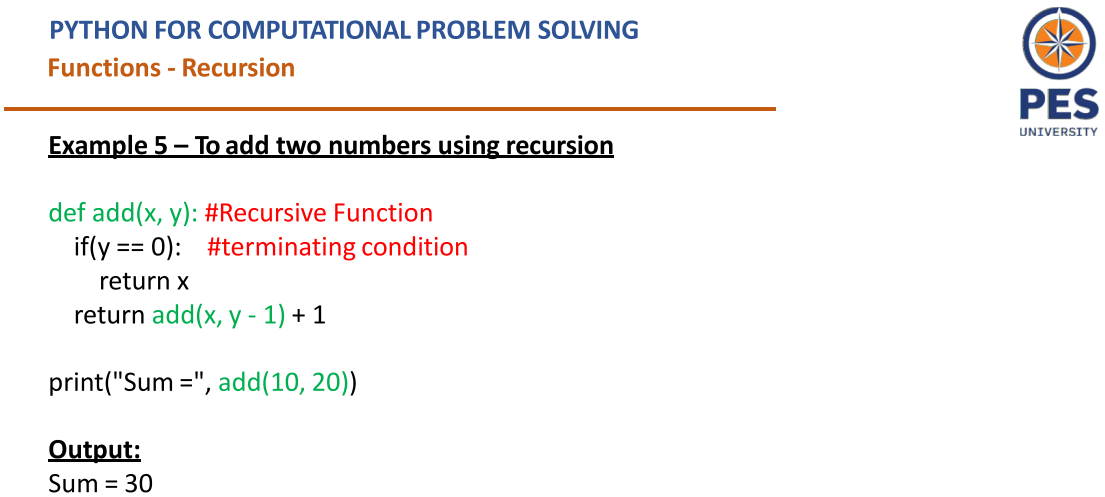

Figure G000017. UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion · source slide page 17.

EV0000167

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

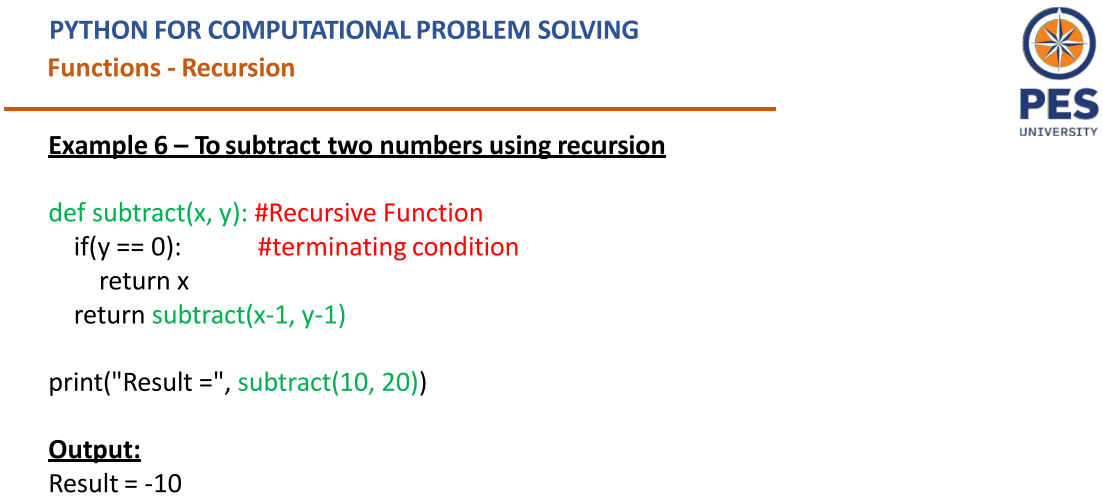

Figure G000018. UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion · source slide page 18.

EV0000170

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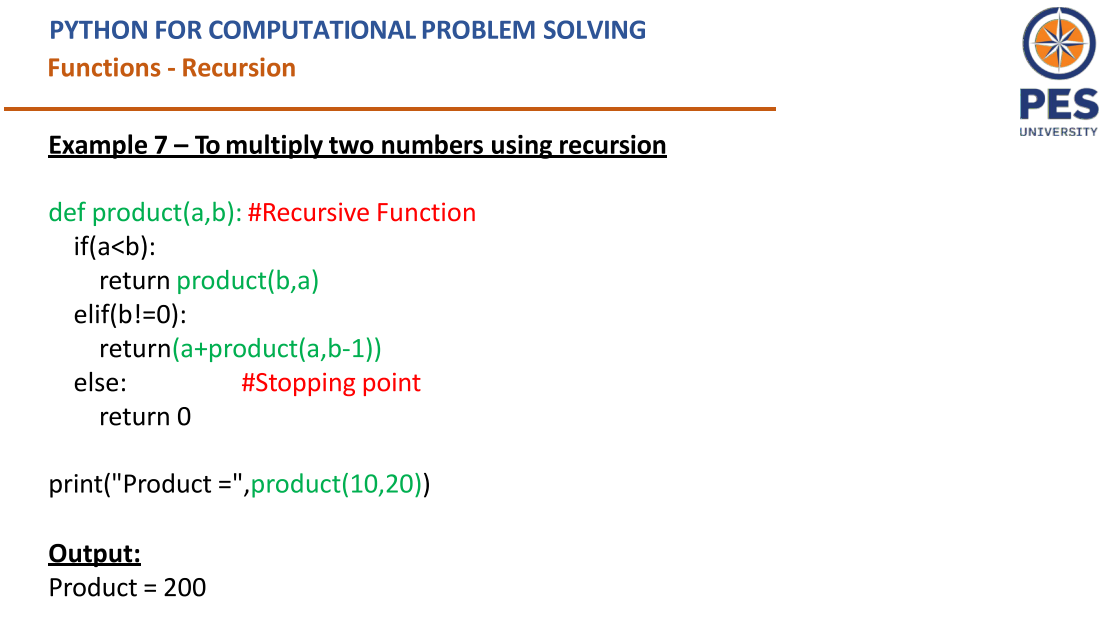

Figure G000019. UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion · source slide page 19.

EV0000173

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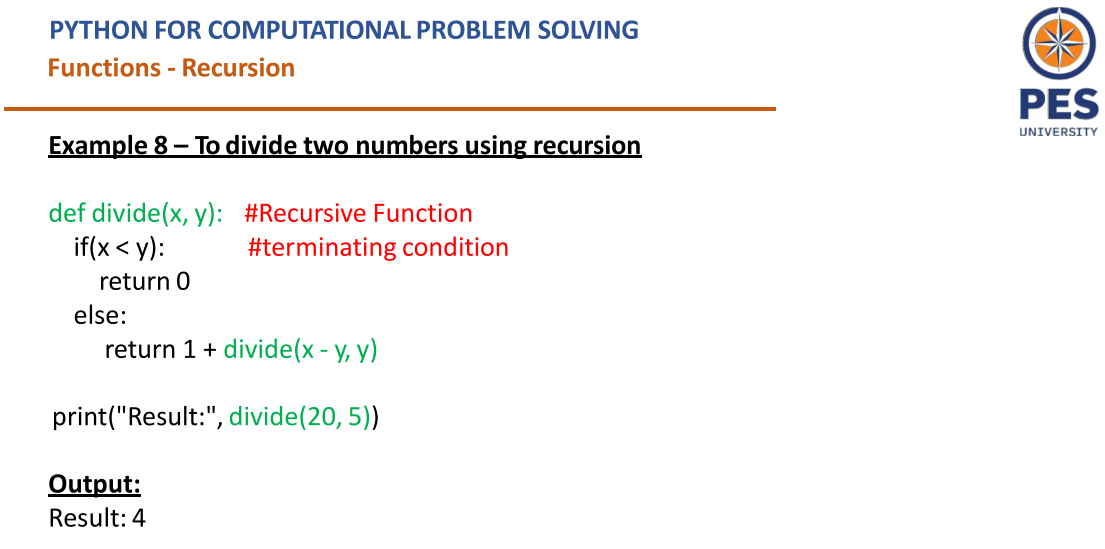

Figure G000020. UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion · source slide page 20.

EV0000176

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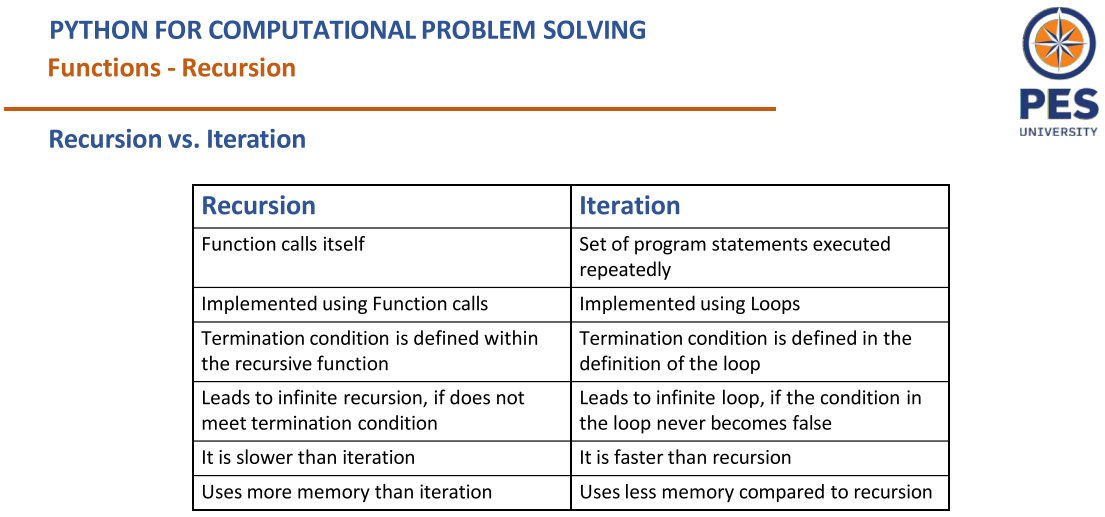

PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Functions - Recursion

PES UNIVERSITY

Recursion vs. Iteration

Recursion	Iteration
Function calls itself	Set of program statements executed repeatedly
Implemented using Function calls	Implemented using Loops
Termination condition is defined within the recursive function	Termination condition is defined in the definition of the loop
Leads to infinite recursion, if does not meet termination condition	Leads to infinite loop, if the condition in the loop never becomes false
It is slower than iteration	It is faster than recursion
Uses more memory than iteration	Uses less memory compared to recursion

Note: When a problem can be solved both recursively and iteratively with similar programming effort, it is generally best to use an iterative approach.

Figure G000021. UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion · source slide page 21.

EV0000179

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

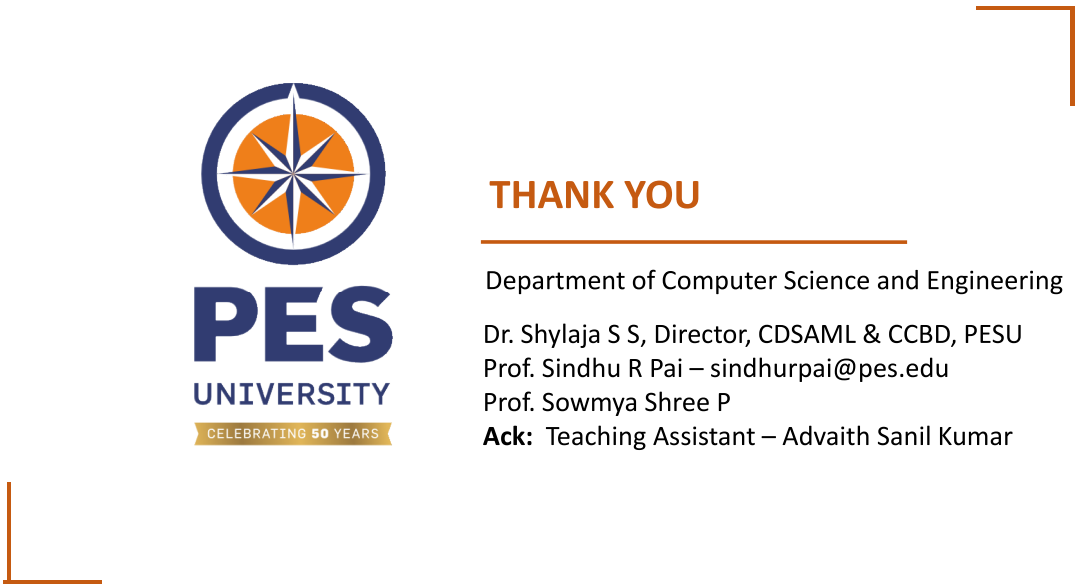

Figure G000022. UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion · source slide page 22.

EV0000182

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Recursion Prof. Sindhu R Pai PCPS Theory Anchor - 2024 Department of Computer Science and Engineering EV0000118 · EV0000119 · EV0000120
2. Source slide page 2 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Recursion Recursion • There are many problems for which the solution can be expressed in terms of the problem itself. • The problem is solved by breaking it down into simpler FunctionFunction AA versions of itself and working on those, until you get to a “base case” or stopping point. .... • In programming, this is represented by a function calling Call Function A ..... itself over and over until it solves a particular problem. Terminatingcondition Recursive Function Definition EV0000121 · EV0000122 · EV0000123

3. Source slide page 3 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Recursion Recursion Every recursive problem can be divided into a base case and a recursive case. • Base Case: This is the simplest subproblem the problem can be reduced to. In a recursive solution, the base case is the one case you know the direct solution for. • Recursive Case: Any case that isn't the base case is automatically a recursive case. We don't know the solutions for these cases; instead, we split them further into simpler cases until we reach the base case. EV0000124 · EV0000125 · EV0000126
4. Source slide page 4 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Recursion Recursion – Basic Example Take the example of opening a set of Russian nesting dolls. Base Case: The doll doesn't open. This means we've reached the smallest doll, and represents the stopping condition for our recursion. Recursive Case: The doll opens. This means we still haven't finished opening the set, and we need to perform some action here called the recursive step. Recursive Step: If you have a doll that can open (i.e., recursive case), then open it and see if the doll inside it can open. This is the action we perform on the recursive case to get it closer to the base case. EV0000127 · EV0000128 · EV0000129
5. Source slide page 5 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Recursion Recursion – Basic Example We can illustrate this using pseudocode as follows: function OpenDoll(doll): # Base Case ifif ddollll iis ththe smallestll t ddoll:ll print("This is the smallest doll. Stop opening.") # Recursive Case else: print("Opening doll to find a smaller one inside.") OpenDoll(smaller_doll_inside) # Recursive Step # Start the recursive process by opening the largest doll OpenDoll(largest_doll) EV0000130 · EV0000131 · EV0000132
6. Source slide page 6 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Recursion Need for Recursion • Used to simplify complex problems into smaller, easier problems • Solving problems with repeated patterns • Handling unknown levels of depth (number of steps to a problem) • Mathematical and algorithmic problems Criteria for Implementing Recursion • Recursive functions must have a base case or stopping condition • Every recursive call to the function must take a step towards reaching the base case EV0000133 · EV0000134 · EV0000135
7. Source slide page 7 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Recursion Example: Computation of factorial of a number factorial(0)=1 The complete definition of the factorial function is,factorial(1)=1*1 factorial(2)=2*1 1 if n=0factorial(3)=3*2*1 factorial(n)=factorial(n)=factorial(4)=4*3*2*1 n*factorial(n-1) otherwise. . . factorial(n)=n*(n-1)*(n-2)*.....*1 factorial(n)=n * factorial(n-1) EV0000136 · EV0000137 · EV0000138
8. Source slide page 8 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Recursion Implementation of Recursion • Recursion is implemented using stack because activation records are to be stored in LIFO order (last in first out). • An activation record of a function call contains arguments,g , return address and local variables of the function. • Stack is a linear data structure in which elements are inserted to the top and deleted from the top. EV0000139 · EV0000140 · EV0000141

9. Source slide page 9 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Recursion Implementation of Recursion • Consider the example of computing the factorial of a number. def fact(n): #Recursive Function ifif nn ==== 00 :: #terminating#terminating conditioncondition res = 1 else: res = n * fact(n - 1) return res • Suppose we want to compute fact(3). The above function gets executed using a stack. EV0000142 · EV0000143 · EV0000144
10. Source slide page 10 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Recursion • When the function call is made recursively, the activation record for each call will be placed on the top of the stack. • Initially, fact(3) is called which recursively calls fact(2), fact(1) and fact(0); the activation record for each of these calls gets inserted to top of the stack. EV0000145 · EV0000146 · EV0000147
11. Source slide page 11 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Recursion • For n=0 i.e. base case (Termination Condition), the function call stops and fact(0) is popped out from the top of the stack. • It returns 1, then the recursion backtracks and solves the pending function calls. These are popped out of the stack as they get computed. EV0000148 · EV0000149 · EV0000150
12. Source slide page 12 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Recursion Stack Memory Allocation • In Python, function calls and the references are stored in stack memory. • Allocation happens on contiguous blocks of memory – referred as Function call Stack. • The size of memory to be allocated is known to the compiler and whenever a function is called, its variables get memory allocated on the stack. • Any local memory assignments such as variable initializations inside the particular functions are stored temporarily on the function call stack, where it is deleted once the function returns. • This allocation onto a contiguous block of memory is handled by the compiler using predefined routines. EV0000151 · EV0000152 · EV0000153
13. Source slide page 13 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Recursion Example 1– Compute factorial of a number def fact(n): #Recursive Function if n == 0 : #terminating condition res = 1 lelse: res = n * fact(n - 1) return res print(fact(5)) print(fact(0)) Output: 120 1 EV0000154 · EV0000155 · EV0000156
14. Source slide page 14 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Recursion Example 2 – Compute GCD of two numbers def gcd(m, n): #Recursive Function if m == n : #terminating condition res = m eliflif m > n : res = gcd(m -n, n) else: res = gcd(m, n - m) return res print(“GCD : ", gcd(65, 91)) Output: GCD : 13 EV0000157 · EV0000158 · EV0000159
15. Source slide page 15 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Recursion Example 3 – Generate Fibonacci Series upto n terms def fib(n): #Recursive Function if n <= 1: Output: #terminatin Enter the number of terms for g conditiondi i Fibonacci Series return n 5 else: 0 return(fib(n-1) + fib(n-2)) 1 n_terms=int(input("Enter the number of terms for 1 2 Fibonacci Series\n")) 3 for i in range(n_terms): print(fib(i)) EV0000160 · EV0000161 · EV0000162

16. Source slide page 16 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Recursion Example 4 – Solving Tower of Hanoi Puzzle def TowerOfHanoi(n , src, aux, dest): #Recursive Function if n==1: #terminating condition print ("Move disk 1 from source",src,"to destination",dest) returnreturn Output: TowerOfHanoi(n-1, src, dest, aux) Enter number print ("Move disk",n,"from source",src,"to destination",dest) of disks 3 TowerOfHanoi(n-1, aux, src, dest) Move disk 1 from source A to destination C Move disk 2 from source A to destination B n=int(input("Enter number of disks\n")) Move disk 1 from source C to destination B Move disk 3 from source A to destination CTowerOfHanoi(n,'A','B','C') Move disk 1 from source B to destination A Move disk 2 from source B to destination C Move disk 1 from source A to destination C EV0000163 · EV0000164 · EV0000165

17. Source slide page 17 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Recursion Example 5 – To add two numbers using recursion def add(x, y): #Recursive Function if(y == 0): #terminating condition return x return add(x,dd(y - 1)) + 1 print("Sum =", add(10, 20)) Output: Sum = 30 EV0000166 · EV0000167 · EV0000168

18. Source slide page 18 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Recursion Example 6 – To subtract two numbers using recursion def subtract(x, y): #Recursive Function if(y == 0): #terminating condition return x return subtract(x-1,b (y-1)) print("Result =", subtract(10, 20)) Output: Result = -10 EV0000169 · EV0000170 · EV0000171

19. Source slide page 19 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Recursion Example 7 – To multiply two numbers using recursion def product(a,b): #Recursive Function if(a<b): return product(b,a) elif(b!=0):lif(b! 0) return(a+product(a,b-1)) else: #Stopping point return 0 print("Product =",product(10,20)) Output: Product = 200 EV0000172 · EV0000173 · EV0000174

20. Source slide page 20 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Recursion Example 8 – To divide two numbers using recursion def divide(x, y): #Recursive Function if(x < y): #terminating condition return 0 else:l return 1 + divide(x - y, y) print("Result:", divide(20, 5)) Output: Result: 4 EV0000175 · EV0000176 · EV0000177

21. Source slide page 21 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Recursion Recursion vs. Iteration Recursion Iteration Function calls itself Set of program statements executed repeatedly ImplementedImplemented usingusing FunctionFunction callscalls ImplementedImplemented usingusing LoopsLoops Termination condition is defined within Termination condition is defined in the the recursive function definition of the loop Leads to infinite recursion, if does not Leads to infinite loop, if the condition in meet termination condition the loop never becomes false It is slower than iteration It is faster than recursion Uses more memory than iteration Uses less memory compared to recursion Note: When a problem can be solved both recursively and iteratively with similar programming effort, it is generally best to use an iterative approach. EV0000178 · EV0000179 · EV0000180

22. Source slide page 22 (extracted text): THANK YOU Departmentp of Computerp Science and Engineeringg g Dr. Shylaja S S, Director, CDSAML & CCBD, PESU Prof. Sindhu R Pai – sindhurpai@pes.edu Prof. Sowmya Shree P Ack: Teaching Assistant – Advaith Sanil Kumar EV0000181 · EV0000182 · EV0000183

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.828] Hello, all, welcome back, Python for computational problem solving. In today's class, we are going to be looking into the concept called as recursion. [00:00:18.404] So let's get started and try to understand what is recursion, why, and where do I use a recursion along with an example. [00:00:30.180] Recursion is basically a process where a function calls itself in order to solve a problem. So I keep calling a same function again and again. Okay? So function [00:00:46.404] It's called [00:00:49.956] By itself. [00:00:54.372] This is called as a recursion. So why do I basic need to make use of a recursion? There are few recursive functions or sometimes uh recursive functions are so useful that they make it very easy for us to write a code. [00:01:10.500] And there are few algorithms where it is very easy to express in terms of a recursive paradigm rather than writing it as an uh iterative process. [00:01:20.960] So, and uh there is no recursive function that cannot be uh rewritten in an iterative fashion. So a problem given, either you can write it in an iterative fashion or you can write it in an recursive manner. [00:01:36.484] But make sure that whatever is required and the best approach in order to solve your requirement, make use of it. Fine. So in order to uh write a recursive problem, what do I need it? [00:01:51.952] In order to write a recursive program, I need a recursive case. I say recursive case. [00:02:02.340] I have to mention what is my recursive case. At the same time, I also need to specify something called as a a base case. Hopefully don't mind with my uh spelling mistakes, base case, or I can also call it as a termination case. [00:02:22.116] Also called as a termination case. What will happen if I do not have a base case? If you do not mention where this particular function should stop executing, if you do not mention the function gets called again and again and you end up in an infinite loop. [00:02:39.376] That's the reason why whenever you are writing a recursive function, make sure you write a recursive case at the same time, a very important one which is called as a base case. Now, what are the advantages of a recursion? The main advantages of recursion is it is simple in nature. [00:02:59.428] And it is simple in nature in the sense the code looks so simple. Code looks simple. At the same time it is the moment it is simple in nature, it also looks elegant. [00:03:16.260] And the second advantage is basically your task can be broken down into a simpler subtlask. Okay? Task is broken down. [00:03:31.524] [ASR low confidence; check audio] The set task. [00:03:35.524] Is broken down into subtask. These are the advantages of your recursion, and the disadvantages are. [00:03:47.428] Disadvantages are. Let me just say this as advantage so it is easy for us to uh go through. Okay, hope you don't mind this. And what are the disadvantages? The disadvantages are [00:04:06.404] Lots of memory. It consumes lots of memory. It it becomes inefficient. At the same time, it also consumes lots of time. [00:04:22.596] Right? And if you do a mistake, very tough to debug. [00:04:30.436] Because you never know where the mistake is happening. [00:04:33.956] And uh understanding uh the logic behind the recursion is not so easy. So these are the advantages as well as the disadvantages. Let me uh quickly iterate over the advantages. Code looks simple and elegant. [00:04:50.916] Task is broken down into the subtasks. Disadvantages is it takes lots of time, at the same time it also consumes lots of memory. Right? And uh programs written uh using recursion are very hard to debug and understanding the logic behind recursive program is not easy.

EV0000002 · EV0000003 · EV0000004 · EV0000005

Transcript block 2 (source interval 000300-000599 seconds): [00:05:10.692] Okay, so let's uh quickly write a simple program in order to understand uh the concept of recursion. [00:05:22.948] Let's get started. Very famous problem we usually write. So let me take up the same example. Okay. So let me push it up so that it is easy for us to write the program. Factorial program. [00:05:41.124]

Okay. So in order to do this, let me define a function called as factorial. What is this what do I want to find out? I want to find out what is a factorial of a particular number. So I say it as factorial of x. [00:06:00.836] I will say if the number that I give you if it is equal to zero, which means you want to find a factorial of zero. So I will say return one. Because factorial of uh zero is one. [00:06:21.156] Otherwise, what I will say else, I will also say it as what? [00:06:29.092] Instead of doing that, perform X into fact of X minus one. [00:06:39.108] Okay. Every time you're trying to find out uh the factorial, what I'm trying to basically do, I am trying to uh find the value and the previous one, x minus one. Okay. I have to return this, so I will say it as what? Return. [00:06:58.564] factorial x into factorial of x minus one. My function is ready. What's the next thing that I have to do? I have to call this function. So by this you would have understood how do you call the function and on? So I will say factorial of [00:07:17.572] You will do the every time okay, maybe we will take a simple one, factorial of three. Fine. So now in order to do factorial of three, what do I uh basically do? [00:07:30.180] Three into factorial of two minus one I will do, then followed by two into factorial of the previous number. So that's what we try to do. So I am saying three factorial is what I want to find out. Sorry. [00:07:46.660] Okay, so this is what I am uh try to do. Let's quickly execute this program and find out what is the answer for this. Okay. I've kept everything ready so that uh one second I will [00:08:09.124] Any time in this fine. So let me execute this program. Let's find out are there any errors or there are no errors, we'll find out. So if you see factorial of three is what? Six. Let me change the number. Factorial of five. [00:08:27.300] And the expected answer is 120. Okay. So this is how uh we try to find out or uh write a recursive program. And if you can see this is a name of a function, and this function factorial is again calling itself. [00:08:44.960] This is fine. And I call this as a base case because when I say x becomes equal to 0, what it has to do? It has to return one. To whom it has to return? It has to return to a calling function. Okay. So what I will do is I will help you out by copying this. [00:09:04.196] and we'll use a Python tutor. Okay, it's an online tutor that is available which helps us to visualize what we are trying to do. Okay so I've just copy pasted uh my program [00:09:18.864] And then you have to say it as visualize the execution. Hope you have already learned this. Just go through this. And if you see the number of lines of code that I have written is 1, 2, 3, 4, 5, and 6. And how many number of steps that are involved here? 26 different steps are involved. [00:09:36.560] Involved in order to uh execute the factorial program. That is what uh I said. It takes the amount of time it takes in order to execute is more. At the same time, internally, so many states have to be stored. So uh the memory consumption is also more in this. So let me uh help you out with this. [00:09:57.348] Right? Factorial function because if you can if you had seen your cursor, let me show this. The cursor is pointing at the beginning, the function header, okay? It basically processed the function header.

EV0000006 · EV0000007 · EV0000008 · EV0000009

Transcript block 3 (source interval 000600-000899 seconds): [00:10:10.064] I will get into the function body only when a particular function is called. So this concept is also known to you. So let me uh do this. [00:10:19.172] Fine. Fine. And I'm now I'm trying to call what? Factorial of five. So it basically checks is it equal to zero? No, it is not equal to zero. If it is not equal to zero, what do I need to do? Go to your else part. [00:10:33.120] Okay, in the L's part you say return x into x minus 1. What is x? X is 5, factorial of x minus 1 would be the previous number. 4 my sorry 5 minus 1, which is going to be 4. Okay. So go ahead with the next one and see. Now x becomes 4 in this case. [00:10:52.324] Fine. Similarly, same thing. X becomes three. X becomes two. Now.

[00:11:03.748] X becomes 1. The next time your X becomes 0. Okay? The moment your X becomes a 0, what will happen? It returns 1. It returns 1 to whom? To the one that has called. So the to the previous one. So it has returned 1. [00:11:21.220] Fine. Now it is returning to back. 2 into 3 was 6. Now 6 into 4 is 24. 24 into 5 would become 120. So that is how the factorial of a given number, which is 5 in this case, is equal to 120. [00:11:40.352] Okay, so hope uh you have uh understood uh uh the basic concept of uh recursion. I will tell you any number of programs that you do until and unless you code uh the problems. [00:11:56.356] Uh writing a recursive programs would be definitely tougher. So I request you all to uh try with a few more uh recursive programs, writing Fibonacci numbers and uh [00:12:09.408] uh the other one finding a G C D all those things you can definitely try it out. Fine. Hope you have understood the uh basic concept and uh in the uh actual classes we will try to work with a few other programs. Thank you.

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Full transcript: 1687 normalized words, 50 timestamped segments, 747.178667 seconds duration. Low-confidence segments flagged: 1. Review locators: 00:03:31.524

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3 Callback

This lesson is indexed from the source video titled Callback. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000014 · EV0000015

Graphs and diagrams

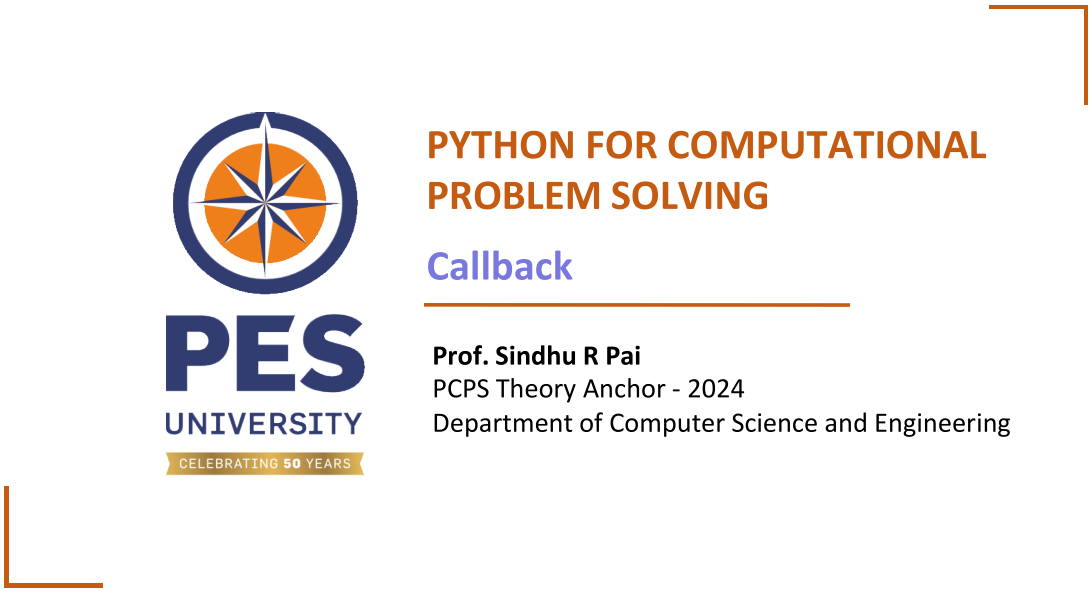

Figure G000023. UE24CS151A PCPS Unit 3 Sindhu Class 3 4 Callback · source slide page 1.

EV0000185

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

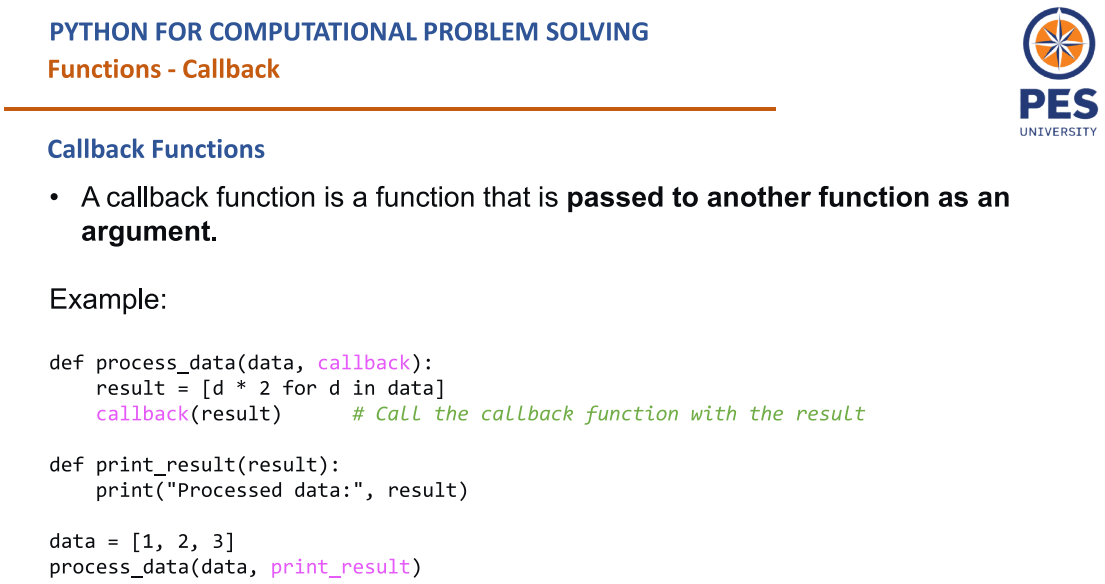

Figure G000024. UE24CS151A PCPS Unit 3 Sindhu Class 3 4 Callback · source slide page 2.

EV0000188

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Functions - Callback

PES UNIVERSITY

Need for Callback in Functions

- **Used in event-driven programming**, where a function is called in response to a specific event or action, such as a button press or the completion of a network request.
- **Also used in functional programming**, where a function is passed as an argument to another function to be used as a "hook" for performing specific operations.
- **Helps to separate functions' functionality** and make code more reusable and modular.

Figure G000025. UE24CS151A PCPS Unit 3 Sindhu Class 3 4 Callback · source slide page 3.

EV0000191

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

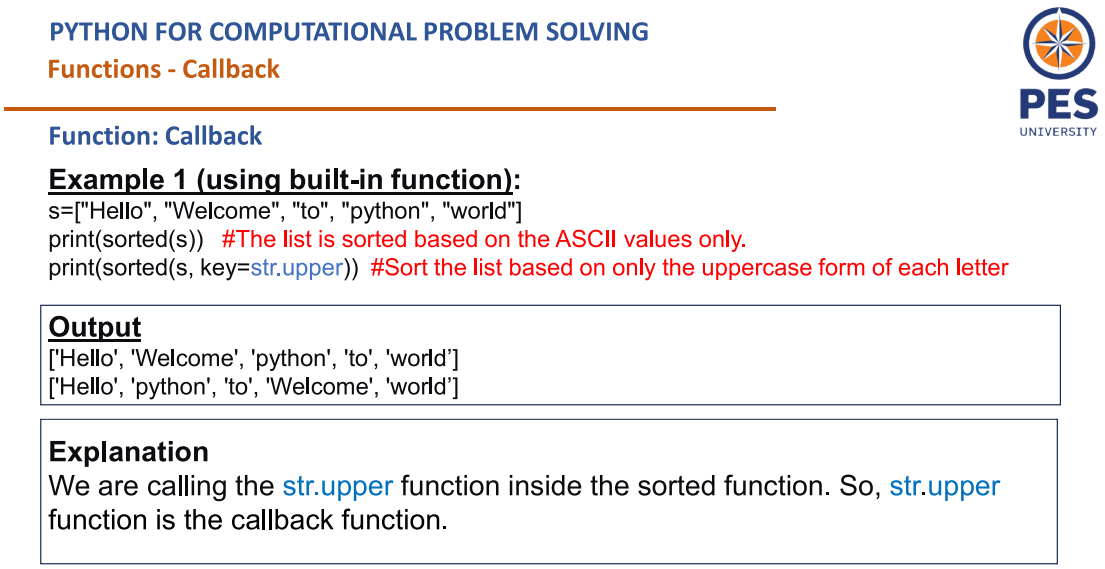

Figure G000026. UE24CS151A PCPS Unit 3 Sindhu Class 3 4 Callback · source slide page 4.

EV0000194

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

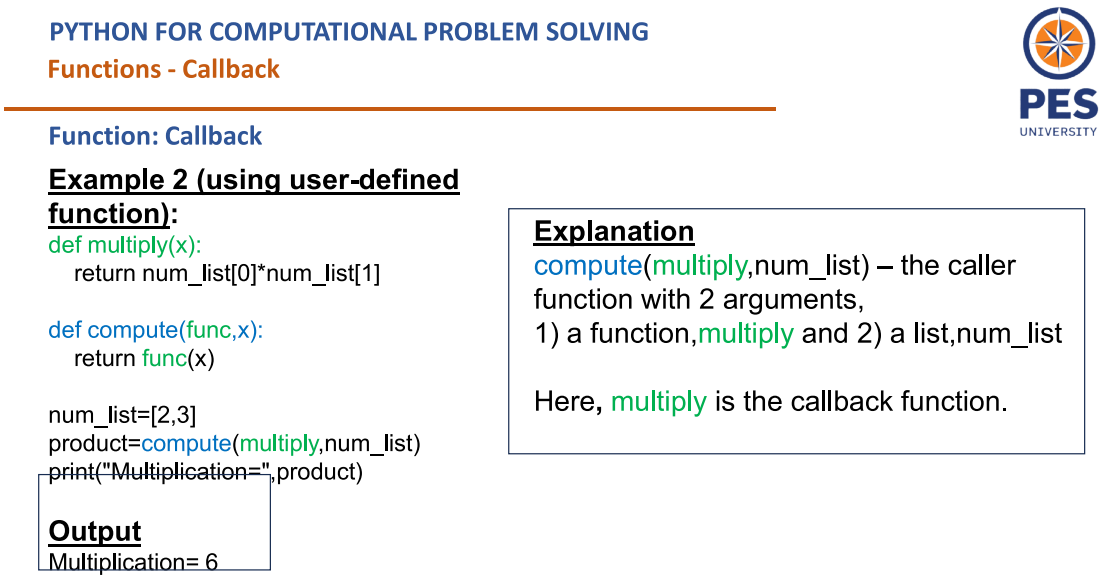

Figure G000027. UE24CS151A PCPS Unit 3 Sindhu Class 3 4 Callback · source slide page 5.

EV0000197

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

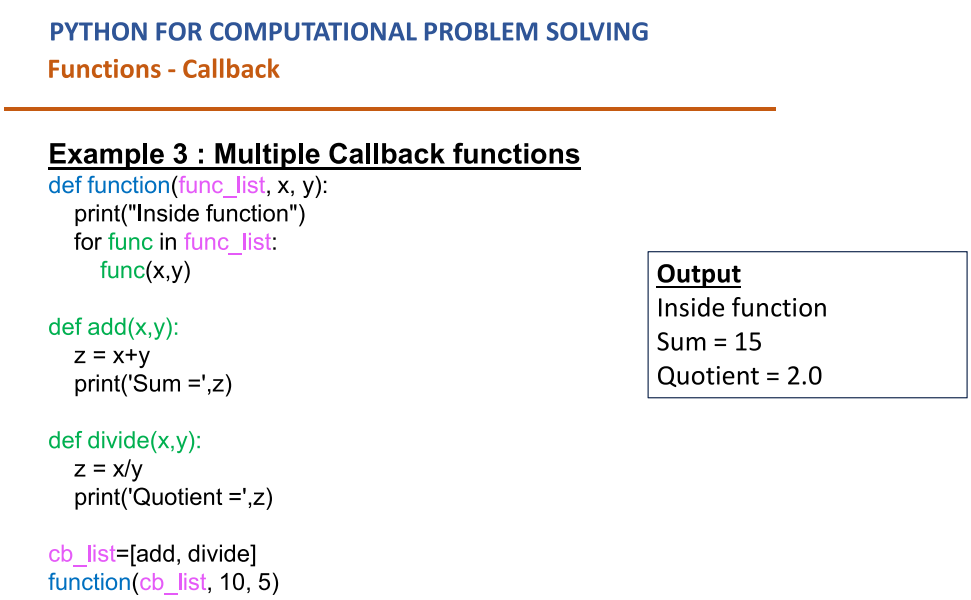

Figure G000028. UE24CS151A PCPS Unit 3 Sindhu Class 3 4 Callback · source slide page 6.

EV0000200

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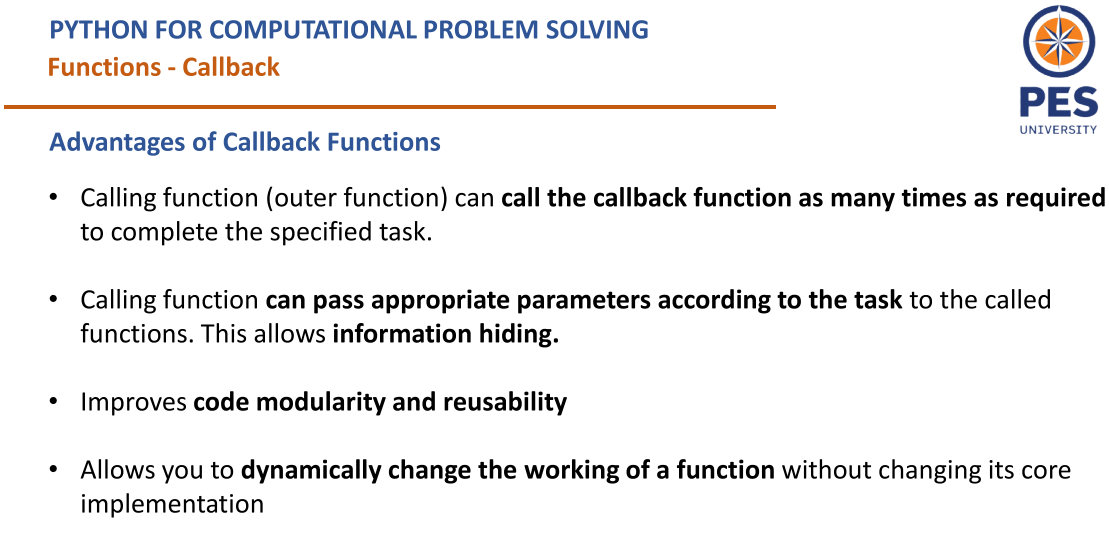

Figure G000029. UE24CS151A PCPS Unit 3 Sindhu Class 3 4 Callback · source slide page 7.

EV0000203

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

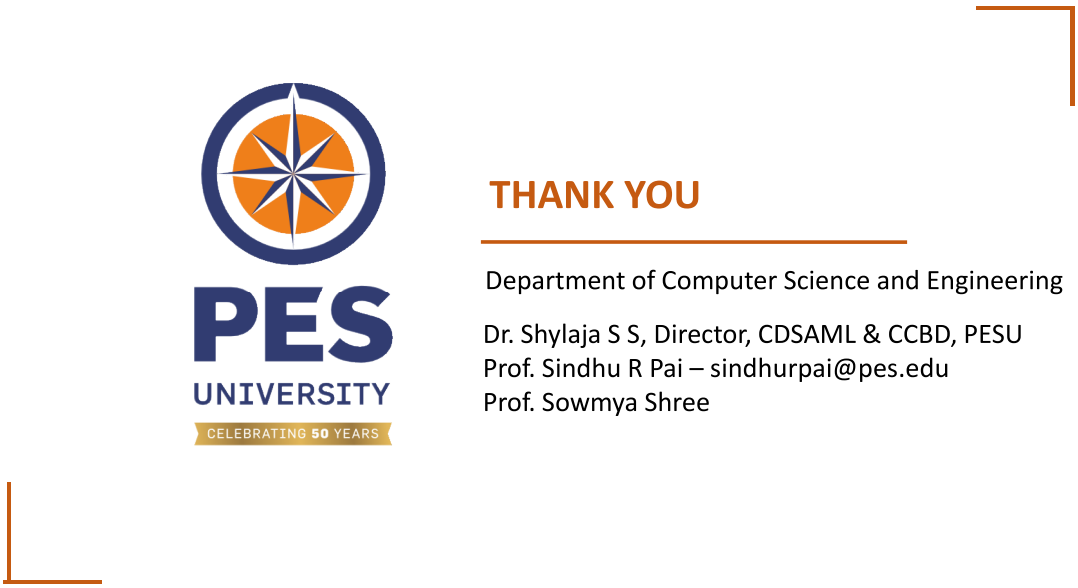

Figure G000030. UE24CS151A PCPS Unit 3 Sindhu Class 3 4 Callback · source slide page 8.

EV0000206

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Callback Prof. Sindhu R Pai PCPS Theory Anchor - 2024 Department of Computer Science and Engineering EV0000184 · EV0000185 · EV0000186
2. Source slide page 2 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Callback Callback Functions • A callback function is a function that is passed to another function as an argument. Example:Example: def process_data(data, callback): result = [d * 2 for d in data] callback(result) # Call the callback function with the result def print_result(result): print("Processed data:", result) data = [1, 2, 3] process_data(data, print_result) EV0000187 · EV0000188 · EV0000189

3. Source slide page 3 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Callback Need for Callback in Functions • Used in event-driven programming, where a function is called in response to a specific event or action, such as a button press or the completionp of a network request.q • Also used in functional programming, where a function is passed as an argument to another function to be used as a "hook" for performing specific operations. • Helps to separate functions' functionality and make code more reusable and modular. EV0000190 · EV0000191 · EV0000192
4. Source slide page 4 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Callback Function: Callback Example 1 (using built-in function): s=["Hello", "Welcome", "to", "python", "world"] print(sorted(s)) #The list is sorted based on the ASCII values only. print(sorted(s, key=str.upper)) #Sort the list based on only the uppercase form of each letter Output ['Hello', 'Welcome', 'python', 'to', 'world'] ['Hello', 'python', 'to', 'Welcome', 'world'] Explanation We are calling the str.upper function inside the sorted function. So, str.upper function is the callback function. EV0000193 · EV0000194 · EV0000195
5. Source slide page 5 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Callback Function: Callback Example 2 (using user-defined function): Explanationdef multiply(x): compute(multiply,num_list) – the caller return num_list[0]*num_list[1] functionfunction withwith 22 argumentsarguments, def compute(func,x): 1) a function,multiply and 2) a list,num_list return func(x) Here, multiply is the callback function.num_list=[2,3] product=compute(multiply,num_list) print("Multiplication=",product) Output Multiplication= 6 EV0000196 · EV0000197 · EV0000198
6. Source slide page 6 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Callback Example 3 : Multiple Callback functions def function(func_list, x, y): print("Inside function") for func in func_list: func(x,y) Output InsideInside functionfunction def add(x,y): Sum = 15 z = x+y Quotient = 2.0 print('Sum =',z) def divide(x,y): z = x/y print('Quotient =',z) cb_list=[add, divide] function(cb_list, 10, 5) EV0000199 · EV0000200 · EV0000201
7. Source slide page 7 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Callback Advantages of Callback Functions • Calling function (outer function) can call the callback function as many times as required to complete the specified task. • CallingCalling functionfunction cancan passpass appropriateappropriate parametersparameters accordingaccording toto thethe tasktask toto thethe calledcalled functions. This allows information hiding. • Improves code modularity and reusability • Allows you to dynamically change the working of a function without changing its core implementation EV0000202 · EV0000203 · EV0000204
8. Source slide page 8 (extracted text): THANK YOU Departmentp of Computerp Science and Engineeringg g Dr. Shylaja S S, Director, CDSAML & CCBD, PESU Prof. Sindhu R Pai – sindhurpai@pes.edu Prof. Sowmya Shree EV0000205 · EV0000206 · EV0000207

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.420] Hello everyone. Welcome back to the lecture series on Python for computational problem solving. I am Nita Anne Jacob, Assistant Professor from the Department of Computer Science and Engineering. [00:00:13.156] In this video, we're going to continue our discussion with functions, and what we will explore is the concept of callback. Now, what exactly is a function callback? We know that you can pass any kind

of [00:00:30.084] arguments to functions as input. But when a function is passed to another function as an argument, which is in turn invoked inside the outer function to complete a specific kind of task, it is known as a function callback. [00:00:48.452] So let me demonstrate this with an example to help you understand it better. Right? So I'm just going to move on to uh the ideal interpreter. And [00:00:59.428] Show you how exactly it works and what the concept really is. So first what I need to do is type I'm going to create two different functions, say f1 and f2, and in the print [00:01:14.212] function I'm going to say um hello from f1 okay I'm gonna copy the same thing again I hope that's okay so instead of typing it again I'm just going to create another function f2 [00:01:32.644] And in the print statement, I'm going to say hello from F2. So these are two different functions. And if I were to invoke F1, okay, let's go over to the interpreter, type Python. [00:01:47.876] functions dot py you can see it says print hello from f one. If I want to change this to F two, it's going to say [00:01:58.308] Hello from F2, right? So we know how to create functions and how to invoke them individually. Now what I'm going to do is define another function and I'm just going to call it as interface, right? And the input to the interface is just one [00:02:16.032] parameter which I'm just going to name as Fn the leader and internally [00:02:23.044] and the body of the function, I'm going to just say fn and a function call. So what exactly is interface going to do? So when I call interface, it does expect a parameter, but as you can see [00:02:38.212] Whatever parameter or argument you pass here, it's going to be invoked as a function, which means the value of fn must be a function object. So what are the function objects that I have in this particular program? [00:02:54.288] The function objects I have are F1 and F2. So when I call interface of F1, what is the output I'm going to get? Let's look at the command prompt. [00:03:07.876] You can see that it prints hello from F1. And just by changing the argument to F2 in interface, if I were to run the code again, it's going to say hello from F2. So what happens here is [00:03:23.268] The functions F1 and F2 are called through the interface. Let me rephrase that. Fn is a function object that calls back the functions F1 or F2. [00:03:40.036] respectively based on the argument value that I have given. Right? So my function definition remains the same. I'm not going to change F1 or F2 here, but rather just pass the name of the function object. [00:03:54.116] Right? So what if I would pass anything that's not a function object? Like for example the number two. Let's look at the interpreter. I'm going to clear the screen so you can see the error clearly. It says int object is not callable, which means in callback, the parameter. [00:04:13.956] must receive a function object. Okay? So this is going to pan across the line. Let's just add this here. Okay? So [00:04:29.796] This is just the simple example of callback where you pass the parameter or the argument as a function object, and within the body of the function, I am going to invoke any function that is being passed as the parameter, right? [00:04:47.204] So what exactly is the advantage of using function callback? A calling function can call the callback function as many times required, and all you have to do is just call the callback function once. That is, for example,

EV0000014 · EV0000015 · EV0000016 · EV0000017

Transcript block 2 (source interval 000300-000599 seconds): [00:05:01.184] Within the interface function, I could have put a loop of say calling it four times, right? Similarly, the calling or alternatively, the calling function can pass appropriate parameters according to the task, and this allows information hiding. [00:05:18.116] So what exactly is information hiding? We're trying to just show the user what exactly the user needs to see. He doesn't need to know what are the intricacies of the application. [00:05:31.780] One such example is that of your Google Maps. Okay. So I'm going to write uh remove this entire code and emulate an example of how we would use

callback in Google Maps. Now all of you have used the uh [00:05:48.036] famous cab booking websites or applications like Uber and Ola. And let's say I want to just uh book a cab from some source to a destination. Okay yeah I was supposed to put a bracket there [00:06:07.236] Okay, and I'm just going to print booked an Uber from [00:06:16.260] Source? [00:06:20.132] Two [00:06:23.300] Sorry, to destination. Okay, so this is just the body of my function, right? For Uber. So every time I call Uber and just pass the source and destination, it's going to book. [00:06:38.820] just say printing booked in Uber. I'm going to repeat the same task for Ola and create another function, Ola. Right? So like you have on your phones, you can book a cab directly from Uber or directly from Ola, or [00:06:56.612] If you open your Google Maps application and type a particular destination, apart from just the directions and how much time it takes from point A to point B, [00:07:09.284] via car, bus, bike, walking, it also gives you an option to hire a cab. So what it does is internally it calls back the applications Uber and Ola, but all the [00:07:22.400] client does or the user does is just enter the source and destination and click a button. Why not exiting from Google and then going to Ola or Uber? Right? So this is a classic example of callback. So I'm going to define Google Maps G Maps as [00:07:40.356] an application or a function, right? So I need to specify the source and the destination, but I also need to specify the app through which I want to book the cab service. So since app is going to be [00:07:59.364] the name of my function objects Uber and Ola. When I invok app, it must get source and destination as the parameters as well. [00:08:11.492] Okay, so let's create a user interface where the user can enter which app he or she wants to use. So I'm just going to create um an input function where I will take the source, the destination, and the [00:08:30.596] app that I want to use as input. Okay, so input I'm gonna say enter source destination and app [00:08:46.180] Okay, so we're just gonna tell the user to input it and I'm going to call the split function so that when the user inputs the three values separated by a space, it's going to split it and store it in the values S T and App. So how exactly will I now call the [00:09:05.988] Google Maps function. So since it's going to all return strings, obviously when I pass a string to Google Maps, app will say that string is not a callable object. Right? So I'm just going to write a simple if statement. If the value in app is equal to Uber. [00:09:25.092] Okay, so uh I'm not taking care of case sensitivity or insensitivity. You can do that when you try this out. I if it's Uber, what I'm going to do is call G maps with source, destination, and app. [00:09:42.148] Right? But what will the app be? It's going to be a string, so I'm going to pass the function object Uber. Right? Next. I've taken care of Uber. I have to talk about Ola instead. Okay, I'm not taking care of any uh user.

EV0000018 · EV0000019 · EV0000020 · EV0000021

Transcript block 3 (source interval 000600-000899 seconds): [00:10:01.604] input errors like say else if and then again an else enter a correct app I'm just going to assume that my user is going to use the correct input right in [00:10:15.108] Ideal in an ideal world, this is an incorrect form of programming where I'm not taking care of the mistakes a user would make. But this is just to illustrate the concept of callbacks. So when you do write a program, make sure that you write else if. [00:10:35.108] It's not else if you have to give LF app equal to hola. Okay G maps and else we will write print [00:10:53.540] And valid application. [00:10:59.620] Right? So this completes our program. Now let's move on to the execution and see how this would work. So python functions.py enter source destination and app, let's put call mangler. [00:11:17.636] Space Electronic City. Space Ola. So it's going to say booked anola from Kor Mangla to Electronic City. Now I'm going to say bookanola from EC to Kor Mangla. [00:11:31.780] via Uber in uppercase letters, it's going to say

enter valid application because we've only taken care of Uber and Ola in terms of case sensitive strings. I hope you have understood the concept of callback. So here Google Maps [00:11:48.400] gets Uber the function object as a parameter. App will be replaced with Uber and you're saying Uber of Source and Destination. I hope you've understood this concept. Do try this entire program out on Python Tutor to understand it further. [00:12:06.532] Thank you everyone. I hope you will practice this and also try all the worksheets that are available on Pesso Academy. Thank you, everyone.

EV0000022 · EV0000023 · EV0000024 · EV0000025

Full transcript: 1672 normalized words, 49 timestamped segments, 738.837333 seconds duration. Low-confidence segments flagged: 0.

EV0000014 · EV0000015 · EV0000016 · EV0000017

4 Functional and object oriented programming

This lesson is indexed from the source video titled Functional and object oriented programming. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. No slide deck is assigned to this video; candidate slide sources remain separate in the unit master.

EV0000026 · EV0000027

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.932] Hello, welcome back a Python for computational problem solving. In today's class, we are going to be looking into the concept of functional programming. [00:00:17.252] Before uh I start uh um with actual functional programming, let us uh try to write very simple programs uh and then uh understand what those programs are doing. At the same time we will also un uh try to understand how uh [00:00:33.568] Python provides uh the different functions in order to work with the functional programming. Above all this, I really want to uh thank uh Professor Anis Kumar for providing us a beautiful examples in order to understand uh the concept of functional programming in an easy [00:00:51.216] So thank you so much, sir. Uh so with this acknowledgement, we will uh try to uh continue our discussion. [00:01:00.260] Okay, so let me uh start writing the uh program. I want you people to let me know what exactly uh these functions are trying to do. Okay, so let me define a function, define a function one. [00:01:17.636] Fine, and there goes a parameter here, and then C. [00:01:23.044] Define a list, an empty list in this case. So I'll say for I in X. What is this X? I will let you know. So do not worry at this particular point. I'll say result.appen [00:01:39.076] Hope you know the functionality of an append with respect to a list. I'll say l of i. And after doing a computation, return the result for me. Having said this, let me create [00:02:00.228] This a list C B E S University [00:02:09.284] Fine. Um our very own uh Python. [00:02:17.476] You can also write anything of your choice. Computer science. [00:02:26.180] Maybe more also, but I'll stop at this particular point. F1 and I will pass A as an argument here. I'll capture the result and then print it. [00:02:42.820] Okay. I hope by this time you should have known what I'm trying to do here. So let me execute this program. [00:02:50.852] And if you can see the result is fourteen, six and sixteen.

What exactly is this? If you can see count this three, four, five, six, seven, eight, nine, ten, [00:03:03.232] 10 plus 4, 14. So I'm trying to find the length of every element that is present in a list. Okay. So let me do one more uh simple example for you. Let me create a function. In this case, my function will be F2. There's got to be a parameter here also, which is x. [00:03:23.332] Fine. I will just copy the same thing. But what I'll do is I'll do some sort of changes. Okay. Hope uh this is okay with you all. Fine. And let me comment this so that you will not have any confusions here. [00:03:42.372] Okay. So give me time so that I will remove your confusions. Okay? Yeah, here we go. Now what I'll do is I'll just change this particular function instead of len of i, what I will replace is I will replace it with [00:04:00.516] STR dot upper. STR dot upper. Fine. So let me execute this. I'll have to change this because this is function two. And let me execute. [00:04:15.652] Fine. When I execute this program, if you can see that all the uh elements of a list have been converted to its upper case. [00:04:28.420] Hope this is clear. Now, in both the examples, in the example one, where uh we had the function f1, and in this case this is my function f2, there are few things that we need to observe. The observation goes like this. [00:04:46.980] The observation is some initialization has been done. [00:04:54.468] I have initialized certain things. Fine. And we are traversing through all the elements of an I trouble in this case. Is this fine? We have gone through all the elements of a list. List is an I tribble for you.

EV0000026 · EV0000027 · EV0000028 · EV0000029

Transcript block 2 (source interval 000300-000599 seconds): [00:05:10.596] Okay. So now after iterating what did I do? We are trying to apply some function to this. We have applied some function. Okay, some operation on every element that is present in our list. So the result is collected. [00:05:28.036] Okay, where is the result that is being collected? In this case, the result is collected again in terms of a list. And then followed by the list has been returned. [00:05:39.396] Okay. In this example, at the same time in the previous example also we have done this. Fine. So now we what let me repeat it again. We did some initialization. Okay? Some initialization have been done. I am walking through every element of an I trouble. [00:05:57.092] Okay, so I'll just say walk through. [00:06:03.172] I take every element from an I tribble and apply some operation. And result is collected again in terms of an I triple itself, but in this case it is a list and I will return the list. [00:06:18.596] You can see the previous example, aren't we doing the same thing here? The same thing we are trying to do here also. Okay, so now, so now what we are going to be doing next. [00:06:31.172] In this case, I will rewrite the same set of programs in a different manner. Just look at how I will write the program. Okay. So let me start here. I will uh use the same uh list. [00:06:48.356] Fine, I'll copy paste it here. Let me remove this. Hope this is not confusing to you. Just give me a time. I will uh remove this. And I will comment it till here. [00:07:06.660] Okay, rest entire rest other things have been commented out. I do not have anything else here. Okay, so now this is my list. This is my list. Okay, so now to this list I will try to do some things. I'll say A is equal to map. [00:07:26.948] Okay, A is equal to map. Map of what? Map of Len, comma, A. Hope this is clear to you. Hope this is fine. Okay? So now [00:07:46.372] Fine. Let me see print A. [00:07:52.644] It's just fine. What am I trying to do? I'm making use of a function. In this case, map is a built-in function. Please let me remind you, map is a built-in function. Now for this map, what did I do? For this particular map, I have [00:08:11.492] given an I trouble fine and I have called a particular function len in this case. So thus map is a built-in function. Map is a built-in function which takes a callable [00:08:31.076] Along with that, it also takes an eye trouble. [00:08:38.756] And hope I am not confusing you. So in this case if you can see I have a map. I have

a map. For this map [00:08:45.744] What did I do? An itable. So a list is an itable. So now I am passing a list here. And then I am calling a particular function. What is this function? This function is a callable function. I can call it as and when it is required. [00:09:02.256] Okay, so now what I will do is hope this is clear. Hope there are no more confusions. Okay. As I mentioned, map takes two arguments, a callable and an iterable. Okay. Now [00:09:14.400] What am I trying to do? What am I trying to do? Before me explaining, let me get the output first and then I will try to explain it. Okay? And if you can see, you're getting some map objects. Okay? You're getting a map object. [00:09:28.944] What I will do is I will just change something like this, I'll say list of this. [00:09:36.836] Is this fine? The reason is, the main reason is conceptually, whatever it is happening, in general what is happening, I take every element in the previous example, I'm taking every element from a list and apply this function. [00:09:52.708] Just fine. So in case of the map function also, I'm doing the same thing. So I will take every element that is there in the list and I will apply a particular function.

EV0000030 · EV0000031 · EV0000032 · EV0000033

Transcript block 3 (source interval 000600-000899 seconds): [00:10:03.408] That is what I'm trying to do. So now why am I not getting an output previously when I executed? The reason is map is lazy. Map is lazy. Let me write map is lazy. [00:10:19.236] Okay, until and unless you call it explicitly, it will not give you the required result. Because of which what did I do? I said list of this. Okay. So now let me uh execute this. [00:10:36.932] Right and if you can see you're still getting 14, 6 and 16. So the same thing you got when you executed a program by writing a define. [00:10:47.588] Okay, so now map function causes an iteration through an it tribble, applies the callable on each element of an itable. [00:10:59.844] If you could not understand that particular statement, don't worry, you have an item, element, on this particular element, okay, on this particular element, what am I doing? I will apply this particular function. That's it. Nothing else we are trying to do. [00:11:16.544] Okay, it's lazy, so you have to explicitly ask it to print it to you. Okay? And one thing you should always remember when you work with your map function, when you work with your map function, the number of input elements [00:11:31.872] will be equal to the number of output elements. That is the very important thing that you should always remember. Okay? Number of input elements will always be equal to the number of output elements. [00:11:48.080] Okay, suppose if I have to uh change the previous um uh examples program that we have written here. Now can you tell me what is the new what is the only one change that I'll have to do? The one change is instead of len, I will say string dot upper. And that's it. [00:12:08.836] Is this fine? So what does the map will do? You're saying take every element in an I tribble and apply a callable on it. That's what you are trying to do. So let me go back here. Hope our pro uh problem is saved. Program is saved, and if you can see they've been converted into capital letters. [00:12:28.708] Right. So hope uh the concept of map is clear to you. Work out on uh on the similar grounds so that uh uh you can uh understand the things in an easier manner. Hop you enjoyed the class. Thank you.

EV0000034 · EV0000035 · EV0000036 · EV0000037

Full transcript: 1759 normalized words, 53 timestamped segments, 767.68 seconds duration. Low-confidence segments flagged: 0.

EV0000026 · EV0000027 · EV0000028 · EV0000029

5 Lambda Function

This lesson is indexed from the source video titled Lambda Function. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000038 · EV0000039

Graphs and diagrams

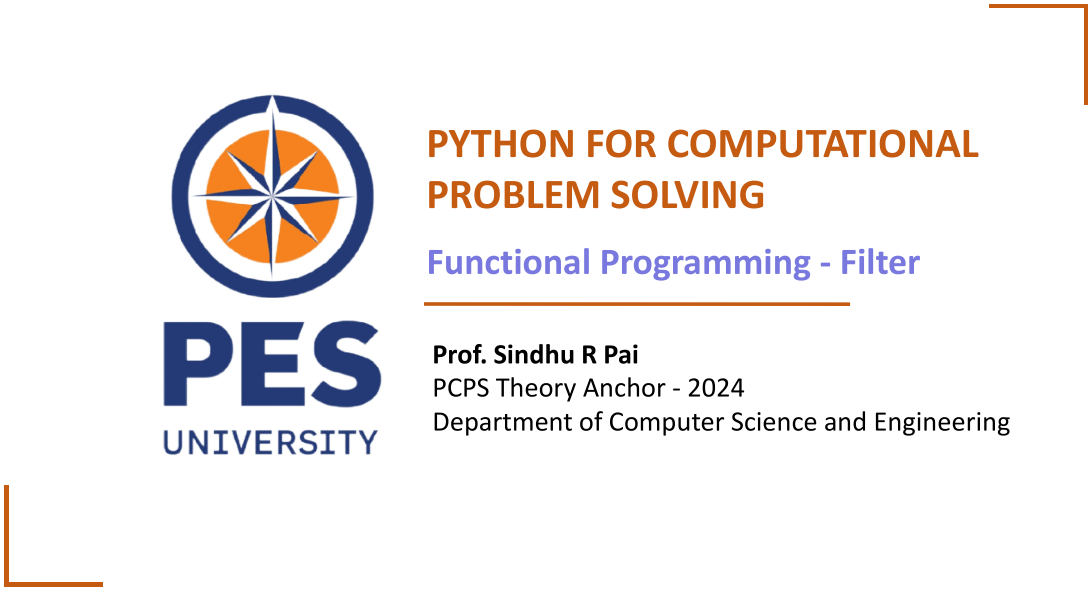

Figure G000067. UE24CS151A PCPS Unit 4 Sindhu Class 2 Functional programming filter · source slide page 1.

EV0000317

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Filter() Function

PES UNIVERSITY

•Filter function filters the given sequence with the help of a function that tests each element in the sequence to be true or not.

•There are cases where we want to remove a few elements of the input iterable. Then we make use of the filter function.

•Syntax: filter(*function*, *sequence/iterable*)
•***function:*** function that tests if each element of a sequence is true or not.

•***sequence:*** The sequence which needs to be filtered, it can be sets, lists, tuples, or containers of any iterators.

•***Returns:*** an iterator that is already filtered.

1

Figure G000068. UE24CS151A PCPS Unit 4 Sindhu Class 2 Functional programming filter · source slide page 2.

EV0000320

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Filter() Function

The filter function has the following characteristics.

- Input : an iterable of some number of elements (say n)
- Output: a lazy iterable of 0 to n elements (between 0 and n)
- Elements in the output: apply the callback on each element of the iterable – if the function returns true, then the input element is selected otherwise the input element is rejected.

2

Figure G000069. UE24CS151A PCPS Unit 4 Sindhu Class 2 Functional programming filter · source slide page 3.

EV0000323

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

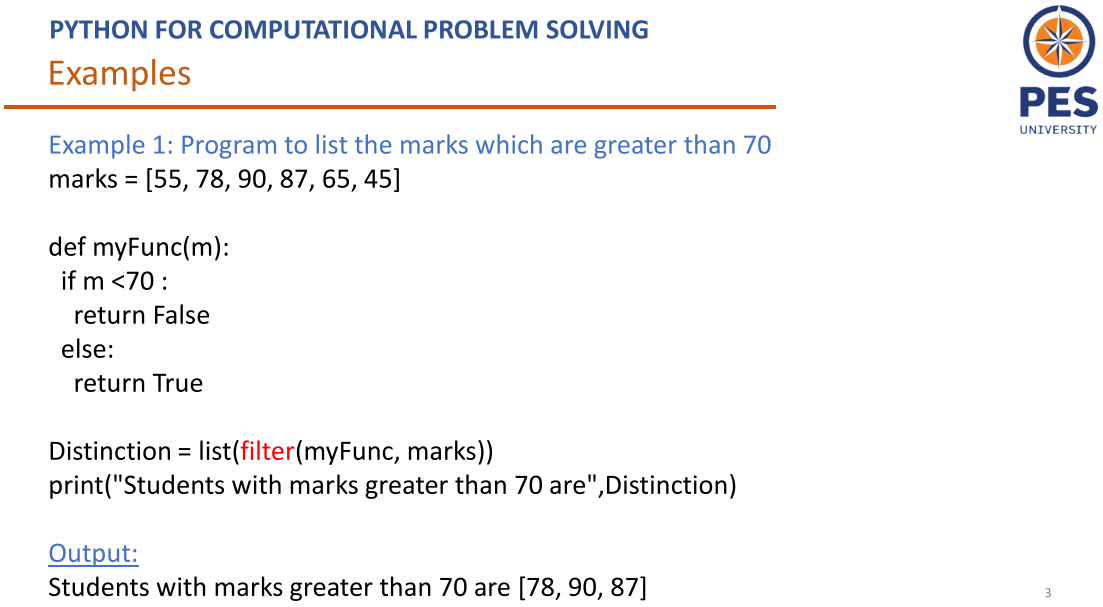

Figure G000070. UE24CS151A PCPS Unit 4 Sindhu Class 2 Functional programming filter · source slide page 4.

EV0000326

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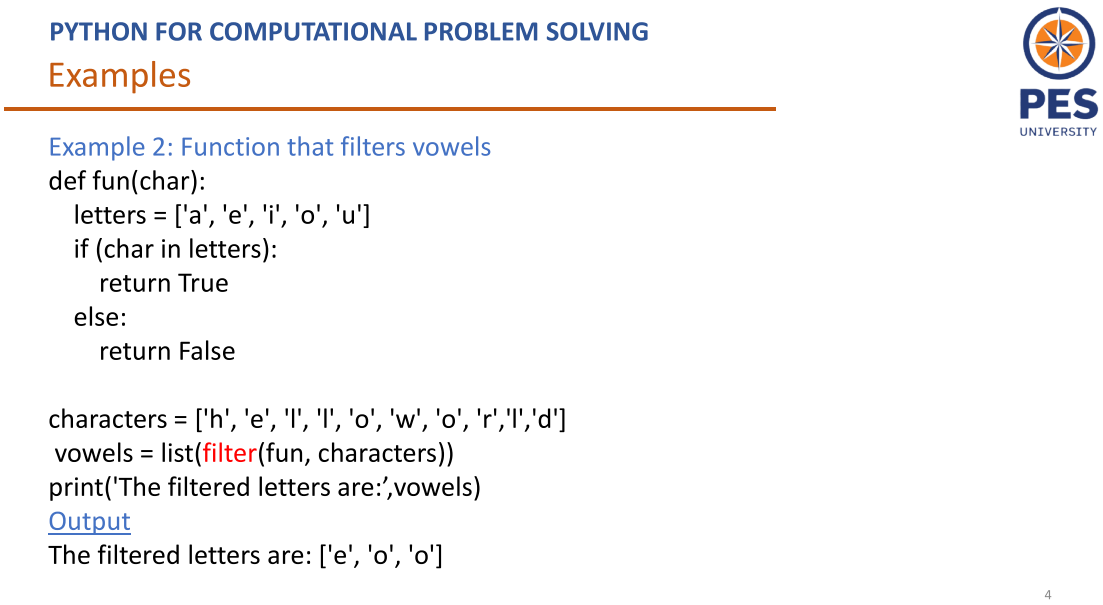

Figure G000071. UE24CS151A PCPS Unit 4 Sindhu Class 2 Functional programming filter · source slide page 5.

EV0000329

> ***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

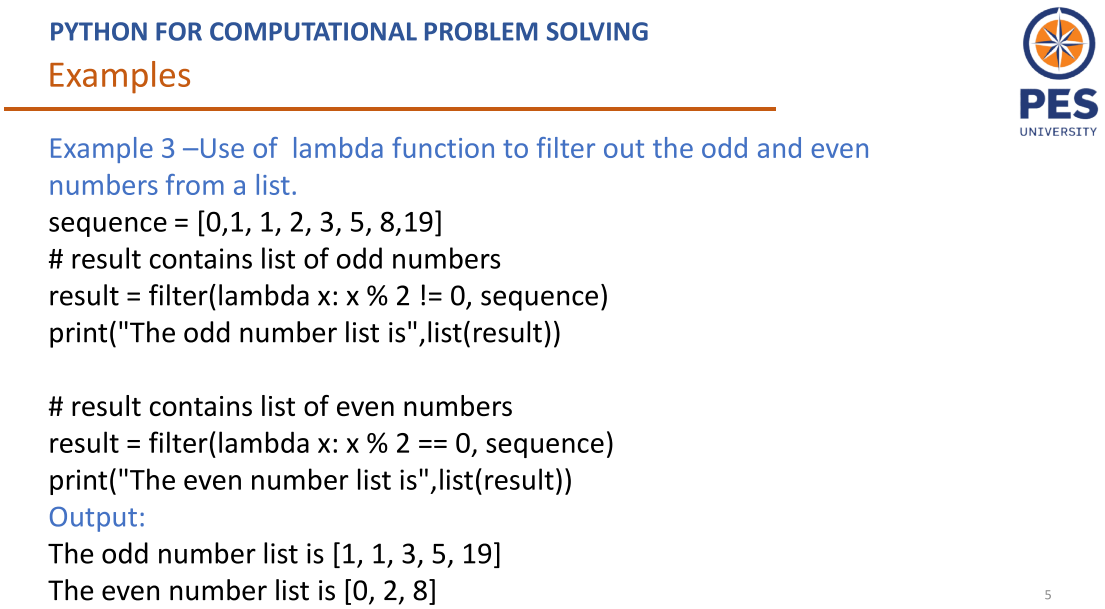

Figure G000072. UE24CS151A PCPS Unit 4 Sindhu Class 2 Functional programming filter · source slide page 6.

EV0000332

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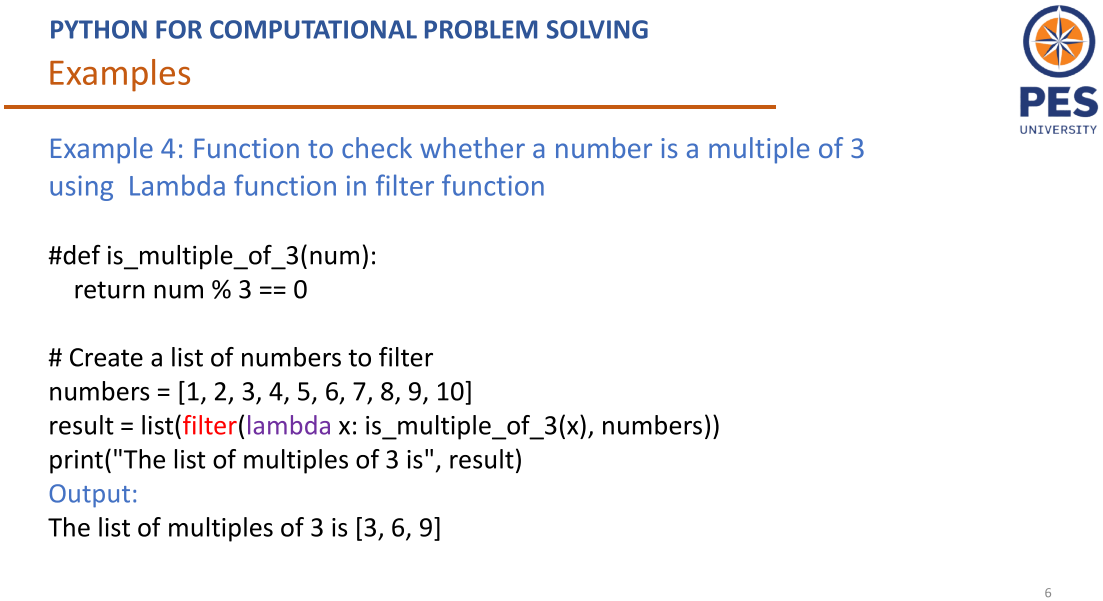

Figure G000073. UE24CS151A PCPS Unit 4 Sindhu Class 2 Functional programming filter · source slide page 7.

EV0000335

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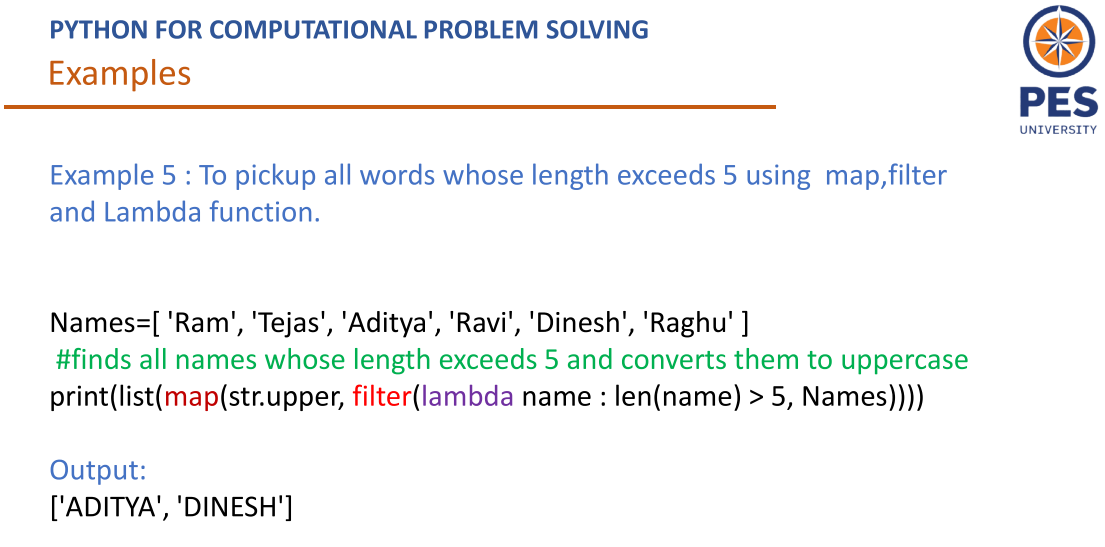

Figure G000074. UE24CS151A PCPS Unit 4 Sindhu Class 2 Functional programming filter · source slide page 8.

EV0000338

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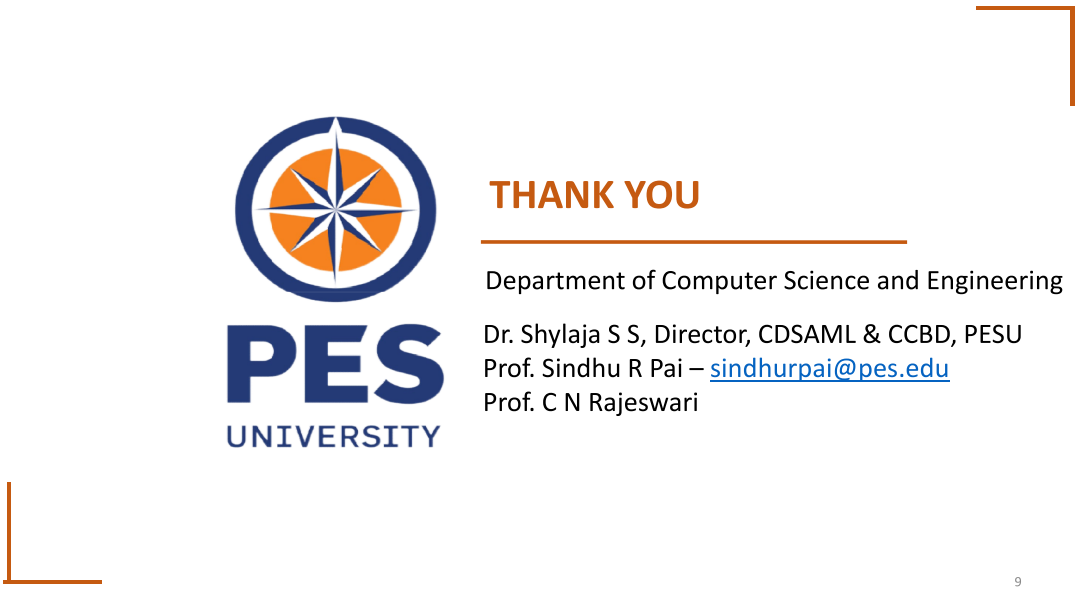

Figure G000075. UE24CS151A PCPS Unit 4 Sindhu Class 2 Functional programming filter · source slide page 9.

EV0000341

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functional Programming - Filter Prof. Sindhu R Pai PCPS Theory Anchor - 2024 Department of Computer Science and Engineering EV0000316 · EV0000317 · EV0000318
2. Source slide page 2 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Filter() Function •Filter function filters the given sequence with the help of a function that tests each element in the sequence to be true or not. •There are cases where we want to remove a few elements of the input iterable. ThenThen wewe makemake useuse ofof thethe filterfilter functionfunction. •Syntax: filter(function, sequence/iterable) •function: function that tests if each element of a sequence is true or not. •sequence: The sequence which needs to be filtered, it can be sets, lists, tuples, or containers of any iterators. 1•Returns: an iterator that is already filtered.
   EV0000319 · EV0000320 · EV0000321

3. Source slide page 3 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Filter() Function The filter function has the following characteristics. • Input : an iterable of some number of elements (say n) • Output: a lazy iterable of 0 to n elements (between 0 and n) • Elements in the output: apply the callback on each element of the iterable – if the function returns true, then the input element is selected otherwise the input element is rejected. 2 EV0000322 · EV0000323 · EV0000324

4. Source slide page 4 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples Example 1: Program to list the marks which are greater than 70 marks = [55, 78, 90, 87, 65, 45] def myFunc(m): if m <70 : return False else: return True Distinction = list(filter(myFunc, marks)) print("Students with marks greater than 70 are",Distinction) Output: Students with marks greater than 70 are [78, 90, 87] 3 EV0000325 · EV0000326 · EV0000327

5. Source slide page 5 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples Example 2: Function that filters vowels def fun(char): letters = ['a', 'e', 'i', 'o', 'u'] if (char in letters): return True else: return False characters = ['h', 'e', 'l', 'l', 'o', 'w', 'o', 'r','l','d'] vowels = list(filter(fun, characters)) print('The filtered letters are:',vowels) Output The filtered letters are: ['e', 'o', 'o'] 4 EV0000328 · EV0000329 · EV0000330

6. Source slide page 6 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples Example 3 –Use of lambda function to filter out the odd and even numbers from a list. sequence = [0,1, 1, 2, 3, 5, 8,19] # result contains list of odd numbers resultresult = filter(lambdafilter(lambda x:x: xx %% 2 !=! 0,0, sequence)sequence) print("The odd number list is",list(result)) # result contains list of even numbers result = filter(lambda x: x % 2 == 0, sequence) print("The even number list is",list(result)) Output: The odd number list is [1, 1, 3, 5, 19] 5The even number list is [0, 2, 8] EV0000331 · EV0000332 · EV0000333

7. Source slide page 7 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples Example 4: Function to check whether a number is a multiple of 3 using Lambda function in filter function #def is_multiple_of_3(num): return num % 3 == 0 # Create a list of numbers to filter numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10] result = list(filter(lambda x: is_multiple_of_3(x), numbers)) print("The list of multiples of 3 is", result) Output: The list of multiples of 3 is [3, 6, 9] 6 EV0000334 · EV0000335 · EV0000336

8. Source slide page 8 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples Example 5 : To pickup all words whose length exceeds 5 using map,filter and Lambda function. Names=['Ram', 'Tejas', 'Aditya', 'Ravi', 'Dinesh', 'Raghu'] #finds all names whose length exceeds 5 and converts them to uppercase print(list(map(str.upper, filter(lambda name : len(name) > 5, Names)))) Output: ['ADITYA', 'DINESH'] EV0000337 · EV0000338 · EV0000339

9. Source slide page 9 (extracted text): THANK YOU Departmentp of Computerp Science and Engineeringg g Dr. Shylaja S S, Director, CDSAML & CCBD, PESU Prof. Sindhu R Pai – sindhurpai@pes.edu Prof. C N Rajeswari 9 EV0000340 · EV0000341 · EV0000342

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.796] Hello all, welcome back a Python for computational problem solving. In today's uh class, we are going to be looking into the concept of a lambda. [00:00:17.636] Let's try to understand what exactly is a lambda, then we will uh also learn the usage of lambda. Lambda is basically a function without any name. Okay? There

will not be any name for the functions. So a function without name. [00:00:38.884] function without name. They're also called as they're famously called as anonymous functions. They're called as anonymous functions. [00:00:54.820] Anonymous functions. They are also called as throw away functions. [00:01:11.556] Lambda is not a statement, but it's an expression. So let me say it's an expression. [00:01:23.364] An expression. [00:01:27.012] And one important thing that you should always uh know with respect to lambda is lambda's body does not contain a block of statement. You will not have a block of statement. Instead, uh it's going to be a single expression. So let me say that it's a single expression. [00:01:46.596] It's a single expression. [00:01:54.788] I will basically use a lambda function when I do not want to use a function again and again. For example, in the previous sessions you have already seen how to create a function, and we have also given a name for a function, which means that I can reuse a function as and when it [00:02:12.560] Is required. But in this particular case, if I make use of lambda, which means that I do not want to use this function again and again. Fine. And I will also make use of a lambda function whenever I need a function object. [00:02:30.884] Okay, so use when a function object is required. [00:02:44.068] And a function object is required. Fine. Um the today when I uh use a lambda function, you may not appreciate uh the usage of the lambda functions. [00:02:55.396] Okay. So the uh these functions are uh very useful whenever uh uh you work with the map reduce and filter, which you're going to be looking in the other topic. So one important thing, use when [00:03:15.300] You want to use it for a shorter duration. [00:03:24.580] So we will now start and try to understand how do I uh write a lambda function. So the syntax goes like this. [00:03:36.644] The syntax for the lambda function is you will have lambda, it's a keyword. Lambda is a keyword which is used to define or declare an anonymous function. So lambda is a keyword, followed by argument. [00:03:55.076] And I'll say either it you can have a single argument or you can have a number of arguments. [00:04:01.636] This is fine. Okay, if you have more than one argument, then what do you need to do? They all needs to be separated by a comma. Followed by a colon. A colon is a must. And then it's an expression. You will have an expression. [00:04:19.364] Fine. So now the expression gets evaluated. Fine. And the results are returned back. [00:04:27.684] So I will add one more statement. Whenever you're working with a lambda function, you should never return a value. Do not return anything because it has an implicit written. [00:04:45.284] Okay, so now the syntax is lambda, a keyword, you can have n number of arguments or n single argument followed by a single expression. If you cannot write a function using a lambda, then the regular way of writing you should always make use.

EV0000038 · EV0000039 · EV0000040 · EV0000041

Transcript block 2 (source interval 000300-000599 seconds): [00:05:02.432] Right. If your intention is to work with an inline function, [00:05:06.372] Fine, a single line of code, then always better to use a lambda function. Okay, so let's quickly uh understand and try to write a lambda function. So I'll just define a function. Okay, now this is not a lambda function, this is a regular function. [00:05:23.424] X, Y, and Z is what I will try to write. And then I will say return. [00:05:33.796] X plus Y plus Z. A simple one. Okay. X plus Y plus Z. This is what I'm trying to do. Okay, as I have a written, let me capture this in some variable. I will call the function. Fine. Two, three, and four. [00:05:52.132] I'll say print key. [00:05:56.900] So this is what I will try to do. So let me execute this. I know it is uh the same program that you have already done. Okay, but still see you can see that uh the final answer is nine in this case. What we will try to do is we will try to convert this function into a lambda function. [00:06:15.680] Okay, so this is a regular function. Let me convert

this into a lambda function. So as you all know, I should make use of what? A lambda as a keyword. So I will say lambda five. You can have a single [00:06:32.420] Argument. If you have a single argument, you will specify it like this, then followed by a colon. [00:06:39.408] And then followed by your expression. But in this case, if you can see that there are how many number of arguments are there? There are basically three arguments are there. So I will mention all the three arguments, and if you have more than one argument, it should be separated by a comma. [00:06:57.220] Fine. So hope this is clear to you. And what are we trying to do? We are trying to perform addition in this case. So I'll say X plus Y plus Z. [00:07:08.964] This is what I will try to do. Okay. Now how do I call this function? So in order to call this function, I need to have an object. So I will say function object F. Okay, so now I'll say print F of [00:07:26.692] Two, three, comma four. This is how you try to write a lambda function. Okay. F is basically getting a function object. Okay. This function object is when a lambda expression creates it. [00:07:47.396] Define also will work in a similar fashion, okay? DEF when we use it, it works in a similar fashion. But our assignment is going to happen automatically. In this case, the function object is available only when a lambda expression is created. [00:08:04.932] Okay so let me execute this particular program and we'll see. Let me go here execute both of them. So I should get nine and nine here and if you can see we've got the answer which is nine and nine. [00:08:23.876] Hope the concept of uh lambda is clear to you. And uh maybe we'll take up one more uh simple example and then we'll try to write. Okay. I'll write a program and I want you people to find out what I'm trying to do. So I'll say lambda. [00:08:41.956] And an argument is gonna be a single argument. I'll say A star A star A. So what am I doing? I'm trying to find a cube of a particular number. Fine? That's it. Nothing else I will have to do. So I'll say print. [00:09:01.124] X of five. Okay. And you may also specify as it is a print statement Q value is Okay, so this is what something I will do. [00:09:16.420] Okay. Function object X. X is a function object in this case. So let me uh write a program. Sorry, let me execute this program. [00:09:26.820] And C nine nine over the previous uh programs output because it's not been commented, and then followed by uh the Q value of five is what we were trying to find out. [00:09:40.324] Okay, so that's all about lambda. I request you to uh solve some of the other examples in order to understand the concept of lambda in a better way. Hope you enjoyed. Thank you.

EV0000042 · EV0000043 · EV0000044 · EV0000045

Full transcript: 1303 normalized words, 41 timestamped segments, 596.309333 seconds duration. Low-confidence segments flagged: 0.

EV0000038 · EV0000039 · EV0000040 · EV0000041

6 List Comprehension

This lesson is indexed from the source video titled List Comprehension. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000046 · EV0000047

Graphs and diagrams

Figure G000083. UE24CS151A PCPS Unit 4 SindhuRPai class 6 list comprehension · source slide page 1.

EV0000365

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

List Comprehension

Introduction

- List comprehension is a concise way of defining and creating a list.
- It is used to create functionality within a single line of code.
- Return value is always a new list obtained by evaluating the expression in the context of for and if clauses which follows it.
- List comprehension is faster in processing than a list using for loop.

2

Figure G000084. UE24CS151A PCPS Unit 4 SindhuRPai class 6 list comprehension · source slide page 2.

EV0000368

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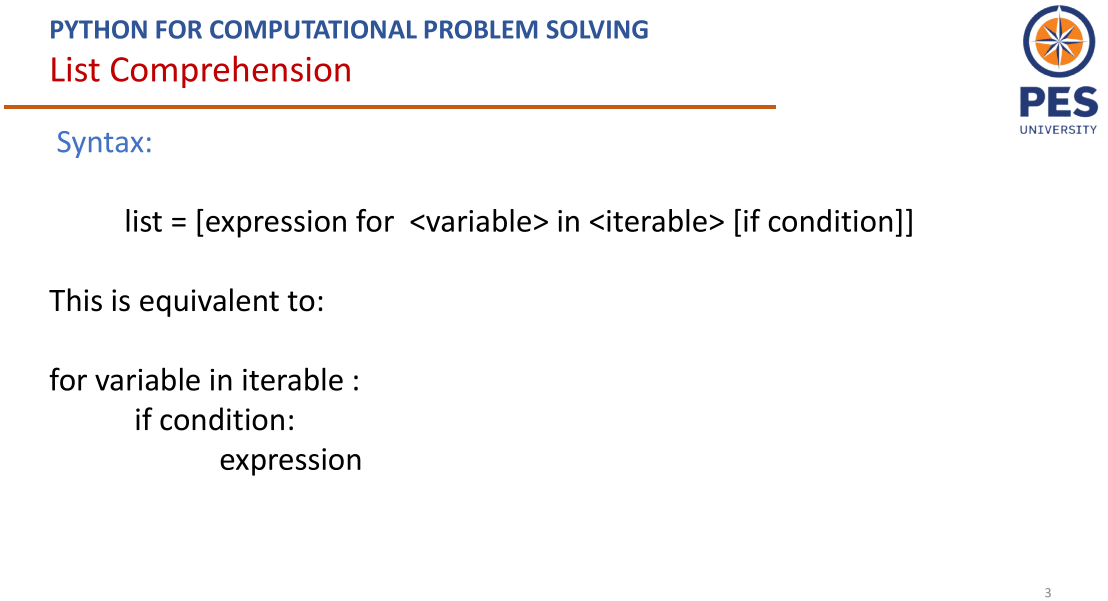

Figure G000085. UE24CS151A PCPS Unit 4 SindhuRPai class 6 list comprehension · source slide page 3.

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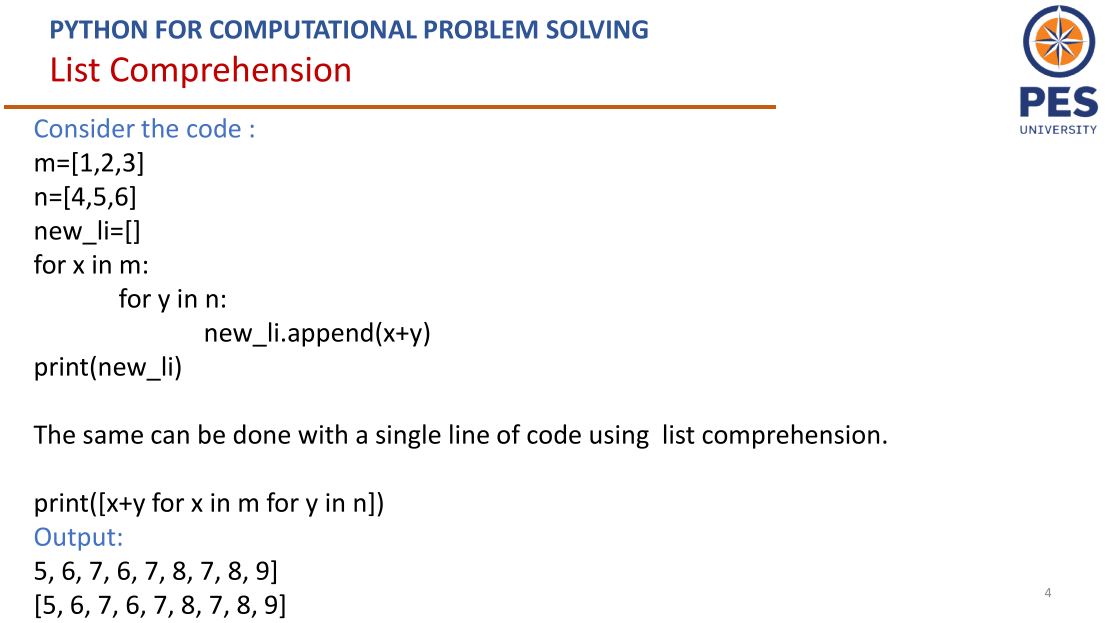

Figure G000086. UE24CS151A PCPS Unit 4 SindhuRPai class 6 list comprehension · source slide page 4.

EV0000374

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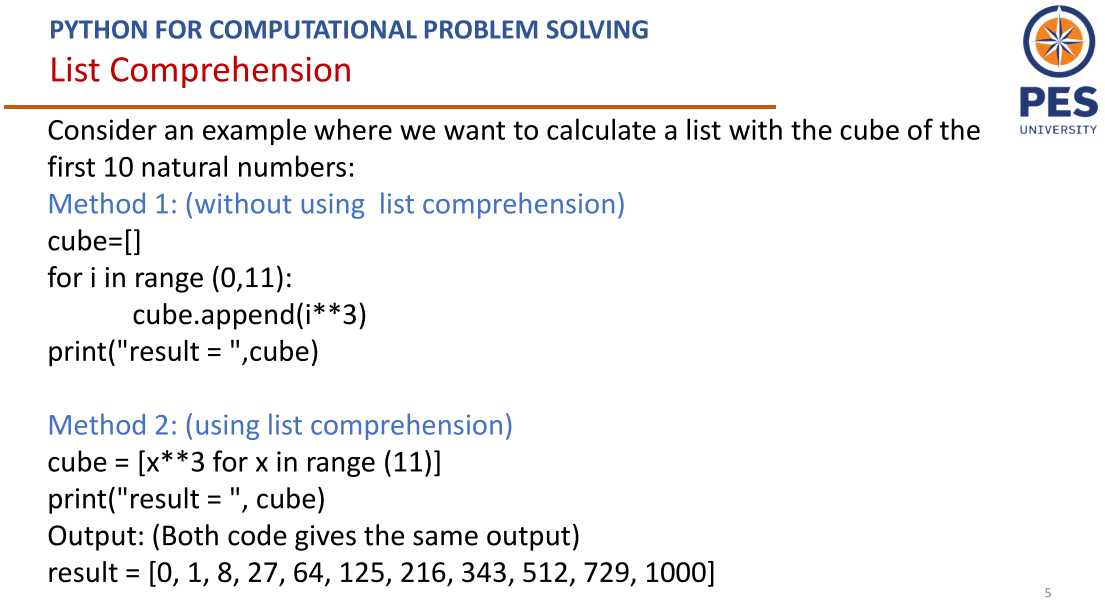

Figure G000087. UE24CS151A PCPS Unit 4 SindhuRPai class 6 list comprehension · source slide page 5.

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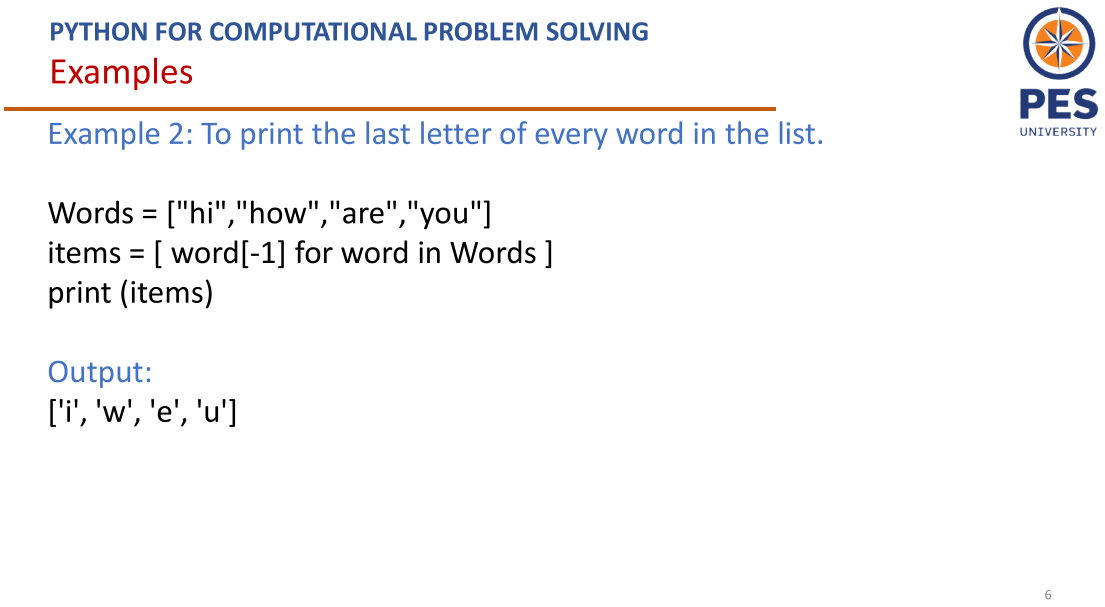

Figure G000088. UE24CS151A PCPS Unit 4 SindhuRPai class 6 list comprehension · source slide page 6.

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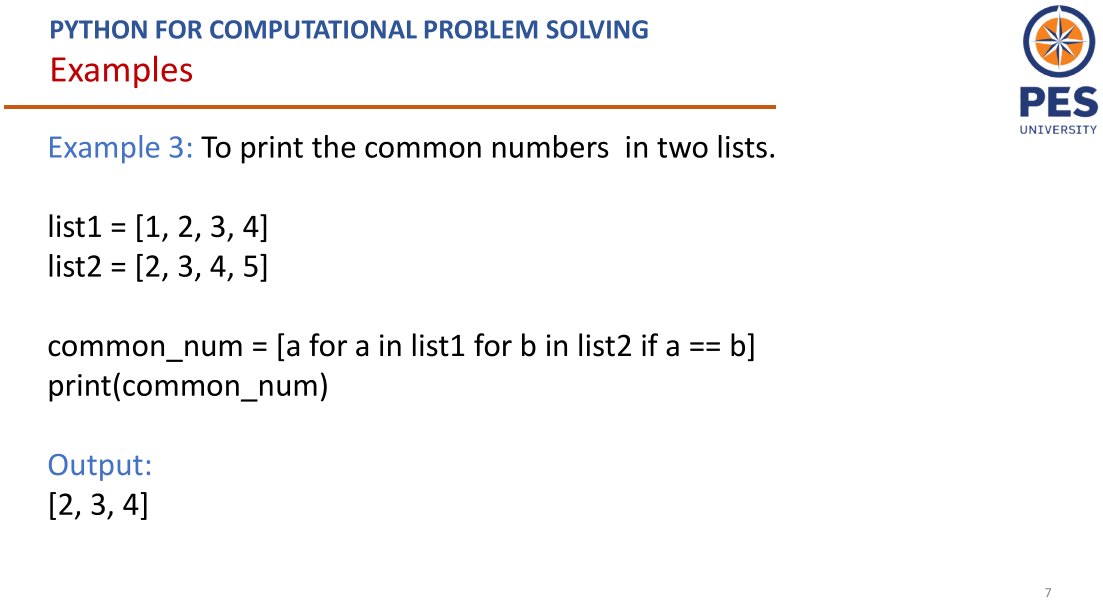

Figure G000089. UE24CS151A PCPS Unit 4 SindhuRPai class 6 list comprehension · source slide page 7.

EV0000383

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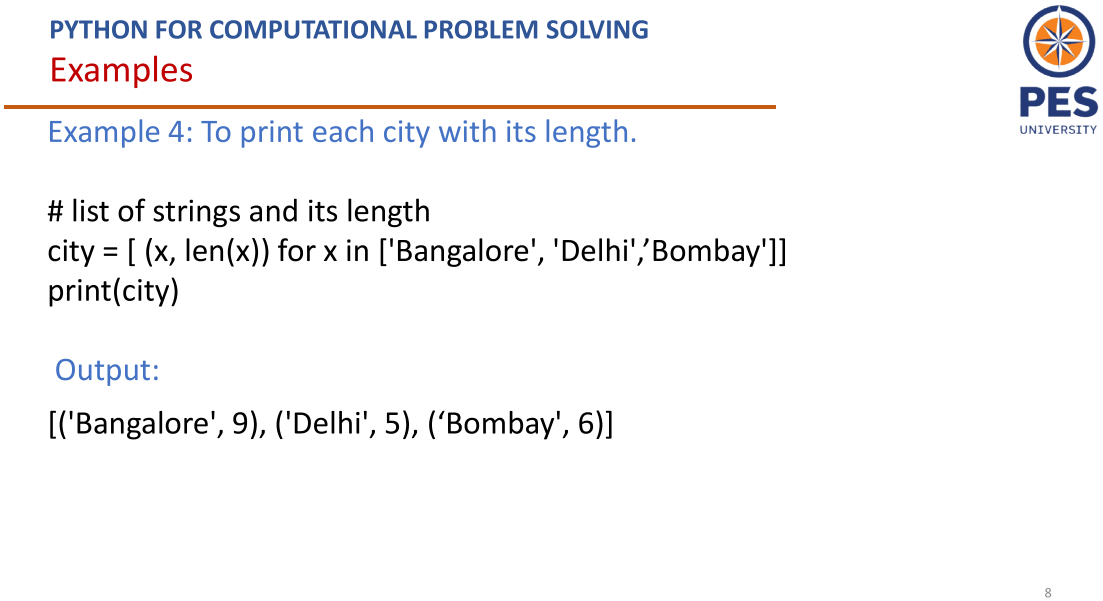

Figure G000090. UE24CS151A PCPS Unit 4 SindhuRPai class 6 list comprehension · source slide page 8.

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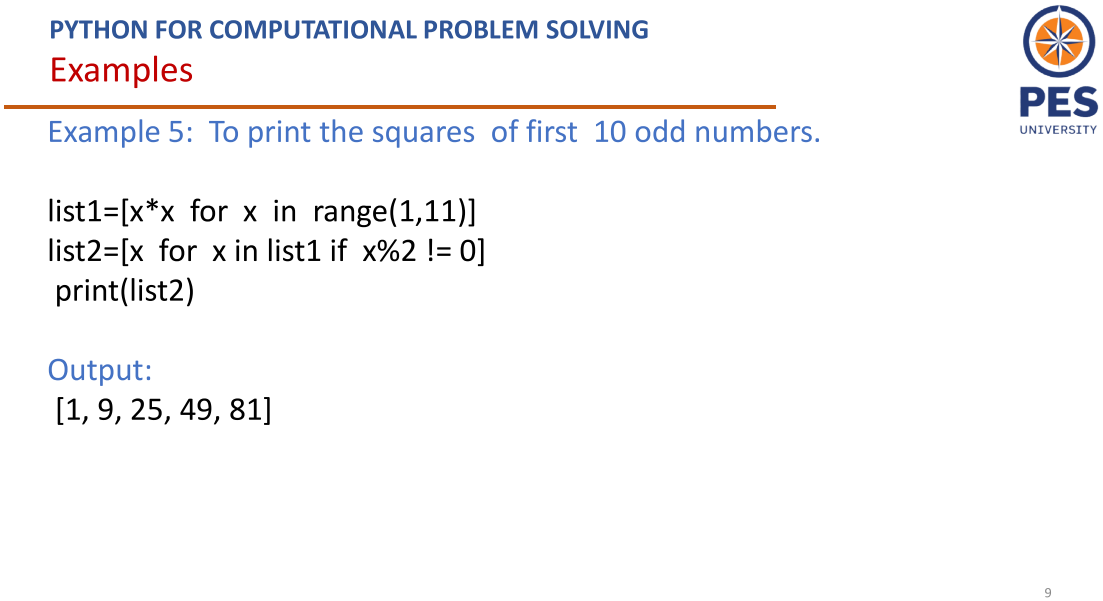

Figure G000091. UE24CS151A PCPS Unit 4 SindhuRPai class 6 list comprehension · source slide page 9.

EV0000389

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

List comprehension –Practice programs

PES UNIVERSITY

1. Using list comprehension find the transpose of a given matrix.

```
matrix = [[1, 2], [3,4], [5,6], [7,8]]
transpose = [[row[i] for row in matrix] for i in range(2)]
print (transpose)
```

Output:

```
[[1, 3, 5, 7], [2, 4, 6, 8]]
```

10

Figure G000092. UE24CS151A PCPS Unit 4 SindhuRPai class 6 list comprehension · source slide page 10.

EV0000392

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

List comprehension –Practice programs

PES UNIVERSITY

2. Using list comprehension print all the numbers which are divisible by both 2 and 5 in the range of 0 to 100

```
num_list = [y for y in range(100) if y % 2 == 0 if y % 5 == 0]   #Use of nested if
print(num_list)
```

Output:

```
[0, 10, 20, 30, 40, 50, 60, 70, 80, 90]
```

11

Figure G000093. UE24CS151A PCPS Unit 4 SindhuRPai class 6 list comprehension · source slide page 11.

EV0000395

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

List comprehension –Practice programs

PES UNIVERSITY

3.Using list comprehension, find all of the numbers that have a 6 in them.

```
nums=input("Enter the number")
List_with_6 = [num for num in nums if  "6"  in  str(num)]
print(List_with_6)
```

Output:

```
Enter the number76548
['6']
```

12

Figure G000094. UE24CS151A PCPS Unit 4 SindhuRPai class 6 list comprehension · source slide page 12.

EV0000398

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

List comprehension –Practice programs

PES UNIVERSITY

4.Using list comprehension count the number of spaces in a string

```
string=input("Enter the string:  ")
len_spaces = len([char for char in string if char == " "])
print("The length of the spaces is::",len_spaces)
```

Output:

```
Enter the string:  This is a test string
The length of the spaces is:: 4
```

13

Figure G000095. UE24CS151A PCPS Unit 4 SindhuRPai class 6 list comprehension · source slide page 13.

EV0000401

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

List comprehension –Practice programs

PES UNIVERSITY

5.Using list comprehension, remove all of the vowels present in the input string.

```
string=input("Enter the string::")
str_no_vowel = "".join([char for char in string.lower() if char not in ["a","e","i","o","u"]])
print("The  string without vowel::",str_no_vowel)
```

Output:

```
Enter the string:: All welcome
The  string without vowel::  ll wlcm
```

14

Figure G000096. UE24CS151A PCPS Unit 4 SindhuRPai class 6 list comprehension · source slide page 14.

EV0000404

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

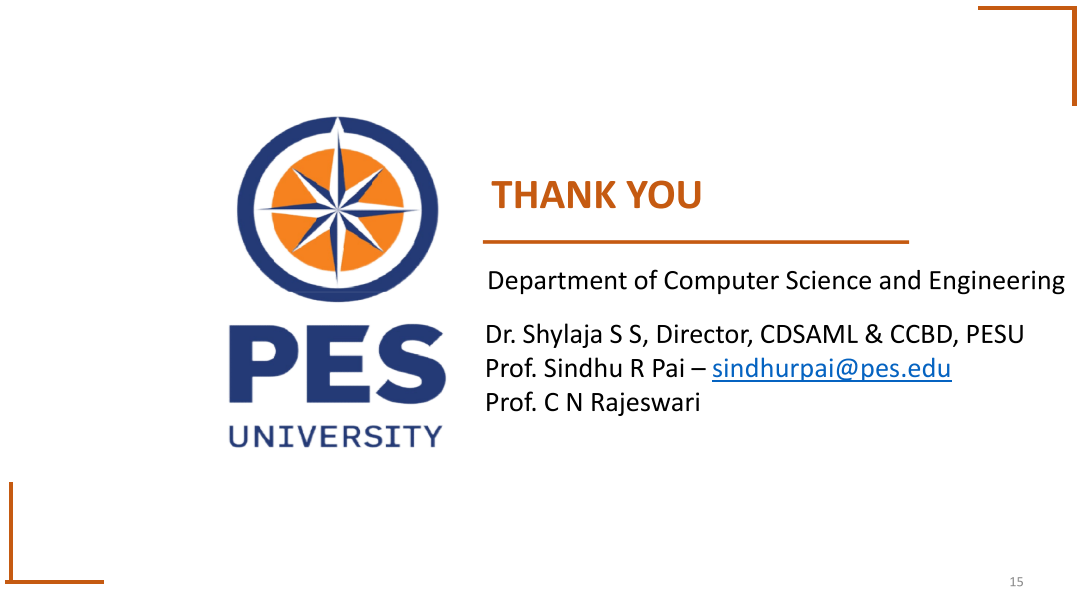

Figure G000097. UE24CS151A PCPS Unit 4 SindhuRPai class 6 list comprehension · source slide page 15.

EV0000407

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING List Comprehension Prof. Sindhu R Pai PCPS Theory Anchor - 2024 Department of Computer Science and Engineering EV0000364 · EV0000365 · EV0000366
2. Source slide page 2 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING List Comprehension Introduction • List comprehension is a concise way of defining and creating a list. •• ItIt isis usedused toto createcreate functionalityfunctionality withinwithin aa singlesingle lineline ofof codecode. • Return value is always a new list obtained by evaluating the expression in the context of for and if clauses which follows it. • List comprehension is faster in processing than a list using for loop. 2 EV0000367 · EV0000368 · EV0000369
3. Source slide page 3 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING List Comprehension Syntax: list = [expression for <variable> in <iterable> [if condition]] ThisThis isis equivalentequivalent to:to: for variable in iterable : if condition: expression 3 EV0000370 · EV0000371 · EV0000372

4. Source slide page 4 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING List Comprehension Consider the code : m=[1,2,3] n=[4,5,6] new_li=[] for x in m: forfor yy inin n:n: new_li.append(x+y) print(new_li) The same can be done with a single line of code using list comprehension. print([x+y for x in m for y in n]) Output: 5, 6, 7, 6, 7, 8, 7, 8, 9] [5, 6, 7, 6, 7, 8, 7, 8, 9] 4 EV0000373 · EV0000374 · EV0000375
5. Source slide page 5 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING List Comprehension Consider an example where we want to calculate a list with the cube of the first 10 natural numbers: Method 1: (without using list comprehension) cube=[] for i in range (0,11): cube.append(i**3)b d(**) print("result = ",cube) Method 2: (using list comprehension) cube = [x**3 for x in range (11)] print("result = ", cube) Output: (Both code gives the same output) result = [0, 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000] 5 EV0000376 · EV0000377 · EV0000378
6. Source slide page 6 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples Example 2: To print the last letter of every word in the list. Words = ["hi","how","are","you"] items = [word[-1] for word in Words] printp (items)( ) Output: ['i', 'w', 'e', 'u'] 6 EV0000379 · EV0000380 · EV0000381
7. Source slide page 7 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples Example 3: To print the common numbers in two lists. list1 = [1, 2, 3, 4] list2 = [2, 3, 4, 5] common_num = [a for a in list1 for b in list2 if a == b] print(common_num) Output: [2, 3, 4] 7 EV0000382 · EV0000383 · EV0000384
8. Source slide page 8 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples Example 4: To print each city with its length. # list of strings and its length city = [(x, len(x)) for x in ['Bangalore', 'Delhi','Bombay']] print(city)p (y) Output: [('Bangalore', 9), ('Delhi', 5), (‘Bombay', 6)] 8 EV0000385 · EV0000386 · EV0000387
9. Source slide page 9 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples Example 5: To print the squares of first 10 odd numbers. list1=[x*x for x in range(1,11)] list2=[x for x in list1 if x%2 != 0] print(list2)p ( ) Output: [1, 9, 25, 49, 81] 9 EV0000388 · EV0000389 · EV0000390
10. Source slide page 10 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING List comprehension –Practice programs 1. Using list comprehension find the transpose of a given matrix. matrix = [[1, 2], [3,4], [5,6], [7,8]] transpose = [[row[i] for row in matrix] for i in range(2)] printprint (transpose)(transpose) Output: [[1, 3, 5, 7], [2, 4, 6, 8]] 10 EV0000391 · EV0000392 · EV0000393
11. Source slide page 11 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING List comprehension –Practice programs 2. Using list comprehension print all the numbers which are divisible by both 2 and 5 in the range of 0 to 100 num_list = [y for y in range(100) if y % 2 == 0 if y % 5 == 0] #Use of nestedd iff print(num_list) Output: [0, 10, 20, 30, 40, 50, 60, 70, 80, 90] 11 EV0000394 · EV0000395 · EV0000396
12. Source slide page 12 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING List comprehension –Practice programs 3.Using list comprehension, find all of the numbers that have a 6 in them. nums=input("Enter the number") ListList_with_6with 6 == [num[num forfor numnum inin numsnums ifif "6"6 inin str(num)]str(num)] print(List_with_6) Output: Enter the number76548 ['6'] 12 EV0000397 · EV0000398 · EV0000399

13. Source slide page 13 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING List comprehension –Practice programs 4.Using list comprehension count the number of spaces in a string string=input("Enter the string: ") len_spaces = len([char for char in string if char == " "]) print("Theprint(The lengthlength ofof thethe spacesspaces is::"is:: ,len_spaces)len spaces) Output: Enter the string: This is a test string The length of the spaces is:: 4 13 EV0000400 · EV0000401 · EV0000402
14. Source slide page 14 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING List comprehension –Practice programs 5.Using list comprehension, remove all of the vowels present in the input string. string=input("Enter the string::") str_ no _ vowel = "".join([charj ([for char in string.lower()g () if char not in ["a","e","i","o","u"]]) print("The string without vowel::",str_no_vowel) Output: Enter the string:: All welcome The string without vowel:: ll wlcm 14 EV0000403 · EV0000404 · EV0000405
15. Source slide page 15 (extracted text): THANK YOU Departmentp of Computerp Science and Engineeringg g Dr. Shylaja S S, Director, CDSAML & CCBD, PESU Prof. Sindhu R Pai – sindhurpai@pes.edu Prof. C N Rajeswari 15 EV0000406 · EV0000407 · EV0000408

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.348] Hello, welcome back a Python for computational problem solving. Till now, uh we have understood how to create a list. So just an example to show it to you, this is what uh we would have done. [00:00:19.364] Fine. A list can contain integers, it can be a float point, floating values, it can be strings, fine. In general, a list can be heterogeneous in nature. And this way of creating a list we call it as enumeration. [00:00:37.604] And in today's class we will uh try to understand how to create list by using some rules. [00:00:46.820] So we are going to be learning to create a list using rules. Okay. So if you create list using rules, then it is called as list comprehension. Hope this is clear to you. Creating a list using rules is called as list comprehension. [00:01:12.932] List comprehension. Right? So uh we'll try to look into a few examples uh to understand how exactly uh list comprehension can be done. There is also a dictionary comprehension, set comprehension, but they are all uh [00:01:28.032] uh out of the scope for us so we are going to be learning only the concept of list comprehension. So let me uh write a simple code for you. So I'll say for I in uh range five okay and I'll print [00:01:47.524] Print Python. [00:01:51.588] So as you know, uh it's a follow and uh how many number of times Python gets printed here uh it it gets printed five times. Okay. So let's uh look into the output. Okay, and uh sorry. I in range. [00:02:11.140] For I in range. Uh while speaking I said it as five I in while I in range, but uh while typing I missed it. Okay. So if you can see uh it is printed for five times. Now [00:02:25.264] Can we really convert this into list comprehension? Can I write this using a rule? Yes, certainly. So I'll try to do list one. This is equal to [00:02:40.004] Because it's a list comprehension, I have to use a brackets here. Right. Let me write for I in range. [00:02:50.820] Oh five. Okay, so a for loop constraint has been a written here. Now what is the next thing that I'll have to do? I'll have to print the values. So in general, the print I will try to do it here. So what is [00:03:05.252] Need what needs to be printed? Python. So I'll say it as B Y T H O N here. Okay. So let's execute this. It would have returned an object and I'll tell uh print this. Okay. So let's execute this first and then uh we'll continue. [00:03:25.028] Okay. So I'll say L C dot P Y and Python has been printed. The Python is printed for this. Let me comment on this. [00:03:36.548] So now let me execute and no output because it has returned an object for us. We'll try to say print L1. So let's execute this again. [00:03:53.380] Fine, and if you

can see it has printed five times and within a list. Okay. So this way of writing we call it as list comprehension. Fine. So the general form of list comprehension is [00:04:10.180] And the general form. You have to write like this. So let me say it as general form. General form is this. Okay, it is less comprehension. So as usual I'll write for. As usual, because it's a for loop variable. [00:04:32.068] In eye trouble. [00:04:36.932] for I variable in I triple. Okay so followed by one main thing that I'll have to write is in this case you wanted to print Python. Okay so in general this is called as what? This is called as an expression. [00:04:55.204] So uh the general form of list comprehension is expression for variable in I triple. Okay? So in uh semantically this is equivalent to me telling that create an empty list.

EV0000046 · EV0000047 · EV0000048 · EV0000049

Transcript block 2 (source interval 000300-000599 seconds): [00:05:12.676] Execute the for a loop, fine, which is present as a part of uh list comprehension, execute this. Then every time you execute for loop, evaluate this expression. [00:05:28.740] Okay, and then append it to the list that you have specified. So let me uh explain it to you. [00:05:36.832] So you have a list in this particular case and you have a for loop. Execute for loops. Every time you execute a for loop, you will have to evaluate this particular expression. Okay? And thus the r uh resultant is a list that is getting created for us. [00:05:54.180] Fine. So evalu sorry, execute for loop, evaluate the expression and append the result into the particular list. Okay, what is this list? This is an empty list, L1, that it's getting created at the beginning. Okay, so hope uh this is clear to you. [00:06:14.084] Fine. So shall we just quickly look into few examples for us to understand uh the list comprehension in a better way. Okay so now I want to uh perform a square of a numbers. Fine. I want to do square of a numbers. [00:06:34.564] Square of numbers from [00:06:40.164] Um maybe one to ten? Okay. So one to ten. In the sense uh uh one to nine I want to do, do not consider ten. Or if you want to consider ten, you do it like this. Okay. So this is a program that I want to do. What do I need to do? [00:06:55.312] Write a list comprehension, equivalentless comprehension. So as usual, you write four I in range and leven. Because the the moment I um write level, it also includes a zero. So I can specify [00:07:15.268] 1 comma 11. Okay? And then do this. What I need to do? I need to perform the square of a number. So I'll say I star I. Okay. So this is what I will try to do and say print L2. [00:07:34.244] Let's execute this and check. Okay, if you don't have any issues, I will comment it till here. [00:07:43.236] Let's find out what is going to happen. Right? And if you can see square of the numbers from one to ten has happened. Maybe uh we will look into one more uh example, maybe one or two examples. [00:08:00.036] So understand less comprehension. For example, if I'm uh if I'm going to tell uh perform Q, then I into I into I similarly. Okay. So let me take up one more example. I want to uh print [00:08:17.508] The string length. Okay. Suppose if you have uh a list which has some strings, I want to find out uh what is the length of the string. Along with that, I may also uh say Bangalore followed by the length. So something of this kind if I want to do. So I'll just tell it as list. [00:08:36.996] of strings and its length. Okay. So let me uh write I'll say L three is equal to before this uh let's create a list. So I will say this [00:08:55.396] Uh Python. [00:08:59.428] [ASR low confidence; check audio] And I'll see PES University. [00:09:10.852] Something of this kind. This is what uh I want to do and uh let me uh write a less comprehension. So L3 is equal to [00:09:28.932] Okay. So I will say Alice Comprehension mention this. Or I in range. [00:09:39.780] A for I in range. Sorry, not range. For I in A, because uh I already know I want to find out a string. [00:09:49.988] and its length. Okay, I should also get a string and its

length. So now where is the uh string that is available? It is available in I and I will say L of I. Okay, hope uh this is uh uh clear to you

EV0000050 · EV0000051 · EV0000052 · EV0000053

Transcript block 3 (source interval 000600-000899 seconds): [00:10:09.028] Fine for I in a it should be fine. So let me uh print print L tree. [00:10:18.980] Execute this, I'm going to comment the next one also because I know what is the output here. Let me execute. [00:10:29.956] Fine. Python six, PS University 14, language is eight, and so on. Please um remember whenever you try to find the uh length, it also considers the space. Okay so [00:10:45.860] I'll use the same list. Let's use the same list. And I want to uh convert all of them to uppercase. Fine, all of them to uppercase. So how will you do it? I'll say L4 is equal to [00:11:02.308] for INA because I'm still considering A itself. What do I need to do? I need to find the uh uh convert all of them to uppercase. So I'll say I dot upper [00:11:17.988] You've already used uh this particular function in case of the uh strings request you to go back and check it out. So I'll say print L4. [00:11:32.484] Okay, so let me execute and if you see these were all small letters and now it's been converted to capital letters. [00:11:43.556] There are uh uh so many things that you could do. You can also say that um uh the one sort of uh whose length is greater than six, uh print them and so on. Okay you can uh try out with any number of uh uh programs. [00:11:59.332] I hope uh you have uh understood the concept of uh less comprehension. Thank you.

EV0000054 · EV0000055 · EV0000056 · EV0000057

Full transcript: 1558 normalized words, 49 timestamped segments, 726.186667 seconds duration. Low-confidence segments flagged: 1. Review locators: 00:08:59.428

EV0000046 · EV0000047 · EV0000048 · EV0000049

7 Closure

This lesson is indexed from the source video titled Closure. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000058 · EV0000059

Graphs and diagrams

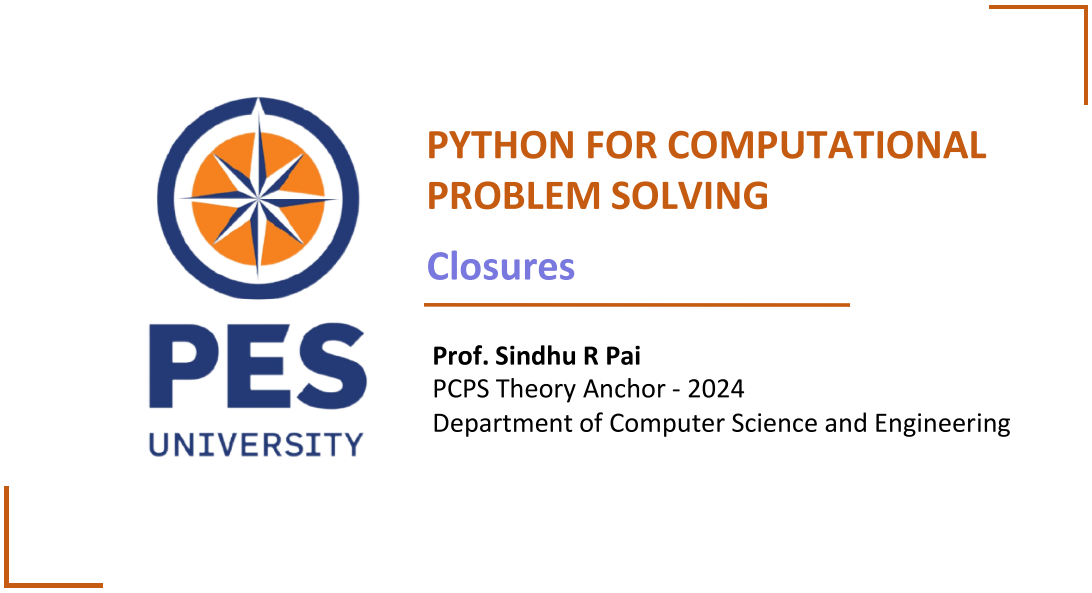

Figure G000121. UE24CS151A PCPS Unit 3 Sindhu Class 5 Closures · source slide page 1.

EV0000479

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

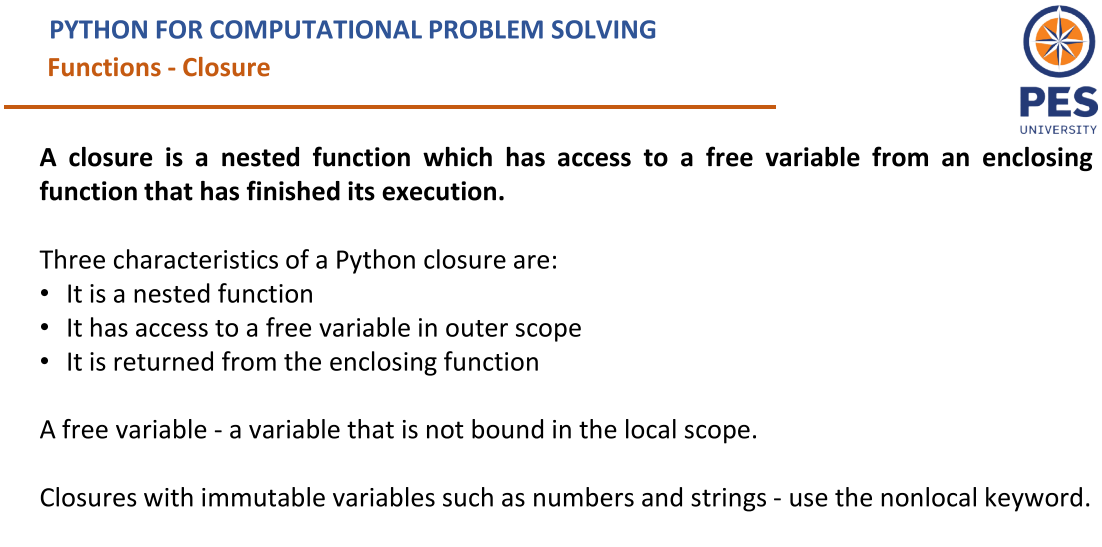

Figure G000122. UE24CS151A PCPS Unit 3 Sindhu Class 5 Closures · source slide page 2.

EV0000482

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

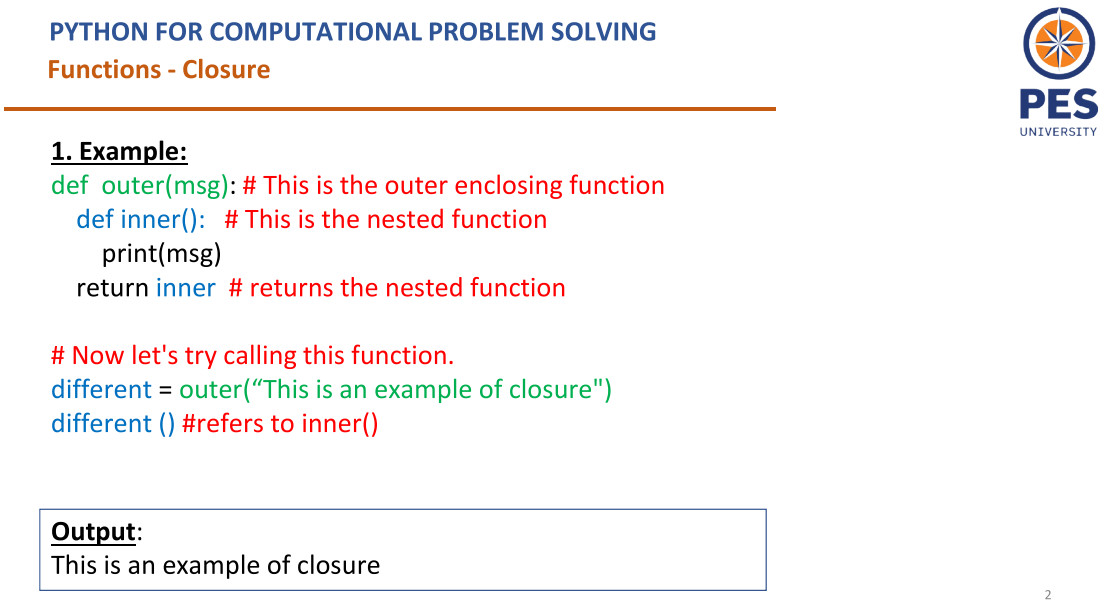

Figure G000123. UE24CS151A PCPS Unit 3 Sindhu Class 5 Closures · source slide page 3.

EV0000485

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

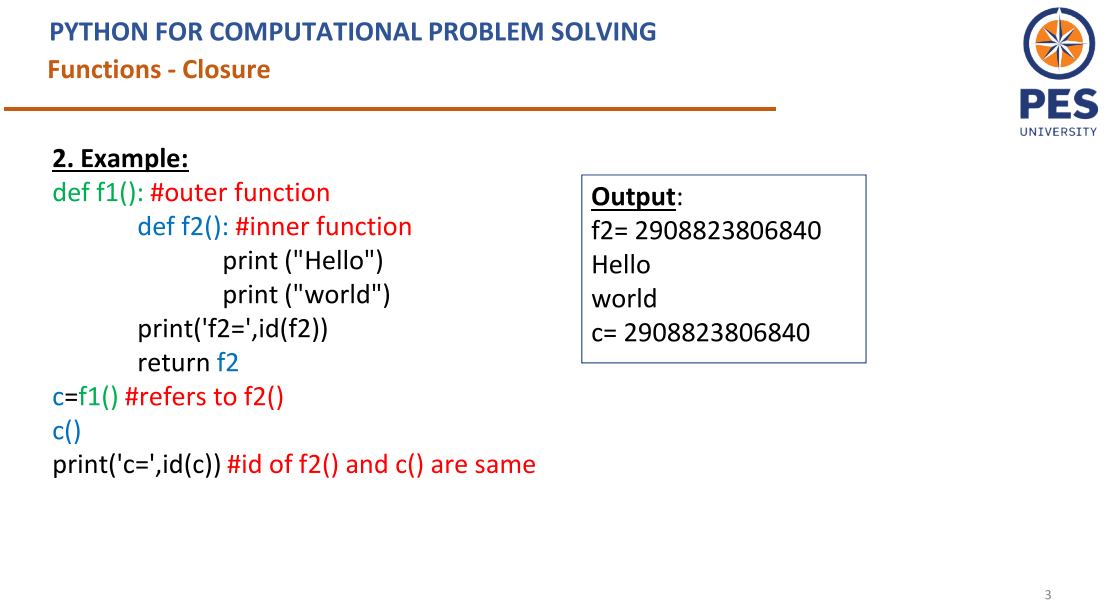

Figure G000124. UE24CS151A PCPS Unit 3 Sindhu Class 5 Closures · source slide page 4.

EV0000488

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

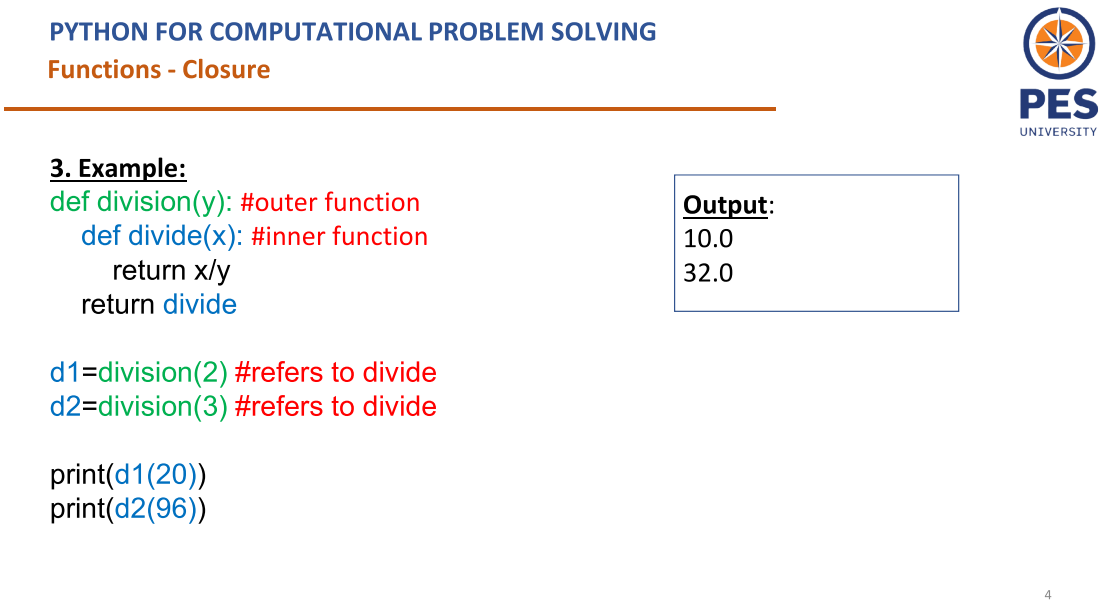

Figure G000125. UE24CS151A PCPS Unit 3 Sindhu Class 5 Closures · source slide page 5.

EV0000491

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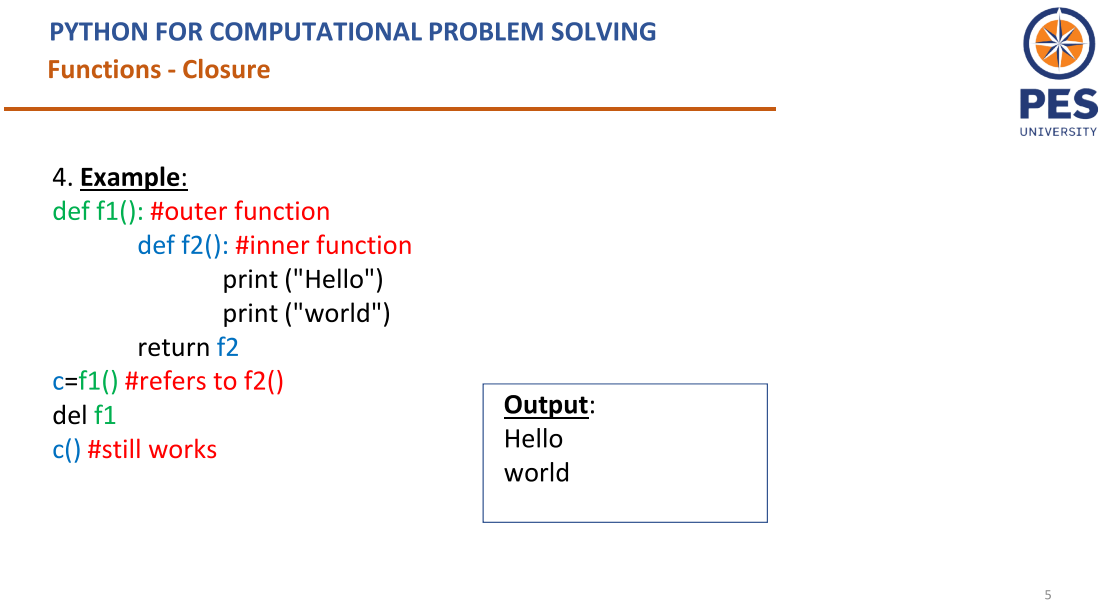

Figure G000126. UE24CS151A PCPS Unit 3 Sindhu Class 5 Closures · source slide page 6.

EV0000494

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

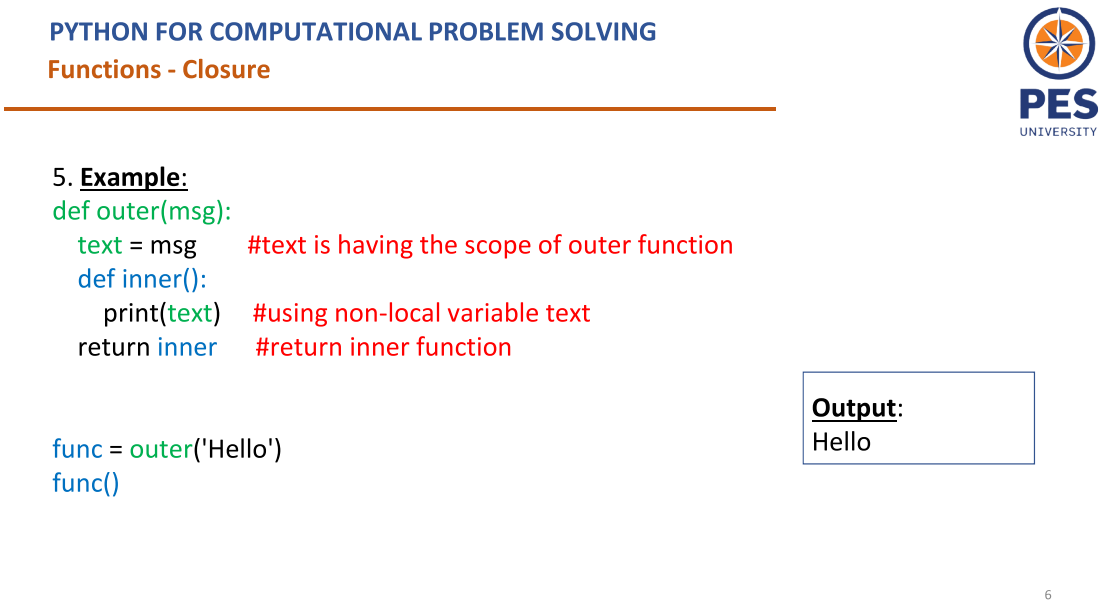

Figure G000127. UE24CS151A PCPS Unit 3 Sindhu Class 5 Closures · source slide page 7.

EV0000497

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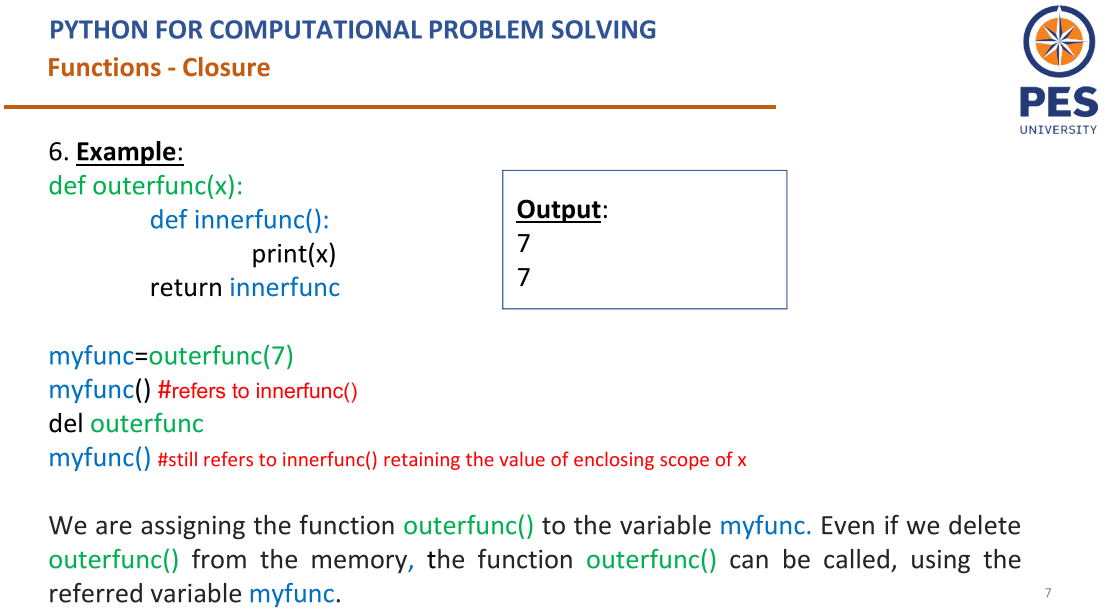

Figure G000128. UE24CS151A PCPS Unit 3 Sindhu Class 5 Closures · source slide page 8.

EV0000500

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

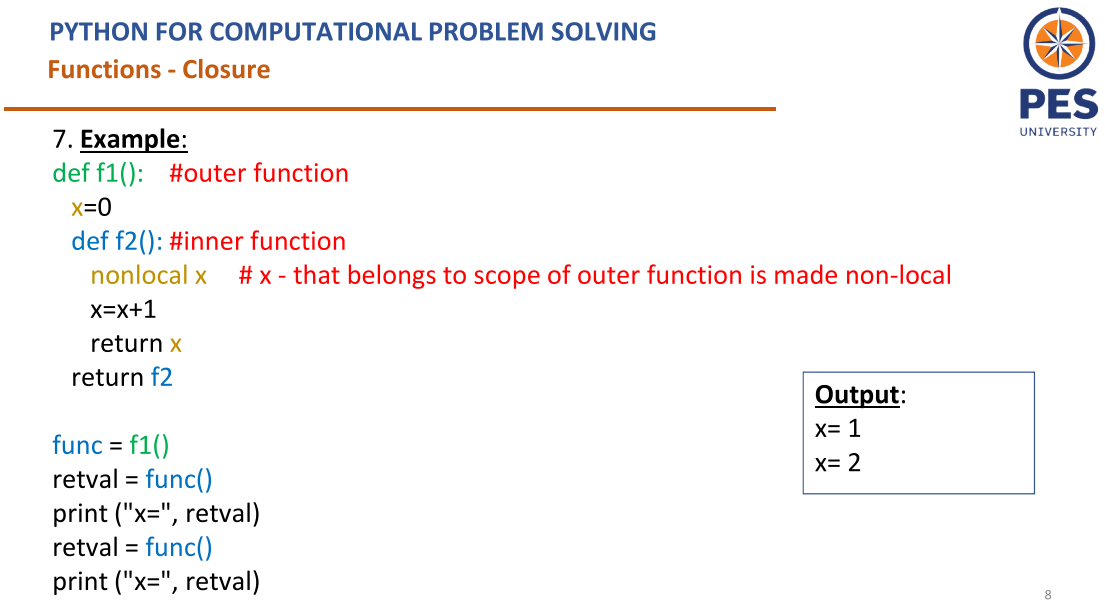

Figure G000129. UE24CS151A PCPS Unit 3 Sindhu Class 5 Closures · source slide page 9.

EV0000503

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Functions - Closure

PES UNIVERSITY

Function Closure vs. Nested function

- Not all nested functions are closures.
- For a nested function to be a closure, the following conditions need to be satisfied:

1. The inner function has access to the non-local variables or local variables of the outer function.

2. The outer function must return the inner function.

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Figure G000130. UE24CS151A PCPS Unit 3 Sindhu Class 5 Closures · source slide page 10.

EV0000506

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

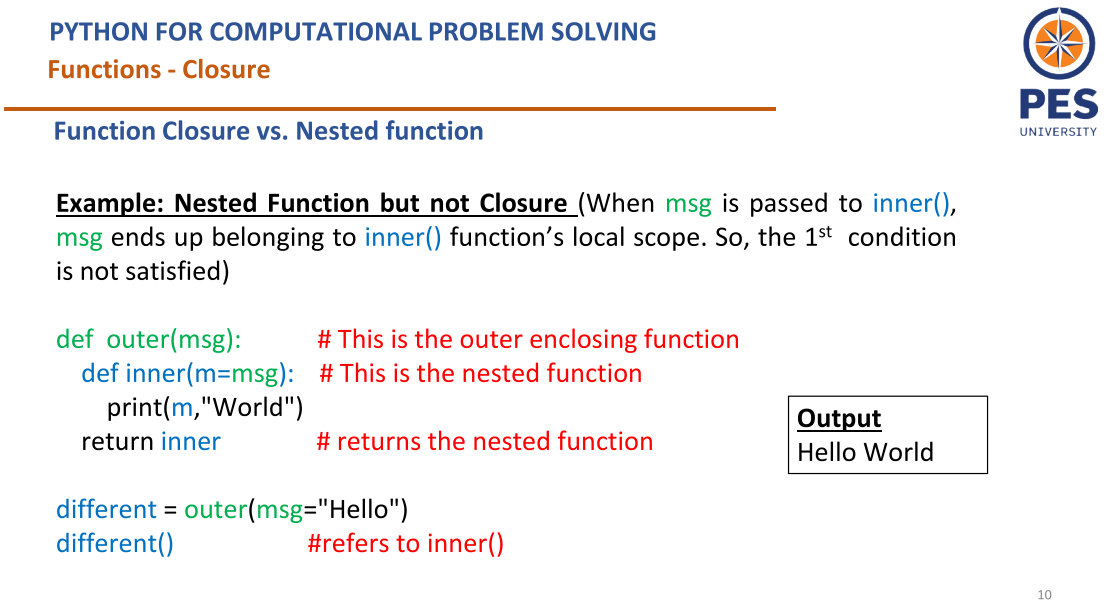

Figure G000131. UE24CS151A PCPS Unit 3 Sindhu Class 5 Closures · source slide page 11.

EV0000509

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Functions - Closure

PES UNIVERSITY

Function Closure: Summary

- A function object that remembers values in enclosing scopes even when the variable goes out of scope.
- Python closures help avoiding the usage of global values and provide some form of data hiding. They are used in Python decorators.

11

Figure G000132. UE24CS151A PCPS Unit 3 Sindhu Class 5 Closures · source slide page 12.

EV0000512

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

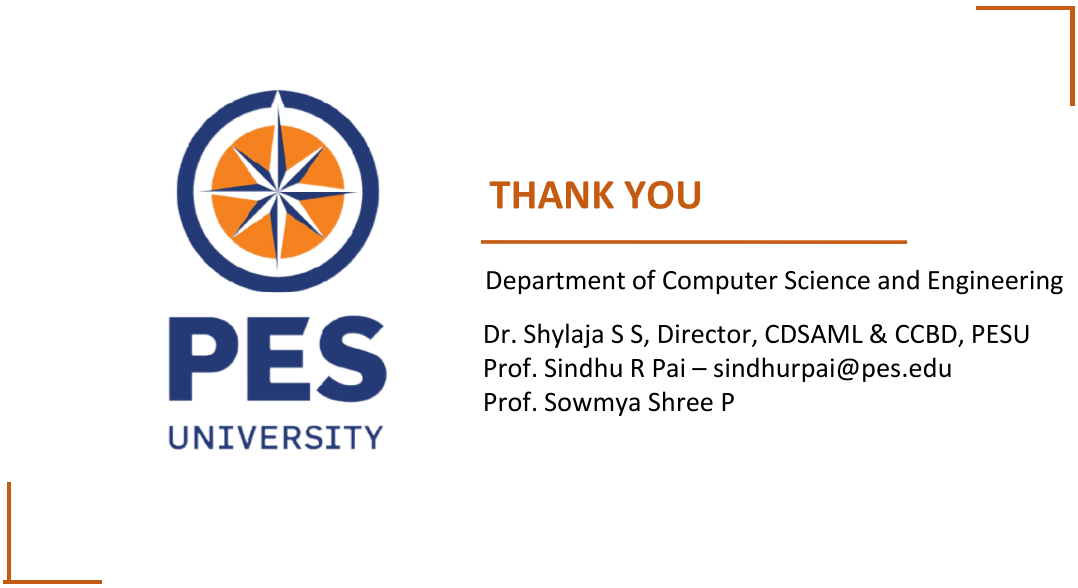

Figure G000133. UE24CS151A PCPS Unit 3 Sindhu Class 5 Closures · source slide page 13.

EV0000515

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Closures Prof. Sindhu R Pai PCPS Theory Anchor - 2024 Department of Computer Science and Engineering EV0000478 · EV0000479 · EV0000480
2. Source slide page 2 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Closure A closure is a nested function which has access to a free variable from an enclosing function that has finished its execution. Three characteristics of a Python closure are: • It is a nested function • It has access to a free variable in outer scope • It is returned from the enclosing function A free variable - a variable that is not bound in the local scope. Closures with immutable variables such as numbers and strings - use the nonlocal keyword. 1 EV0000481 · EV0000482 · EV0000483
3. Source slide page 3 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Closure 1. Example: def outer(msg): # This is the outer enclosing function def inner(): # This is the nested function print(msg) return inner # returns the nested function # Now let's try calling this function. different = outer(“This is an example of closure") different () #refers to inner() Output: This is an example of closure 2 EV0000484 · EV0000485 · EV0000486

4. Source slide page 4 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Closure 2. Example: def f1(): #outer function Output: def f2(): #inner function f2= 2908823806840 print ("Hello") Hello printp ("world")( ) worldworld print('f2=',id(f2)) c= 2908823806840 return f2 c=f1() #refers to f2() c() print('c=',id(c)) #id of f2() and c() are same 3 EV0000487 · EV0000488 · EV0000489
5. Source slide page 5 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Closure 3. Example: def division(y): #outer function Output: def divide(x): #inner function 10.0 return x/y 32.0 returnreturn dividedivide d1=division(2) #refers to divide d2=division(3) #refers to divide print(d1(20)) print(d2(96)) 4 EV0000490 · EV0000491 · EV0000492
6. Source slide page 6 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Closure 4. Example: def f1(): #outer function def f2(): #inner function print ("Hello") printi ("("world")ld") return f2 c=f1() #refers to f2() Output:del f1 Hello Hello c() #still works world world 5 EV0000493 · EV0000494 · EV0000495
7. Source slide page 7 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Closure 5. Example: def outer(msg): text = msg #text is having the scope of outer function def inner(): print(text)i t(t t) ##usingi non-locall l variablei bl ttextt return inner #return inner function Output:Hello world Hellofunc = outer('Hello') func() 6 EV0000496 · EV0000497 · EV0000498
8. Source slide page 8 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Closure 6. Example: def outerfunc(x): Output: def innerfunc(): Hello 7 print(x) world 7 return innerfunc myfunc=outerfunc(7) myfunc() #refers to innerfunc() del outerfunc myfunc() #still refers to innerfunc() retaining the value of enclosing scope of x We are assigning the function outerfunc() to the variable myfunc. Even if we delete outerfunc() from the memory, the function outerfunc() can be called, using the 7referred variable myfunc. EV0000499 · EV0000500 · EV0000501
9. Source slide page 9 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Closure 7. Example: def f1(): #outer function x=0 def f2(): #inner function nonlocal x # x - that belongs to scope of outer function is made non-local xx=x+1x+1 return x return f2 Output: x= 1 Hellofunc = f1() world x= 2retval = func() print ("x=", retval) retval = func() print ("x=", retval) 8 EV0000502 · EV0000503 · EV0000504
10. Source slide page 10 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Closure Function Closure vs. Nested function • Not all nested functions are closures. • For a nested function to be a closure, the following conditions need to be satisfied: 1. The inner function has access to the non-local variables or local variables of the outer function. 2. The outer function must return the inner function. 9 EV0000505 · EV0000506 · EV0000507
11. Source slide page 11 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Closure Function Closure vs. Nested function Example: Nested Function but not Closure (When msg is passed to inner(), msg ends up belonging to inner() function's local scope. So, the 1st condition is not satisfied) def outer(msg): # This is the outer enclosing function def inner(m=msg): # This is the nested function print(m,"World") Output return inner # returns the nested function Hello World different = outer(msg="Hello") different() #refers to inner() 10 EV0000508 · EV0000509 · EV0000510

12. Source slide page 12 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Closure Function Closure: Summary • A function object that remembers values in enclosing scopes even when the variable goes out of scope. • Python closures help avoiding the usage of global values and provide some form of data hiding. They are used in Python decorators. 11 EV0000511 · EV0000512 · EV0000513

13. Source slide page 13 (extracted text): THANK YOU Departmentp of Computerp Science and Engineeringg g Dr. Shylaja S S, Director, CDSAML & CCBD, PESU Prof. Sindhu R Pai – sindhurpai@pes.edu Prof. Sowmya Shree P EV0000514 · EV0000515 · EV0000516

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.964] Hello, all, welcome back to Python for computational problem solving. In today's session, let's understand the concept of closures. [00:00:12.708] Closure is applicable to nested functions. Nested functions have already been covered. Please uh uh look at uh the concept of nested functions once again. [00:00:27.076] As I mentioned, uh the closure is applicable to nested functions. So basically you will have an inner function, along with that you will also have an outer functions. So I will have an inner function and I will have an outer functions. [00:00:45.348] So these inner functions, these inner functions have an accessibility. They can access the variables declared or the ones that are initialized inside this outer functions. So let me repeat. [00:01:01.632] Inner functions can access the variables that are initialized or declared within this outer functions. So till what point of time I can access? [00:01:13.572] This accessing can be done even though the outer functions have been removed completely from the memory. Okay? So [00:01:23.088] The inner functions have an access to the variables declared or initialized inside an outer function even after the outer functions have been completely executed or it has been removed completely from the memory. [00:01:37.440] Okay. So hope you have uh understood this concept. The uh the main intention is that the state information would be still retained even after the outer function has been removed completely from the memory. [00:01:50.400] Okay. So what we will do is we'll quickly uh code this to understand the concept of closures. Okay. So let me do this. I will say it as an outer itself because I want to um I should have an nested function. [00:02:06.192] Fine. So I will also have a define innermost function. Okay. So what do I do? I'll say print some message. Okay. Print some message. [00:02:22.884] Hope this is clear and I will say return the innermost functions [00:02:31.108] Uh address where it is present. Okay. So this is what I am trying to do. So let me call this function. I will say it as x is equal to outer. I will say I will have to pass some uh message. So I will say hello. [00:02:50.500] Closure. I'm passing some message. What should happen for me? I should be able to receive this. Okay. So I will also say it as a message. So what I will do is I will try to access this. [00:03:11.108] Is this fine? So now what basically is happening, we have uh defined a function called as an outer function. This is an outer enclosing function. [00:03:20.996] And uh your inner is an uh nested function in this case. We're trying to print the message and return the inner. So the address at which inner function is present, that will be returned. [00:03:34.304] To which one it is going to be returned? It is going to be returned here, X. So the moment you call over this X, what is basically happening? You are trying to call the innermost function. [00:03:47.104] Okay, the output that we are expecting at this particular point is hello closure. So let's quickly uh run this. Hope it's not doing this. Okay. So I've already executed once, let me do it. And uh [00:04:02.308] One second. Give me time. We are uh doing this. So let me uh say it as sprint. [00:04:19.908] See that was not

saved. Let me check it once again. Okay. Sorry, uh sorry for the previous one. The program uh was not saved. Okay the last line when I typed it was not saved because of which uh you were not getting the output. So hope it is clear to you. [00:04:37.936] What have we done? There are uh two functions, outer functions and inner functions. We have returned the inner functions address and I'm trying to uh call the function where it has been captured. Okay so [00:04:55.204] Continue with uh one more example because uh when I was uh discussing I said

EV0000058 · EV0000059 · EV0000060 · EV0000061

Transcript block 2 (source interval 000300-000599 seconds): [00:05:01.700] Even after the outer function has been removed completely, I can still access uh the inner function. That itself is a concept of closure. This is just to explain you that uh uh the how inner functions and outer functions can be accessed. Okay. So let me quickly uh write one more program. [00:05:19.472] To help you out in this case. So I will say define F one and I will say define F two. Okay. It is done in the similar manner. Fine. Instead of passing some message, I will say print same thing, hello closure. [00:05:41.476] Hello closure. What do we do? We will do it in the similar fashion. Return F2. So I will say X is equal to the function F1. [00:05:59.108] I've called this particular function F1. [00:06:03.428] At this point I will say delete f1. Okay, at this point I'm uh deleting the f1. But what I will do is I will use this x and then call this function. So let's try to understand. I have defined a function f1. [00:06:21.060] Nested function is F2. This nested function has a print statement. We are returning F2. The address of F2 is returned at this point. So what does X contain? X contains the address of F2. Now [00:06:35.620] I'm trying to delete F1. Okay, F1 is your outer function. The moment I delete an outer function, what we basically thought is we cannot access an F2. But in general, it is not true. I can still access uh the [00:06:50.352] Uh F2, the innermost function because it's a state information is stored in X. So that is why I call it as X, which means you're trying to call the innermost function which is F2. Let's uh quickly uh try to do this. Let me comment on this. [00:07:09.540] So we will not have uh any sort of confusions. Okay. So let me execute this. I'll say clear screen. [00:07:20.804] And uh execute. If you can see hello closure. Maybe you are getting confused with the previous output. So maybe I will add few dots to tell that we are getting the output for the same program. Okay so let's execute this once more, and if you can see hello closure. [00:07:38.468] I hope uh you have understood. In order to uh make things a little more clear to you, as usual, I will copy paste it onto the uh our uh Python uh tutor. [00:07:55.940] The visualization tool that is there. Easy for you to understand. Let's go ahead with this. Number of steps involved is uh 11. Okay. Quickly look at what is happening. Headers are processed. Fine, only one header. [00:08:13.348] The function has been called. The moment you call a function, you enter into the body of the function, fine, define F2. So F2 has been created. And [00:08:26.660] Again, when do you uh get into uh the F2? Only when F2 is being called. We haven't called F2 at this point. So return F2. The moment you say return F2, to which one it is returning? So it is returning. [00:08:41.120] From where you have been called. Okay. So now you say delete F1. Just look at the visualization. F1 needs to be deleted. So the existence of F1 is not there. But if you can see parent f2's parent is still F1. Okay. And now you are calling F1. [00:09:00.420] Sorry, now you're calling F2 through X where it is stored, and because of which we are able to enter into your function F2, and I'm going to be getting the output which is called as hello closure. [00:09:15.268] Fine. So hope uh you have understood this particular concept. Okay. Try uh solving uh few more uh problems or programs on this in order to have a better understanding. Hope you have uh understood. Thank you.

EV0000062 · EV0000063 · EV0000064 · EV0000065

Full transcript: 1310 normalized words, 38 timestamped segments, 575.018667 seconds duration. Low-confidence segments flagged: 0.

EV0000058 · EV0000059 · EV0000060 · EV0000061

8 Decorators

This lesson is indexed from the source video titled Decorators. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000066 · EV0000067

Graphs and diagrams

Figure G000134. UE24CS151A PCPS Unit 3 Sindhu Class 6 Decorators · source slide page 1.

EV0000518

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Functions - Decorators

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- A powerful and useful tool in Python since it allows programmers to modify the behavior of function or class.
- Decorators **wrap a function and modify its behavior in one or the other way, without changing the source code** of the function being decorated.
- In Decorators, functions are taken as the argument into another function and then called inside the wrapper function.

1

Figure G000135. UE24CS151A PCPS Unit 3 Sindhu Class 6 Decorators · source slide page 2.

EV0000521

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Functions - Decorators

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Function Decorators are used

- When we need to change the behavior of a function without modifying the function itself.
  Eg: logging, test performance, verify permissions and so on.

- When we need to run the same code on multiple functions. This avoids writing duplicate code.

2

Figure G000136. UE24CS151A PCPS Unit 3 Sindhu Class 6 Decorators · source slide page 3.

EV0000524

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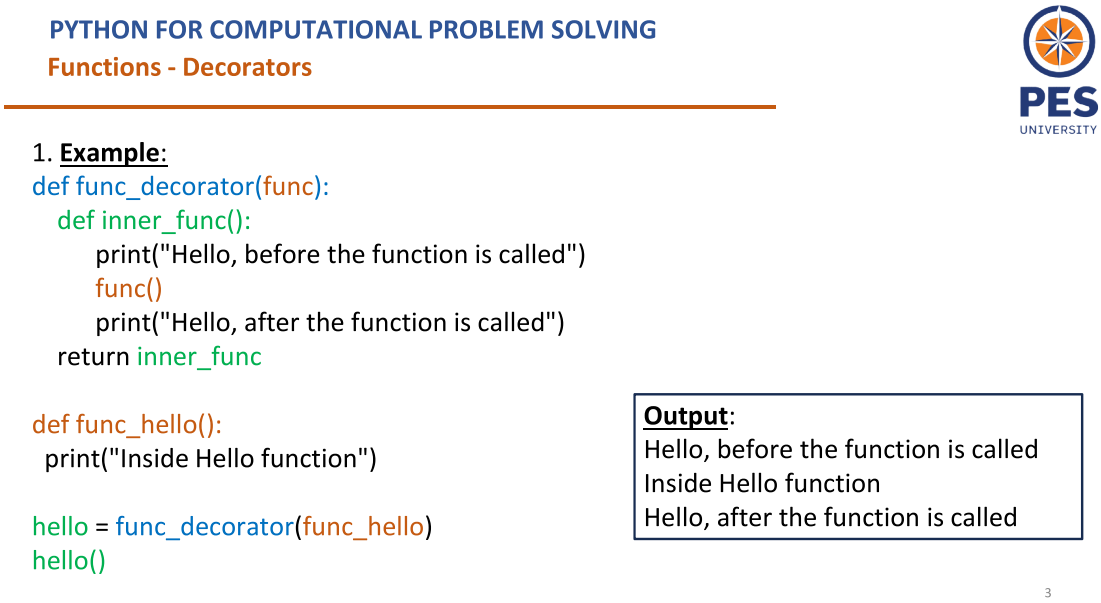

Figure G000137. UE24CS151A PCPS Unit 3 Sindhu Class 6 Decorators · source slide page 4.

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Functions - Decorators

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- ***func_decorator*** is the decorator function, accepts another function as an argument and "decorates it".
- ***func_hello*** is an ordinary function that we need to decorate.
- ***inner_func*** is the wrapper function, that is actually decorating the ***func_hello*** function. In this example, all it does is print a simple statement before and after ***func_hello***.

The function decorator in the above example can also be implemented in other way. (See Example 2)
By using @ symbol

4

Figure G000138. UE24CS151A PCPS Unit 3 Sindhu Class 6 Decorators · source slide page 5.

EV0000530

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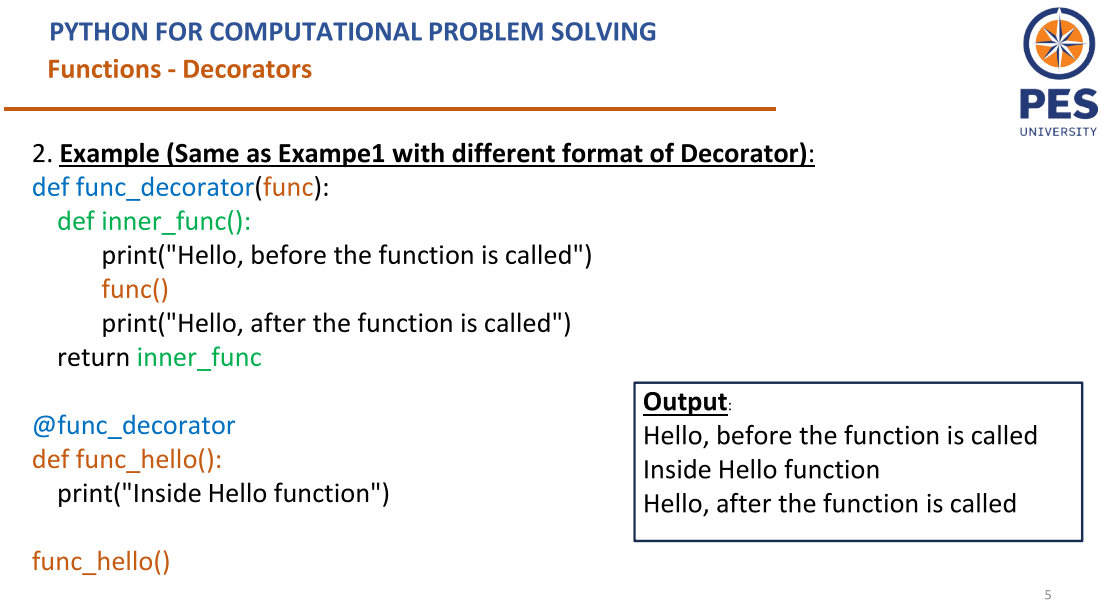

Figure G000139. UE24CS151A PCPS Unit 3 Sindhu Class 6 Decorators · source slide page 6.

EV0000533

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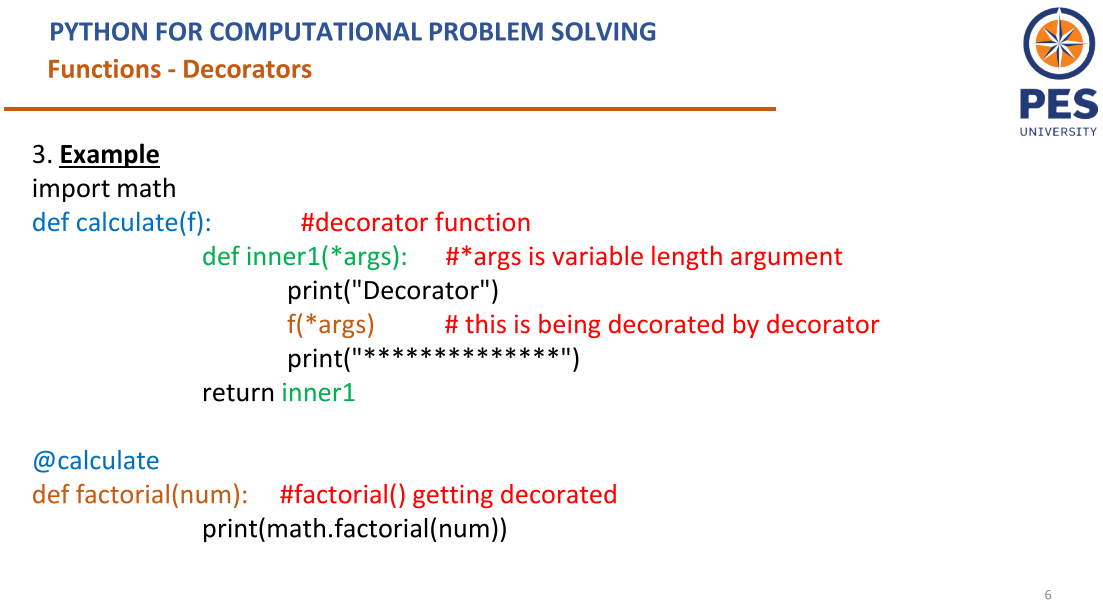

Figure G000140. UE24CS151A PCPS Unit 3 Sindhu Class 6 Decorators · source slide page 7.

EV0000536

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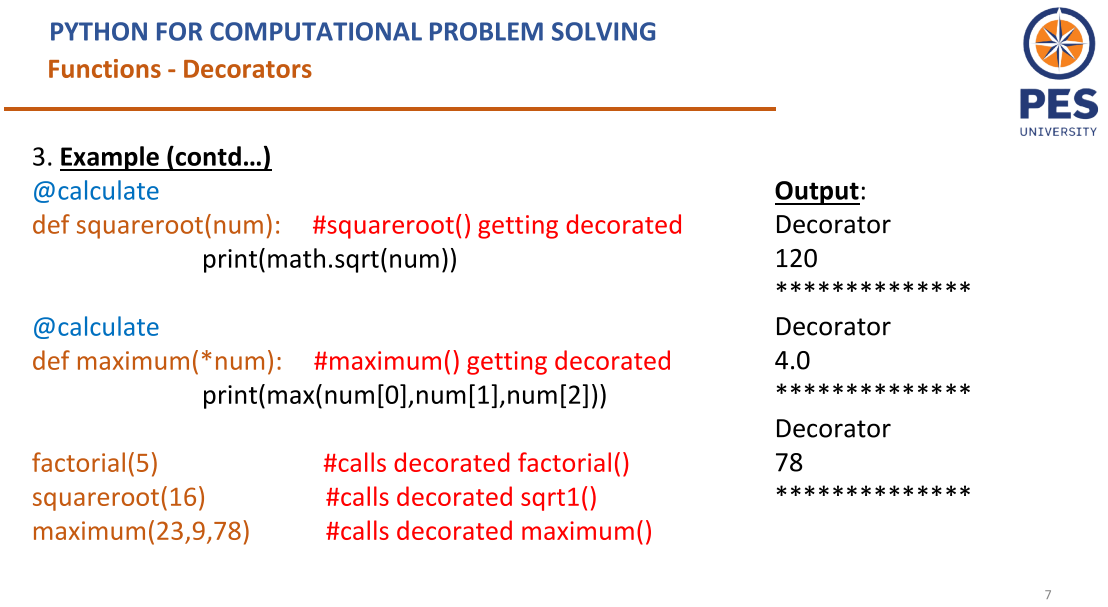

Figure G000141. UE24CS151A PCPS Unit 3 Sindhu Class 6 Decorators · source slide page 8.

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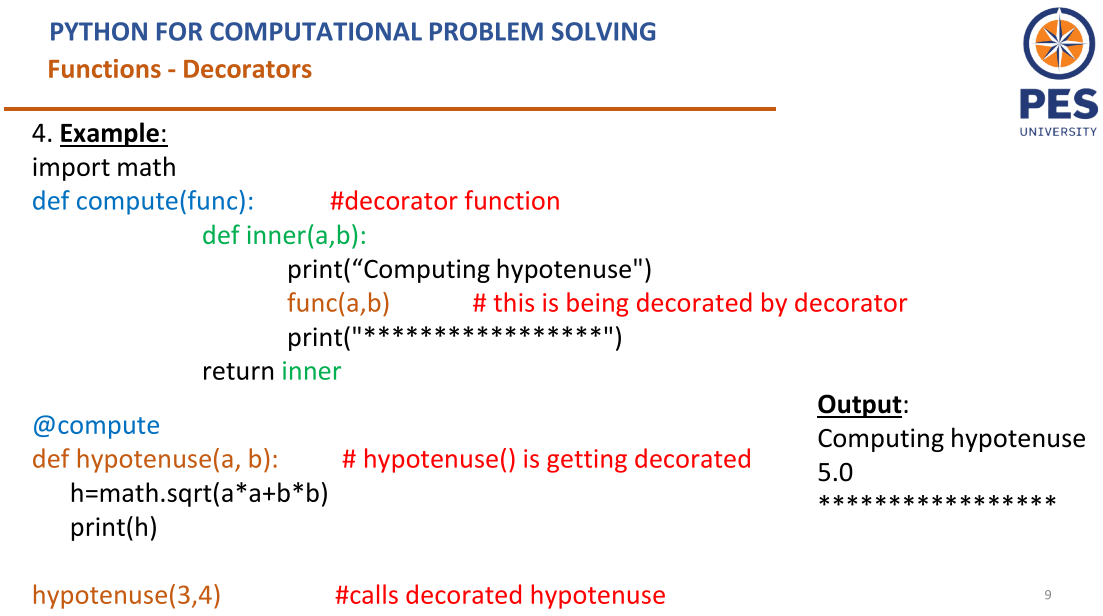

Figure G000142. UE24CS151A PCPS Unit 3 Sindhu Class 6 Decorators · source slide page 9.

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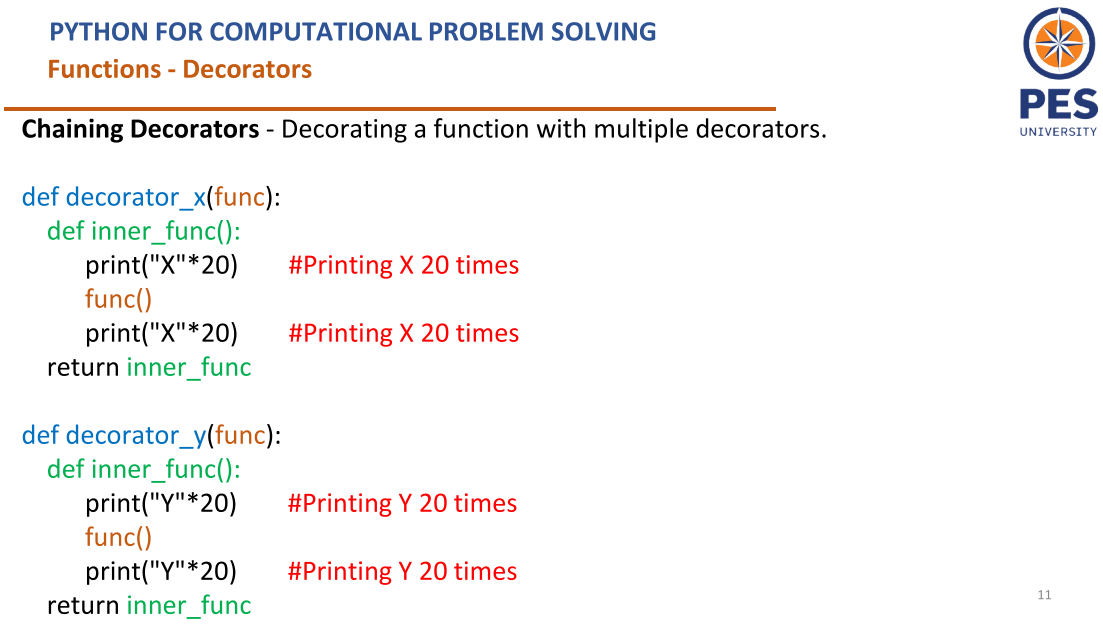

Figure G000143. UE24CS151A PCPS Unit 3 Sindhu Class 6 Decorators · source slide page 10.

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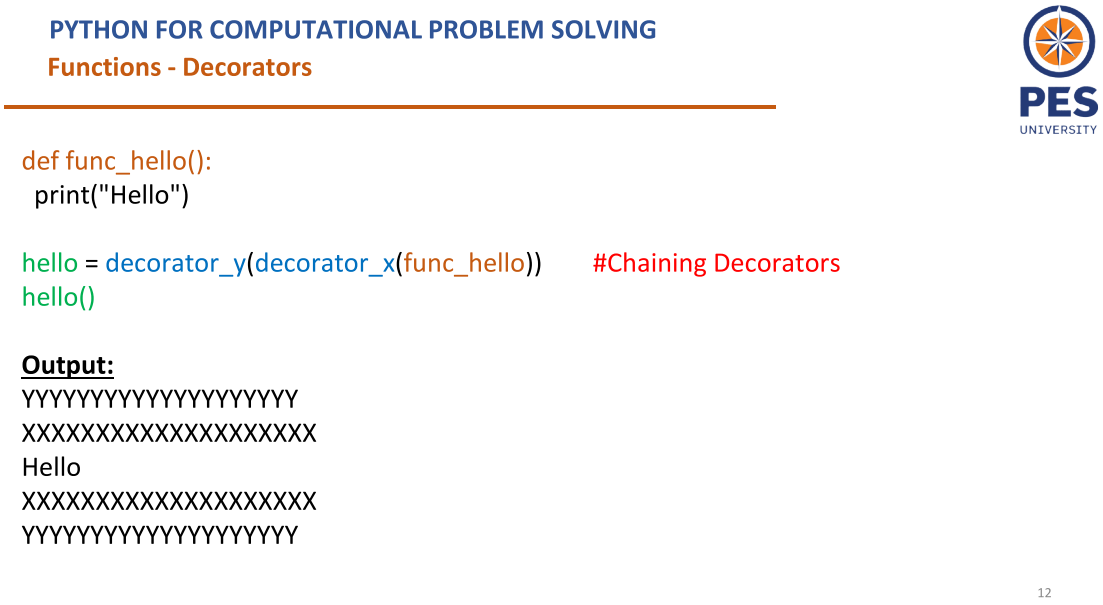

Figure G000144. UE24CS151A PCPS Unit 3 Sindhu Class 6 Decorators · source slide page 11.

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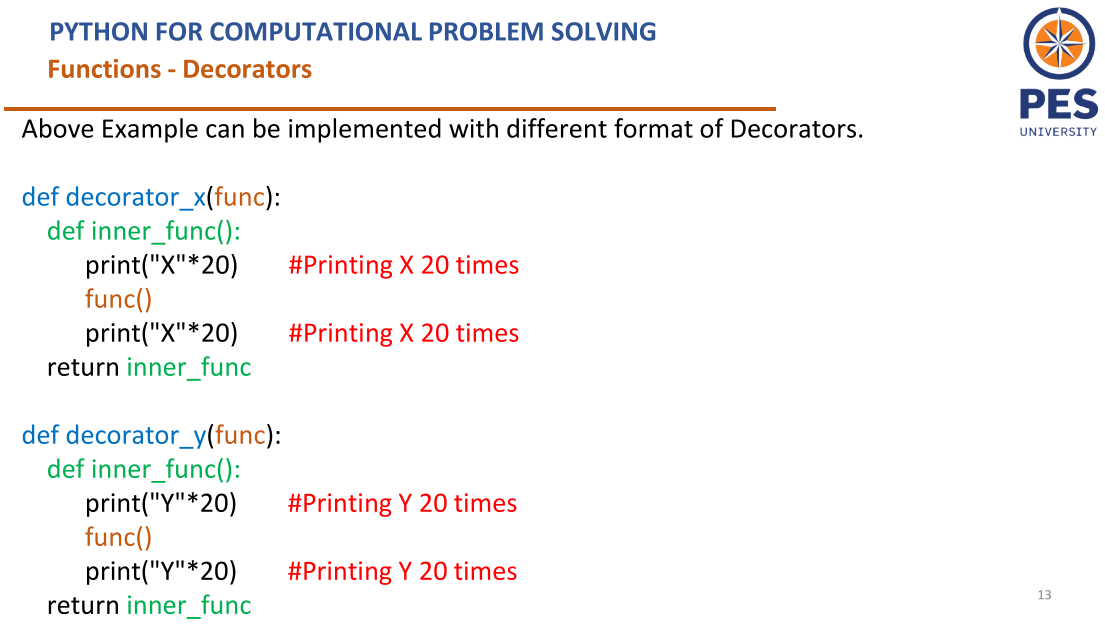

Figure G000145. UE24CS151A PCPS Unit 3 Sindhu Class 6 Decorators · source slide page 12.

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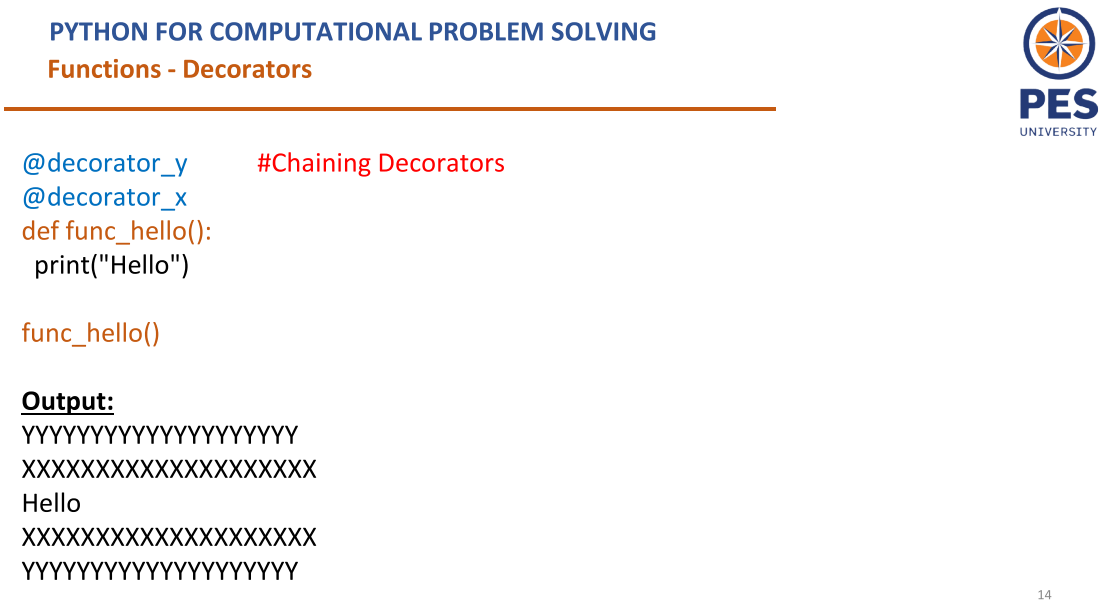

Figure G000146. UE24CS151A PCPS Unit 3 Sindhu Class 6 Decorators · source slide page 13.

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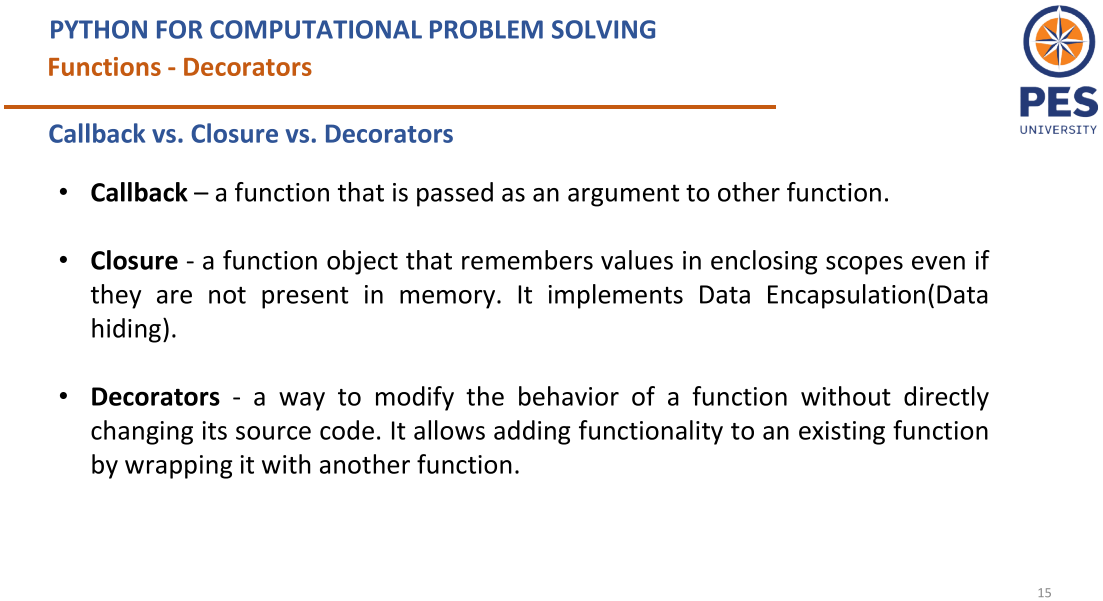

Figure G000147. UE24CS151A PCPS Unit 3 Sindhu Class 6 Decorators · source slide page 14.

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Functions - Decorators

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Functions Decorators: Summary

- A Decorator is just a function that takes another function as an argument and extends its behavior without explicitly modifying it.
- Decorators allow us to wrap another function in order to extend the behavior of wrapped function, without permanently modifying it.
- Using decorators, we can extend the features of different functions in a common way.

16

Figure G000148. UE24CS151A PCPS Unit 3 Sindhu Class 6 Decorators · source slide page 15.

EV0000560

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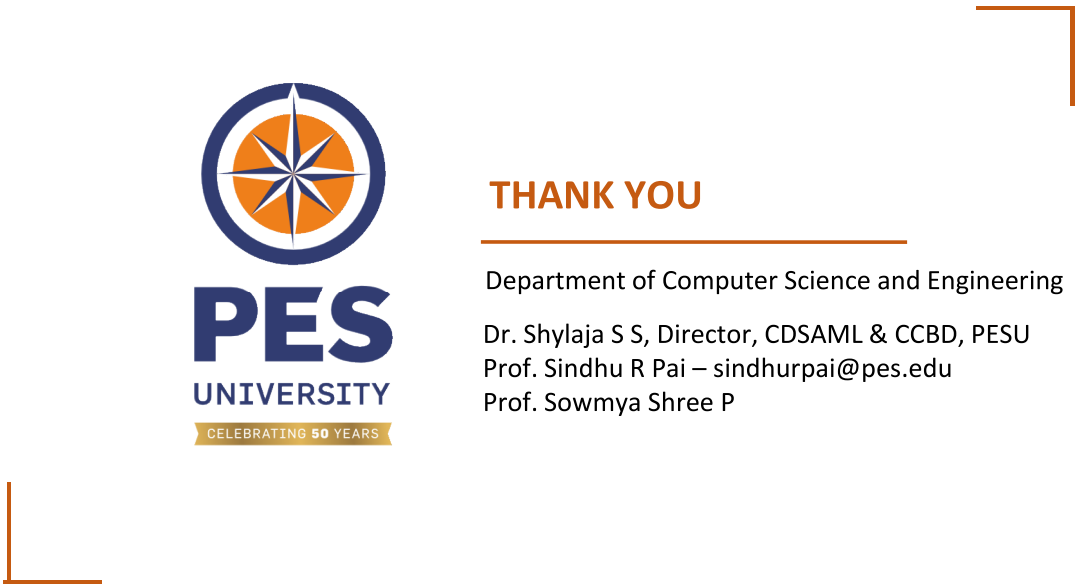

Figure G000149. UE24CS151A PCPS Unit 3 Sindhu Class 6 Decorators · source slide page 16.

EV0000563

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Decorators Prof. Sindhu R Pai PCPS Theory Anchor - 2024 Department of Computer Science and Engineering EV0000517 · EV0000518 · EV0000519
2. Source slide page 2 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Decorators • A powerful and useful tool in Python since it allows programmers to modify the behavior of function or class. • Decorators wrap a function and modify its behavior in one or the other way, without changing the source code of the function being decorated. • In Decorators, functions are taken as the argument into another function and then called inside the wrapper function. 1 EV0000520 · EV0000521 · EV0000522
3. Source slide page 3 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Decorators Function Decorators are used • WhenWhen wewe needneed totto changechange thethe behaviorbehavior ofof aa functionfunction withoutwithout modifyingmodifying thethe functionfunction itself. Eg: logging, test performance, verify permissions and so on. • When we need to run the same code on multiple functions. This avoids writing duplicate code. 2 EV0000523 · EV0000524 · EV0000525

4. Source slide page 4 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Decorators 1. Example: def func_decorator(func): def inner_func(): print("Hello, before the function is called") func() print("Hello, after the function is called") return inner_func Output:def func_hello(): Hello, before the function is called print("Inside Hello function") Inside Hello function Hello, after the function is calledhello = func_decorator(func_hello) hello() 3 EV0000526 · EV0000527 · EV0000528

5. Source slide page 5 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Decorators • func_decorator is the decorator function, accepts another function as an argument and "decorates it". • func_hello is an ordinary function that we need to decorate. • inner_func is the wrapper function, that is actually decorating the func_hello function. In this example, all it does is print a simple statement before and after func_hello. The function decorator in the above example can also be implemented in other way. (See Example 2) By using @ symbol 4 EV0000529 · EV0000530 · EV0000531

6. Source slide page 6 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Decorators 2. Example (Same as Exampe1 with different format of Decorator): def func_decorator(func): def inner_func(): print("Hello, before the function is called") func() print("Hello, after the function is called") return inner_func Output: @func_decorator Hello, before the function is called def func_hello(): Inside Hello function print("Inside Hello function") Hello, after the function is called func_hello() 5 EV0000532 · EV0000533 · EV0000534

7. Source slide page 7 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Decorators 3. Example import math def calculate(f): #decorator function def inner1(*args): #*args is variable length argument print("Decorator") f(*args) # this is being decorated by decorator print("**********") return inner1 @calculate def factorial(num): #factorial() getting decorated print(math.factorial(num)) 6 EV0000535 · EV0000536 · EV0000537

8. Source slide page 8 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Decorators 3. Example (contd...) @calculate Output: def squareroot(num): #squareroot() getting decorated Decorator print(math.sqrt(num)) 120 ********** @calculate Decorator def maximum(*num): #maximum() getting decorated 4.0 print(max(num[0],num[1],num[2])) ********** Decorator factorial(5) #calls decorated factorial() 78 squareroot(16) #calls decorated sqrt1() ********** maximum(23,9,78) #calls decorated maximum() 7 EV0000538 · EV0000539 · EV0000540

9. Source slide page 9 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Decorators 4. Example: import math def compute(func): #decorator function def inner(a,b): print("Computing hypotenuse") func(afunc(a,b)b) ## thisthis isis beingbeing decorateddecorated byby decoratordecorator print("**********") return inner Output: @compute Computing hypotenuse def hypotenuse(a, b): # hypotenuse() is getting decorated 5.0 h=math.sqrt(a*a+b*b) ********** print(h) 9hypotenuse(3,4) #calls decorated hypotenuse EV0000541 · EV0000542 · EV0000543

10. Source slide page 10 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Decorators Chaining Decorators - Decorating a function with multiple decorators. def decorator_x(func): def inner_func(): print("X"*20) #Printing X 20 times func()func() print("X"*20) #Printing X 20 times return inner_func def decorator_y(func): def inner_func(): print("Y"*20) #Printing Y 20 times func() print("Y"*20) #Printing Y 20 times return inner_func 11 EV0000544 · EV0000545 · EV0000546
11. Source slide page 11 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Decorators def func_hello(): print("Hello") hello = decorator_y(decorator_x(func_hello)) #Chaining Decorators hello()hello() Output: YYYYYYYYYYYYYYYYYYY XXXXXXXXXXXXXXXXXXX Hello XXXXXXXXXXXXXXXXXXX YYYYYYYYYYYYYYYYYYY 12 EV0000547 · EV0000548 · EV0000549
12. Source slide page 12 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Decorators Above Example can be implemented with different format of Decorators. def decorator_x(func): def inner_func(): print("X"*20) #Printing X 20 times func()func() print("X"*20) #Printing X 20 times return inner_func def decorator_y(func): def inner_func(): print("Y"*20) #Printing Y 20 times func() print("Y"*20) #Printing Y 20 times return inner_func 13 EV0000550 · EV0000551 · EV0000552
13. Source slide page 13 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Decorators @decorator_y #Chaining Decorators @decorator_x def func_hello(): print("Hello") func_hello() Output: YYYYYYYYYYYYYYYYYYY XXXXXXXXXXXXXXXXXXX Hello XXXXXXXXXXXXXXXXXXX YYYYYYYYYYYYYYYYYYY 14 EV0000553 · EV0000554 · EV0000555
14. Source slide page 14 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Decorators Callback vs. Closure vs. Decorators • Callback – a function that is passed as an argument to other function. • Closure - a function object that remembers values in enclosing scopes even if theythey areare notnot presentpresent inin memory.memory. ItIt implementsimplements DataData Encapsulation(DataEncapsulation(Data hiding). • Decorators - a way to modify the behavior of a function without directly changing its source code. It allows adding functionality to an existing function by wrapping it with another function. 15 EV0000556 · EV0000557 · EV0000558
15. Source slide page 15 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Decorators Functions Decorators: Summary • A Decorator is just a function that takes another function as an argument and extends its behavior without explicitly modifying it. • Decorators allow us to wrap another function in order to extend the behavior of wrapped function, without permanently modifying it. • Using decorators, we can extend the features of different functions in a common way. 16 EV0000559 · EV0000560 · EV0000561
16. Source slide page 16 (extracted text): THANK YOU Departmentp of Computerp Science and Engineeringg g Dr. Shylaja S S, Director, CDSAML & CCBD, PESU Prof. Sindhu R Pai – sindhurpai@pes.edu Prof. Sowmya Shree P EV0000562 · EV0000563 · EV0000564

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.932] Hello all, welcome back. Python for computational problem solving. In today's class, we are going to be looking at the concept called as decorators. [00:00:13.956] In order to understand uh decorators, there are a few other things that are required. And uh just to remind you, these things have already been the concepts that I'm going to be doing now have already been covered in the previous videos. I kindly request you to check. The ones that are required um uh are [00:00:32.224] Will be uh executed by me now so that it is going to be easy for you to understand the concept of decorators. So let's get started. Let me create a function called as uh [00:00:47.300] Okay, a function F1 in this case, and I want to pass some message to this particular function. And what I really want to do with this is to print this message. [00:00:59.876] Just fine. So similarly, what I will try to do is I will call this function with uh something called as hello. Let it be as simple as possible. Fine. So now header is processed until you call a function, body of the loop will never get executed. Fine. So now [00:01:19.428] Let's quickly execute this. [00:01:24.772] Right, decorators.py and if you can see hello is what we are getting. I will also do something like this, F2 is equal to F1, right? And I will say F2, call this F2, hello decorators. [00:01:45.252] The message that I'm trying to pass now is call as decorator. Fine. Let's quickly execute this and check it out. And if you can see, hello decorators. At this point [00:01:56.672] Uh F1 and F2 are pointing to the same object internally. So that's a reason why you're trying to, even though you're calling with F2, still you're able to access this particular function. Okay. So let's quickly uh go ahead with the other concept. Let's let me comment this. [00:02:17.284] in order to make our life easy. Okay? So what is the next thing that I'm going to be doing is when we uh passed the arguments to a function, we passed some message and all. Now what we will do is we will write a simple code where I am going to be passing [00:02:34.064] one more function as an argument to the other function. Okay? Function as an argument at this particular point. Let me uh define define a function increment [00:02:51.044] Hope it is okay. So then followed by what exactly I really want to do with this. So now in this case I will say a return [00:03:03.268] x plus one because the uh incrementation operation and I will write yet another function called as decrement. [00:03:13.060] Fine. And for the decrement also, I will pass the value, any value that of your choice. So I will say return x minus 1. Fine. Now let me create one more function called as operation. [00:03:30.788] Right. And for this particular function, I will be passing one function as its argument along with it, and I also need to specify what is the [00:03:43.088] value that I am going to be giving. So it's going to be x in this particular case. Just look at what I'm going to be doing now. Now I will call this particular function with this value. [00:03:56.900] Fine and I will retain its return value in this case. [00:04:05.508] So I will say this. Fine. And then followed by this. Say return. [00:04:18.916] The result. Return results. [00:04:31.076] This is uh uh fine for you. Hope there are no confusions here. This is uh clear? So let me draw it. [00:04:44.292] operations of let me call increment and the value that I will give is ten. [00:04:54.628] Let's um quickly uh look into this to find out what is the answer that we are going to be getting because as there is no print statement, let me make use of print so that it gives me the required value. Okay.

EV0000066 · EV0000067 · EV0000068 · EV0000069

Transcript block 2 (source interval 000300-000599 seconds): [00:05:09.476] We quickly execute. If there are any errors in this program, let's try to identify what is an error in this case. Fine. And if you see there are no errors, and we are getting the uh result, which is 11. So now let me quickly help you out what exactly has happened. Now [00:05:28.996] Body is processed, sorry, header is

processed every time, sorry, it's a header. Okay, second header gets processed, third header gets processed, fine, and there is a function call. The function call is for this particular function. Now in this case, what are you trying to pass it as an argument to this particular function? [00:05:49.092] An increment. Now what is an increment? An increment is yet another function. Is this fine? And this x fine will get the value called as 10. So now we are trying to say increment of 10. [00:06:03.684] Fine, because this function is increment and x value is 10. Now you're telling increment of 10. What it gets returned, it returns x plus 1. [00:06:14.628] Fine. So now the result is having what? X plus 1. You are returning this particular result. So now to whom you are returning? You are returning it to this place where it was called. So now what it has returned? It has returned 11, and then you are trying to print a value called as 11. Okay. [00:06:32.400] Okay, so in this case what we have basically done is we have used a function and for this function we have passed one more function as its argument. Okay, hope this is uh clear to you all. [00:06:51.268] Fine. It's all about uh just understanding how things are going on, nothing uh uh much, just um [00:07:00.548] Okay. So now let me write one more function here. Define one more function which is going to be called. Fine, a function called. [00:07:18.692] Then followed by I will define one more function, function returned. [00:07:28.100] Hope this is fine for you. Now I'll see and [00:07:36.964] Is it confusing? [00:07:45.188] The question mark because I don't know. I'm unable to identify whether you're able to uh understand or not. So I will say this. Fine. And I will say return. Function returns. Function underscore returned. [00:08:06.500] Hope uh this is fine for you. Fine. I'll just come back here and then say something new is equal to the function that is called. [00:08:22.468] What is the output that it gives us? And now I will say new of this. [00:08:33.796] That's fine. Function call. Let's find out. These things have already been done up to you. So you don't need to um get confused with this. Okay. You're calling a function, I get into the uh function. [00:08:48.420] Okay, this is an header that gets processed. The moment the headers are processed, what is the next thing that I have to do? I'll have to wait for the function to be called. So now in this case, a function is being called. So I will go here. [00:09:01.284] And then I will process this particular function. Will you get it into the body of this function return? No, I won't get it until it has been called. Now, instead of me calling, I am returning this particular function. The object has been returned. Where it is returned, it is returned. [00:09:19.056] To this particular point. Now, what does it have returned? It has returned the address where this particular function is available. So I am using that for address in order to call the function. So now what did we do? [00:09:33.636] Where uh this program basically illustrates that a function can return another function. Okay? A function can return another function is what we have done or we have understood in this particular example. Fine. [00:09:50.256] So let's uh go ahead and then try to understand our actual concept that is there. Our actual concept is decorator. So now what does a decorator basically do is decorator is a function.

EV0000070 · EV0000071 · EV0000072 · EV0000073

Transcript block 3 (source interval 000600-000899 seconds): [00:10:06.692] One. So this particular function, a decorator function, takes yet another function. It takes one more function. Okay, and for this particular function, it adds [00:10:23.492] Some other functionality. Let me repeat. It adds some other functionality. [00:10:34.948] Okay. Hopefully it is clear and it reser returns the result. [00:10:44.932] So what is a decorator is doing? Decorator is a function, it takes another function, it adds some other functionality with the existing functionality and gives you the result. It is as similar as you wrapping a gift. [00:11:00.420] Fine. You have a gift inside a box. What you basically

do is you try to decorate your gift box in such a way that it is so nice. Fine? Whatever you try to do, you're trying to do it for your gift box and the object within that gift box remains the same. [00:11:16.352] This is fine. So same thing we are also trying to do here. Is this fine? Um so let's quickly try to understand uh by writing a program for the decorator. Fine. So in general, um to make it very simple, what exactly is a decorator? [00:11:34.212] You add some functionality to the existing code. Fine. The original code can be reused again and again according to your requirement because you are trying to use the same function, but add a few other things around it. [00:11:48.260] That's called as a decorator. Fine. So now quickly uh we will code in order to understand uh the decorator. [00:11:55.524] Fine. Let me remind you, I'm taking uh uh the same example uh which is available on the internet itself because this example is very nice to understand. This example is from the internet material itself. [00:12:10.852] So let me uh take up this. I say define make a pretty [00:12:20.612] [ASR low confidence; check audio] Right? So. [00:12:24.964] Let me repeat, this is from the um internet example itself. [00:12:33.124] So I will see and then uh followed by let me define one more function, inner function. [00:12:47.780] [ASR low confidence; check audio] We do this. Let's say print. [00:12:58.052] I look nice after decoration. [00:13:07.204] I look nice after decoration. So now what I will do is I'll say a function and call this okay. This is my function. I'm going to be calling this function. So here [00:13:26.724] We'll see return function. [00:13:35.012] Okay, return inner function. Hope uh this is uh clear to you. So let me uh write one more code. This is not confusing you. [00:13:53.476] So let me define uh one more function for you. Define ordinary [00:14:00.804] So now in this case [00:14:05.988] I think then. [00:14:09.796] Let's see, I look very simple. [00:14:21.924] What do I do? Nothing else. Okay. So now what we will do is we'll say ordinary. We'll try to call the function. Find out are there any errors in my program. So I'll just call ordinary as you know. [00:14:37.924] Header gets processed, body never gets in, this header gets processed, but I am trying to call this function. Okay. So what is great in this at this point? No, nothing. Just to find out are there any errors in our program. [00:14:51.556] Fine. And it says that I look very uh simple. Fine. So now what is the thing I'm going to be doing is anyway, I don't need this. Let me comment or uh let me remove this so that you will not have any confusions for you. Okay. So next thing that I'm going to be doing is pretty.

EV0000074 · EV0000075 · EV0000076 · EV0000077

Transcript block 4 (source interval 000900-001199 seconds): [00:15:12.964] Okay, make a pretty [00:15:21.220] And for this I possess an ordinary. [00:15:26.564] I pass on ordinary. This is fine. Clear? Hope uh this is fine. Don't get confused, so I'll see pretty. [00:15:38.532] Okay, so let's uh quickly execute this. Don't worry, I will show it on Python Tutor in order to understand. Okay, so now let me uh quickly execute it to you. Fine. So I look nice after declaration. I look very simple now. [00:15:57.024] Okay, so now what has happened? You can uh change the statements. I look very simple now. After decoration, I look uh beautiful, all those things you can uh just do the changes and then uh work it out. Right? So now [00:16:10.464] What are we trying to do? We are trying to call make pretty. This is a place where decoration is taking place. Is this fine? This is a place where decoration is taking place. So I say make pretty fine. Make pretty, you're calling a function, and then you call ordinary. [00:16:29.488] Is this fine? So you're passing ordinary. At this point, what does function contains? Your function contains ordinary. This is a point where your ordinary function has been called. Is this fine? So now you're trying to call the function, you process this header, you don't get into this, but you [00:16:47.152] Get a return address. So now where does a return address is present? The return address is present at pretty. [00:16:54.352] Fine.

Now what am I trying to do with this return address? I am calling the function. So now what does pretty has? Pritty has inner function. So now you're calling the inner function. The moment you call the inner function, what I'm doing trying to do, I look nice after decoration. When you uh and then [00:17:12.016] And followed by you have a function here. So now the moment you have a function, which function I'm talking about? Ordinary function. So I say I look very simple now. Previously I was looking very simple. After decoration that took place, I look [00:17:28.324] beautiful. Okay so that is what it is. Now these two statements are making your ordinary function look beautiful. So this is the concept of decorators. [00:17:42.656] Okay. So hope this is clear. There is a few other things. We'll I'll just quickly uh do this. Let me eliminate these two sentences. Right? I'm not going to be uh using this. Fine. I'll just call ordinary. [00:18:00.004] Okay, I'll just call ordinary. This is fine. Nothing else I'm calling. The moment I call ordinary, it becomes like I look very simple. But now did I not eliminate the concept of decorator here? Yes, I have eliminated. Now which was a function that was uh making [00:18:18.884] An ordinary function pretty the make a pretty function. Okay. So now yet another way of writing a decorator is I can use at the rate symbol, and which was the uh function that was making me uh pretty, a make pretty function. [00:18:35.968] So let me use make underscore pretty. [00:18:42.180] And I'm making this as what? A prettier one. Okay. So now previous two statements that eliminated which can be uh which is equivalent to this particular statement. [00:18:56.324] This is fine. The last two statements that I eliminated, which is equal to these two. Okay. So let's quickly execute this and find out. One second, give me uh time. Yeah. So I have to call the function. Yes. So I have to call [00:19:16.100] Otherwise it just becomes an header processing without a function being called. Okay? And if you can see I look nice after decoration, I was looking very simple. I look very simple now. [00:19:29.072] Hope uh this concept is clear to you. Previously, if you have seen our output remained the same. Right? The last two statements I eliminated and uh uh this has been uh uh [00:19:42.628] Be placed with at the rate symbol followed by make written. So as I promised, I'm going to take you to the tutor and then help you out. Okay let me paste this code. [00:19:59.780] Fine. Fourteen lines are there. Fine. Let's quickly get started. Function was processed. Fine. Make pretty. Define the inner function. Return. Fine. The moment you return, what are you trying to do?

EV0000078 · EV0000079 · EV0000080 · EV0000081

Transcript block 5 (source interval 001200-001499 seconds): [00:20:19.748] Where it has returned, it has returned to ordinary. Then now go back to where it has called. Then I look nice after decoration. And you're calling an ordinary function here. You come back here and then print I look very simple now. [00:20:38.660] Okay. So this is what uh the concept of uh decorators, right? Hope you have understood. Um in the uh [00:20:47.584] theory class when we do we will be looking uh into deeper into these things uh in order to understand this concept in a better way. Hope you are all uh fine over this. I request you to uh check it take out few more programs and uh work on the same. Thank you.

EV0000082 · EV0000083 · EV0000084 · EV0000085

Full transcript: 2769 normalized words, 91 timestamped segments, 1267.882667 seconds duration. Low-confidence segments flagged: 2. Review locators: 00:12:20.612, 00:12:47.780

EV0000066 · EV0000067 · EV0000068 · EV0000069

9 Generators

This lesson is indexed from the source video titled Generators. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000086 · EV0000087

Graphs and diagrams

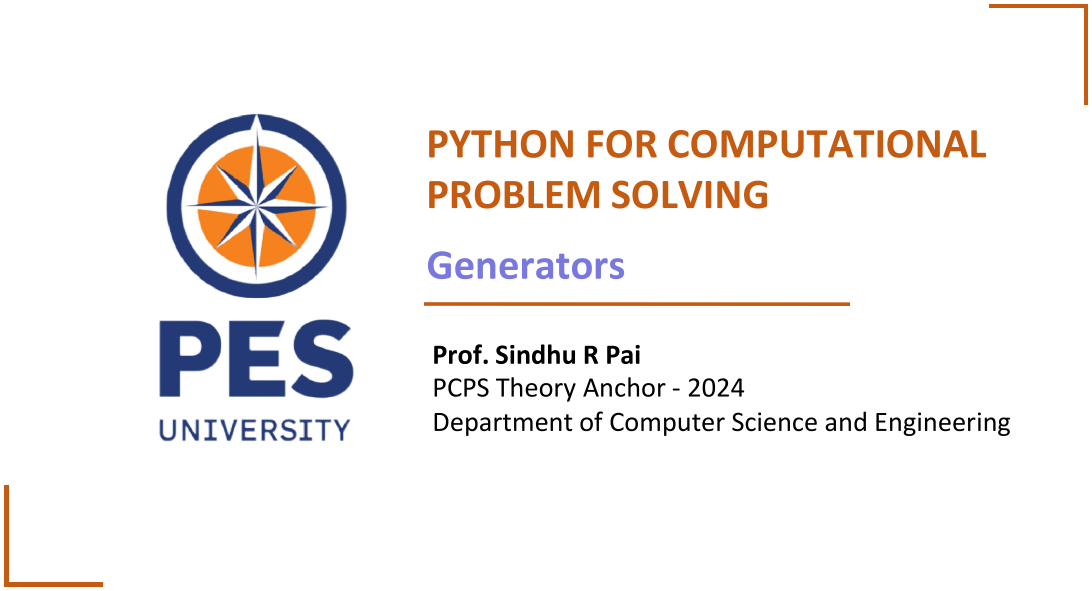

Figure G000150. UE24CS151A PCPS Unit 3 Sindhu Class 7 Generators · source slide page 1.

EV0000566

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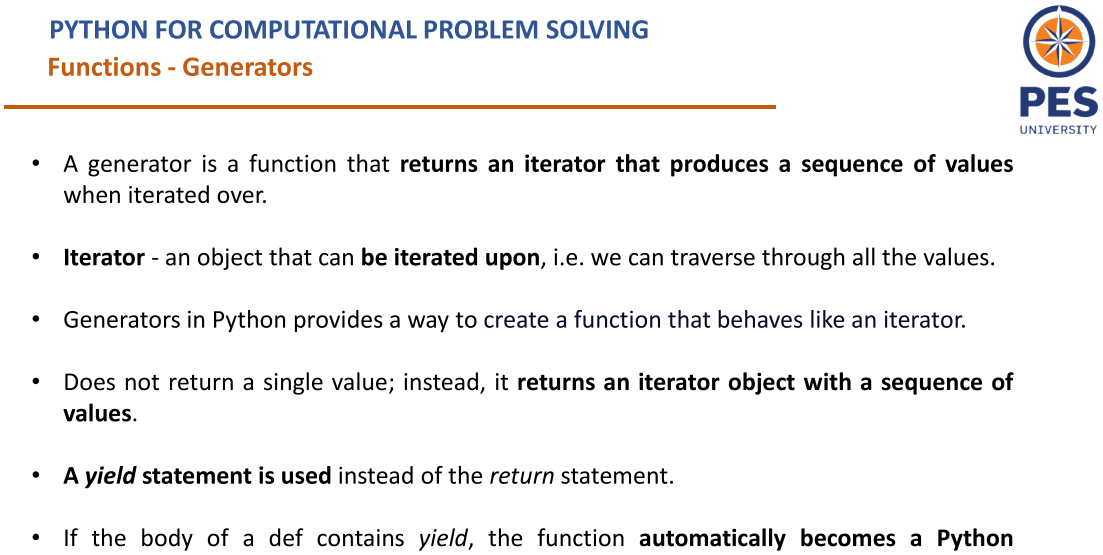

Figure G000151. UE24CS151A PCPS Unit 3 Sindhu Class 7 Generators · source slide page 2.

EV0000569

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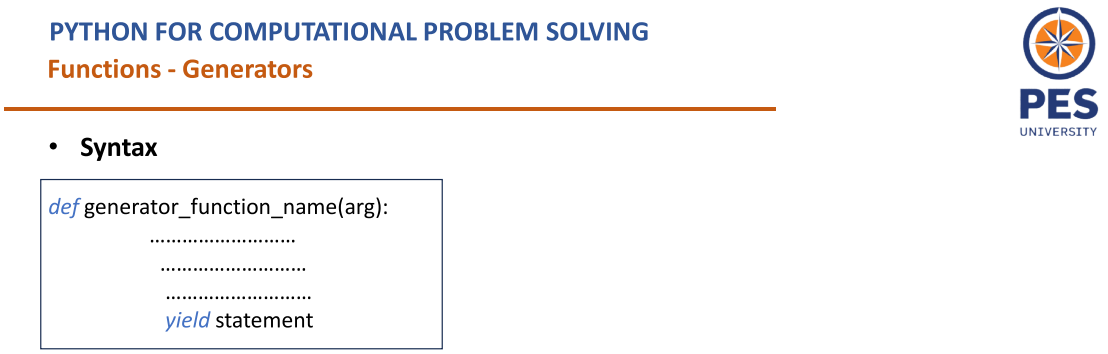

Figure G000152. UE24CS151A PCPS Unit 3 Sindhu Class 7 Generators · source slide page 3.

EV0000572

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING
Functions - Generators

PES UNIVERSITY

Generator Object

- Python Generator functions return a generator object that is iterable (used as an iterator).

- Generator objects are accessed
  - by calling the next method of the generator object (Refer Example 2)

    or
  - using the generator object in a “for” loop (Refer Example 1)

Figure G000153. UE24CS151A PCPS Unit 3 Sindhu Class 7 Generators · source slide page 4.

EV0000575

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

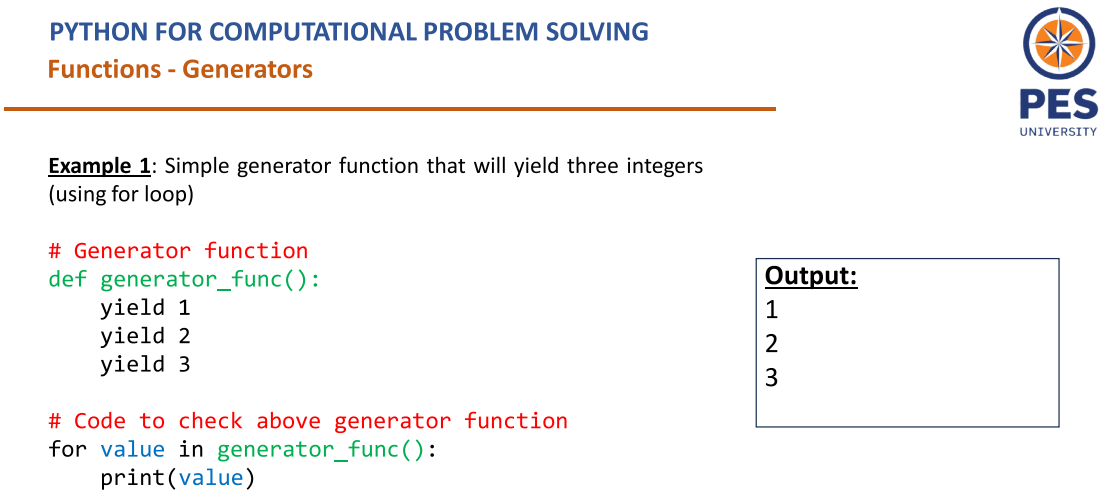

Figure G000154. UE24CS151A PCPS Unit 3 Sindhu Class 7 Generators · source slide page 5.

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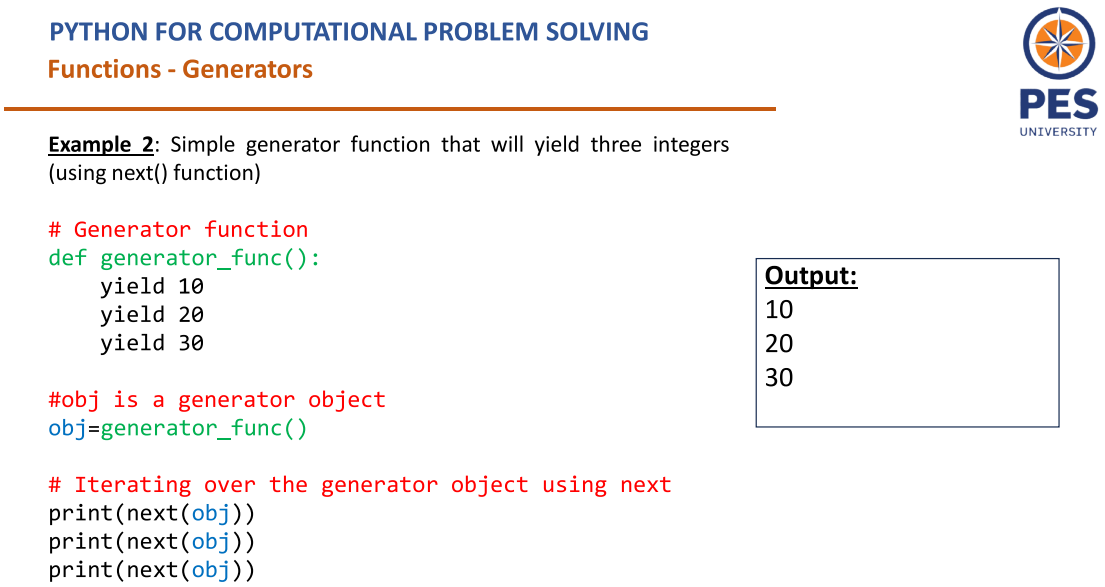

Figure G000155. UE24CS151A PCPS Unit 3 Sindhu Class 7 Generators · source slide page 6.

EV0000581

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Functions - Generators

PES UNIVERSITY

Generator Expression

- Generator expression is another way of writing the generator function.
- Similar to list comprehension technique but instead of storing the elements in a list in memory, it creates generator objects.
- **Syntax**:

```
(expression for element in iterable)
```

Figure G000156. UE24CS151A PCPS Unit 3 Sindhu Class 7 Generators · source slide page 7.

EV0000584

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

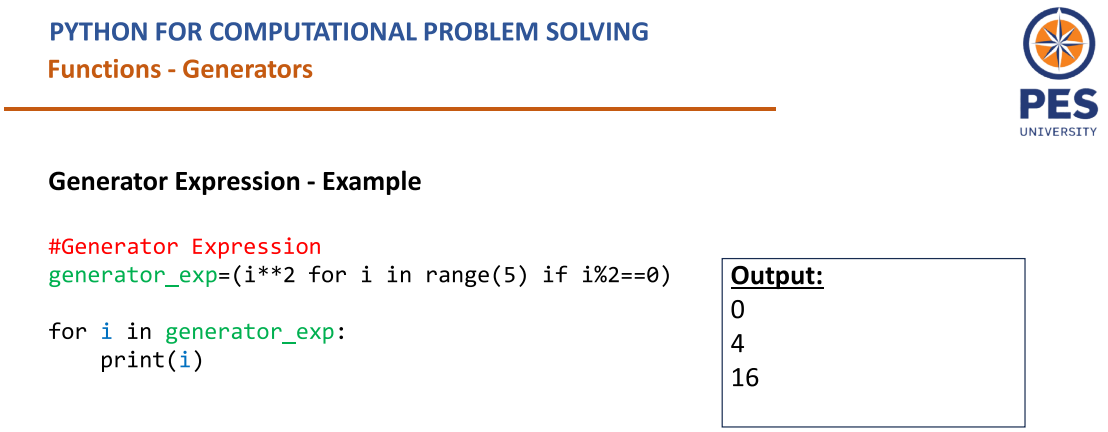

Figure G000157. UE24CS151A PCPS Unit 3 Sindhu Class 7 Generators · source slide page 8.

EV0000587

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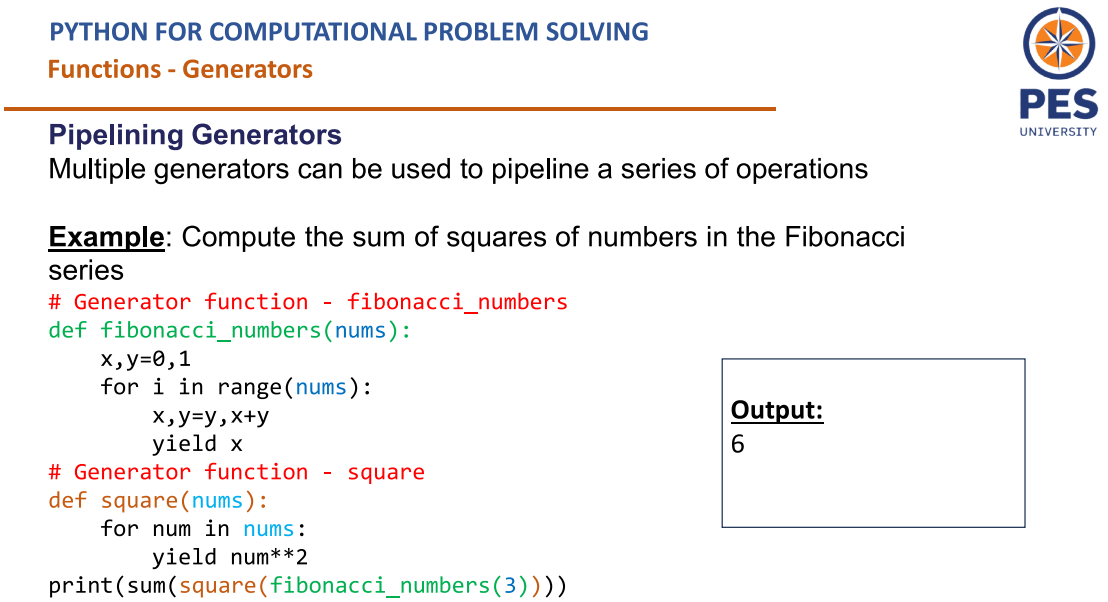

Figure G000158. UE24CS151A PCPS Unit 3 Sindhu Class 7 Generators · source slide page 9.

EV0000590

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Functions - Generators

PES UNIVERSITY

Function Generators: yield vs. return

yield	return
Returns a value and pauses the execution while maintaining the internal states	Returns a value and terminates the execution of the function
Used to convert a regular Python function into a generator	Used to return the result to the caller statement
Used when the generator returns an intermediate result to the caller	Used when a function is ready to send a value
Code written after yield statement execute in next function call	Code written after return statement won't execute
It can run multiple times	It only runs a single time

Note: We can't include *return* inside a generator function. If we do, it will terminate the function.

Figure G000159. UE24CS151A PCPS Unit 3 Sindhu Class 7 Generators · source slide page 10.

EV0000593

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING
Functions - Generators

PES UNIVERSITY

Function Generators: Summary

- Python generator functions allows for the declaration of a function that behaves like an iterator, making it a faster, cleaner and easier way to create an iterator.
- Generators are useful when we want to produce a large sequence of values, but we don't want to store all of them in memory at once.
- The simplification of code is a result of generator function and generator expression support provided by Python.

Figure G000160. UE24CS151A PCPS Unit 3 Sindhu Class 7 Generators · source slide page 11.

EV0000596

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

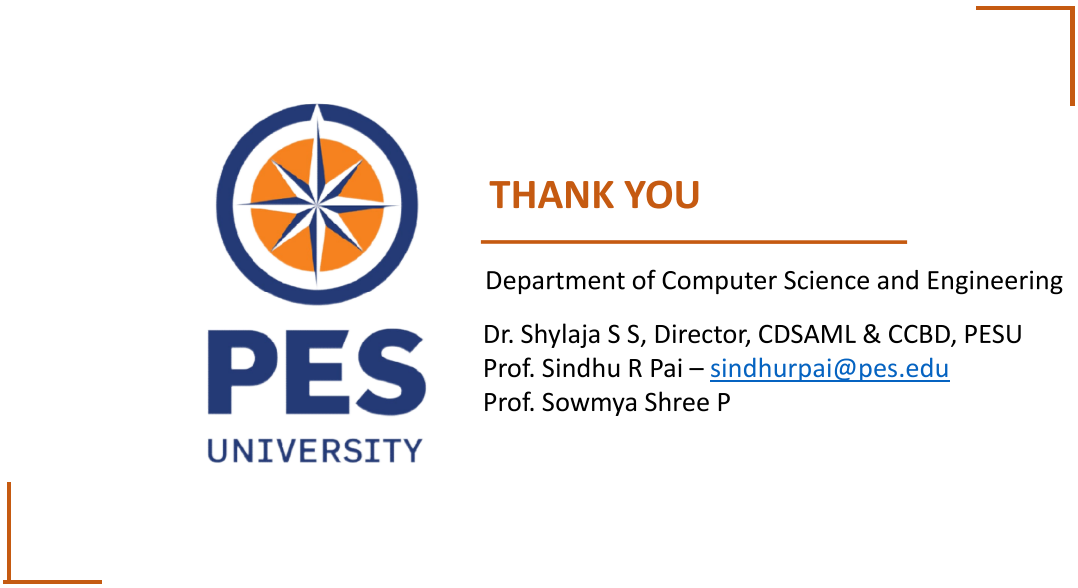

Figure G000161. UE24CS151A PCPS Unit 3 Sindhu Class 7 Generators · source slide page 12.

EV0000599

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Generators Prof. Sindhu R Pai PCPS Theory Anchor - 2024 Department of Computer Science and Engineering EV0000565 · EV0000566 · EV0000567
2. Source slide page 2 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Generators • A generator is a function that returns an iterator that produces a sequence of values when iterated over. • Iterator - an object that can be iterated upon, i.e. we can traverse through all the values. • Generators in Python provides a way to create a function that behaves like an iterator. • Does not return a single value; instead, it returns an iterator object with a sequence of values. • A yield statement is used instead of the return statement. • If the body of a def contains yield, the function automatically becomes a Python generator function. EV0000568 · EV0000569 · EV0000570

3. Source slide page 3 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Generators • Syntax def generator_function_name(arg): .... yield statement • When the generator function is called, it does not execute the function body immediately. Instead, it returns a generator object that can be iterated over to produce the values. EV0000571 · EV0000572 · EV0000573
4. Source slide page 4 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Generators Generator Object • Python Generator functions return a generator object that is iterable (used as an iterator). • Generator objects are accessed • by calling the next method of the generator object (Refer Example 2) or • using the generator object in a “for” loop (Refer Example 1) EV0000574 · EV0000575 · EV0000576
5. Source slide page 5 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Generators Example 1: Simple generator function that will yield three integers (using for loop) # Generator function def generator_func(): Output: yieldyield 11 11 yield 2 2 yield 3 3 # Code to check above generator function for value in generator_func(): print(value) EV0000577 · EV0000578 · EV0000579
6. Source slide page 6 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Generators Example 2: Simple generator function that will yield three integers (using next() function) # Generator function def generator_func(): Output: yield 10 yield 20 1010 yield 30 20 30 #obj is a generator object obj=generator_func() # Iterating over the generator object using next print(next(obj)) print(next(obj)) print(next(obj)) EV0000580 · EV0000581 · EV0000582
7. Source slide page 7 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Generators Generator Expression • Generator expression is another way of writing the generator function. • Similari il to lilist comprehensionh i techniqueh i bbut iinsteadd off storingi theh elementsl iin a list in memory, it creates generator objects. • Syntax: (expression for element in iterable) EV0000583 · EV0000584 · EV0000585
8. Source slide page 8 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Generators Generator Expression - Example #Generator Expression generator_exp= (i**2 for i in range(5) if i%2==0) Output: 00 for i in generator_exp: 4 print(i) 16 EV0000586 · EV0000587 · EV0000588
9. Source slide page 9 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Generators Pipelining Generators Multiple generators can be used to pipeline a series of operations Example: Compute the sum of squares of numbers in the Fibonacci series ## GeneratorGenerator functionfunction - fibonaccifibonacci_numbersnumbers def fibonacci_numbers(nums): x,y=0,1 for i in range(nums): x,y=y,x+y Output: yield x 6 # Generator function - square def square(nums): for num in nums: yield num**2 print(sum(square(fibonacci_numbers(3)))) EV0000589 · EV0000590 · EV0000591

10. Source slide page 10 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Generators Function Generators: yield vs. return yield return Returns a value and pauses the execution Returns a value and terminates the while maintaining the internal states execution of the function UsedUsed toto convertconvert aa regularregular PythonPython UsedUsed toto returnreturn thethe resultresult toto thethe callercaller function into a generator statement Used when the generator returns an Used when a function is ready to send a intermediate result to the caller value Code written after yield statement Code written after return statement execute in next function call won't execute It can run multiple times It only runs a single time Note: We can't include return inside a generator function. If we do, it will terminate the function. EV0000592 · EV0000593 · EV0000594
11. Source slide page 11 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functions - Generators Function Generators: Summary • Python generator functions allows for the declaration of a function that behaves like an iterator, making it a faster, cleaner and easier way to create an iterator. • Generators are useful when we want to produce a large sequence of values, but we don't want to store all of them in memory at once. • The simplification of code is a result of generator function and generator expression support provided by Python. EV0000595 · EV0000596 · EV0000597
12. Source slide page 12 (extracted text): THANK YOU Departmentp of Computerp Science and Engineeringg g Dr. Shylaja S S, Director, CDSAML & CCBD, PESU Prof. Sindhu R Pai – sindhurpai@pes.edu Prof. Sowmya Shree P EV0000598 · EV0000599 · EV0000600

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.060] Hello, all, welcome back. In today's class, we will understand the concepts of generators. [00:00:14.244] Uh in the previous class uh we have uh built our own i traitors. Okay so in order to build an iTrader uh let me give some comments here. Previous class previous class, we have learnt iTraders. [00:00:32.644] Okay, previous class we have learned and and we have built our own iTraders. In order to uh do this or implement our own iTraders, we had to uh use ITE. [00:00:46.500] Okay and then uh followed by we were also forced to make use of a next method. Okay so next method [00:00:57.732] I have to make use of uh next methods. Okay. Um and I have to keep track of the internal states. The reason is the moment I will uh uh end [00:01:08.944] All the values that are present in a particular iterable, I have to stop. Otherwise, what do I need to do? I need to explicitly raise a stop iteration in order to indicate that there are uh no more values that need to be returned. So in such case, uh this is definitely. [00:01:26.672] I have to implement a class, I have to make use of an iter, I have to make use of a method function, then take care of the [00:01:37.824] um count because the moment uh there are no more values to be returned I have to raise a stop iteration. Okay so now is there any method uh which makes our life easier so in such case generators comes into our rescue. [00:01:56.324] Okay, so generators are definitely helpful when I want to um work uh a easier way of building our own i traitors. Okay. Uh we basically make use of a generators in order to uh build uh itrators in a simpler way. [00:02:15.652] Okay what we have already mentioned, okay, uh in the previous class with respect to the iterators. But all these things are uh automatically handled by the generators in [00:02:32.784] Python okay. So let's try to understand how exactly uh I can uh build my own iterators using generators. Okay, so let's uh quickly uh get into this. Let me uh write a simple function, define uh [00:02:51.716] My generator. Okay. I'll say it as what? My generator.

Fine. Nothing more in this case. Fine. I'll just say it as N is equal to one. Let me write the program and then I'll explain it to you. The print [00:03:11.300] This gets printed first. [00:03:18.116] Okay. So this is the first statement that gets printed first. And I will use one new statement called as yield. [00:03:31.108] Okay, I'll use a statement called as an yield. And I'll say it as what uh yield one. Okay. So now when we say [00:03:42.628] That we want to implement generators, definitely. It is very easy because I'm going to be creating my function the way in which we have done now. Okay. Only difference is when I use my uh regular functions, I would use a statement called as return. [00:04:01.872] Okay, but if I want to create my own generators, instead of return, I will make use of something called as an yield statement. Yield represents a generator, return represents a particular function. Okay? If you want [00:04:22.020] To create your own generators, then it should must and should contain at least one yield statement. Okay? If you find an yield statement, then it is easily noticeable that it is your generator function. [00:04:38.084] Okay, now yield as well as return will return some value for you. Yield as well as return gives you some value from a function. The only difference is the moment you come across with a return statement. [00:04:55.024] You're talking about termination of the function entirely. You come out of the function, okay, exit out of the function completely. But with respect to your yields, you're pausing the function.

EV0000086 · EV0000087 · EV0000088 · EV0000089

Transcript block 2 (source interval 000300-000599 seconds): [00:05:08.464] You're going to be saving all its state and then continue its execution. So let me repeat. Return is basically for termination. [00:05:20.288] is to store internal storage, internally store the state where you were basically before. Okay. So let's uh execute this program and then try to understand. Okay. So now I'll try to do n is equal to n plus one [00:05:39.908] Okay, and then I'll say uh if you don't have any issue, I'll copy paste this. Sorry. Okay so let me copy paste this gets printed. Let me change my statement and say second. [00:05:58.692] Okay, so this is the one that is getting printed second. So let me uh do the same thing. [00:06:14.788] Do this gives so [00:06:21.060] Okay. So I say this. [00:06:26.948] Now I'll say n is equal to n plus one itself. Okay. So let me say it as third or fourth or what? I'll say it as last. Okay. This gets printed last and then followed by eight n. [00:06:41.828] And then uh let's call our function a is equal to my generator. Okay, so I'll say it as my generator. Okay so now what I will do is I will try to make use of your [00:06:59.236] Print. I can do this. Print next of A. Okay. So let me just execute this to uh help you out. [00:07:14.500] Unprompts. [00:07:20.900] Ready with this. Python generators.py. See, this gets printed first and first statement is printed. Suppose if I do this. [00:07:33.668] Okay, what is what needs to be printed? One and two needs to be printed. Okay, so hope you have understood what exactly yield is trying to do. Okay, yield statement. [00:07:47.376] pauses the function, saves its states, and later continues from where [00:07:53.956] Uh there is a successive call continues from there to the successive calls. Okay. I will s uh the moment you come across with the yield function here, the state saving everything will happen. It continues [00:08:09.572] Till some point until it come across comes across us with the next yield statement. The moment you come across with the next yield, it returns a value for you. I'll start from here, go till here. Again I have an yield statement. You have to stop at this point. [00:08:24.432] Okay, keep uh doing the same thing. Okay, but now what is the point of me using a next anyway, fine? So instead of doing that, what I can do is I can say for INA. Okay, what is a? A is your generator object, and I'll say [00:08:44.804] Print [00:08:48.708] A print A. So let's sorry, print I. For loop, right? I'm sorry. Print I. So let's uh look into this and find out what happens.

[00:09:04.964] Okay, see here, this gets printed first, this gets printed second, this gets printed last. Okay, and uh I did not use any uh iter method or next method, but still we are able to uh create our own iterators using generators. [00:09:23.760] Okay, I urge you to uh practice few more programs before that. Please try to understand the concept of classes and then come back and understand the concept of I traders. Okay. Hope you uh enjoyed today's class. Thank you.

EV0000090 · EV0000091 · EV0000092 · EV0000093

Full transcript: 1198 normalized words, 40 timestamped segments, 579.008 seconds duration. Low-confidence segments flagged: 0.

EV0000086 · EV0000087 · EV0000088 · EV0000089

10 module

This lesson is indexed from the source video titled module. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000094 · EV0000095

Graphs and diagrams

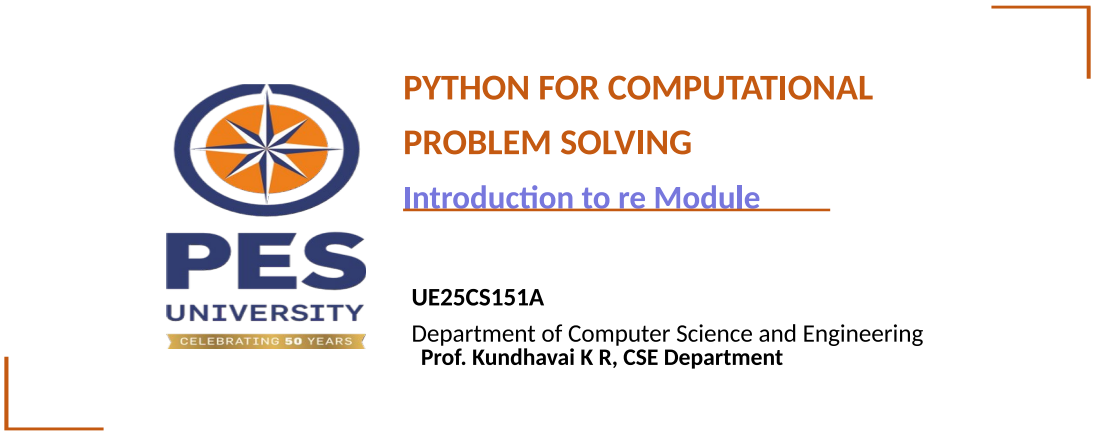

Figure G000209. Re Module · source slide page 1.

EV0000743

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

re Module

- The re module in Python provides support for Regular Expressions (regex). Regex is a powerful text-processing tool used to search, match, and manipulate patterns in strings. Helps automate tasks like validation, extraction, text cleaning, and pattern matching. Need not use pip install, as it is a built-in (standard library) module in Python.
- A **pattern** defines what you want to search in text
- Examples:
  - `r"\d+"` → any sequence of digits
    `r"\b[Pp]\w+"` → words starting with P or p
    `r"\w{3}"` → exactly 3 characters

Why not use string methods?

- `.find()` searches exact text
- RegEx can match **flexible patterns**

Figure G000210. Re Module · source slide page 2.

EV0000746

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Key Metacharacters

Meta	Meaning	Example
`.`	Any character (except newline)	`h.t` → **hat**, **hot**
`\d`	Digit (0–9)	`\d\d` → **12**
`\w`	Word character (a-z, A-Z, 0-9, _)	`\w\w` → **hi**, **A5**
`\s`	Whitespace	`hello\sworld` → **hello world**
`+`	One or more	`\d+` → **123**
`*`	Zero or more	`ab*c` → **ac**, **abc**
`[]`	Character set	`[aeiou]` → any vowel
`^`	Start of string	`^Hello`
`$`	End of string	`world$`

Figure G000211. Re Module · source slide page 3.

EV0000749

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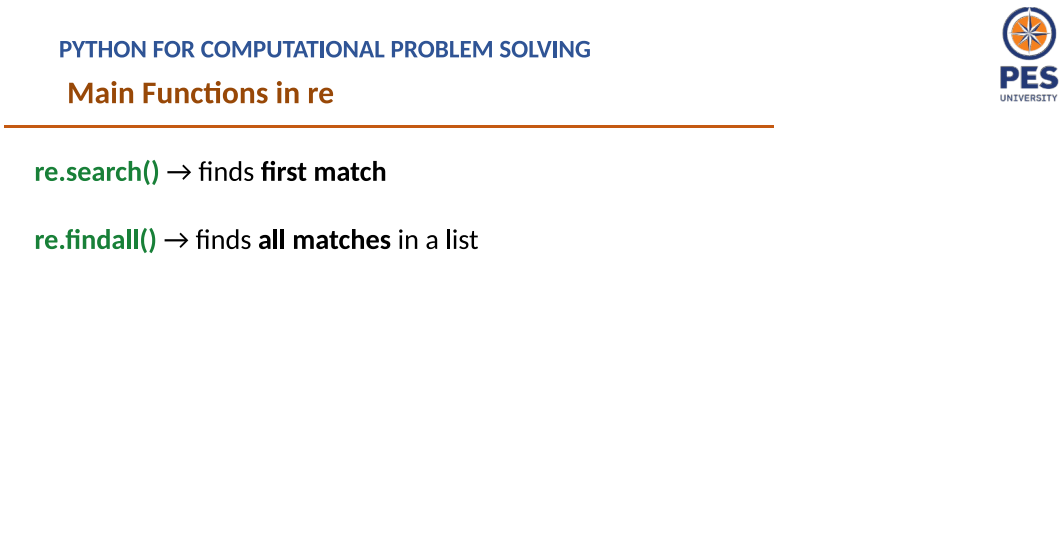

Figure G000212. Re Module · source slide page 4.

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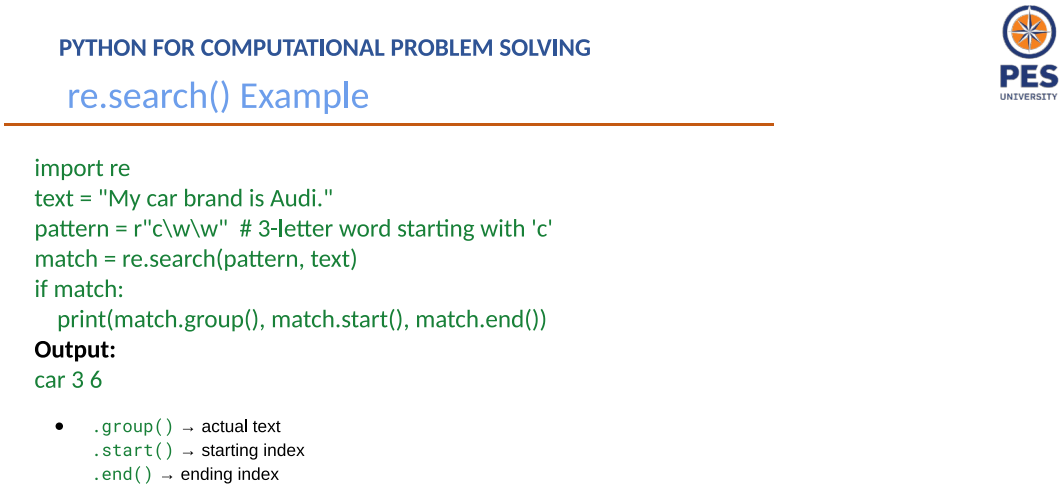

Figure G000213. Re Module · source slide page 5.

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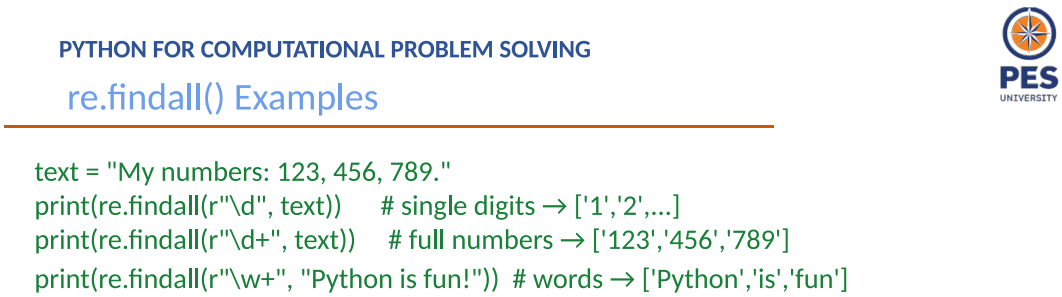

Figure G000214. Re Module · source slide page 6.

EV0000758

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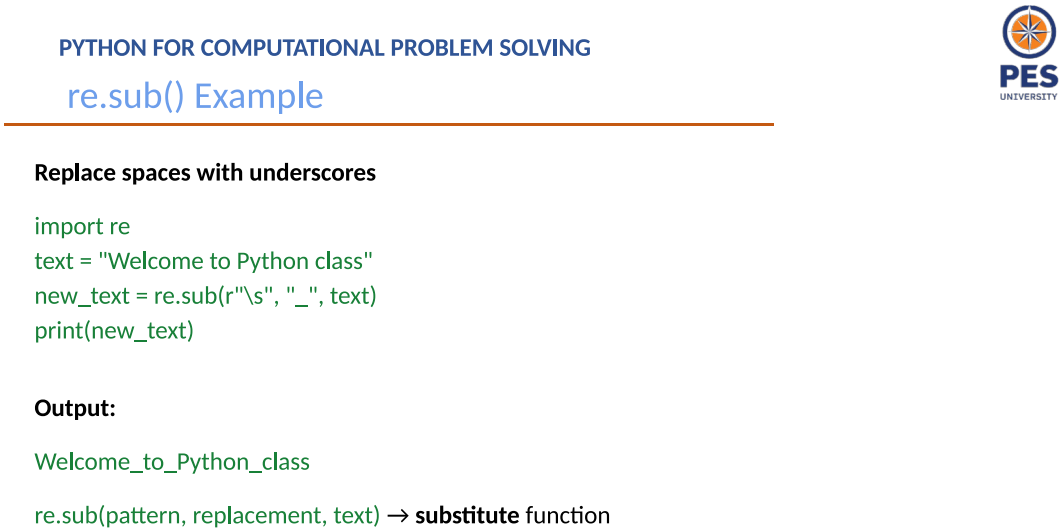

Figure G000215. Re Module · source slide page 7.

EV0000761

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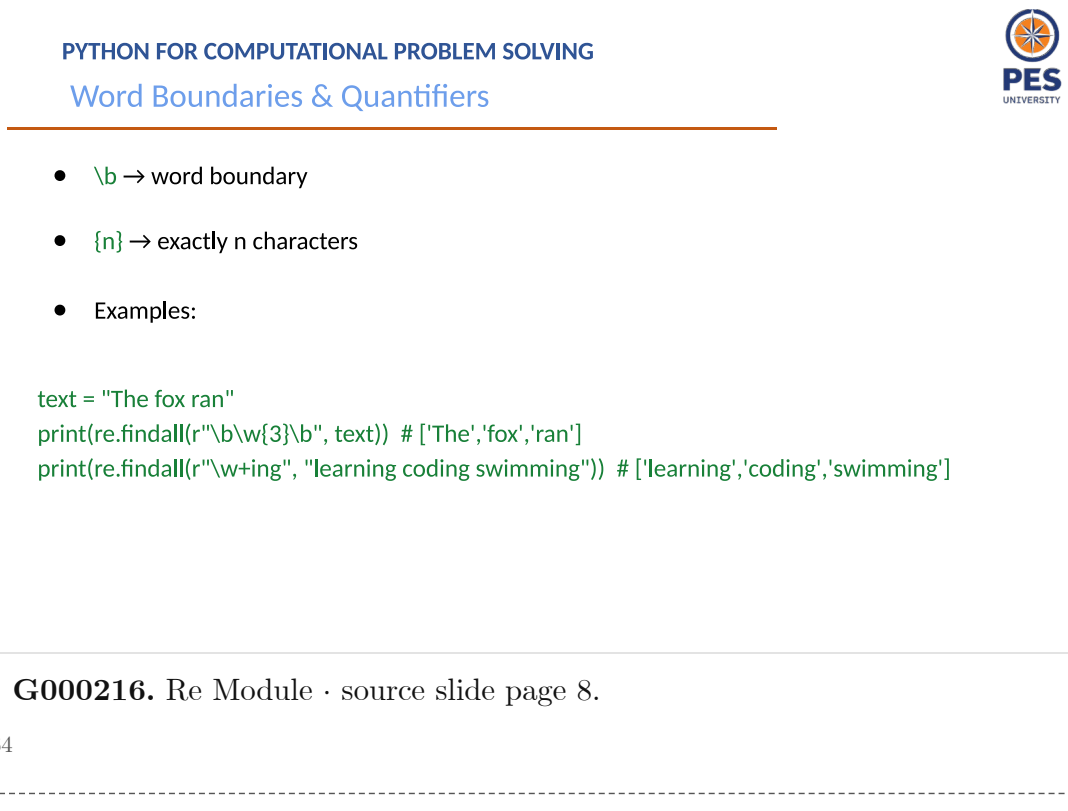

Figure G000216. Re Module · source slide page 8.

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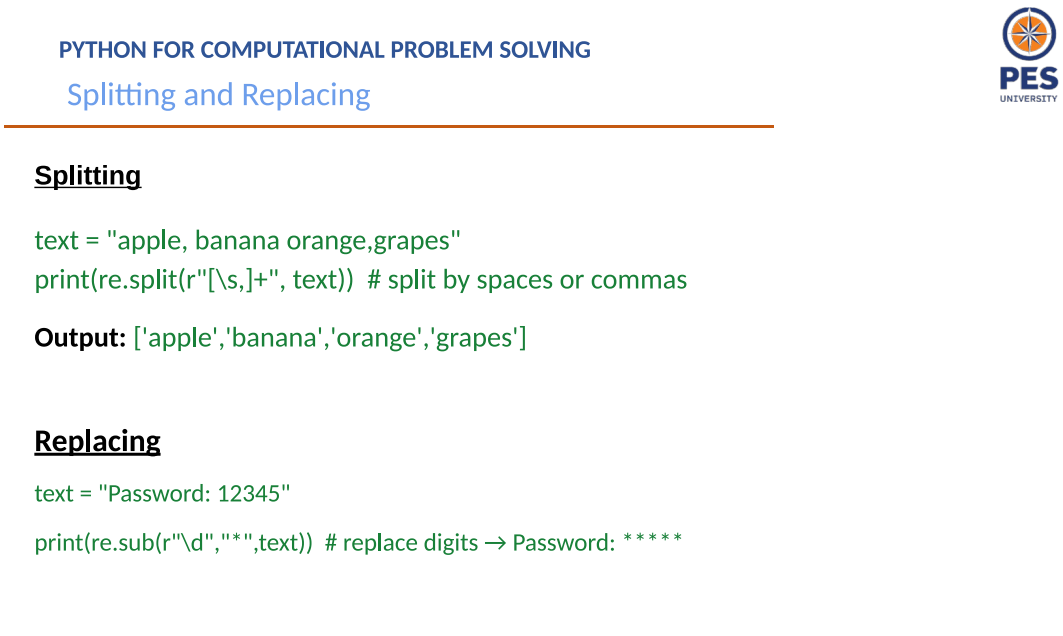

Figure G000217. Re Module · source slide page 9.

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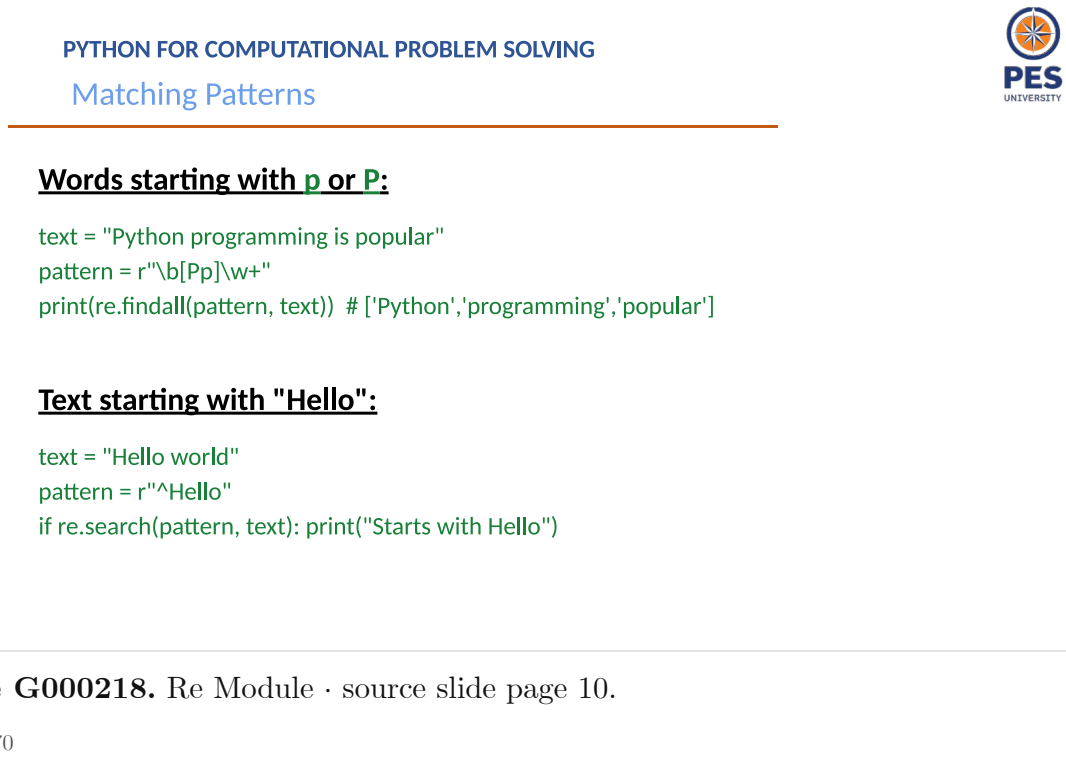

Figure G000218. Re Module · source slide page 10.

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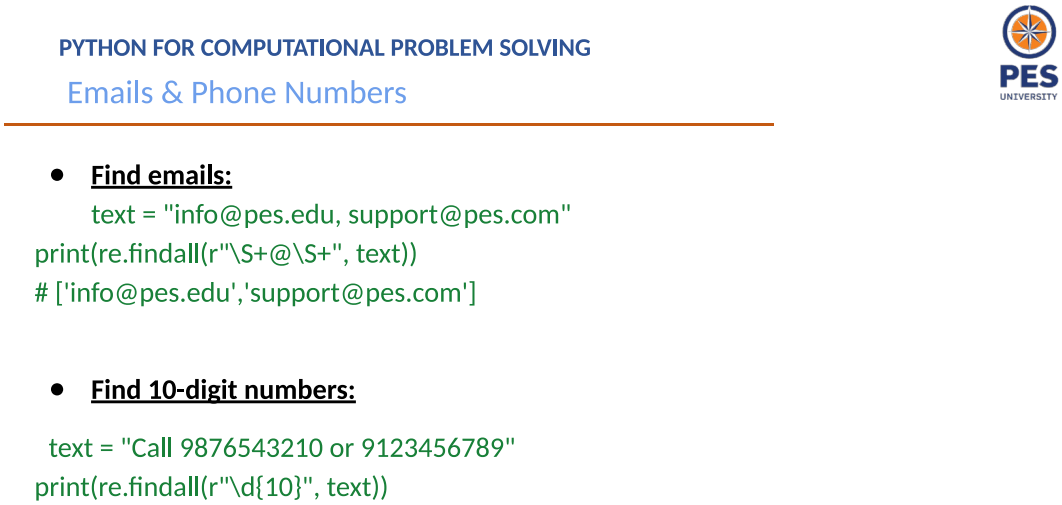

Figure G000219. Re Module · source slide page 11.

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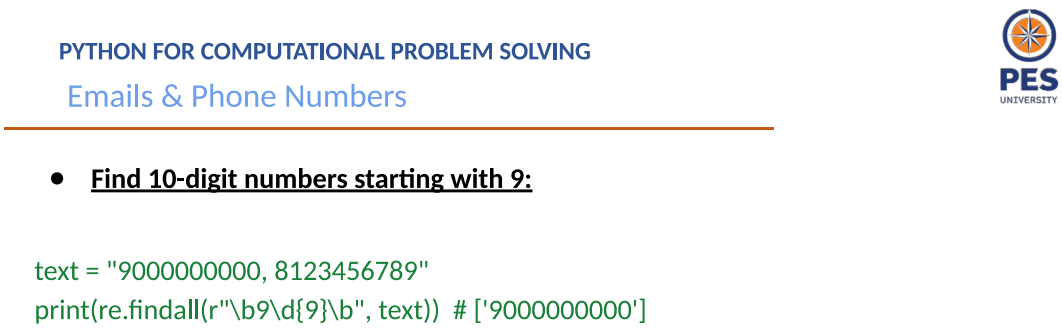

Figure G000220. Re Module · source slide page 12.

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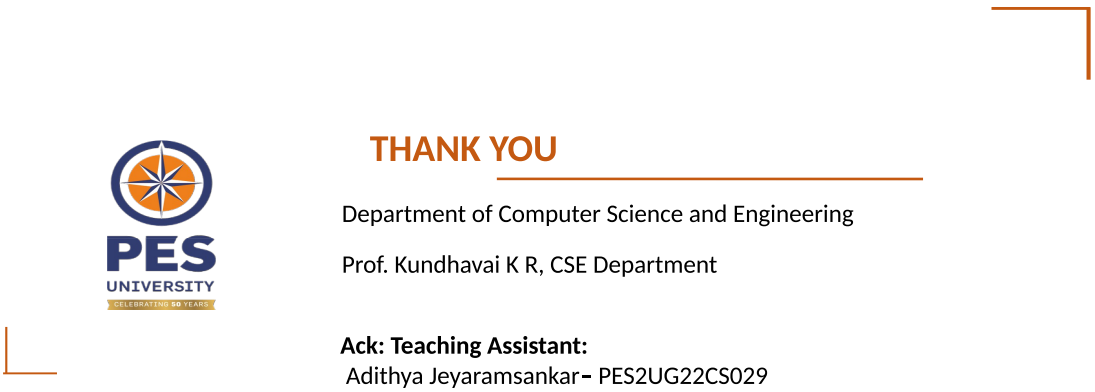

Figure G000221. Re Module · source slide page 13.

EV0000779

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Introduction to re Module UE25CS151A Department of Computer Science and Engineering Prof. Kundhavai K R, CSE Department EV0000742 · EV0000743 · EV0000744
2. Source slide page 2 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING re Module The re module in Python provides support for Regular Expressions (regex). Regex is a powerful text-processing tool used to search, match, and manipulate patterns in strings. Helps automate tasks like validation, extraction, text cleaning, and pattern matching. Need not use pip install, as it is a built-in (standard library) module in Python. A pattern defines what you want to search in text Examples: r"\d+" → any sequence of digits r"\b[Pp]\w+" → words starting with P or p r"\w{3}" → exactly 3 characters Why not use string methods? .find() searches exact text RegEx can match flexible patterns EV0000745 · EV0000746 · EV0000747
3. Source slide page 3 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Key Metacharacters Meta | Meaning | Example . | Any character (except newline) | h.t → hat, hot \d | Digit (0–9) | \d\d → 12 \w | Word character (a-z, A-Z, 0-9, _) | \w\w → hi, A5 \s | Whitespace | hello\sworld → hello world + | One or more | \d+ → 123 * | Zero or more | ab*c → ac, abc [] | Character set | [aeiou] → any vowel ^ | Start of string | ^Hello $ | End of string | world$ EV0000748 · EV0000749 · EV0000750
4. Source slide page 4 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Main Functions in re re.search() → finds first match re.findall() → finds all matches in a list EV0000751 · EV0000752 · EV0000753
5. Source slide page 5 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING re.search() Example import re text = "My car brand is Audi." pattern = r"c\w\w" # 3-letter word starting with 'c' match = re.search(pattern, text) if match: print(match.group(), match.start(), match.end()) Output: car 3 6 .group() → actual text .start() → starting index .end() → ending index EV0000754 · EV0000755 · EV0000756
6. Source slide page 6 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING re.findall() Examples text = "My numbers: 123, 456, 789." print(re.findall(r"\d", text)) # single digits → ['1','2',...] print(re.findall(r"\d+", text)) # full numbers → ['123','456','789'] print(re.findall(r"\w+", "Python is fun!")) # words → ['Python','is','fun'] EV0000757 · EV0000758 · EV0000759
7. Source slide page 7 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING re.sub() Example Replace spaces with underscores import re text = "Welcome to Python class" new_text = re.sub(r"\s", "_", text) print(new_text) Output: Welcome_to_Python_class re.sub(pattern, replacement, text) → substitute function EV0000760 · EV0000761 · EV0000762
8. Source slide page 8 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Word Boundaries & Quantifiers \b → word boundary {n} → exactly n characters Examples: text = "The fox ran" print(re.findall(r"\b\w{3}\b", text)) # ['The','fox','ran'] print(re.findall(r"\w+ing", "learning coding swimming")) # ['learning','coding','swimming'] EV0000763 · EV0000764 · EV0000765

9. Source slide page 9 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Splitting and Replacing Splitting text = "apple, banana orange,grapes" print(re.split(r"[\s,]+", text)) # split by spaces or commas Output: ['apple','banana','orange','grapes'] Replacing text = "Password: 12345" print(re.sub(r"\d","*",text)) # replace digits → Password: ***** EV0000766 · EV0000767 · EV0000768
10. Source slide page 10 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Matching Patterns Words starting with p or P: text = "Python programming is popular" pattern = r"\b[Pp]\w+" print(re.findall(pattern, text)) # ['Python','programming','popular'] Text starting with "Hello": text = "Hello world" pattern = r"^Hello" if re.search(pattern, text): print("Starts with Hello") EV0000769 · EV0000770 · EV0000771
11. Source slide page 11 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Emails & Phone Numbers Find emails: text = "info@pes.edu, support@pes.com" print(re.findall(r"\S+@\S+", text)) # ['info@pes.edu','support@pes.com'] Find 10-digit numbers: text = "Call 9876543210 or 9123456789" print(re.findall(r"\d{10}", text)) EV0000772 · EV0000773 · EV0000774
12. Source slide page 12 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Emails & Phone Numbers Find 10-digit numbers starting with 9: text = "9000000000, 8123456789" print(re.findall(r"\b9\d{9}\b", text)) # ['9000000000'] EV0000775 · EV0000776 · EV0000777
13. Source slide page 13 (extracted text): THANK YOU Department of Computer Science and Engineering Prof. Kundhavai K R, CSE Department Ack: Teaching Assistant: Adithya Jeyaramsankar– PES2UG22CS029 EV0000778 · EV0000779 · EV0000780

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.900] Hello everyone, welcome to the session on Python modules. Today we are going to explore one of the most powerful features of Python, which is modules. [00:00:13.796] Modules help us write a clean, organized and reusable code and then play a major role in real-world software development. So what is exact a module? A module is simply a Python file that contains code [00:00:35.140] functions, classes, or variables. [00:00:42.276] That we can reuse them in other programs. Python already gives us many built-in modules like math. [00:00:52.420] OS, random and date and time, etc. That makes our life easier. And the best part is we can create our own modules. We can create our own modules and even we have third party modules. [00:01:13.220] What are the advantages of modules? Why do we use modules? Let let's look at the advantages. First one is reusability. You write the code once and use it again and again. This saves a huge amount of time. [00:01:28.836] Then modularity. Instead of writing code one, uh writing one long program, we break into smaller parts. Each part becomes easier to develop and debug. Next comes separate namespaces. Every module has its own space for names. So variables and functions won't clash. [00:01:49.476] Better organization, modul us structure a project neatly, especially when it grows larger. In short, modules makes our programs cleaner, smarter, and scalable. [00:02:06.116] Next comes types of modules. We have three types built-in modules, user defined modules, and third party modules. Let's go deeper into each of these types. [00:02:21.124] Before that, let's see what is pip. Pip is a command to install packages. So pip stands for pip installs packages. This is a tool to install and manage the third party Python packages. [00:02:42.500] So some of the common pip commands are pip install and followed by the

package name. Then if you want to uninstall the module, you can give pip uninstall and the module name. And if you want to see what are the modules that you have installed. [00:03:00.516] PIP list gives you a view of what all modules that you have installed. Almost every advanced Python project uses PIP. Next one is [00:03:16.196] We'll talk about the types of modules. Built-in modules. Built-in modules come packaged with Python. You don't need to install anything, just import them and use them. So examples, math, OS, random, date time. [00:03:33.220] These modules for everyday programming needs. So no installation is required. Next comes built built-in modules, ways to import the built-in modules. So if you want to import the entire module, you can use for import keyword. [00:03:51.684] and the module name. In this example it is import math and then you are going to use square root function inside the math module. Import specific attributes. [00:04:03.748] From math import square root. So in this case you need not use math.squire root. It is enough if you use square root, the just the function name. And in this case, [00:04:14.500] When you are importing specific attributes, you can only use those specific ones which you have imported, not all of them. Importing all from math import start. This is not recommended. [00:04:27.524] If you don't if you know the specific attributes or specific functions, please import only them, not import star. [00:04:41.124] Then next one is user defined modules. Now let's talk about user defined modules, these modules that we create. Whenever we write a Python file with some useful functions and save it as.py file, that can be imported to another program. [00:04:59.108] This is an excellent way to organize your own function and reuse them across projects.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:13.956] So [00:05:16.932] Example files that you may create is you can create a.py file of any program and just you need to import use the inp keyword and import the program. [00:05:34.084] It is the same like the built-in modules, but here you have created the own PY5. Then comes third party modules. Finally we have uh third party modules. These are created by Python community and gives us advanced capabilities. [00:05:50.628] Okay, example like NumPai, Skycatlone, Pillow, all of them has advanced capabilities. So these modules are extremely powerful and widely used in industries and research. [00:06:04.740] Some of the third party modules are listed down here. We have NumPy for numerical computing and we have Pandas for data analysis. [00:06:13.572] uh or working with data sets. We have mapplotlib for charts and graphs, basically plotting and visualization. We have requests for fetching data from websites or working with APIs. We have pillow for image processing or image editing. [00:06:29.808] These tools expand Python beyond basic programming and enable the real world application. [00:06:37.924] So understanding the modules is essential for writing efficient, clean and professional Python code. So to summarize, modules are backbone [00:06:48.612] of scalable Python programming. Built-in modules gives us ready-made functionality, user defined modules let us organize our own code, and third party modules bring the power of the entire Python community right into our projects. Thank you.

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Full transcript: 838 normalized words, 29 timestamped segments, 431.68 seconds duration. Low-confidence segments flagged: 0.

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11 Numpy

This lesson is indexed from the source video titled Numpy. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000102 · EV0000103

Graphs and diagrams

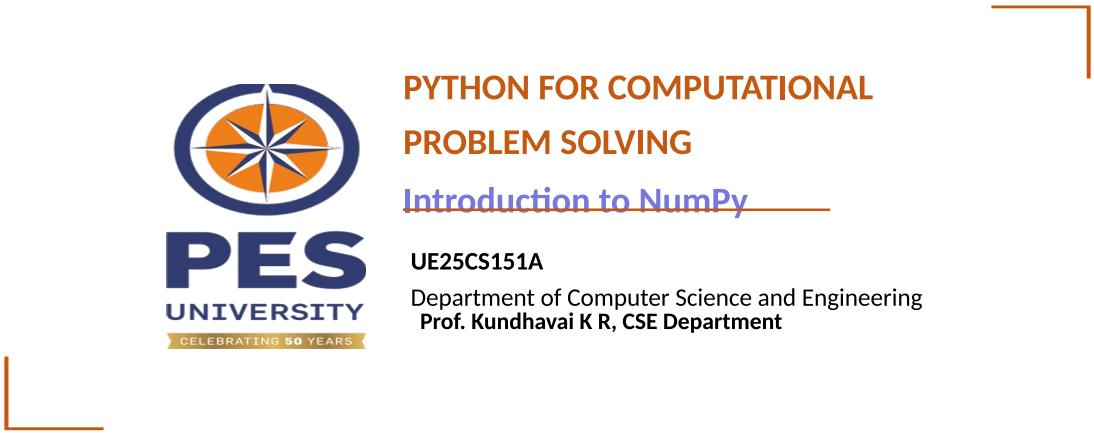

Figure G000174. Numpy · source slide page 1.

EV0000638

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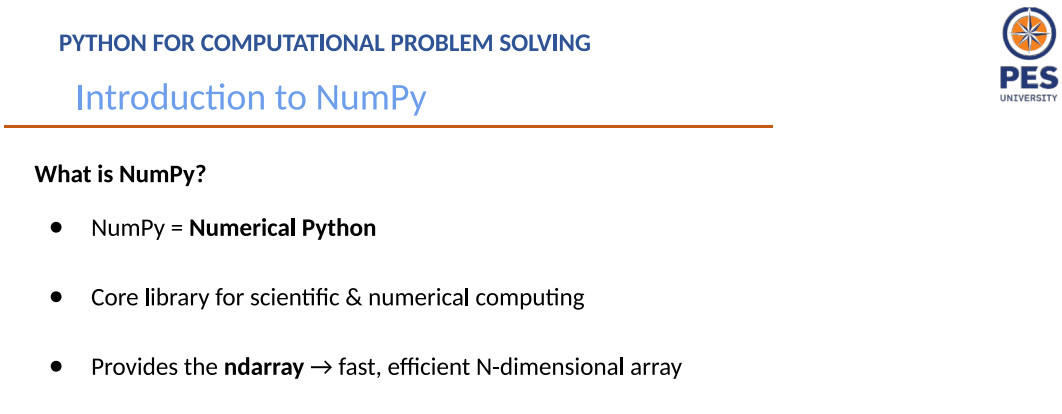

Figure G000175. Numpy · source slide page 2.

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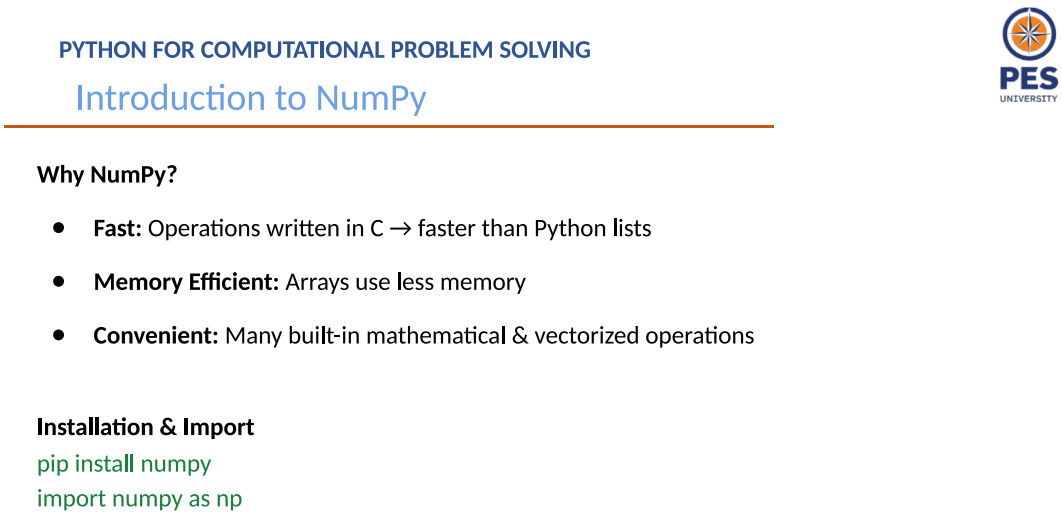

Figure G000176. Numpy · source slide page 3.

EV0000644

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

NumPy Arrays: The Basics

NumPy arrays:

- Store values of **same data type**
- Can be:
  - **1D (Vector)**
  - **2D (Matrix)**
  - **N-Dimensional (Tensor)**

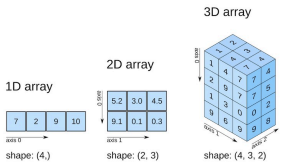

Figure G000177. Numpy · source slide page 4.

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Array Creation and Manipulation

Create Arrays

np.zeros((3,3)) **Output:** [[0. 0. 0.] [0. 0. 0.] [0. 0. 0.]]

np.ones((2,4)) **Output:** [[1. 1. 1. 1.] [1. 1. 1. 1.]]

np.arange(10) **Output:** [0 1 2 3 4 5 6 7 8 9]

Figure G000178. Numpy · source slide page 5.

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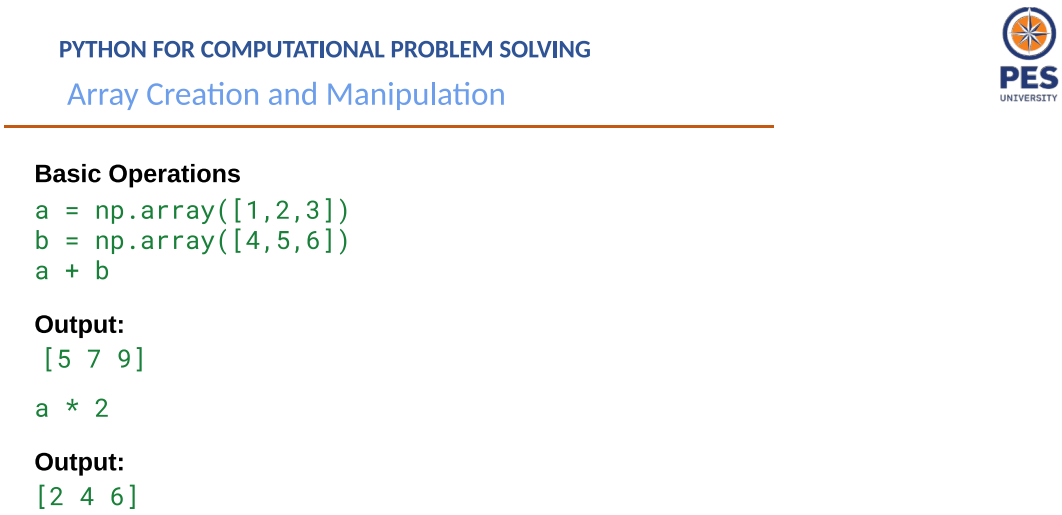

Figure G000179. Numpy · source slide page 6.

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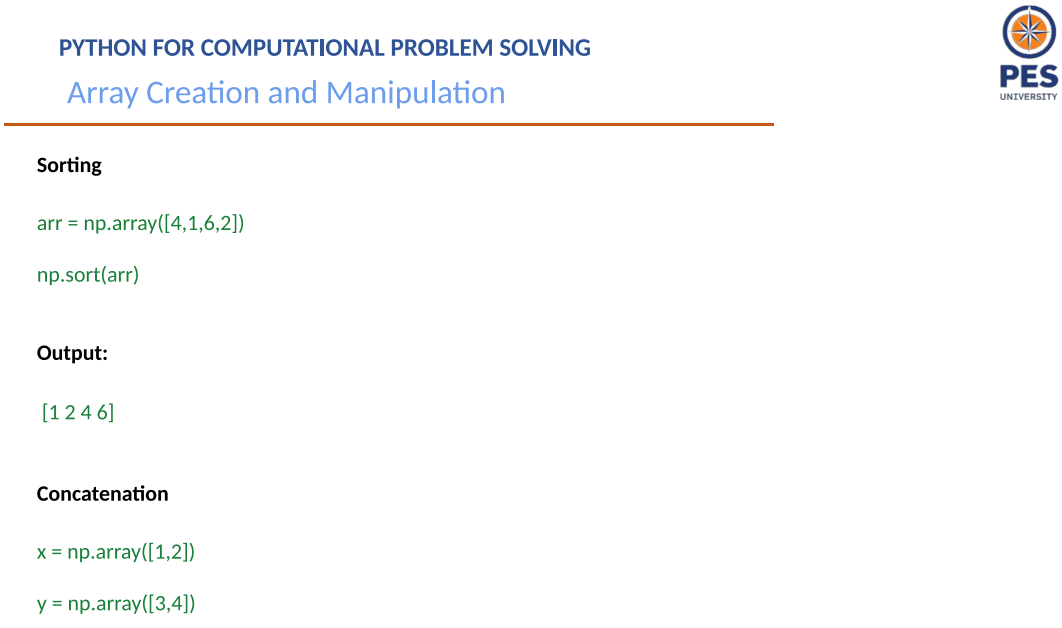

Figure G000180. Numpy · source slide page 7.

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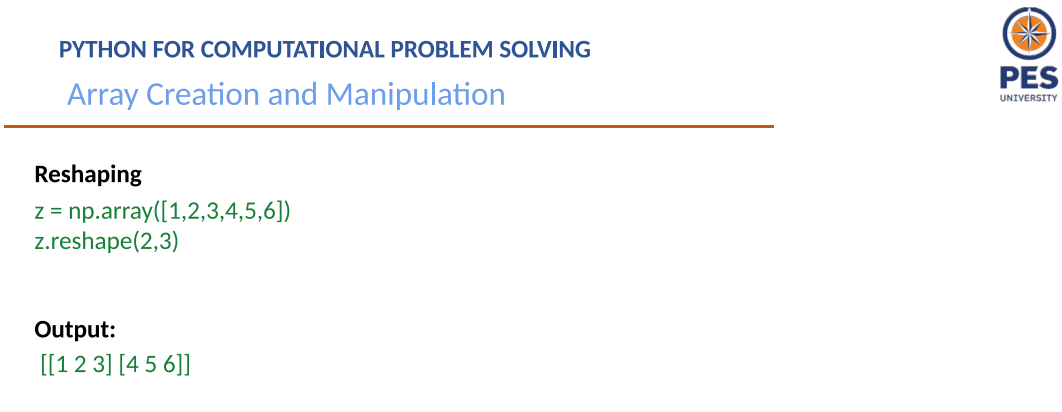

Figure G000181. Numpy · source slide page 8.

EV0000659

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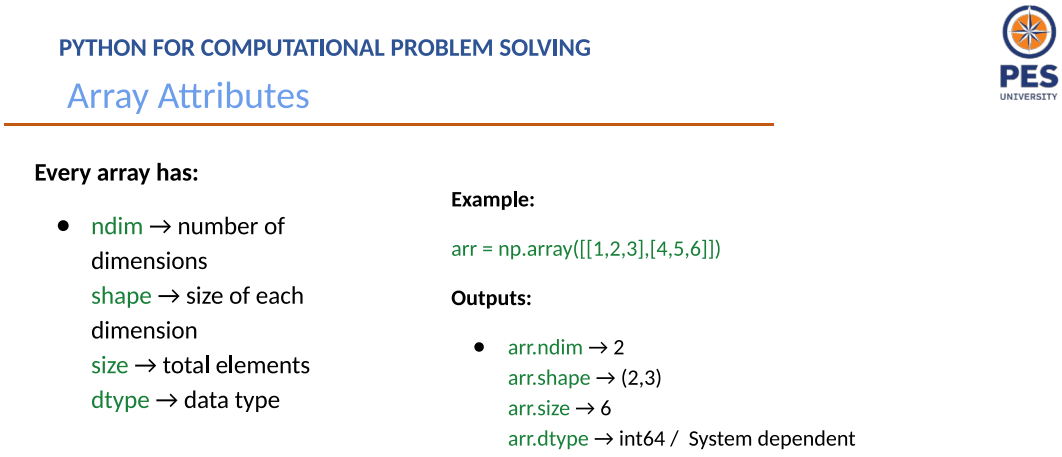

Figure G000182. Numpy · source slide page 9.

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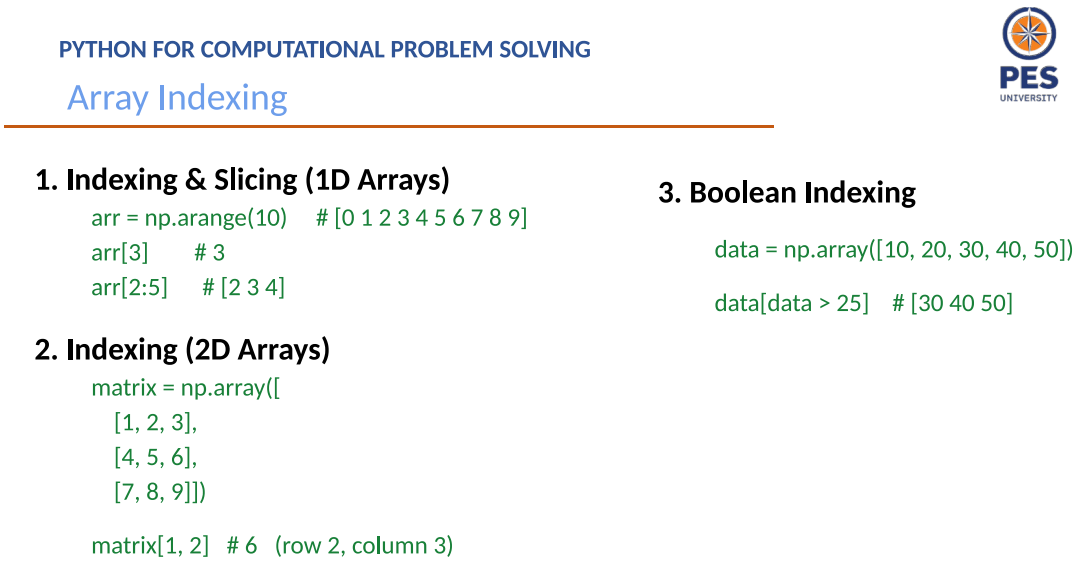

Figure G000183. Numpy · source slide page 10.

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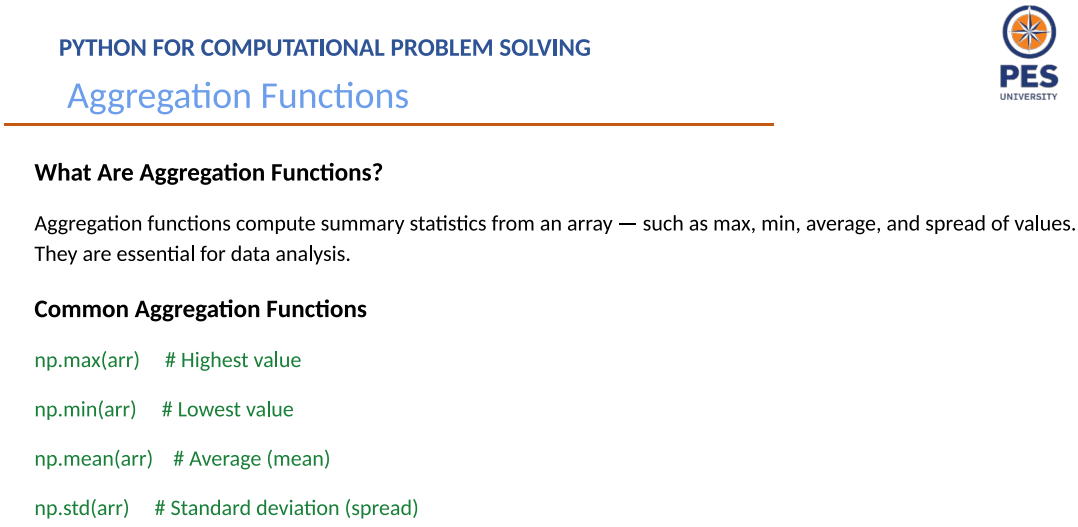

Figure G000184. Numpy · source slide page 11.

EV0000668

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Arrays vs Lists

LISTS	ARRAYS
Can store Heterogeneous types	Usually Homogeneous.
Dynamic in nature. Can grow or shrink in size.	Static and fixed.
Stores elements in separate object references	Stores elements in contiguous memory references.
Slower for large scale numeric operations because they lack vectorization.	Optimized for bulk, element-wise computations and numerical processing.
General Purpose	Math heavy scientific computations.

Figure G000185. Numpy · source slide page 12.

EV0000671

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THANK YOU

Department of Computer Science and Engineering

Prof. Kundhavai K R, CSE Department

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Figure G000186. Numpy · source slide page 13.

EV0000674

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Important points and why

1. Source slide page 1 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Introduction to NumPy UE25CS151A Department of Computer Science and Engineering Prof. Kundhavai K R, CSE Department EV0000637 · EV0000638 · EV0000639
2. Source slide page 2 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Introduction to NumPy What is NumPy? NumPy = Numerical Python Core library for scientific & numerical computing Provides the ndarray → fast, efficient N-dimensional array EV0000640 · EV0000641 · EV0000642
3. Source slide page 3 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Introduction to NumPy Why NumPy? Fast: Operations written in C → faster than Python lists Memory Efficient: Arrays use less memory Convenient: Many built-in mathematical & vectorized operations Installation & Import pip install numpy import numpy as np EV0000643 · EV0000644 · EV0000645
4. Source slide page 4 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING NumPy Arrays: The Basics NumPy arrays: Store values of same data type Can be: 1D (Vector) 2D (Matrix) N-Dimensional (Tensor) [Image text, OCR] 3D array np. arange(3) + 5 0 1 2 5 5 6 7 axis0 np. one s((3, 3)) + np ar ange(3) 2Darray 1 1 1 1 0 1 2 1 2 3 4 4 2 1 1 1 0 2 1 2 3 1D array 9 5 axis 1 1 1 2 1 2 3 1 5.2 3.0 4.5 3 2 np. arange(3).reshape((3, 1)) + np. arange (3) 9 0 0 7 2 9 10 9.1 0.1 0.3 6 8 axis1 0 0 0 1 2 0 1 2 9 9 1 0 2 1 2 3 axis 0 axis 1 axis2 2 2 2 0 2 2 3 4 shape: (4,) shape: (2, 3) shape: (4, 3, 2) EV0000646 · EV0000647 · EV0000648
5. Source slide page 5 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Array Creation and Manipulation Create Arrays np.zeros((3,3)) Output: [[0. 0. 0.] [0. 0. 0.] [0. 0. 0.]] np.ones((2,4)) Output: [[1. 1. 1. 1.] [1. 1. 1. 1.]] np.arange(10) Output: [0 1 2 3 4 5 6 7 8 9] EV0000649 · EV0000650 · EV0000651
6. Source slide page 6 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Array Creation and Manipulation Basic Operations a = np.array([1,2,3]) b = np.array([4,5,6]) a + b Output: [5 7 9] a * 2 Output: [2 4 6] EV0000652 · EV0000653 · EV0000654
7. Source slide page 7 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Array Creation and Manipulation Sorting arr = np.array([4,1,6,2]) np.sort(arr) Output: [1 2 4 6] Concatenation x = np.array([1,2]) y = np.array([3,4]) np.concatenate((x, y)) Output: [1 2 3 4] EV0000655 · EV0000656 · EV0000657
8. Source slide page 8 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Array Creation and Manipulation Reshaping z = np.array([1,2,3,4,5,6]) z.reshape(2,3) Output: [[1 2 3] [4 5 6]] EV0000658 · EV0000659 · EV0000660
9. Source slide page 9 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Array Attributes Every array has: ndim → number of dimensions shape → size of each dimension size → total elements dtype → data type Example: arr = np.array([[1,2,3],[4,5,6]]) Outputs: arr.ndim → 2 arr.shape → (2,3) arr.size → 6 arr.dtype → int64 / System dependent EV0000661 · EV0000662 · EV0000663

10. Source slide page 10 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Array Indexing 1. Indexing & Slicing (1D Arrays) arr = np.arange(10) # [0 1 2 3 4 5 6 7 8 9] arr[3] # 3 arr[2:5] # [2 3 4] 2. Indexing (2D Arrays) matrix = np.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]]) matrix[1, 2] # 6 (row 2, column 3) 3. Boolean Indexing data = np.array([10, 20, 30, 40, 50]) data[data > 25] # [30 40 50] EV0000664 · EV0000665 · EV0000666
11. Source slide page 11 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Aggregation Functions What Are Aggregation Functions? Aggregation functions compute summary statistics from an array — such as max, min, average, and spread of values. They are essential for data analysis. Common Aggregation Functions np.max(arr) # Highest value np.min(arr) # Lowest value np.mean(arr) # Average (mean) np.std(arr) # Standard deviation (spread) EV0000667 · EV0000668 · EV0000669
12. Source slide page 12 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Arrays vs Lists [Image text, OCR] LISTS ARRAYS CanstoreHeterogeneoustypes UsuallyHomogeneous. Dynamic in nature.Can grow or shrink Staticandfixed. in size. Storeselementsinseparateobject Storeselementsincontiguousmemory references references. Slowerforlargescalenumeric Optimizedforbulk,element-wise operationsbecausetheylack computationsandnumerical vectorization. processing. GeneralPurpose Mathheavyscientificcomputations. Every array has: ndim → number of dimensions shape → size of each dimension size → total elements dtype → data type Example: arr = np.array([[1,2,3],[4,5,6]]) Outputs: arr.ndim → 2 arr.shape → (2,3) arr.size → 6 arr.dtype → int64 EV0000670 · EV0000671 · EV0000672
13. Source slide page 13 (extracted text): THANK YOU Department of Computer Science and Engineering Prof. Kundhavai K R, CSE Department Ack: Teaching Assistant: Adithya Jeyaramsankar– PES2UG22CS029 EV0000673 · EV0000674 · EV0000675

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.804] Hello everyone and welcome. Today we are going to explore NumPy, one of the most powerful libraries in Python, especially when it comes to scientific and numerical computing. [00:00:15.556] Imagine you're standing in front of a huge mountain of numbers. Numbers from experiments, data, from marks of students or images filled with pixels. And Python looks at you and says, I can help, but not alone. [00:00:32.260] This is exactly where NumPy enters a story. NumPy is like giving Python a superpower, the power to handle large numerical data elegantly, efficiently, and unbelievingly fast. [00:00:48.260] Today let's take a small and meaningful journey into this magical library called NumPy. NumPy, short for numerical Python, is not just a library, it's the foundation on which most data science and machine learning libraries are built. [00:01:08.356] Its heart is the NB array. Think of it as a disciplined, highly optimized container that holds data in a clean and organized manner. [00:01:23.428] Like a neat cupboard where every item has the same size, same type, and and it is perfectly arranged. Why number? Let me tell you a short story. Imagine having a list of a million numbers and you want to multiply every value by two. [00:01:44.836] If Python lists try to do this, they get tired. They walk slowly. They use a lot of memory on the way. But Numbai, Numbai does it in a single breath. Because the work [00:01:59.364] is done deep inside optimized C code. So NumPy is fast, memory efficient, and convenient to use. [00:02:12.100] So to install numpy, we have to use the command pip install numpy and it can be used as import numpy as mp

np. [00:02:29.636] Think of arrays as building blocks. Let's talk about the basics of arrays. So you can store values of same data type in an array. [00:02:43.332] You can have a 1D array like a simple list of marks. We usually call it as vectors. A 2D array like a table, we represent it as mattress, rows and columns, and then we have an N-dimensional. [00:03:00.836] Array we usually represent call it as tensors, like stacks of images or lay of data. Number arrays keep everything clean, fast and mathematical. [00:03:15.972] Let's now see how to create arrays. Creating arrays feels almost like magic spells. You can see the functions here, np.z, np.1s, and np.a range. [00:03:34.404] NP dot zeros create a clean slate of mattress of zeros. Likewise, np dot one [00:03:45.764] gives you a table full of ones useful for testing. NP.a range gives you numbers from zero to nine with a wave of your hand. With just few lines, massive data sets can be manageable. [00:04:04.420] Imagine having two lists and wanting to add them. Let's now talk about the basic operations on the numpy array. So I have two lists here and I want to add them element by element. In pure Python, you would be probably using a loop. [00:04:21.284] But Numai says why look? When I can do it instantly. So A plus B [00:04:29.700] add both the arrays element wise and gives you the answer. And you have A multiplied by two and you get the multiplication of the A array. So this [00:04:48.260] Number performs lightning fast operations on entire arrays. [00:04:58.532] Let's see, sorting and concatenation on the arrays. If you need to sort your data, NumPy gently rearranges it with the help of the function called sort. And you need to pass the array. If you need to join two arrays,

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:16.964] You can use concatenate function. It connects them smoothly. [00:05:29.156] talking about reshaping arrays, reshaping feels like transforming clay into new shapes without changing the amount of the clay. Take a 1D array twist it and suddenly it becomes a two by three matrix using reshape. [00:05:44.164] So here I use a reshape to transform this 1D array into a two cross three matrix. Reshaping is heavily used in images, ML models and simulations. It's not just technical, it's elegant. [00:06:01.860] Then talking about the array attributes, every array quietly carries important information about itself. So M Dem tells you how many dimensions, shape tells you its structure. [00:06:16.964] Size tells you the total elements, detail tells you what kind of data lives inside. It's like introducing the array itself politely before you start working with it. [00:06:30.404] Next comes indexing and slicing. Indexing in NumPy feels like reading exactly the part of the story you want. You want the fourth number, then you can give array of three because array indexes starts from zero. [00:06:46.788] You want numbers from two to four, then you can use slicing. [00:06:54.404] In a 2D matrix, matrix of one, two is like saying bring me the element in the second row and call third column. So second row, this is row. [00:07:11.556] row 0, row 1 and row 2, column 0, column 1 and column 2, and we are getting this one. Array indexes starts from 0. So this is the first row and the second column. So this is 0, 1 and 2. [00:07:33.028] And add boolean indexing, that's your filter. It quietly handles you only the values that matter. So here I have given I want only the data that is greater than 25 in this array, and you will be able to pull up the answer using Boolean indexing. Next comes [00:07:52.260] What are aggregate functions? Think of aggregation functions as small detectives. They scan your data and report back. Okay? Max gives you the largest value. Min gives you the smallest value in the array. [00:08:10.596] The average of the array. S T D talks about the standard deviation, the average, sorry, the spread or variation in the array. This gives you instant insights with almost no effort. [00:08:27.812] Let's end uh with a simple comparison story. Python lists and python arrays. Python lists are regular school bags, flexible [00:08:37.796] You can put many things

but they are get heavy. But NumPy RAs are like professionally designed travel bag, organized, lightweight, optimised and ready for long journeys. When you deal with serious numerical data, you pick the travel bag. [00:08:55.908] So NumPy isn't just a library, it's a mindset. It teaches y us to think in arrays, to avoid loops, to embrace speed and to enjoy clean mathematical expression. [00:09:11.652] Here is given the difference between the lists and arrays. [00:09:20.708] Thank you all.

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Full transcript: 1062 normalized words, 37 timestamped segments, 568.801567 seconds duration. Low-confidence segments flagged: 0.

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12 PIL

This lesson is indexed from the source video titled PIL. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000110 · EV0000111

Graphs and diagrams

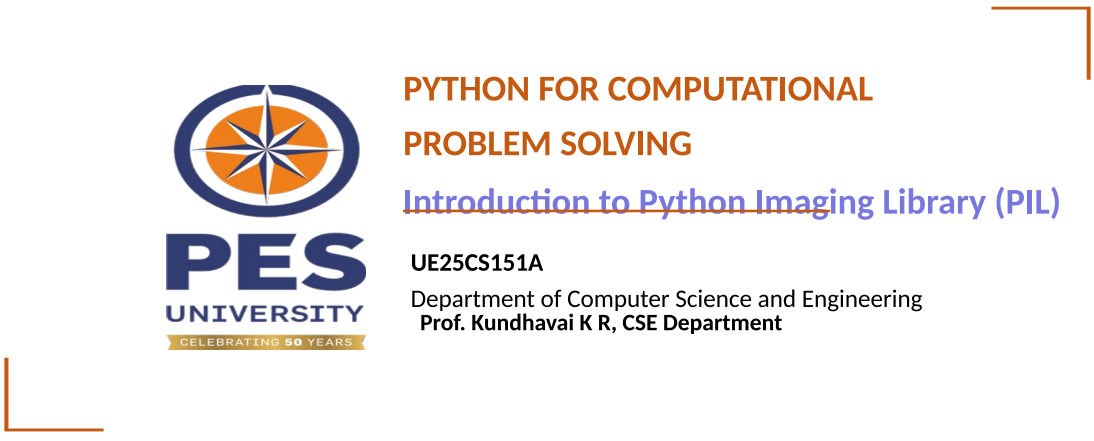

Figure G000200. PIL · source slide page 1.

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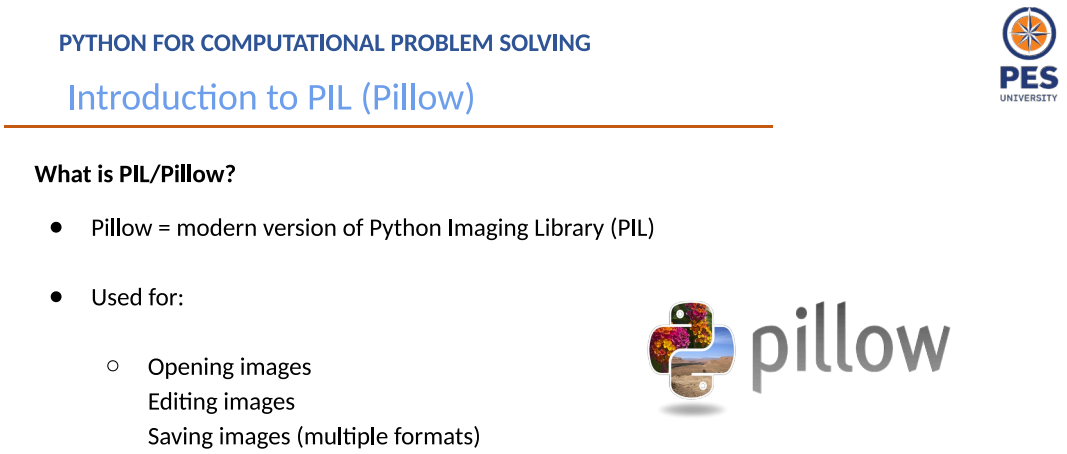

Figure G000201. PIL · source slide page 2.

EV0000719

> ***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

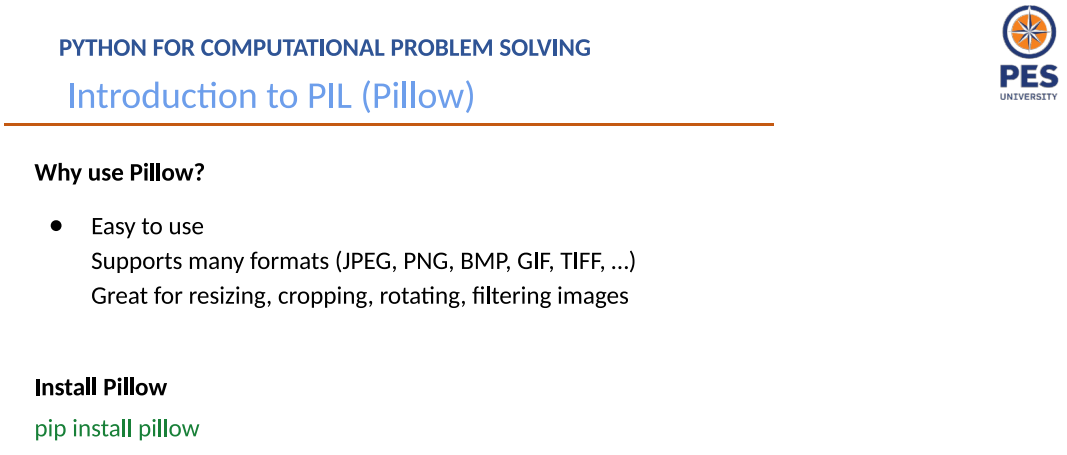

Figure G000202. PIL · source slide page 3.

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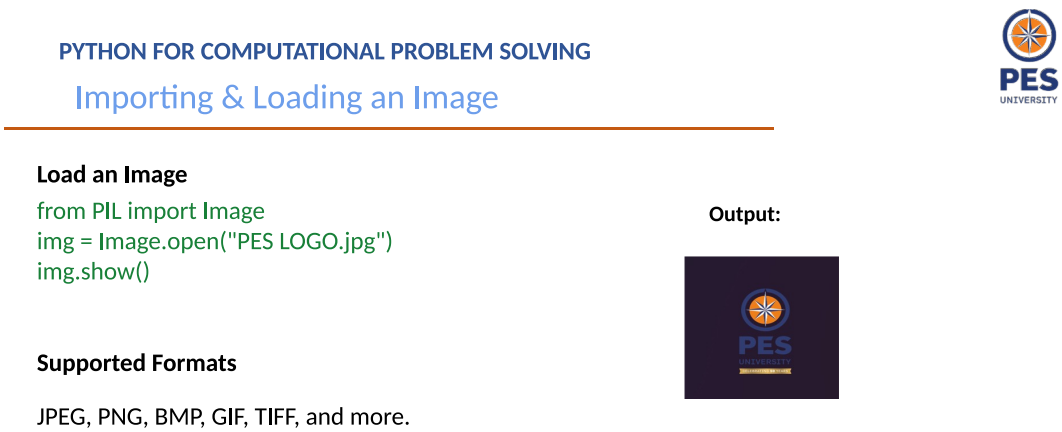

Figure G000203. PIL · source slide page 4.

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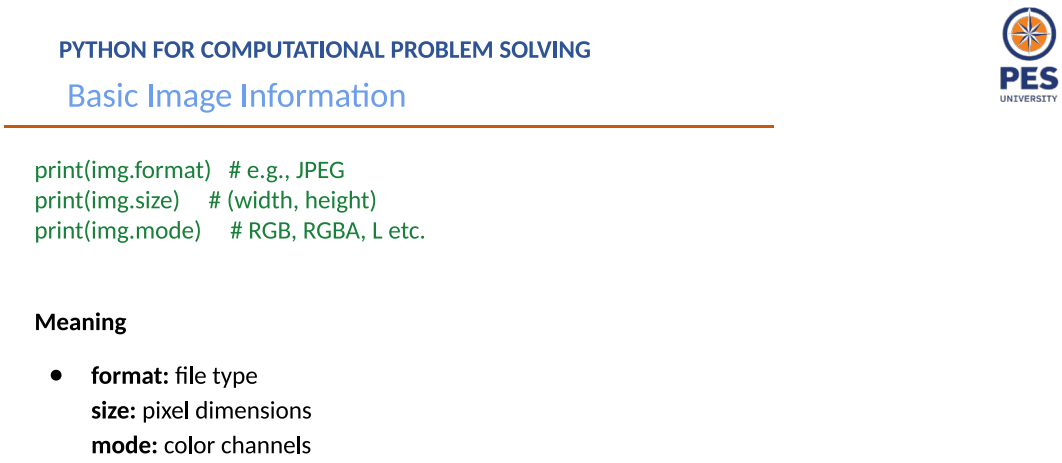

Figure G000204. PIL · source slide page 5.

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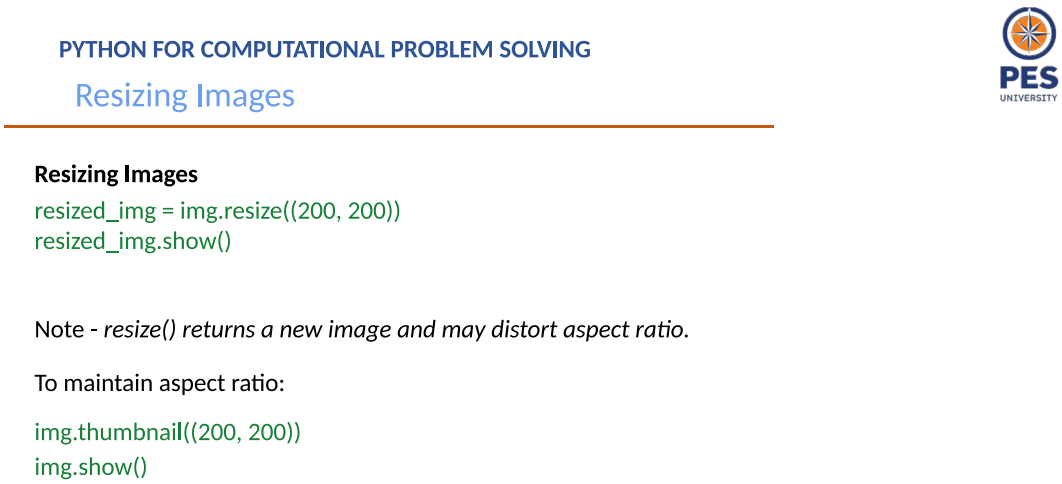

Figure G000205. PIL · source slide page 6.

EV0000731

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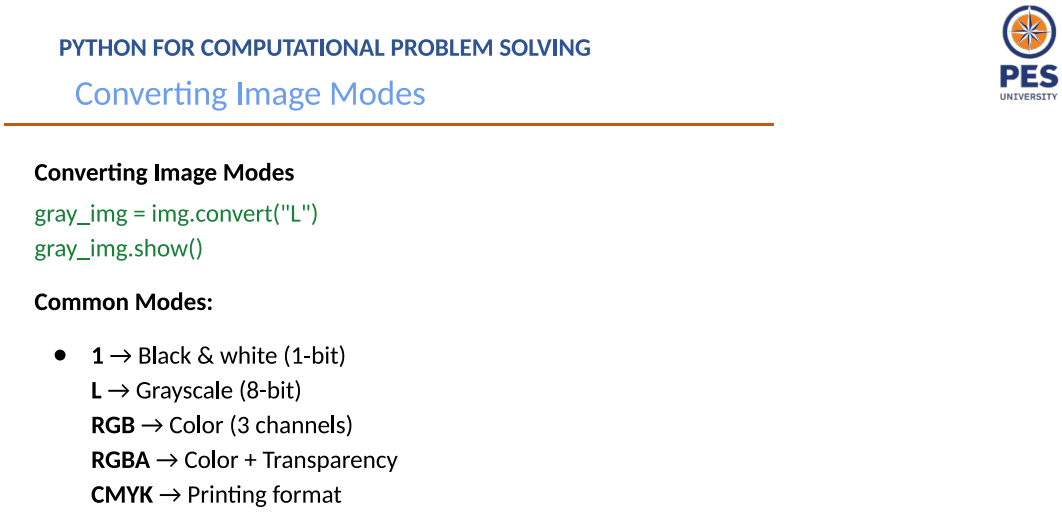

Figure G000206. PIL · source slide page 7.

EV0000734

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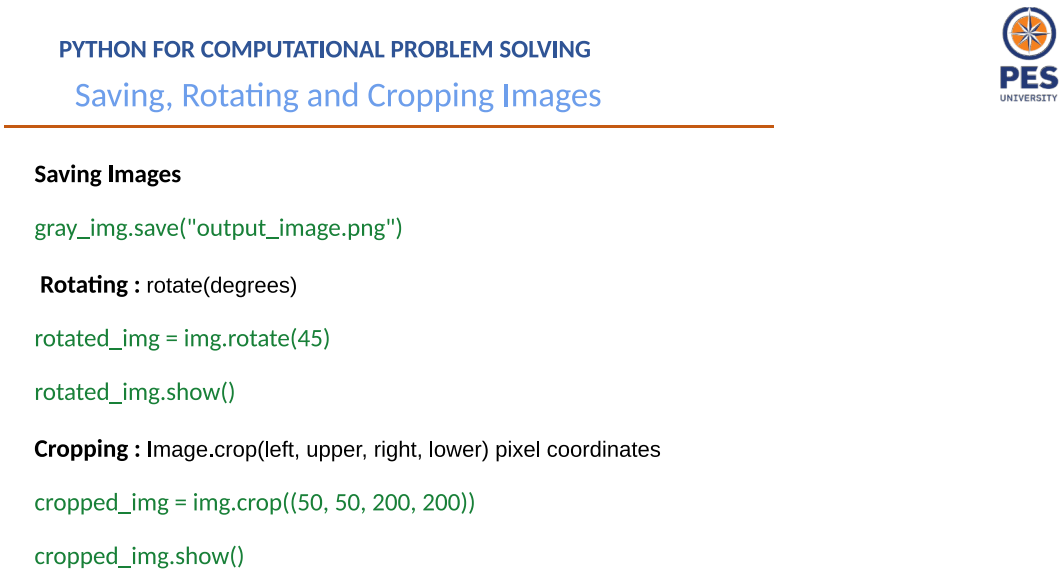

Figure G000207. PIL · source slide page 8.

EV0000737

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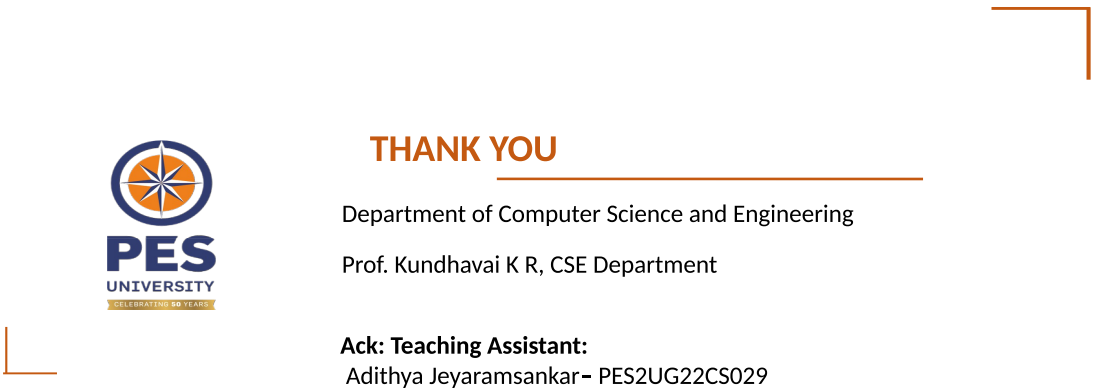

Figure G000208. PIL · source slide page 9.

EV0000740

> ***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Introduction to Python Imaging Library (PIL) UE25CS151A Department of Computer Science and Engineering Prof. Kundhavai K R, CSE Department EV0000715 · EV0000716 · EV0000717
2. Source slide page 2 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Introduction to PIL (Pillow) What is PIL/Pillow? Pillow = modern version of Python Imaging Library (PIL) Used for: Opening images Editing images Saving images (multiple formats) [Image text, OCR] pillow EV0000718 · EV0000719 · EV0000720
3. Source slide page 3 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Introduction to PIL (Pillow) Why use Pillow? Easy to use Supports many formats (JPEG, PNG, BMP, GIF, TIFF, ...) Great for resizing, cropping, rotating, filtering images Install Pillow pip install pillow EV0000721 · EV0000722 · EV0000723
4. Source slide page 4 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Importing & Loading an Image Load an Image from PIL import Image img = Image.open("PES LOGO.jpg") img.show() Supported Formats JPEG, PNG, BMP, GIF, TIFF, and more. Output: EV0000724 · EV0000725 · EV0000726
5. Source slide page 5 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Basic Image Information print(img.format) # e.g., JPEG print(img.size) # (width, height) print(img.mode) # RGB, RGBA, L etc. Meaning format: file type size: pixel dimensions mode: color channels EV0000727 · EV0000728 · EV0000729
6. Source slide page 6 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Resizing Images Resizing Images resized_img = img.resize((200, 200)) resized_img.show() Note - resize() returns a new image and may distort aspect ratio. To maintain aspect ratio: img.thumbnail((200, 200)) img.show() EV0000730 · EV0000731 · EV0000732
7. Source slide page 7 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Converting Image Modes Converting Image Modes gray_img = img.convert("L") gray_img.show() Common Modes: 1 → Black & white (1-bit) L → Grayscale (8-bit) RGB → Color (3 channels) RGBA → Color + Transparency CMYK → Printing format EV0000733 · EV0000734 · EV0000735
8. Source slide page 8 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Saving, Rotating and Cropping Images Saving Images gray_img.save("output_image.png") Rotating : rotate(degrees) rotated_img = img.rotate(45) rotated_img.show() Cropping : Image.crop(left, upper, right, lower) pixel coordinates cropped_img = img.crop((50, 50, 200, 200)) cropped_img.show() EV0000736 · EV0000737 · EV0000738
9. Source slide page 9 (extracted text): THANK YOU Department of Computer Science and Engineering Prof. Kundhavai K R, CSE Department Ack: Teaching Assistant: Adithya Jeyaramsankar– PES2UG22CS029 EV0000739 · EV0000740 · EV0000741

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.836] Hello everyone and welcome. Today we are going to explore Pillow, the modern version of Python Imaging Library, also called PIL. [00:00:15.588] Simple and powerful library that helps us open images, edit images, and save images in multiple formats. [00:00:29.060] It specially deals with image processing tasks.

Let's begin. [00:00:37.796] Pillow is widely used because it is easy to understand, support many image formats, and includes built-in functions for tasks like resizing, cropping, rotating and filtering images. To install Pillow module, we simply use Pip install pillow. [00:01:08.740] To start with below, we first import the image class. So image is a class here. Here's a basic code to open an image and display it on the screen. [00:01:21.572] So for that, image class open function lets you to open the image that is there in the argument. Image.show displays it using your system's default viewer. [00:01:38.116] Pillow supports JPEG, PNG, BMP, JIF, TIF, and many more image formats. So this example is importing the image class and using the functions open and show. [00:01:59.108] Once we load the image, we can check some important details like format, which tells the file type like JPEG, or GIF, or Tiv, etc. Then the size, which gives the width and height in pixels. [00:02:19.524] Then the mode shows the colour representation such as RGB or grayscale. [00:02:29.828] These properties help us understand the image before processing it. [00:02:38.212] Resizing is one of the most common operations in image processing. Here's how we resize an image. [00:02:49.988] So this resize function resizes the image to a given dimension. But it may distort the aspect ratio. [00:03:06.820] To maintain the aspect ratio we can use thumbnail. This ensures the image is resized without stretching or squashing. [00:03:21.572] converting the image modes. You can use convert function [00:03:29.380] to convert to an image to a different colour mode. The common colour modes are one, which is pure black and white, it's just one bit. So zero stands for black, one stands for white. [00:03:49.796] L, which is grayscale. All the shades of gray colour will be represented in an 8-bit representation. Then RGB, red, green, blue colour, three channels, and this is also an 8-bit representation. [00:04:09.732] RGBA. [00:04:13.732] Red, green, blue, and A stands for alpha, which is represents the transparency. Then C M Y K, which stands for Syan, Majenta, Low, and Black. [00:04:31.812] Mostly used in printing. [00:04:35.908] Mode conversion is especially useful in machine learning preprocessing, where special formats are required. [00:04:46.340] Next one, let's take a look into few more essential operations like saving images. To save an image that you have worked on, you can use save function. Then to rotate an image, you can use rotate function and give the degrees as argument.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:05.828] To crop an image, we can use crop function and the syntaxes you can give the pixel coordinates as left, upper, right, lower to extract the specific region from the image. [00:05:24.996] These operations make Pillow extremely useful for automation workflows, image editing tasks, and pre processing pipelines. [00:05:37.700] Thank you. To summarize, Pillow is a very powerful and big enough friendly Python library for image processing. We learned how to use open images, display them, resize, convert color modes, rotate, crop and save images. [00:05:57.156] With these foundations you can build a wide variety of image based applications from simple editors to machine learning pipelines. Thank you for watching and happy coding.

EV0000114 · EV0000115 · EV0000116 · EV0000117

Full transcript: 565 normalized words, 25 timestamped segments, 369.28 seconds duration. Low-confidence segments flagged: 0.

EV0000110 · EV0000111 · EV0000112 · EV0000113

13 Source slide deck: UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda

This entry preserves the complete slide deck titled UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Python for computational Problem Solving, Unit_03, to avoid an unsupported match. EV0000208 · EV0000209

Graphs and diagrams

Figure G000031. UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda · source slide page 1. EV0000209

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

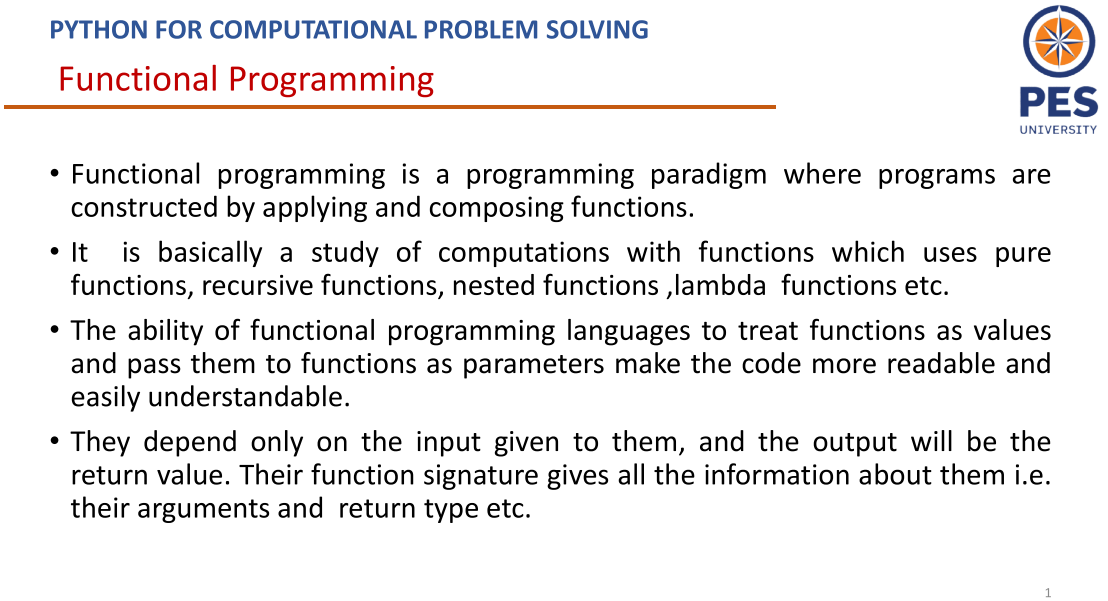

Figure G000032. UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda · source slide page 2.

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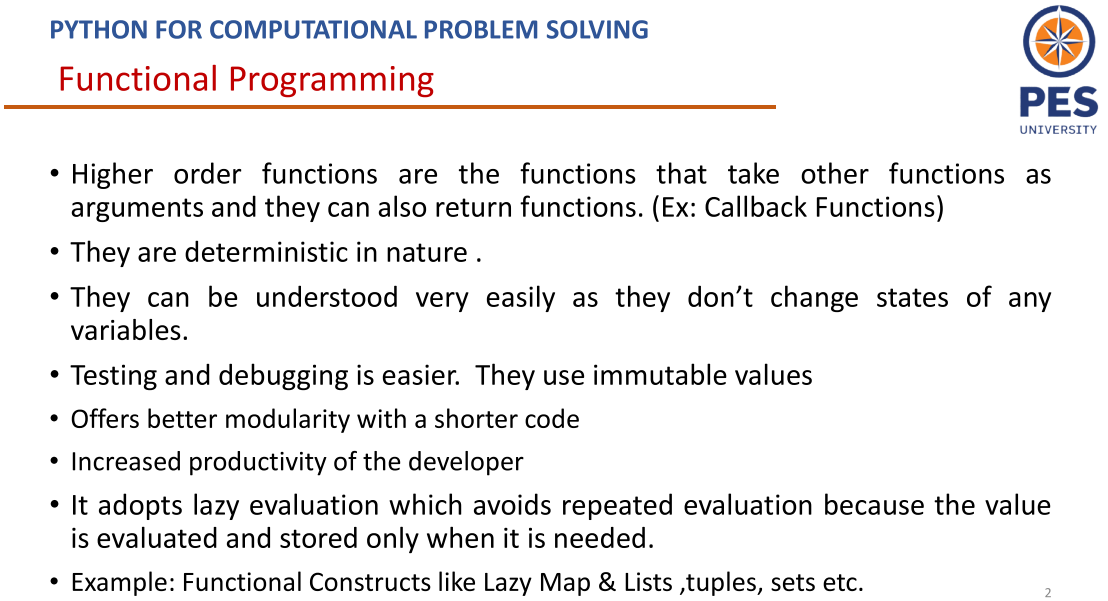

Figure G000033. UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda · source slide page 3.

EV0000215

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

The keyword - lambda

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Lambda (Anonymous function/Nameless Function/ Throw away function creation)

- Lambda functions are functions without the name.
- Its also called as anonymous functions or throw away functions.
- It can have any number of arguments but can have only one expression.
- The expression is executed and returned (Implicit return)
- Lambda functions are used when function objects are required.
- Lambda functions are used when you need a particular function for a short period of time.

3

Figure G000034. UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda · source slide page 4.

EV0000218

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

The keyword - lambda

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Advantages:

•Lambda functions in Python are very useful in defining the in-line functions.

•Most suitable when it is used as a part of another function(especially with map, filter and reduce functions)

•Less Code (Concise code)

4

Figure G000035. UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda · source slide page 5.

EV0000221

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

The keyword - lambda

PES UNIVERSITY

Syntax: **lambda input arguments: expression / Output**

- **lambda** is a keyword used to declare lambda function.
- **Input arguments:** It can have multiple arguments (if there are more than one arguments, then separate the arguments by a comma).
- **Expression or output:** Expression gets evaluated and results are returned

5

Figure G000036. UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda · source slide page 6.

EV0000224

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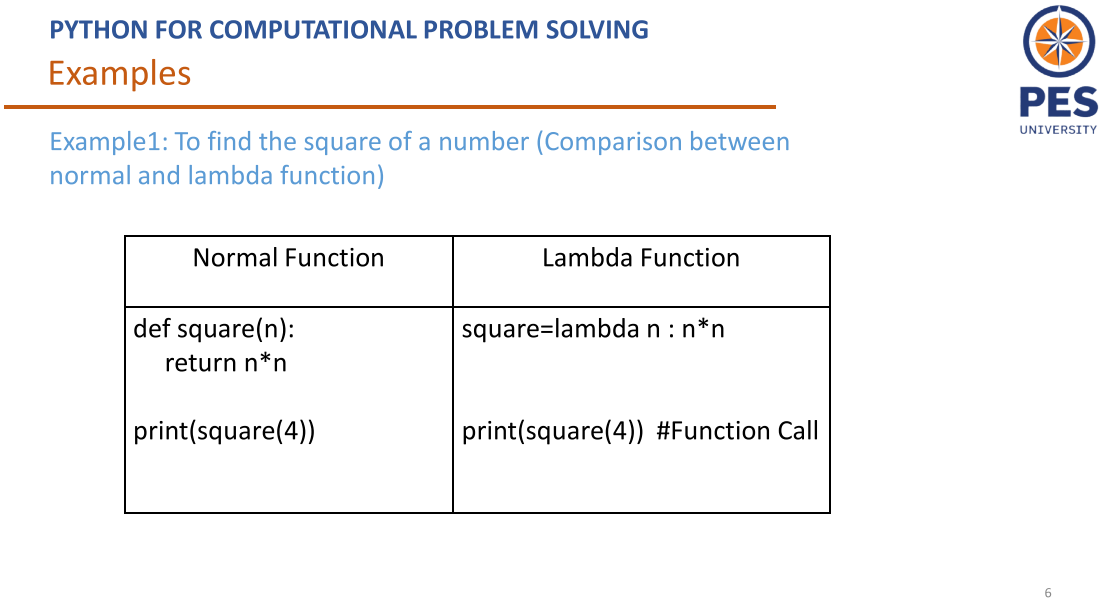

Figure G000037. UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda · source slide page 7.

EV0000227

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Examples

PES UNIVERSITY

Example 2: To find the greatest of two numbers using the relational operator.

```
maximum=lambda a,b : a if a>b else b    #max(a,b) can also be used
print("The maximum is ",maximum(6,7))
print(maximum,type(maximum))
```

Output:

```
The maximum is  7
<function <lambda> at 0x00000217A6F93550> <class 'function'>
```

7

Figure G000038. UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda · source slide page 8.

EV0000230

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Examples

The power of lambda is better understood when you use them as an anonymous function inside another function.

Example 3

Suppose we have a function definition that takes one argument, and that argument will be multiplied with an unknown number.

```
def myfunc(n):
    return lambda a : a * n

double_N = myfunc(2)

print(double_N(12))
```

Output:24

8

Figure G000039. UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda · source slide page 9.

EV0000233

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Examples

We can use the same function definition to make two functions (doube_N and triple_N functions) in the same program.

Example 4

```
def myfunc(n):
    return lambda a : a * n

double_N = myfunc(2)
triple_N = myfunc(3)

print(double_N(15))
print(triple_N(15))
```

Output

```
30
45
```

9

Figure G000040. UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda · source slide page 10.

EV0000236

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Map() Function

PES UNIVERSITY

Python map() function is a built in function which is utilized to apply a function to each element of an iterable (like a list or a tuple) and returns a new iterable object.

Syntax: map(function, iterables)

1. The map() function takes two arguments: a function and iterables.
2. The first argument for the map function should be a callable which takes one argument – could be a free function, function of a type, user defined function, or a lambda function.
3. This function will be applied to every element of the iterable, and creates a map object which is also an iterable.

10

Figure G000041. UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda · source slide page 11.

EV0000239

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Map() Function

PES UNIVERSITY

Working:

- The map function causes iteration through the iterable, and applies the callable on each element of the iterable
- All these happen only conceptually as the map function is lazy and the above said operations happen only after the map object is iterated.
- We can force an iteration of the map object by passing the map object as an argument for the list or set constructor.
- We can pass one or more iterable to the map() function.

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Figure G000042. UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda · source slide page 12.

EV0000242

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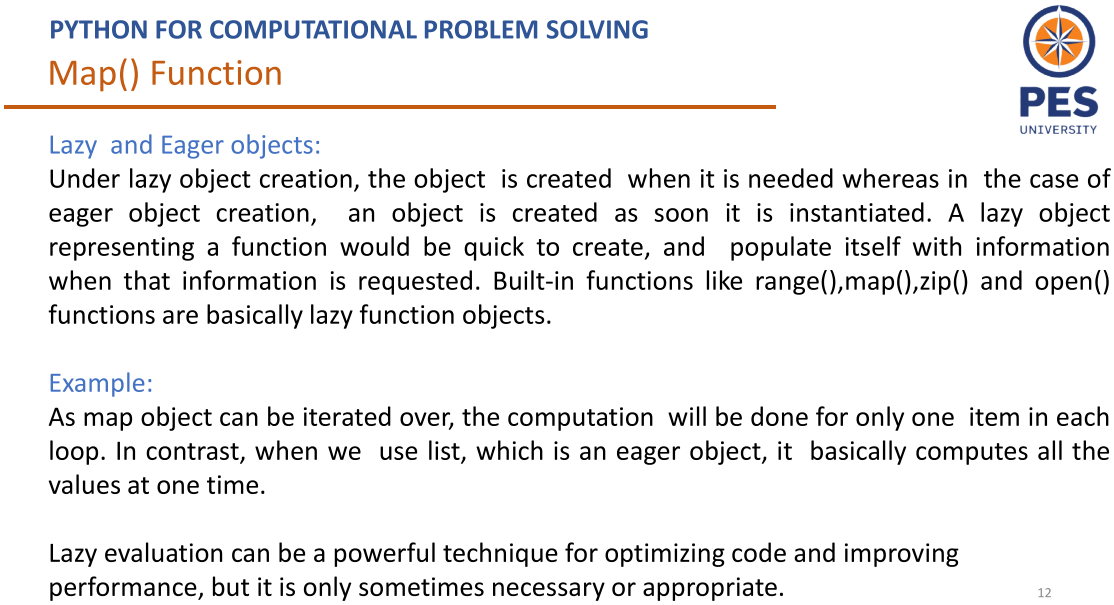

Figure G000043. UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda · source slide page 13.

EV0000245

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Map() Function

PES UNIVERSITY

Choosing between lazy and eager evaluation

- When deciding which evaluation strategy to use for a given programming problem, there is no one-size-fits-all answer. The choice depends on the nature of the problem, the characteristics of the data, the requirements of the solution, and the preferences of the programmer.
- However, some general guidelines can be followed to help make an informed decision. For instance, lazy evaluation should be used when you want to delay or avoid computations that are not needed, save memory by avoiding intermediate results, create and manipulate infinite or recursive data structures, or explore multiple possibilities without committing to one.
- On the other hand, eager evaluation is preferable when you want to perform computations as soon as possible, exploit parallelism or concurrency, or ensure predictable and consistent performance.

13

Figure G000044. UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda · source slide page 14.

EV0000248

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Map() Function

PES UNIVERSITY

Example 1 To double all numbers using map and lambda

```
numbers = (1, 2, 3, 4)
result = map(lambda x: x + x, numbers)
print(list(result))
```

Output:

```
<class 'map'>
[2, 4, 6, 8]
```

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Figure G000045. UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda · source slide page 15.

EV0000251

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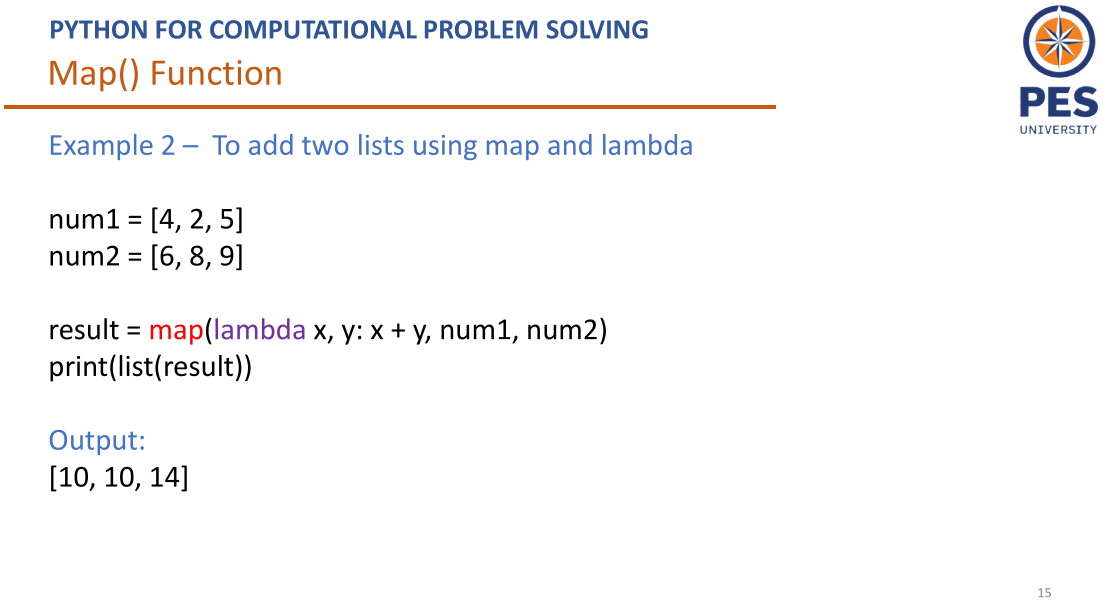

Figure G000046. UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda · source slide page 16.

EV0000254

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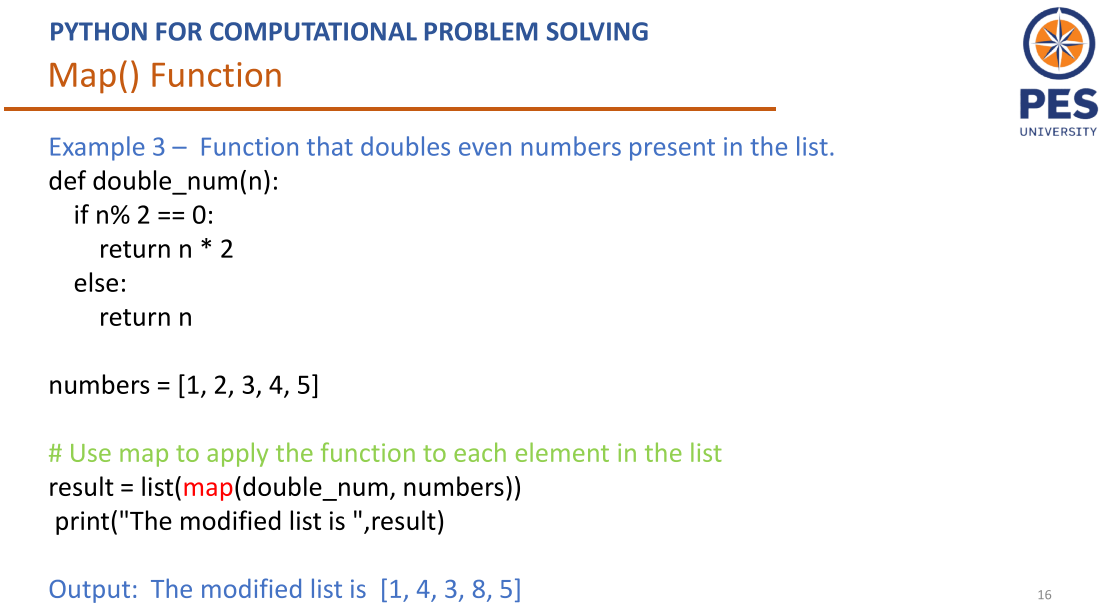

Figure G000047. UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda · source slide page 17.

EV0000257

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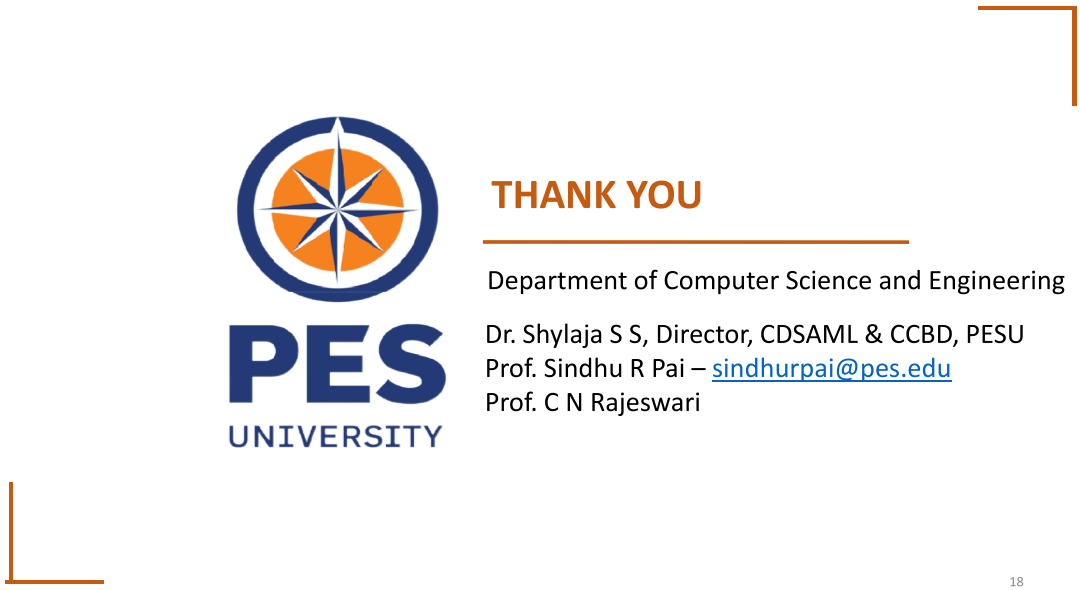

Figure G000048. UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda · source slide page 18. EV0000260

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Important points and why

1. Source slide page 1 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functional Programming-Lambda & Map Prof. Sindhu R Pai PCPS Theory Anchor - 2024 Department of Computer Science and Engineering EV0000208 · EV0000209 · EV0000210
2. Source slide page 2 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functional Programming • Functional programming is a programming paradigm where programs are constructed by applying and composing functions. • It is basically a study of computations with functions which uses pure functions, recursive functions, nested functions ,lambda functions etc. • The ability of functional programming languages to treat functions as values and pass them to functions as parameters make the code more readable and easily understandable. • They depend only on the input given to them, and the output will be the return value. Their function signature gives all the information about them i.e. their arguments and return type etc. 1 EV0000211 · EV0000212 · EV0000213

3. Source slide page 3 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functional Programming • Higher order functions are the functions that take other functions as arguments and they can also return functions. (Ex: Callback Functions) • They are deterministic in nature . • TheyThey cancan bebe understoodunderstood veryvery easilyeasily asas theythey don'tdon t changechange statesstates ofof anyany variables. • Testing and debugging is easier. They use immutable values • Offers better modularity with a shorter code • Increased productivity of the developer • It adopts lazy evaluation which avoids repeated evaluation because the value is evaluated and stored only when it is needed. • Example: Functional Constructs like Lazy Map & Lists ,tuples, sets etc. 2 EV0000214 · EV0000215 · EV0000216
4. Source slide page 4 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING The keyword - lambda Lambda (Anonymous function/Nameless Function/ Throw away function creation) • Lambda functions are functions without the name. • ItIts alsol calledll d as anonymous ffunctionsti or ththrow away ffunctions.ti • It can have any number of arguments but can have only one expression. • The expression is executed and returned (Implicit return) • Lambda functions are used when function objects are required. • Lambda functions are used when you need a particular function for a short period of time. 3 EV0000217 · EV0000218 · EV0000219
5. Source slide page 5 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING The keyword - lambda Advantages: •Lambda functions in Python are very useful in defining the in-line functions. •Most suitable when it is used as a part of another function(especially with map, filter and reduce functions) •Less Code (Concise code) 4 EV0000220 · EV0000221 · EV0000222
6. Source slide page 6 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING The keyword - lambda Syntax: lambda input arguments: expression / Output • lambda is a keyword used to declare lambda function. • Input arguments: It can have multiple arguments (if there are more than one arguments, then separate the arguments by a comma). • Expression or output: Expression gets evaluated and results are returned 5 EV0000223 · EV0000224 · EV0000225
7. Source slide page 7 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples Example1: To find the square of a number (Comparison between normal and lambda function) Normal Function Lambda Function def square(n): square=lambda n : n*n return n*n print(square(4)) print(square(4)) #Function Call 6 EV0000226 · EV0000227 · EV0000228
8. Source slide page 8 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples Example 2: To find the greatest of two numbers using the relational operator. maximum=lambdal bd a,bb : a iff a>bb elsel bb #max(a,b)(b) can alsol bbe usedd print("The maximum is ",maximum(6,7)) print(maximum,type(maximum)) Output: The maximum is 7 <function <lambda> at 0x00000217A6F93550> <class 'function'> 7 EV0000229 · EV0000230 · EV0000231
9. Source slide page 9 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples The power of lambda is better understood when you use them as an anonymous • function inside another function. Example 3 Suppose we have a function definition that takes one argument, and that argument will be multiplied with an unknown number. def myfunc(n): return lambda a : a * n double_N = myfunc(2) print(double_N(12)) Output:24 8 EV0000232 · EV0000233 · EV0000234

10. Source slide page 10 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples We can use the same function definition to make two functions (doube_N and triple_N functions) in the same program. • Example 4 def myfunc(n):y ( ) return lambda a : a * n double_N = myfunc(2) triple_N = myfunc(3) Output 30 print(double_N(15)) 45 print(triple_N(15)) 9
EV0000235 · EV0000236 · EV0000237

11. Source slide page 11 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Map() Function Python map() function is a built in function which is utilized to apply a function to each element of an iterable (like a list or a tuple) and returns a new iterable object. Syntax: map(function, iterables) 1. The map() function takes two arguments: a function and iterables. 2. The first argument for the map function should be a callable which takes one argument – could be a free function, function of a type, user defined function, or a lambda function. 3. This function will be applied to every element of the iterable, and creates a map object which is also an iterable. 10
EV0000238 · EV0000239 · EV0000240

12. Source slide page 12 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Map() Function Working: • The map function causes iteration through the iterable, and applies the callable on each element of the iterable • All these happen only conceptually as the map function is lazy and the above said operations happen only after the map object is iterated. • We can force an iteration of the map object by passing the map object as an argument for the list or set constructor. • We can pass one or more iterable to the map() function. 11 EV0000241 · EV0000242 · EV0000243

13. Source slide page 13 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Map() Function Lazy and Eager objects: Under lazy object creation, the object is created when it is needed whereas in the case of eager object creation, an object is created as soon it is instantiated. A lazy object representing a function would be quick to create, and populate itself with information when that information is requested. Built-in functions like range(),map(),zip() and open() functions are basically lazy function objects. Example: As map object can be iterated over, the computation will be done for only one item in each loop. In contrast, when we use list, which is an eager object, it basically computes all the values at one time. Lazy evaluation can be a powerful technique for optimizing code and improving performance, but it is only sometimes necessary or appropriate. 12
EV0000244 · EV0000245 · EV0000246

14. Source slide page 14 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Map() Function Choosing between lazy and eager evaluation • When deciding which evaluation strategy to use for a given programming problem, there is no one-size-fits-all answer. The choice depends on the nature of the problem, the characteristics of the data, the requirements of the solution, and the preferences of the programmer. • However, some general guidelines can be followed to help make an informed decision. For instance, lazy evaluation should be used when you want to delay or avoid computations that are not needed, save memory by avoiding intermediate results, create and manipulate infinite or recursive data structures, or explore multiple possibilities without committing to one. • On the other hand, eager evaluation is preferable when you want to perform computations as soon as possible, exploit parallelism or concurrency, or ensure predictable and consistent performance. 13 EV0000247 · EV0000248 · EV0000249

15. Source slide page 15 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Map() Function Example 1 To double all numbers using map and lambda numbers = (1, 2, 3, 4) result = map(lambda x: x + x, numbers) print(list(result)) Output: <class 'map'> [2, 4, 6, 8] 14 EV0000250 · EV0000251 · EV0000252
16. Source slide page 16 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Map() Function Example 2 – To add two lists using map and lambda num1 = [4, 2, 5] num2 = [6, 8, 9] result = map(lambda x, y: x + y, num1, num2) print(list(result)) Output: [10, 10, 14] 15 EV0000253 · EV0000254 · EV0000255
17. Source slide page 17 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Map() Function Example 3 – Function that doubles even numbers present in the list. def double_num(n): if n% 2 == 0: return n * 2 else: return n numbers = [1, 2, 3, 4, 5] # Use map to apply the function to each element in the list result = list(map(double_num, numbers)) print("The modified list is ",result) Output: The modified list is [1, 4, 3, 8, 5] 16 EV0000256 · EV0000257 · EV0000258
18. Source slide page 18 (extracted text): THANK YOU Departmentp of Computerp Science and Engineeringg g Dr. Shylaja S S, Director, CDSAML & CCBD, PESU Prof. Sindhu R Pai – sindhurpai@pes.edu Prof. C N Rajeswari 18 EV0000259 · EV0000260 · EV0000261

SHORT-NOTE RECAP

Source deck preserved: UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000208

14 Source slide deck: UE24CS151A PCPS Unit 4 Sindhu Class 1 4 functional programming map lambda

This entry preserves the complete slide deck titled UE24CS151A PCPS Unit 4 Sindhu Class 1 4 functional programming map lambda as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Python for computational Problem Solving, Unit_03, to avoid an unsupported match. EV0000262 · EV0000263

Graphs and diagrams

Figure G000049. UE24CS151A PCPS Unit 4 Sindhu Class 1 4 functional programming map lambda · source slide page 1.

EV0000263

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Functional Programming

PES UNIVERSITY

- Functional programming is a programming paradigm where programs are constructed by applying and composing functions.
- It is basically a study of computations with functions which uses pure functions, recursive functions, nested functions ,lambda functions etc.
- The ability of functional programming languages to treat functions as values and pass them to functions as parameters make the code more readable and easily understandable.
- They depend only on the input given to them, and the output will be the return value. Their function signature gives all the information about them i.e. their arguments and return type etc.

1

Figure G000050. UE24CS151A PCPS Unit 4 Sindhu Class 1 4 functional programming map lambda · source slide page 2.

EV0000266

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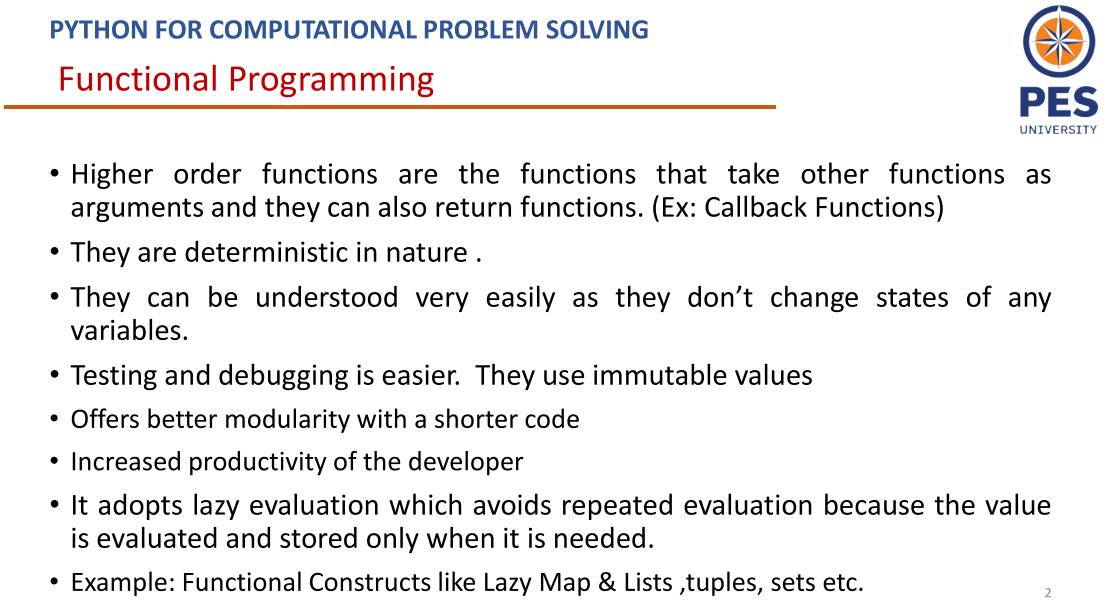

Figure G000051. UE24CS151A PCPS Unit 4 Sindhu Class 1 4 functional programming map lambda · source slide page 3.

EV0000269

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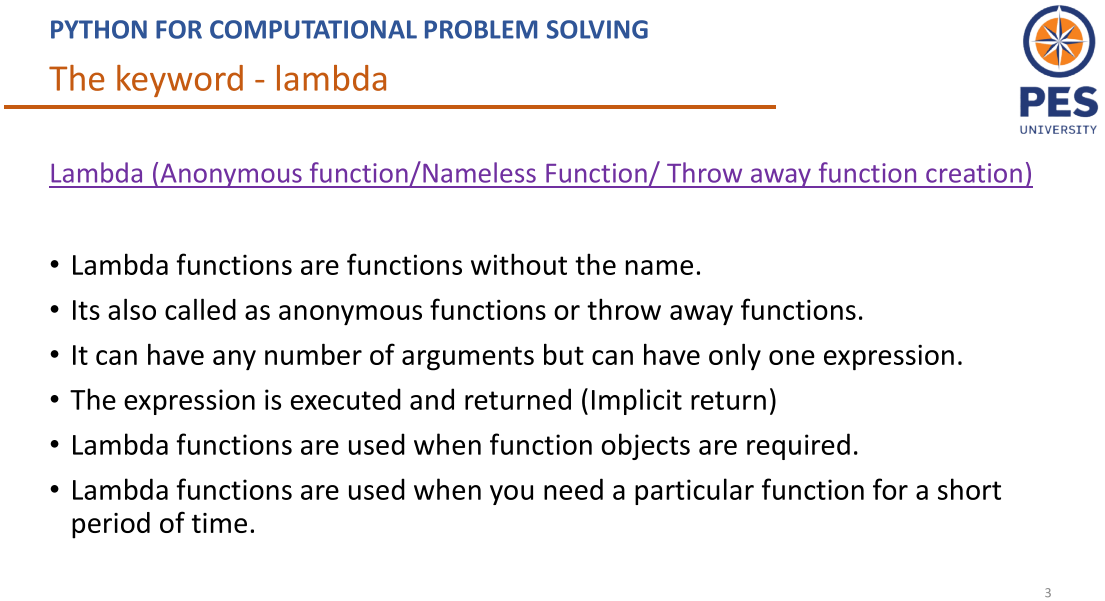

Figure G000052. UE24CS151A PCPS Unit 4 Sindhu Class 1 4 functional programming map lambda · source slide page 4.

EV0000272

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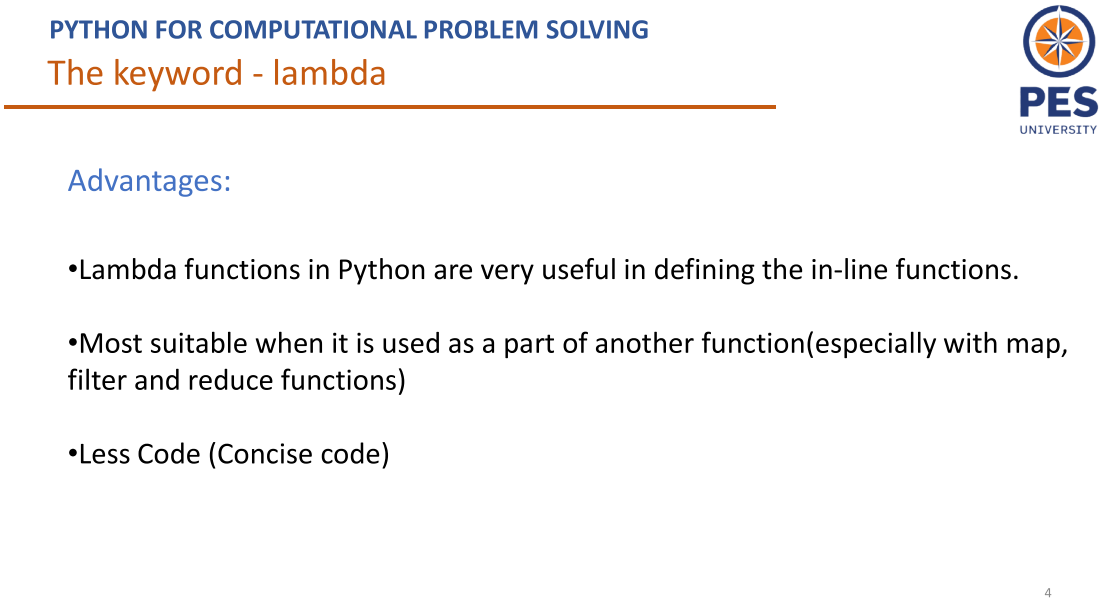

Figure G000053. UE24CS151A PCPS Unit 4 Sindhu Class 1 4 functional programming map lambda · source slide page 5.

EV0000275

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

The keyword - lambda

PES UNIVERSITY

Syntax: **lambda input arguments: expression / Output**

- **lambda** is a keyword used to declare lambda function.
- **Input arguments:** It can have multiple arguments (if there are more than one arguments, then separate the arguments by a comma).
- **Expression or output:** Expression gets evaluated and results are returned

5

Figure G000054. UE24CS151A PCPS Unit 4 Sindhu Class 1 4 functional programming map lambda · source slide page 6.

EV0000278

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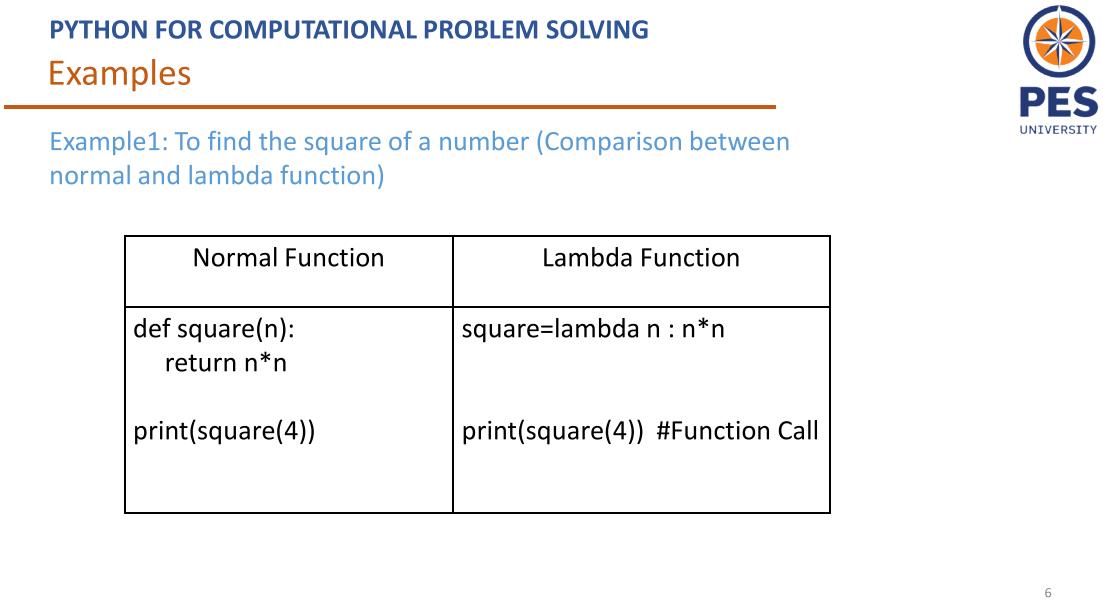

Figure G000055. UE24CS151A PCPS Unit 4 Sindhu Class 1 4 functional programming map lambda · source slide page 7.

EV0000281

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Examples

PES UNIVERSITY

Example 2: To find the greatest of two numbers using the relational operator.

```
maximum=lambda a,b : a if a>b else b    #max(a,b) can also be used
print("The maximum is ",maximum(6,7))
print(maximum,type(maximum))

Output:
The maximum is  7
<function <lambda> at 0x00000217A6F93550> <class 'function'>
```

7

Figure G000056. UE24CS151A PCPS Unit 4 Sindhu Class 1 4 functional programming map lambda · source slide page 8.

EV0000284

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Examples

The power of lambda is better understood when you use them as an anonymous function inside another function.

Example 3

Suppose we have a function definition that takes one argument, and that argument will be multiplied with an unknown number.

```
def myfunc(n):
    return lambda a : a * n

double_N = myfunc(2)

print(double_N(12))
```

Output:24

8

Figure G000057. UE24CS151A PCPS Unit 4 Sindhu Class 1 4 functional programming map lambda · source slide page 9.

EV0000287

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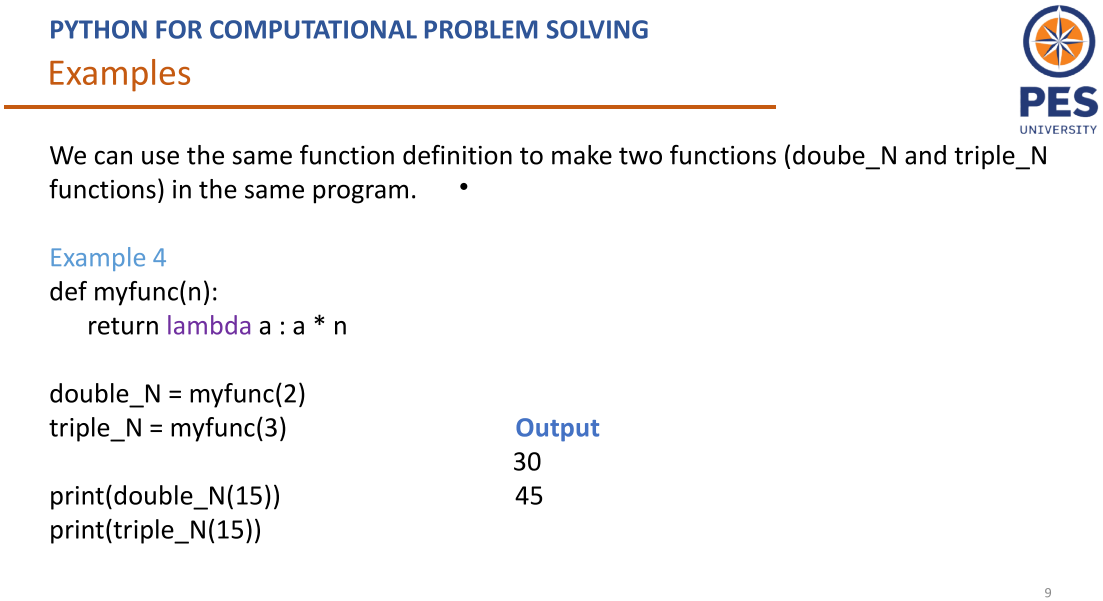

Figure G000058. UE24CS151A PCPS Unit 4 Sindhu Class 1 4 functional programming map lambda · source slide page 10.

EV0000290

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Map() Function

PES UNIVERSITY

Python map() function is a built in function which is utilized to apply a function to each element of an iterable (like a list or a tuple) and returns a new iterable object.

Syntax: map(function, iterables)

1. The map() function takes two arguments: a function and iterables.
2. The first argument for the map function should be a callable which takes one argument – could be a free function, function of a type, user defined function, or a lambda function.
3. This function will be applied to every element of the iterable, and creates a map object which is also an iterable.

10

Figure G000059. UE24CS151A PCPS Unit 4 Sindhu Class 1 4 functional programming map lambda · source slide page 11.

EV0000293

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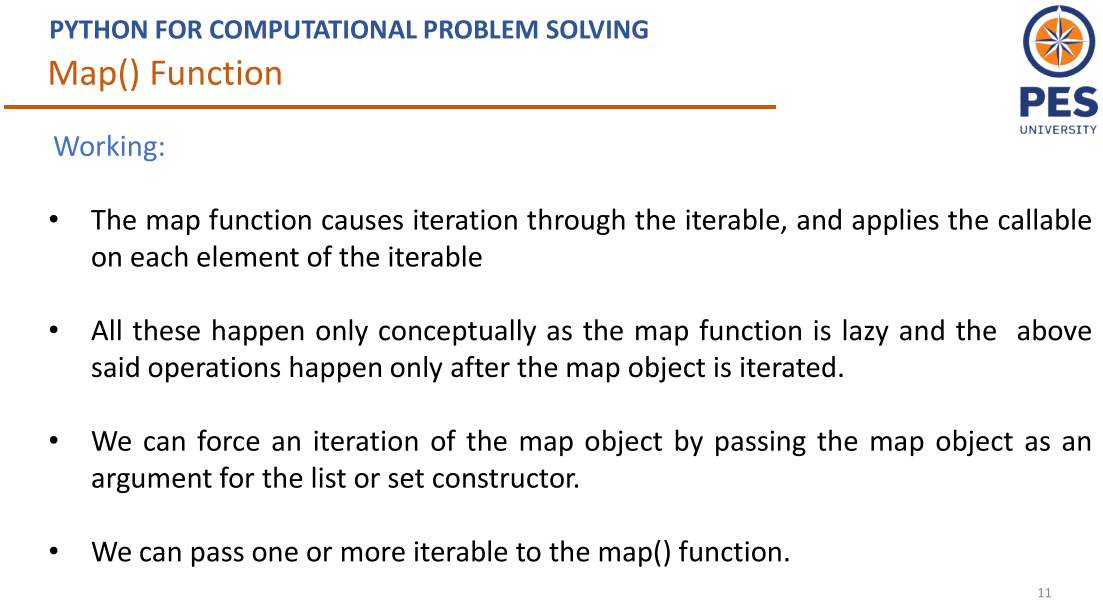

Figure G000060. UE24CS151A PCPS Unit 4 Sindhu Class 1 4 functional programming map lambda · source slide page 12.

EV0000296

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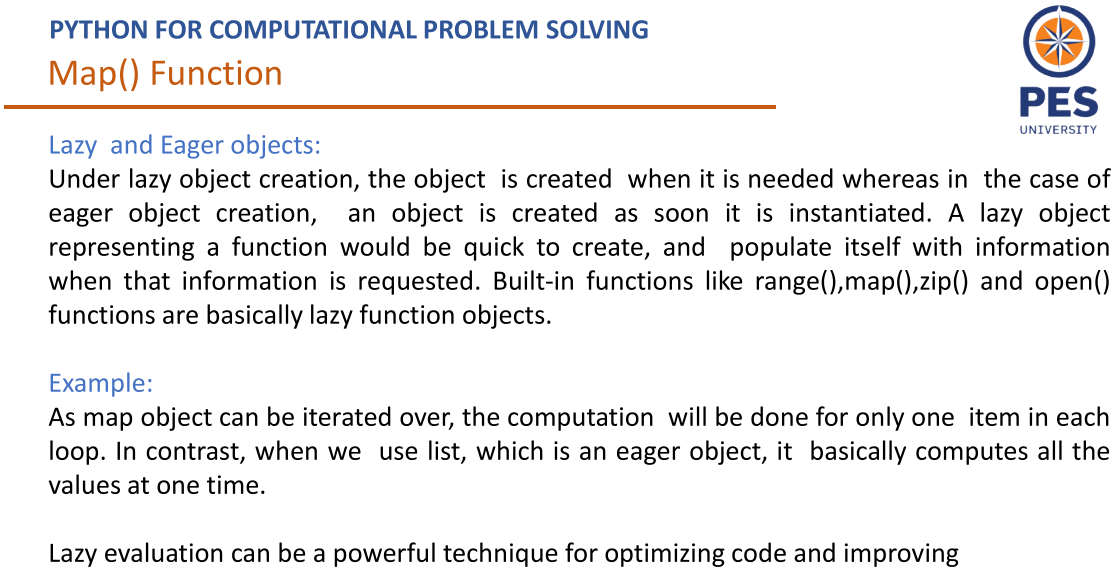

Figure G000061. UE24CS151A PCPS Unit 4 Sindhu Class 1 4 functional programming map lambda · source slide page 13.

EV0000299

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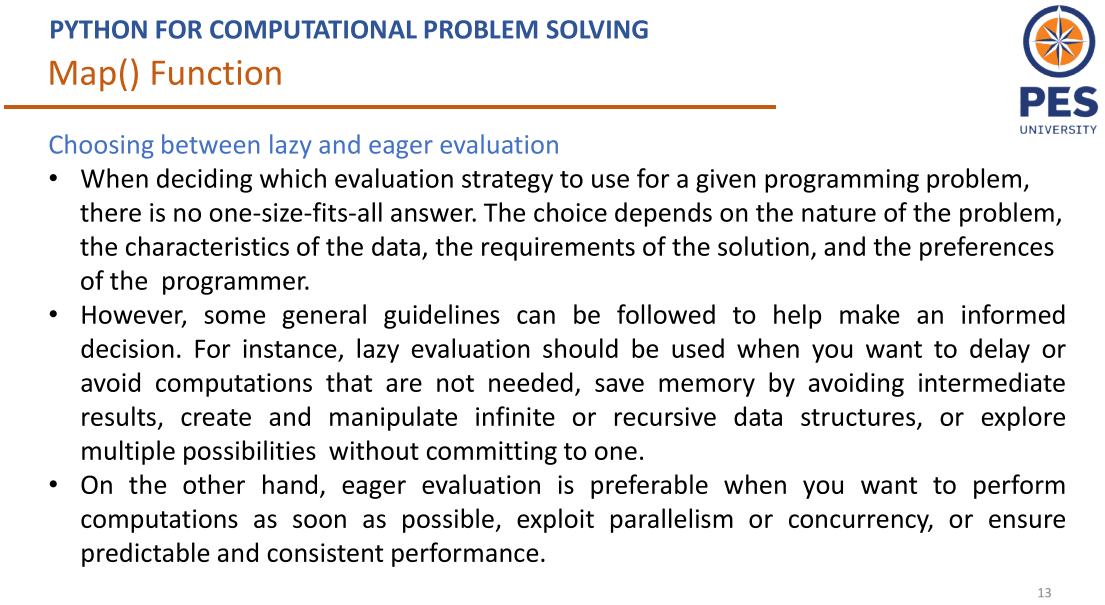

Figure G000062. UE24CS151A PCPS Unit 4 Sindhu Class 1 4 functional programming map lambda · source slide page 14.

EV0000302

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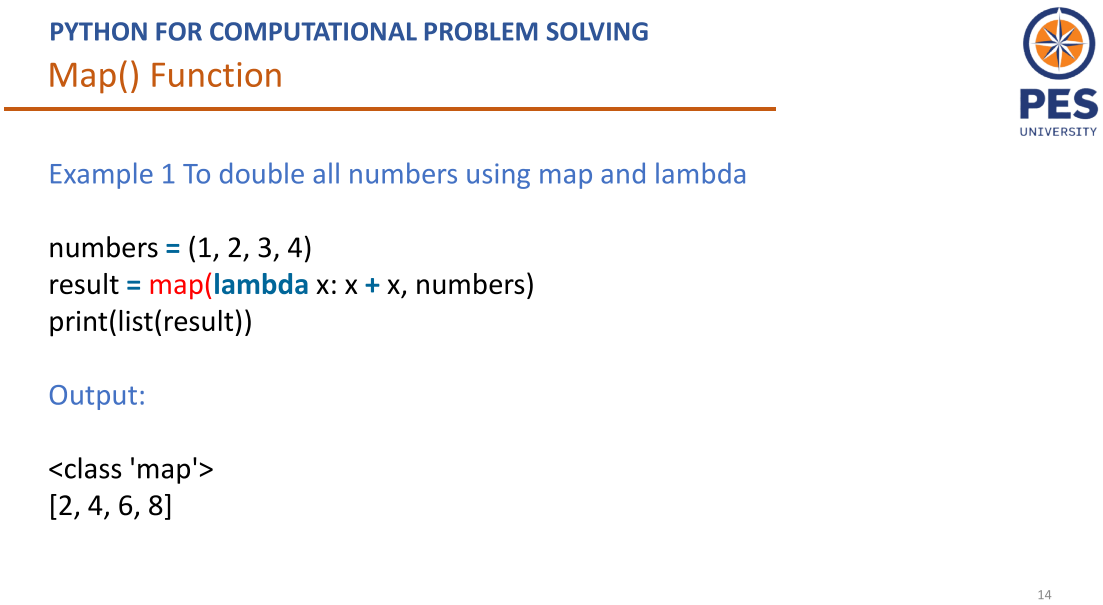

Figure G000063. UE24CS151A PCPS Unit 4 Sindhu Class 1 4 functional programming map lambda · source slide page 15.

EV0000305

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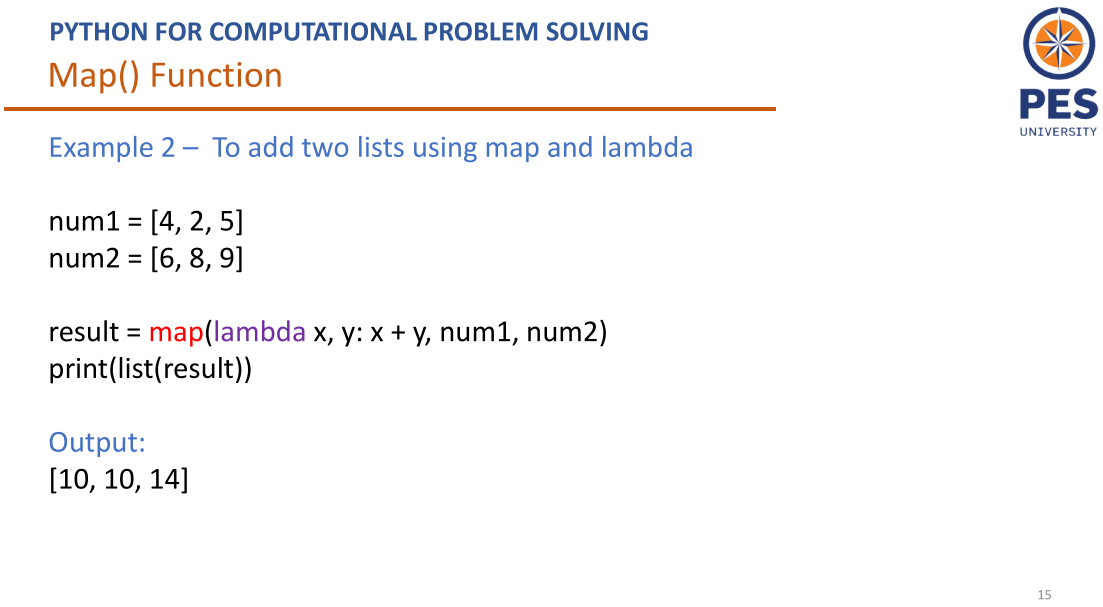

Figure G000064. UE24CS151A PCPS Unit 4 Sindhu Class 1 4 functional programming map lambda · source slide page 16.

EV0000308

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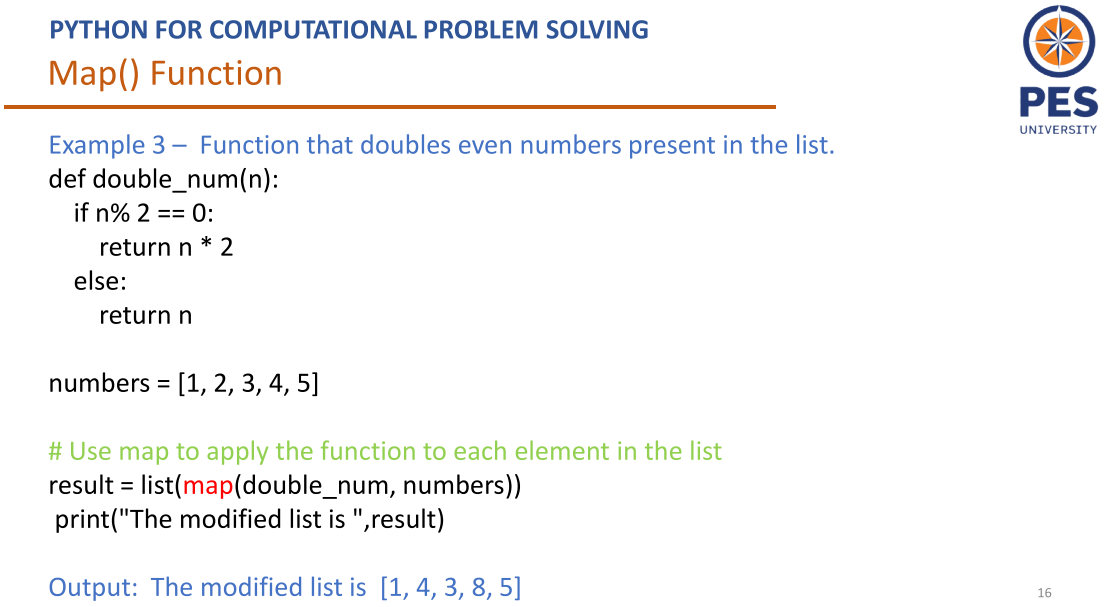

Figure G000065. UE24CS151A PCPS Unit 4 Sindhu Class 1 4 functional programming map lambda · source slide page 17.

EV0000311

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

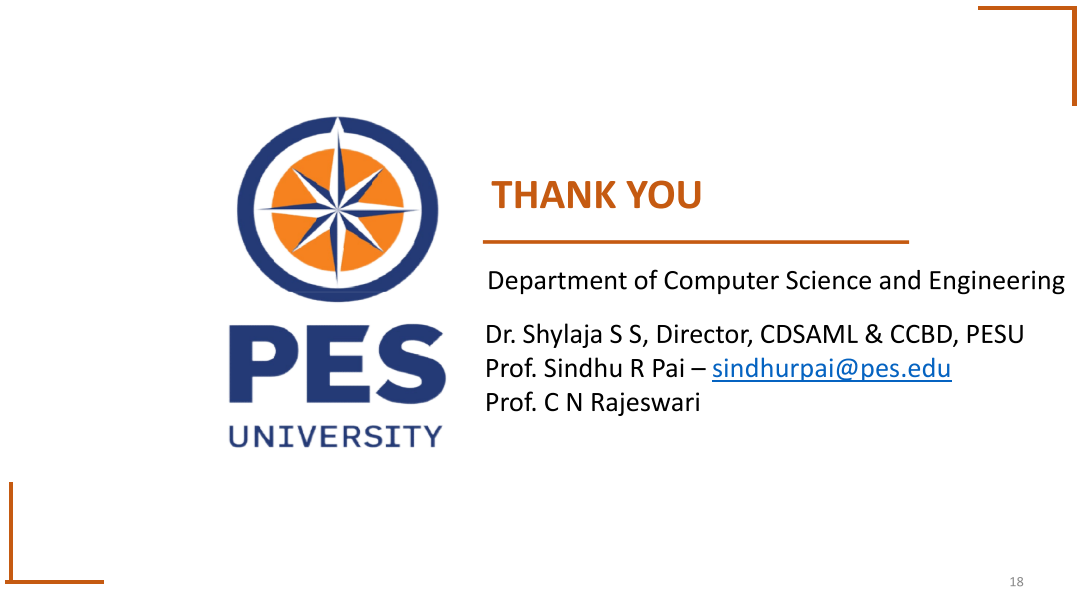

Figure G000066. UE24CS151A PCPS Unit 4 Sindhu Class 1 4 functional programming map lambda · source slide page 18.

EV0000314

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functional Programming-Lambda & Map Prof. Sindhu R Pai PCPS Theory Anchor - 2024 Department of Computer Science and Engineering EV0000262 · EV0000263 · EV0000264
2. Source slide page 2 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functional Programming • Functional programming is a programming paradigm where programs are constructed by applying and composing functions. • It is basically a study of computations with functions which uses pure functions, recursive functions, nested functions ,lambda functions etc. • The ability of functional programming languages to treat functions as values and pass them to functions as parameters make the code more readable and easily understandable. • They depend only on the input given to them, and the output will be the return value. Their function signature gives all the information about them i.e. their arguments and return type etc. 1 EV0000265 · EV0000266 · EV0000267

3. Source slide page 3 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Functional Programming • Higher order functions are the functions that take other functions as arguments and they can also return functions. (Ex: Callback Functions) • They are deterministic in nature . • TheyThey cancan bebe understoodunderstood veryvery easilyeasily asas theythey don'tdon t changechange statesstates ofof anyany variables. • Testing and debugging is easier. They use immutable values • Offers better modularity with a shorter code • Increased productivity of the developer • It adopts lazy evaluation which avoids repeated evaluation because the value is evaluated and stored only when it is needed. • Example: Functional Constructs like Lazy Map & Lists ,tuples, sets etc. 2 EV0000268 · EV0000269 · EV0000270
4. Source slide page 4 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING The keyword - lambda Lambda (Anonymous function/Nameless Function/ Throw away function creation) • Lambda functions are functions without the name. • ItIts alsol calledll d as anonymous ffunctionsti or ththrow away ffunctions.ti • It can have any number of arguments but can have only one expression. • The expression is executed and returned (Implicit return) • Lambda functions are used when function objects are required. • Lambda functions are used when you need a particular function for a short period of time. 3 EV0000271 · EV0000272 · EV0000273
5. Source slide page 5 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING The keyword - lambda Advantages: •Lambda functions in Python are very useful in defining the in-line functions. •Most suitable when it is used as a part of another function(especially with map, filter and reduce functions) •Less Code (Concise code) 4 EV0000274 · EV0000275 · EV0000276
6. Source slide page 6 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING The keyword - lambda Syntax: lambda input arguments: expression / Output • lambda is a keyword used to declare lambda function. • Input arguments: It can have multiple arguments (if there are more than one arguments, then separate the arguments by a comma). • Expression or output: Expression gets evaluated and results are returned 5 EV0000277 · EV0000278 · EV0000279
7. Source slide page 7 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples Example1: To find the square of a number (Comparison between normal and lambda function) Normal Function Lambda Function def square(n): square=lambda n : n*n return n*n print(square(4)) print(square(4)) #Function Call 6 EV0000280 · EV0000281 · EV0000282
8. Source slide page 8 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples Example 2: To find the greatest of two numbers using the relational operator. maximum=lambdal bd a,bb : a iff a>bb elsel bb #max(a,b)(b) can alsol bbe usedd print("The maximum is ",maximum(6,7)) print(maximum,type(maximum)) Output: The maximum is 7 <function <lambda> at 0x00000217A6F93550> <class 'function'> 7 EV0000283 · EV0000284 · EV0000285
9. Source slide page 9 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples The power of lambda is better understood when you use them as an anonymous • function inside another function. Example 3 Suppose we have a function definition that takes one argument, and that argument will be multiplied with an unknown number. def myfunc(n): return lambda a : a * n double_N = myfunc(2) print(double_N(12)) Output:24 8 EV0000286 · EV0000287 · EV0000288

10. Source slide page 10 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples We can use the same function definition to make two functions (doube_N and triple_N functions) in the same program. • Example 4 def myfunc(n):y ( ) return lambda a : a * n double_N = myfunc(2) triple_N = myfunc(3) Output 30 print(double_N(15)) 45 print(triple_N(15)) 9
EV0000289 · EV0000290 · EV0000291

11. Source slide page 11 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Map() Function Python map() function is a built in function which is utilized to apply a function to each element of an iterable (like a list or a tuple) and returns a new iterable object. Syntax: map(function, iterables) 1. The map() function takes two arguments: a function and iterables. 2. The first argument for the map function should be a callable which takes one argument – could be a free function, function of a type, user defined function, or a lambda function. 3. This function will be applied to every element of the iterable, and creates a map object which is also an iterable. 10
EV0000292 · EV0000293 · EV0000294

12. Source slide page 12 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Map() Function Working: • The map function causes iteration through the iterable, and applies the callable on each element of the iterable • All these happen only conceptually as the map function is lazy and the above said operations happen only after the map object is iterated. • We can force an iteration of the map object by passing the map object as an argument for the list or set constructor. • We can pass one or more iterable to the map() function. 11 EV0000295 · EV0000296 · EV0000297

13. Source slide page 13 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Map() Function Lazy and Eager objects: Under lazy object creation, the object is created when it is needed whereas in the case of eager object creation, an object is created as soon it is instantiated. A lazy object representing a function would be quick to create, and populate itself with information when that information is requested. Built-in functions like range(),map(),zip() and open() functions are basically lazy function objects. Example: As map object can be iterated over, the computation will be done for only one item in each loop. In contrast, when we use list, which is an eager object, it basically computes all the values at one time. Lazy evaluation can be a powerful technique for optimizing code and improving performance, but it is only sometimes necessary or appropriate. 12
EV0000298 · EV0000299 · EV0000300

14. Source slide page 14 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Map() Function Choosing between lazy and eager evaluation • When deciding which evaluation strategy to use for a given programming problem, there is no one-size-fits-all answer. The choice depends on the nature of the problem, the characteristics of the data, the requirements of the solution, and the preferences of the programmer. • However, some general guidelines can be followed to help make an informed decision. For instance, lazy evaluation should be used when you want to delay or avoid computations that are not needed, save memory by avoiding intermediate results, create and manipulate infinite or recursive data structures, or explore multiple possibilities without committing to one. • On the other hand, eager evaluation is preferable when you want to perform computations as soon as possible, exploit parallelism or concurrency, or ensure predictable and consistent performance. 13 EV0000301 · EV0000302 · EV0000303

15. Source slide page 15 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Map() Function Example 1 To double all numbers using map and lambda numbers = (1, 2, 3, 4) result = map(lambda x: x + x, numbers) print(list(result)) Output: <class 'map'> [2, 4, 6, 8] 14 EV0000304 · EV0000305 · EV0000306
16. Source slide page 16 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Map() Function Example 2 – To add two lists using map and lambda num1 = [4, 2, 5] num2 = [6, 8, 9] result = map(lambda x, y: x + y, num1, num2) print(list(result)) Output: [10, 10, 14] 15 EV0000307 · EV0000308 · EV0000309
17. Source slide page 17 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Map() Function Example 3 – Function that doubles even numbers present in the list. def double_num(n): if n% 2 == 0: return n * 2 else: return n numbers = [1, 2, 3, 4, 5] # Use map to apply the function to each element in the list result = list(map(double_num, numbers)) print("The modified list is ",result) Output: The modified list is [1, 4, 3, 8, 5] 16 EV0000310 · EV0000311 · EV0000312
18. Source slide page 18 (extracted text): THANK YOU Departmentp of Computerp Science and Engineeringg g Dr. Shylaja S S, Director, CDSAML & CCBD, PESU Prof. Sindhu R Pai – sindhurpai@pes.edu Prof. C N Rajeswari 18 EV0000313 · EV0000314 · EV0000315

SHORT-NOTE RECAP

Source deck preserved: UE24CS151A PCPS Unit 4 Sindhu Class 1 4 functional programming map lambda. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000262

15 Source slide deck: UE24CS151A PCPS Unit 4 SindhuRPai Class 5 reduce

This entry preserves the complete slide deck titled UE24CS151A PCPS Unit 4 SindhuRPai Class 5 reduce as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Python for computational Problem Solving, Unit_03, to avoid an unsupported match. EV0000343 · EV0000344

Graphs and diagrams

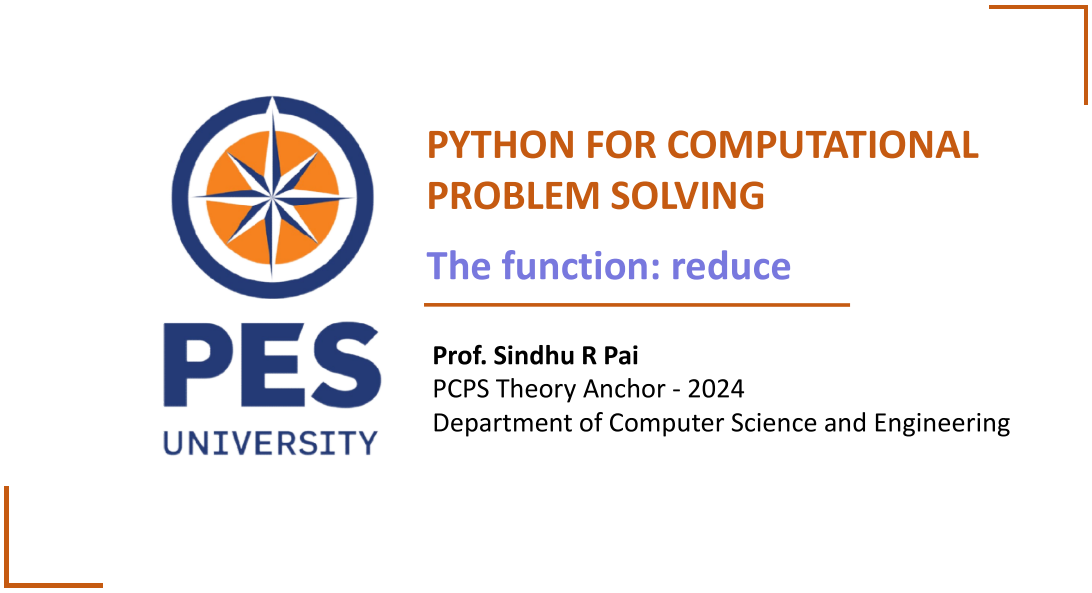

Figure G000076. UE24CS151A PCPS Unit 4 SindhuRPai Class 5 reduce · source slide page 1.

EV0000344

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

reduce() Function

Introduction

The reduce() is a function that applies a given function to the elements of an iterable, reducing them to a single value. This function is defined in "**functools**" module.

Syntax

functools.reduce(function, iterable[, initializer])

- The **function argument** is a function that takes two arguments and returns a single value. The first argument is the accumulated value, and the second argument is the current value from the iterable.
- The **iterable** argument is the sequence of values to be reduced.
- The optional initializer argument is used to provide an initial value. If no initializer is specified, the first element of the iterable is used as the initial value.

1

Figure G000077. UE24CS151A PCPS Unit 4 SindhuRPai Class 5 reduce · source slide page 2.

EV0000347

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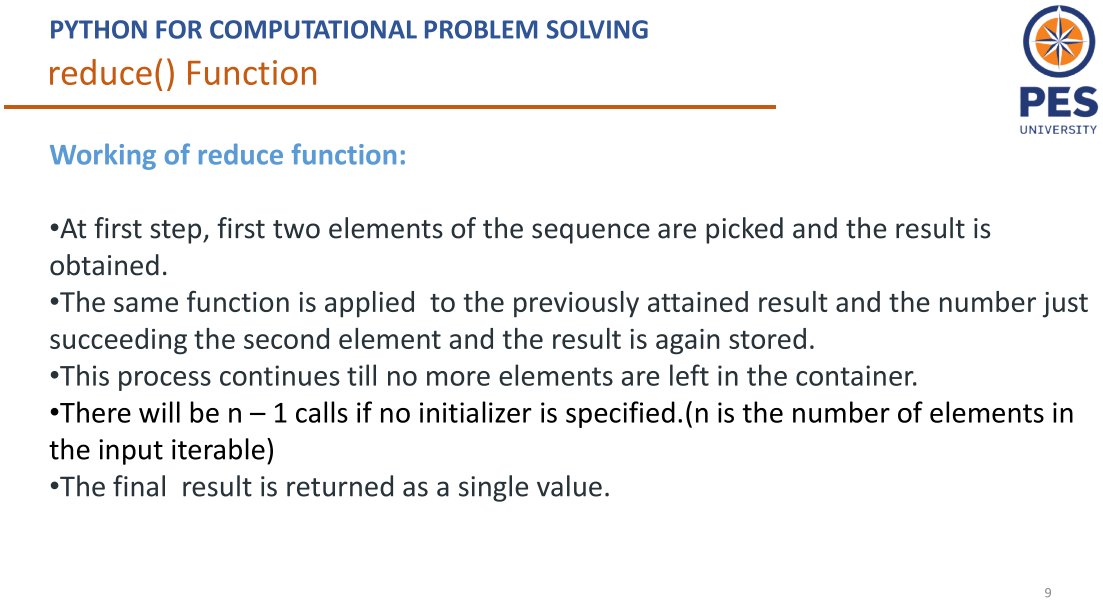

Figure G000078. UE24CS151A PCPS Unit 4 SindhuRPai Class 5 reduce · source slide page 3.

EV0000350

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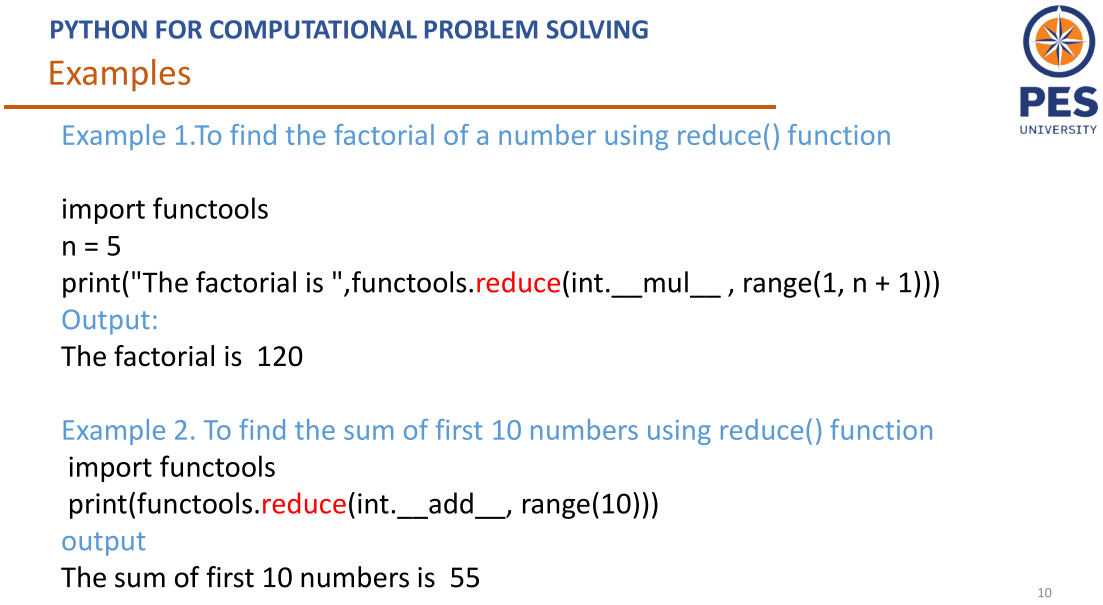

Figure G000079. UE24CS151A PCPS Unit 4 SindhuRPai Class 5 reduce · source slide page 4.

EV0000353

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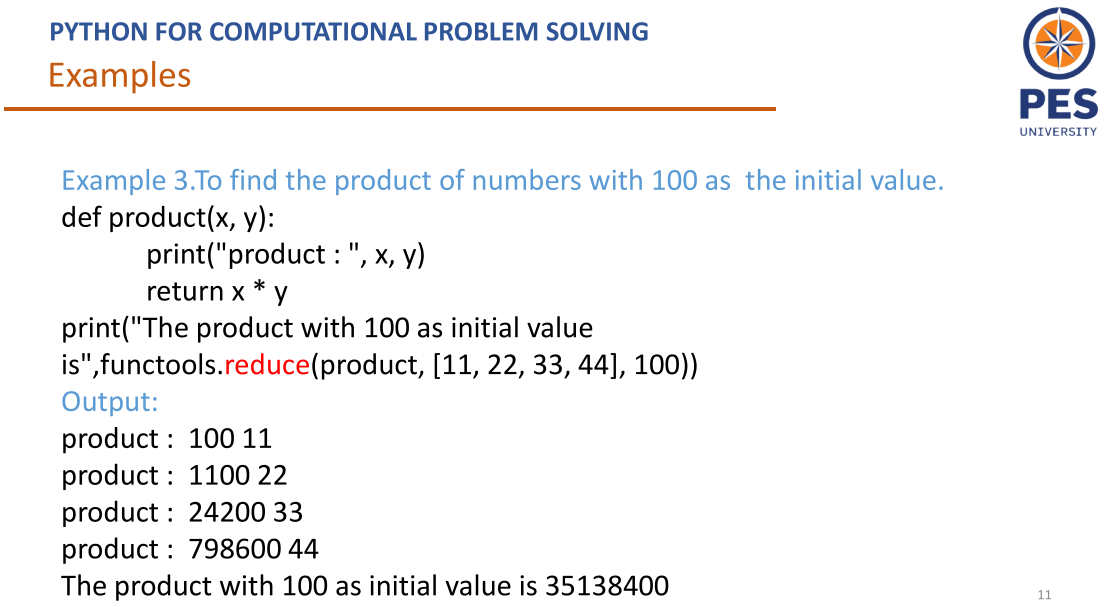

Figure G000080. UE24CS151A PCPS Unit 4 SindhuRPai Class 5 reduce · source slide page 5.

EV0000356

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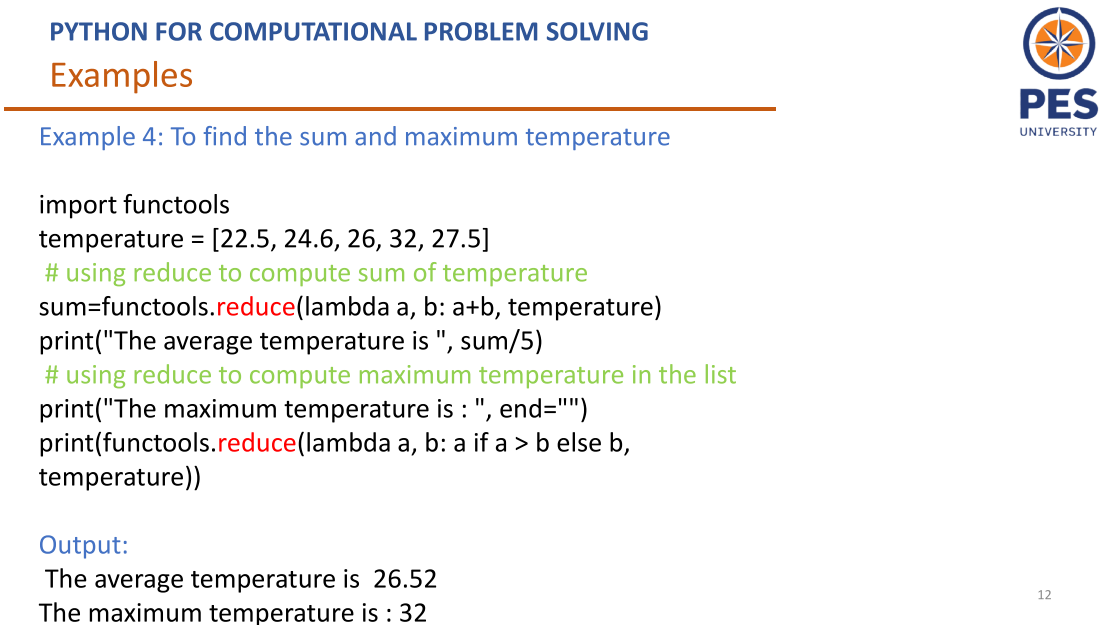

Figure G000081. UE24CS151A PCPS Unit 4 SindhuRPai Class 5 reduce · source slide page 6.

EV0000359

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

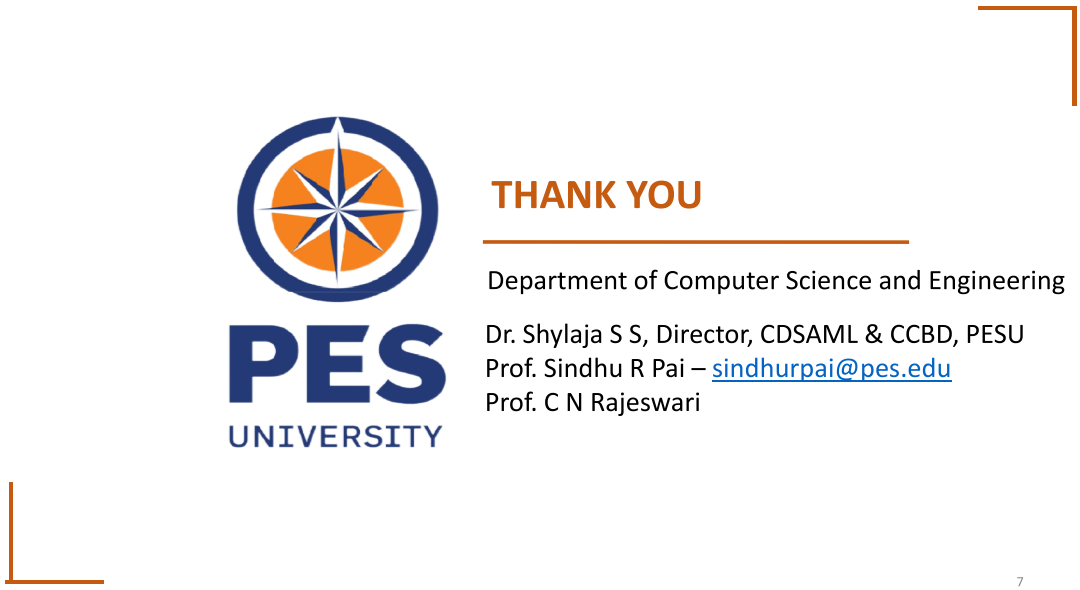

Figure G000082. UE24CS151A PCPS Unit 4 SindhuRPai Class 5 reduce · source slide page 7.

EV0000362

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING The function: reduce Prof. Sindhu R Pai PCPS Theory Anchor - 2024 Department of Computer Science and Engineering EV0000343 · EV0000344 · EV0000345
2. Source slide page 2 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING reduce() Function Introduction The reduce() is a function that applies a given function to the elements of an iterable, reducing them to a single value. This function is defined in “functools” module. Syntax functools.reduce(function, iterable[, initializer]) • The function argument is a function that takes two arguments and returns a single value. The first argument is the accumulated value, and the second argument is the current value from the iterable. • The iterable argument is the sequence of values to be reduced. • The optional initializer argument is used to provide an initial value. If no initializer is specified, the first element of the iterable is used as the initial value. 1 EV0000346 · EV0000347 · EV0000348

3. Source slide page 3 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING reduce() Function Working of reduce function: •At first step, first two elements of the sequence are picked and the result is obtained. •The•The samesame functionfunction isis appliedapplied totothethe previouslypreviously attainedattained resultresult andand thethe numbernumber justjust succeeding the second element and the result is again stored. •This process continues till no more elements are left in the container. •There will be n – 1 calls if no initializer is specified.(n is the number of elements in the input iterable) •The final result is returned as a single value. 9 EV0000349 · EV0000350 · EV0000351

4. Source slide page 4 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples Example 1.To find the factorial of a number using reduce() function import functools n = 5 print("The factorial is ",functools.reduce(int.__mul__ , range(1, n + 1))) Output: The factorial is 120 Example 2. To find the sum of first 10 numbers using reduce() function import functools print(functools.reduce(int.__add__, range(10))) output The sum of first 10 numbers is 55 10 EV0000352 · EV0000353 · EV0000354

5. Source slide page 5 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples Example 3.To find the product of numbers with 100 as the initial value. def product(x, y): print("product : ", x, y) return x * yy print("The product with 100 as initial value is",functools.reduce(product, [11, 22, 33, 44], 100)) Output: product : 100 11 product : 1100 22 product : 24200 33 product : 798600 44 The product with 100 as initial value is 35138400 11 EV0000355 · EV0000356 · EV0000357

6. Source slide page 6 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples Example 4: To find the sum and maximum temperature import functools temperature = [22.5, 24.6, 26, 32, 27.5] # using reduce to compute sum of temperature sum=functoolssum=functools.reduce(lambdareduce(lambda aa, b:b: a+ba+b, temperature)temperature) print("The average temperature is ", sum/5) # using reduce to compute maximum temperature in the list print("The maximum temperature is : ", end="") print(functools.reduce(lambda a, b: a if a > b else b, temperature)) Output: The average temperature is 26.52 12 The maximum temperature is : 32 EV0000358 · EV0000359 · EV0000360

7. Source slide page 7 (extracted text): THANK YOU Departmentp of Computerp Science and Engineeringg g Dr. Shylaja S S, Director, CDSAML & CCBD, PESU Prof. Sindhu R Pai – sindhurpai@pes.edu Prof. C N Rajeswari 7 EV0000361 · EV0000362 · EV0000363

SHORT-NOTE RECAP

Source deck preserved: UE24CS151A PCPS Unit 4 SindhuRPai Class 5 reduce. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000343

16 Source slide deck: UE24CS151A PCPS Unit 4 SindhuRPai class 6 zip

This entry preserves the complete slide deck titled UE24CS151A PCPS Unit 4 SindhuRPai class 6 zip as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Python for computational Problem Solving, Unit_03, to avoid an unsupported match. EV0000409 · EV0000410

Graphs and diagrams

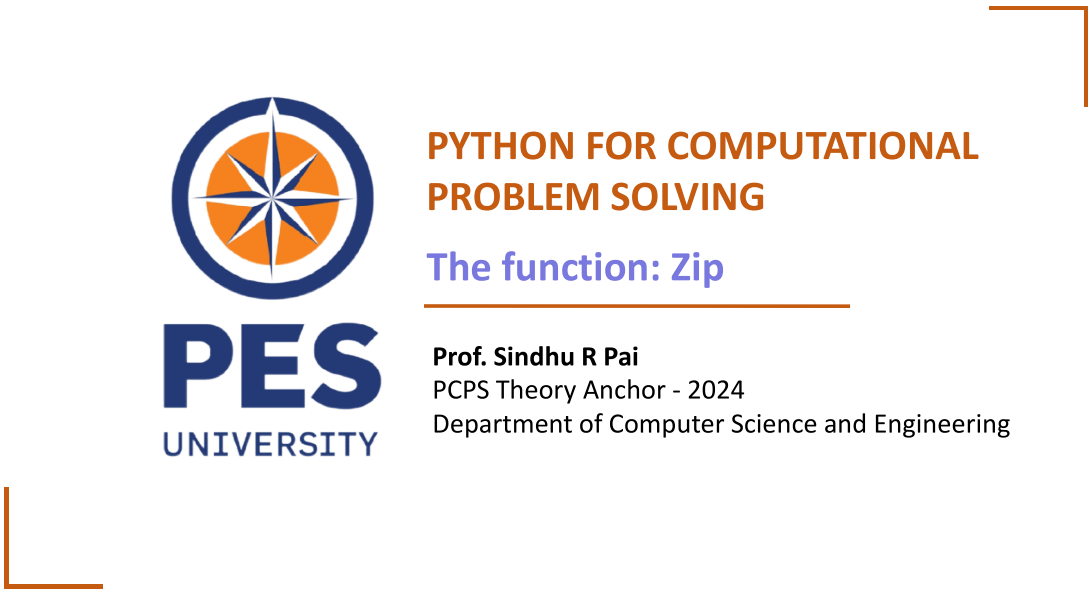

Figure G000098. UE24CS151A PCPS Unit 4 SindhuRPai class 6 zip · source slide page 1. EV0000410

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Zip() function

PES UNIVERSITY

- Python zip() method takes iterable containers and returns a single iterator object, having mapped values from all the containers.
- It is used to map the similar index of multiple containers so that they can be used just using a single entity.
- The function zip is used to associate the corresponding elements of two or more iterables into a single lazy iterable of tuples.
- It does not have any callback function.

Syntax : zip(*iterators)

2

Figure G000099. UE24CS151A PCPS Unit 4 SindhuRPai class 6 zip · source slide page 2.

EV0000413

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PYTHON FOR COMPUTATIONAL PROBLEM SOLVING

Zip() function

PES UNIVERSITY

Working

- The zip() function is used to combine two or more lists (or any other iterables) into a single iterable.
- Elements from corresponding positions are paired together.
- The resulting iterable contains tuples, where the first element from each list is paired together, the second element from each list is paired together, and so on.

3

Figure G000100. UE24CS151A PCPS Unit 4 SindhuRPai class 6 zip · source slide page 3.

EV0000416

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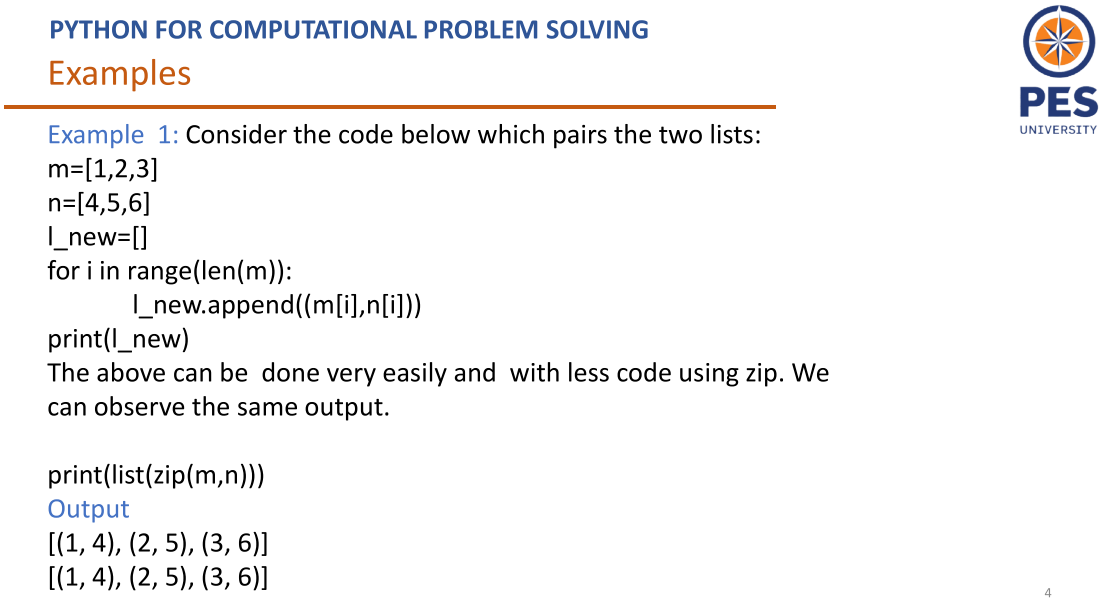

Figure G000101. UE24CS151A PCPS Unit 4 SindhuRPai class 6 zip · source slide page 4.

EV0000419

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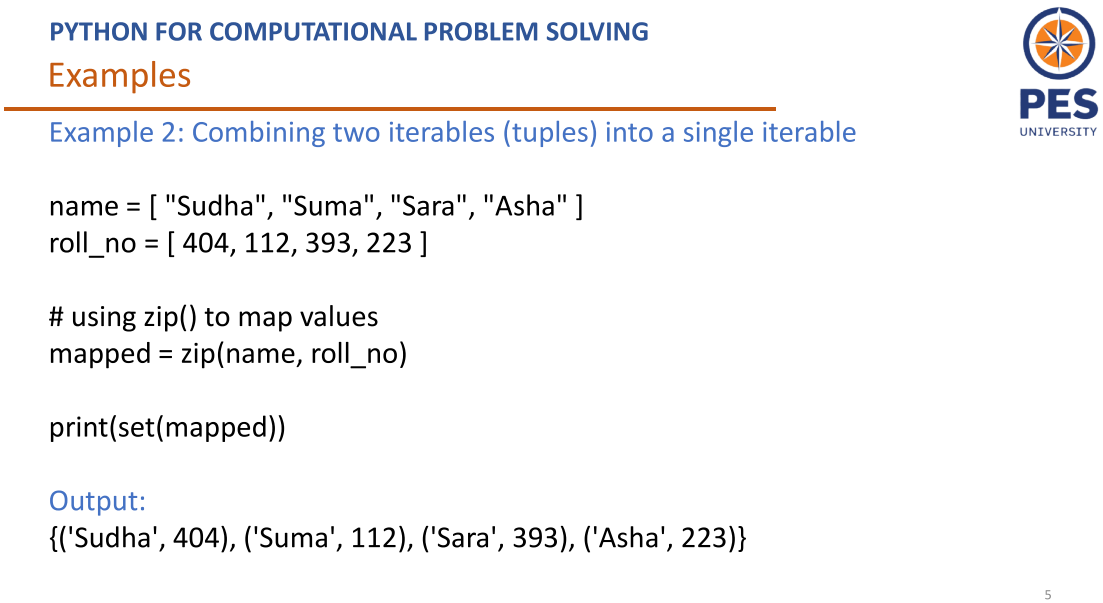

Figure G000102. UE24CS151A PCPS Unit 4 SindhuRPai class 6 zip · source slide page 5.

EV0000422

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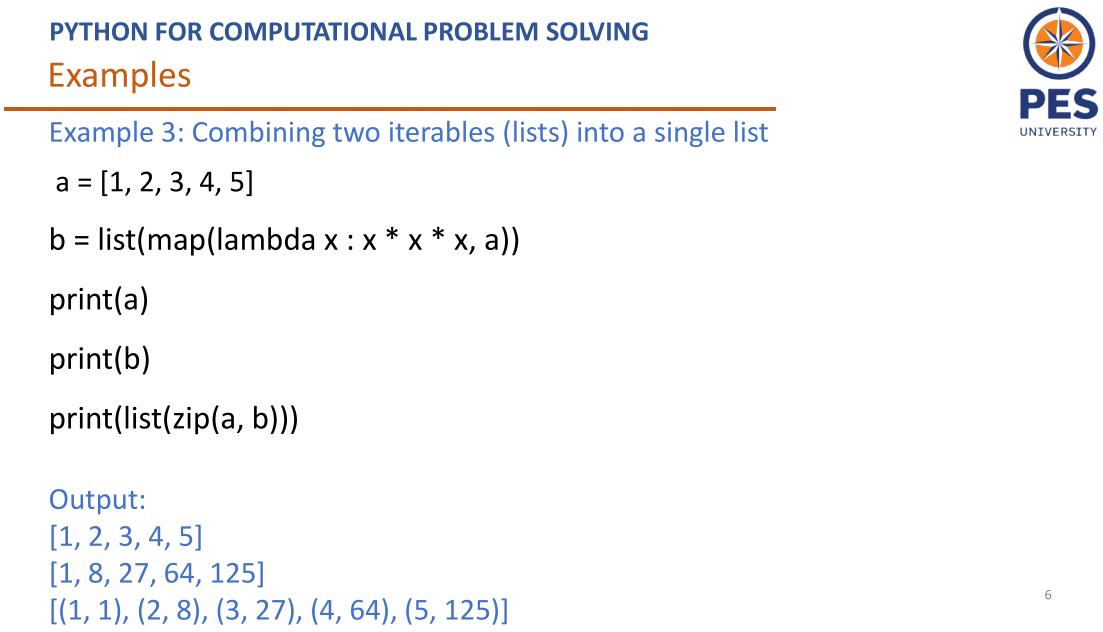

Figure G000103. UE24CS151A PCPS Unit 4 SindhuRPai class 6 zip · source slide page 6.

EV0000425

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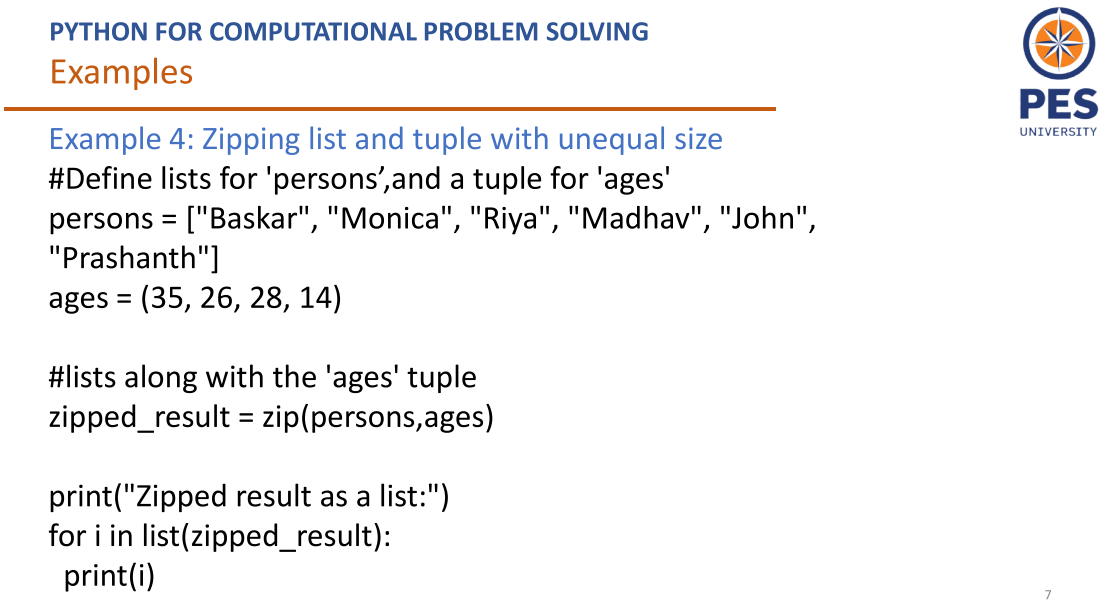

Figure G000104. UE24CS151A PCPS Unit 4 SindhuRPai class 6 zip · source slide page 7.

EV0000428

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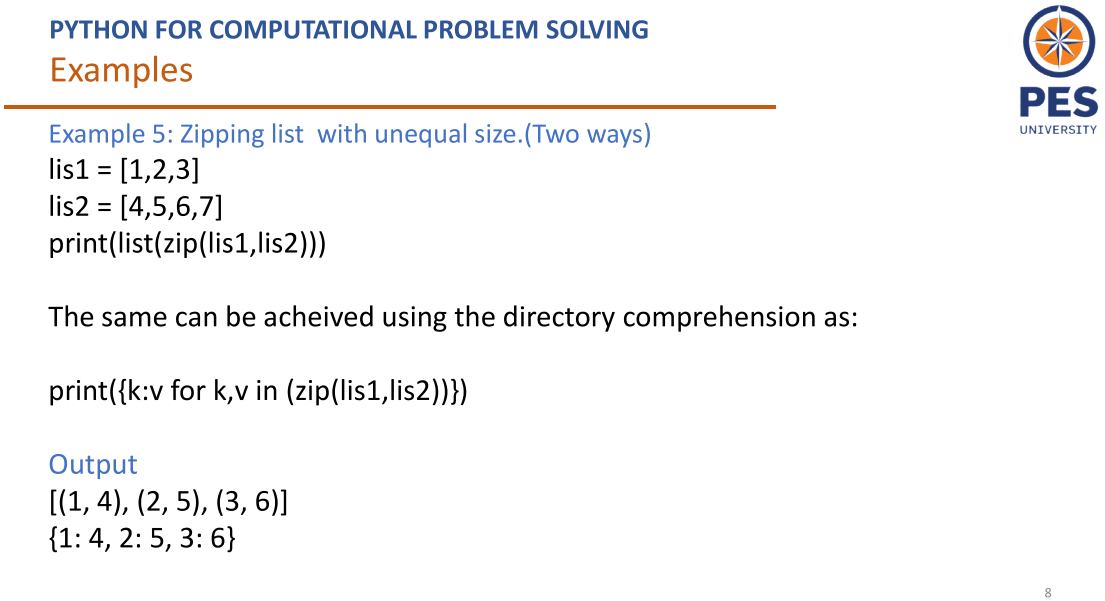

Figure G000105. UE24CS151A PCPS Unit 4 SindhuRPai class 6 zip · source slide page 8.

EV0000431

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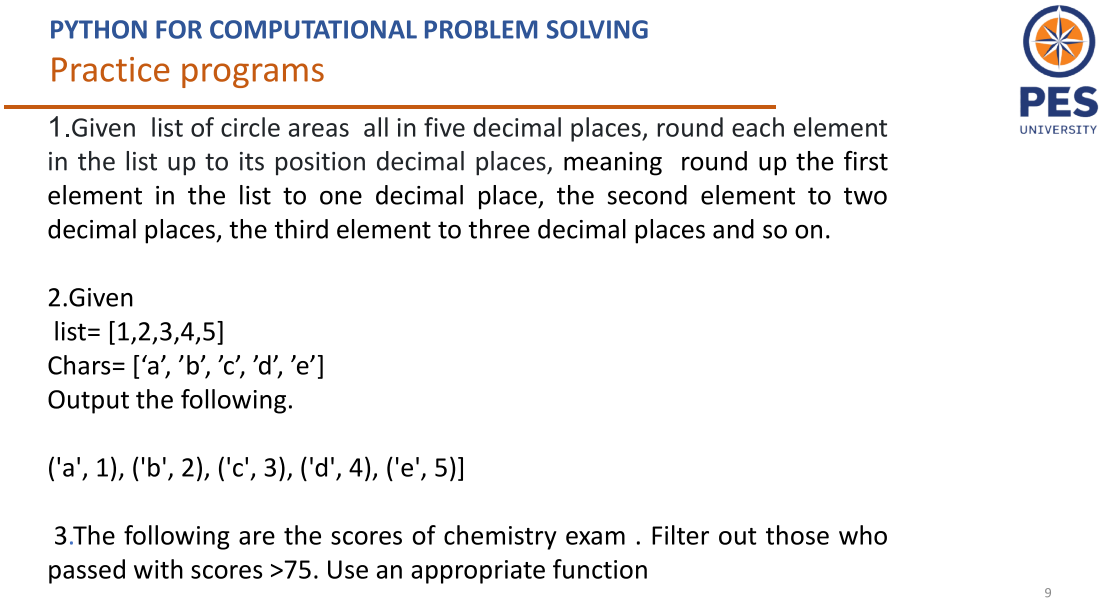

Figure G000106. UE24CS151A PCPS Unit 4 SindhuRPai class 6 zip · source slide page 9.

EV0000434

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

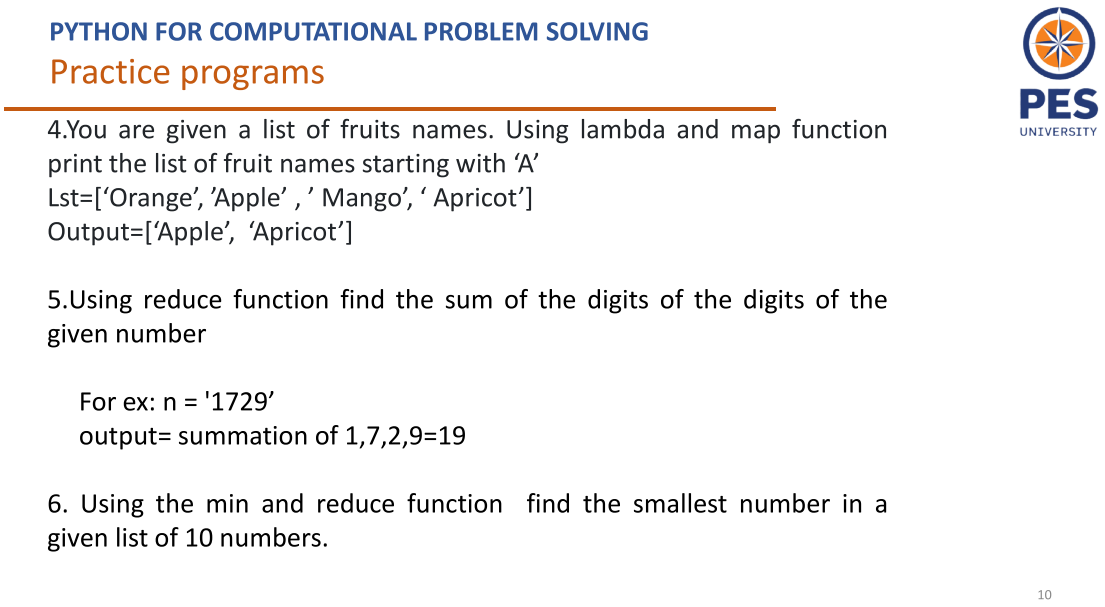

Figure G000107. UE24CS151A PCPS Unit 4 SindhuRPai class 6 zip · source slide page 10.

EV0000437

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

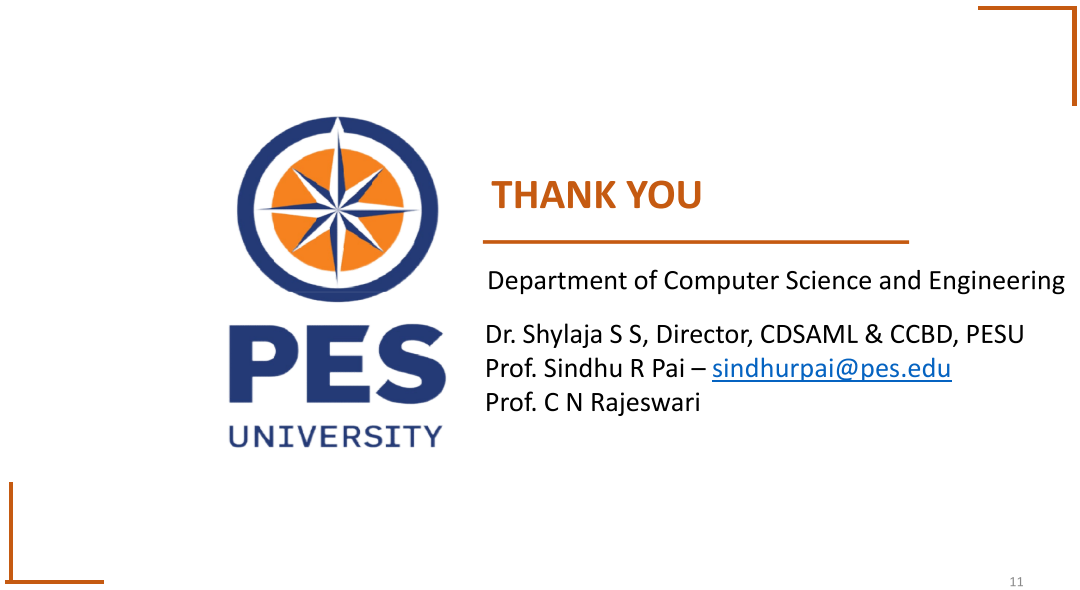

Figure G000108. UE24CS151A PCPS Unit 4 SindhuRPai class 6 zip · source slide page 11.

EV0000440

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING The function: Zip Prof. Sindhu R Pai PCPS Theory Anchor - 2024 Department of Computer Science and Engineering EV0000409 · EV0000410 · EV0000411
2. Source slide page 2 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Zip() function • Python zip() method takes iterable containers and returns a single iterator object, having mapped values from all the containers. • It is used to map the similar index of multiple containers so that they can be used just using a single entity. • The function zip is used to associate the corresponding elements of two or more iterables into a single lazy iterable of tuples. • It does not have any callback function. Syntax : zip(*iterators) 2 EV0000412 · EV0000413 · EV0000414
3. Source slide page 3 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Zip() function Working • The zip() function is used to combine two or more lists (or anyany otherother iterables)iterables) intointo aa singlesingle iterableiterable. • Elements from corresponding positions are paired together. • The resulting iterable contains tuples, where the first element from each list is paired together, the second element from each list is paired together, and so on. 3 EV0000415 · EV0000416 · EV0000417

4. Source slide page 4 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples Example 1: Consider the code below which pairs the two lists: m=[1,2,3] n=[4,5,6] l_new=[] for i in range(len(m)): ll_new.append((m[i],n[i]))new append((m[i] n[i])) print(l_new) The above can be done very easily and with less code using zip. We can observe the same output. print(list(zip(m,n))) Output [(1, 4), (2, 5), (3, 6)] [(1, 4), (2, 5), (3, 6)] 4 EV0000418 · EV0000419 · EV0000420
5. Source slide page 5 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples Example 2: Combining two iterables (tuples) into a single iterable name = ["Sudha", "Suma", "Sara", "Asha"] roll_no = [404, 112, 393, 223] # using zip()() to map valuesl mapped = zip(name, roll_no) print(set(mapped)) Output: {('Sudha', 404), ('Suma', 112), ('Sara', 393), ('Asha', 223)} 5 EV0000421 · EV0000422 · EV0000423
6. Source slide page 6 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples Example 3: Combining two iterables (lists) into a single list a = [1, 2, 3, 4, 5] b = list(map(lambda x : x * x * x, a)) print(a)print(a) print(b) print(list(zip(a, b))) Output: [1, 2, 3, 4, 5] [1, 8, 27, 64, 125] 6 [(1, 1), (2, 8), (3, 27), (4, 64), (5, 125)] EV0000424 · EV0000425 · EV0000426
7. Source slide page 7 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples Example 4: Zipping list and tuple with unequal size #Define lists for 'persons',and a tuple for 'ages' persons = ["Baskar", "Monica", "Riya", "Madhav", "John", "Prashanth"] agesages = (35, (35, 26,26, 28,28, 14)14) #lists along with the 'ages' tuple zipped_result = zip(persons,ages) print("Zipped result as a list:") for i in list(zipped_result): print(i) 7 EV0000427 · EV0000428 · EV0000429
8. Source slide page 8 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Examples Example 5: Zipping list with unequal size.(Two ways) lis1 = [1,2,3] lis2 = [4,5,6,7] print(list(zip(lis1,lis2))) The same can be acheived using the directory comprehension as: print({k:v for k,v in (zip(lis1,lis2))}) Output [(1, 4), (2, 5), (3, 6)] {1: 4, 2: 5, 3: 6} 8 EV0000430 · EV0000431 · EV0000432
9. Source slide page 9 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Practice programs 1.Given list of circle areas all in five decimal places, round each element in the list up to its position decimal places, meaning round up the first element in the list to one decimal place, the second element to two decimal places, the third element to three decimal places and so on. 2.Given2.Given list= [1,2,3,4,5] Chars= [‘a’, ’b’, ’c’, ’d’, ’e’] Output the following. ('a', 1), ('b', 2), ('c', 3), ('d', 4), ('e', 5)] 3.The following are the scores of chemistry exam . Filter out those who passed with scores >75. Use an appropriate function 9 EV0000433 · EV0000434 · EV0000435
10. Source slide page 10 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Practice programs 4.You are given a list of fruits names. Using lambda and map function print the list of fruit names starting with ‘A’ Lst=[‘Orange’, ’Apple’ , ’ Mango’, ‘ Apricot’] Output=[‘Apple’, ‘Apricot’] 5.Using5.Using reducereduce functionfunction findfind thethe sumsum ofof thethe digitsdigits ofof thethe digitsdigits ofof thethe given number For ex: n = '1729’ output= summation of 1,7,2,9=19 6. Using the min and reduce function find the smallest number in a given list of 10 numbers. 10 EV0000436 · EV0000437 · EV0000438
11. Source slide page 11 (extracted text): THANK YOU Departmentp of Computerp Science and Engineeringg g Dr. Shylaja S S, Director, CDSAML & CCBD, PESU Prof. Sindhu R Pai – sindhurpai@pes.edu Prof. C N Rajeswari 11 EV0000439 · EV0000440 · EV0000441

SHORT-NOTE RECAP

Source deck preserved: UE24CS151A PCPS Unit 4 SindhuRPai class 6 zip. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000409

17 Source slide deck: Max Min

This entry preserves the complete slide deck titled Max Min as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Python for computational Problem Solving, Unit_03, to avoid an unsupported match. EV0000442 · EV0000443

Graphs and diagrams

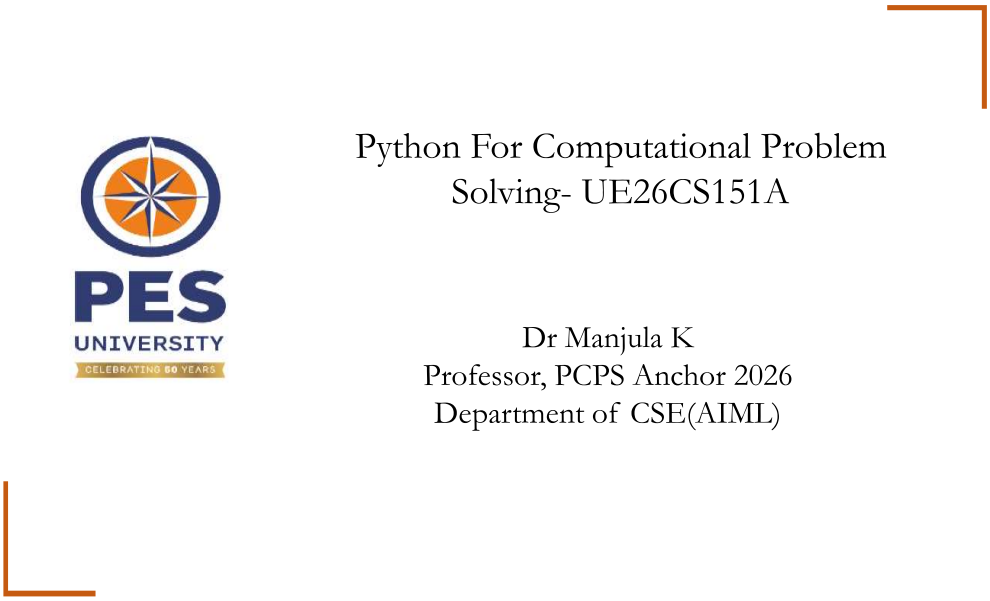

Figure G000109. Max Min · source slide page 1.

EV0000443

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

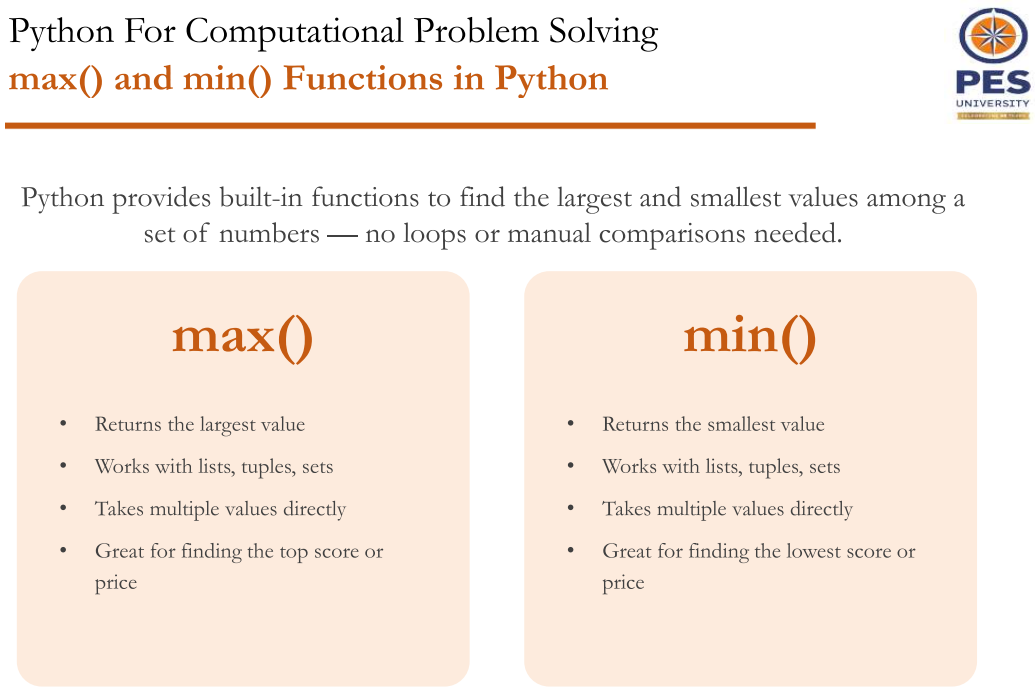

Figure G000110. Max Min · source slide page 2.

EV0000446

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

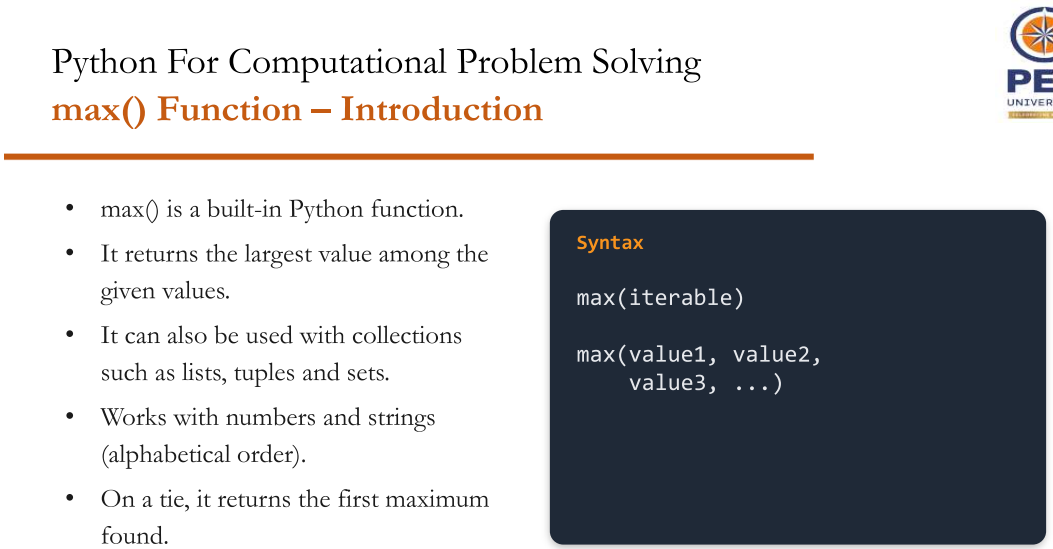

Figure G000111. Max Min · source slide page 3.

EV0000449

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

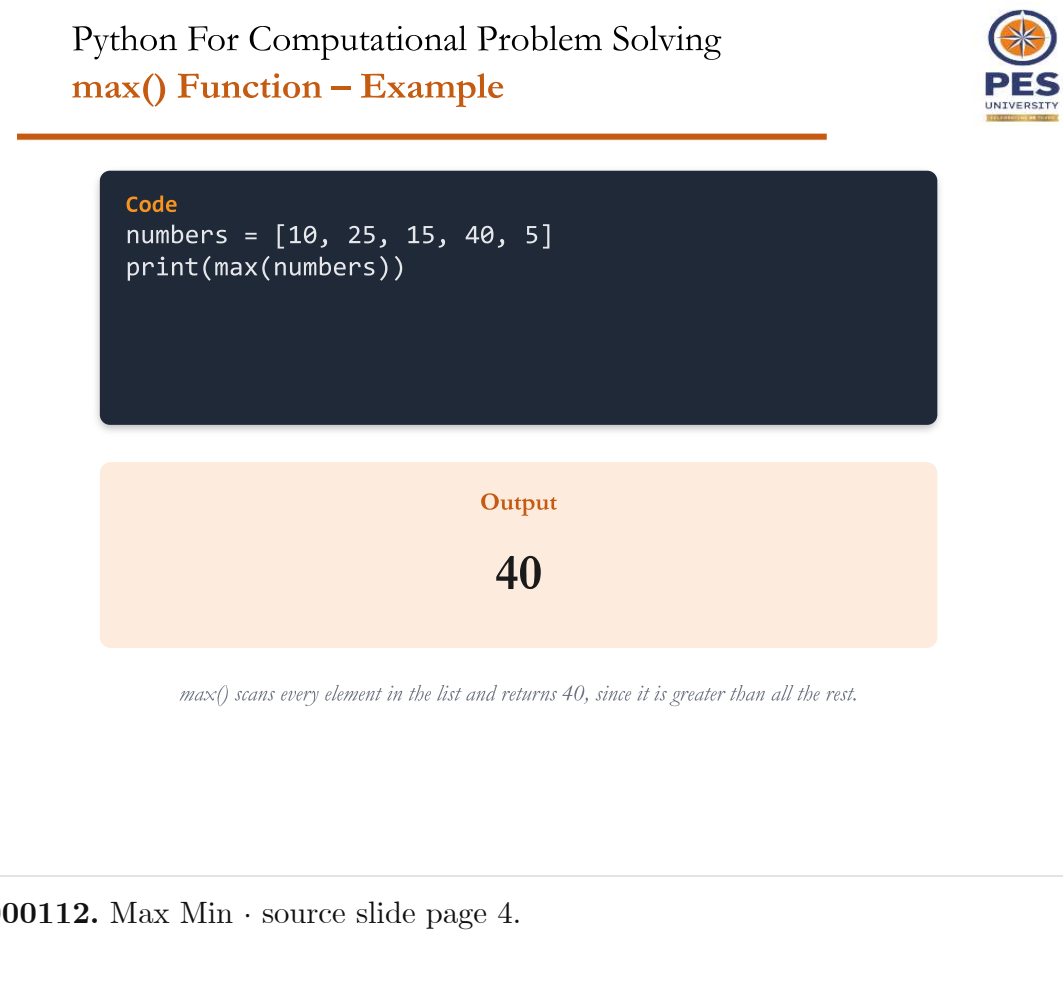

Figure G000112. Max Min · source slide page 4.

EV0000452

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

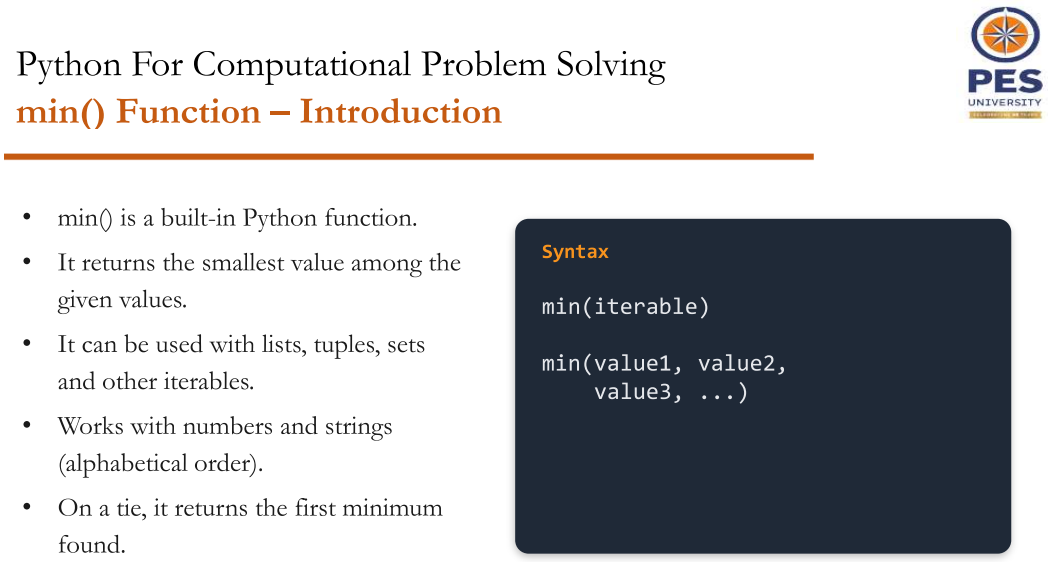

Figure G000113. Max Min · source slide page 5.

EV0000455

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

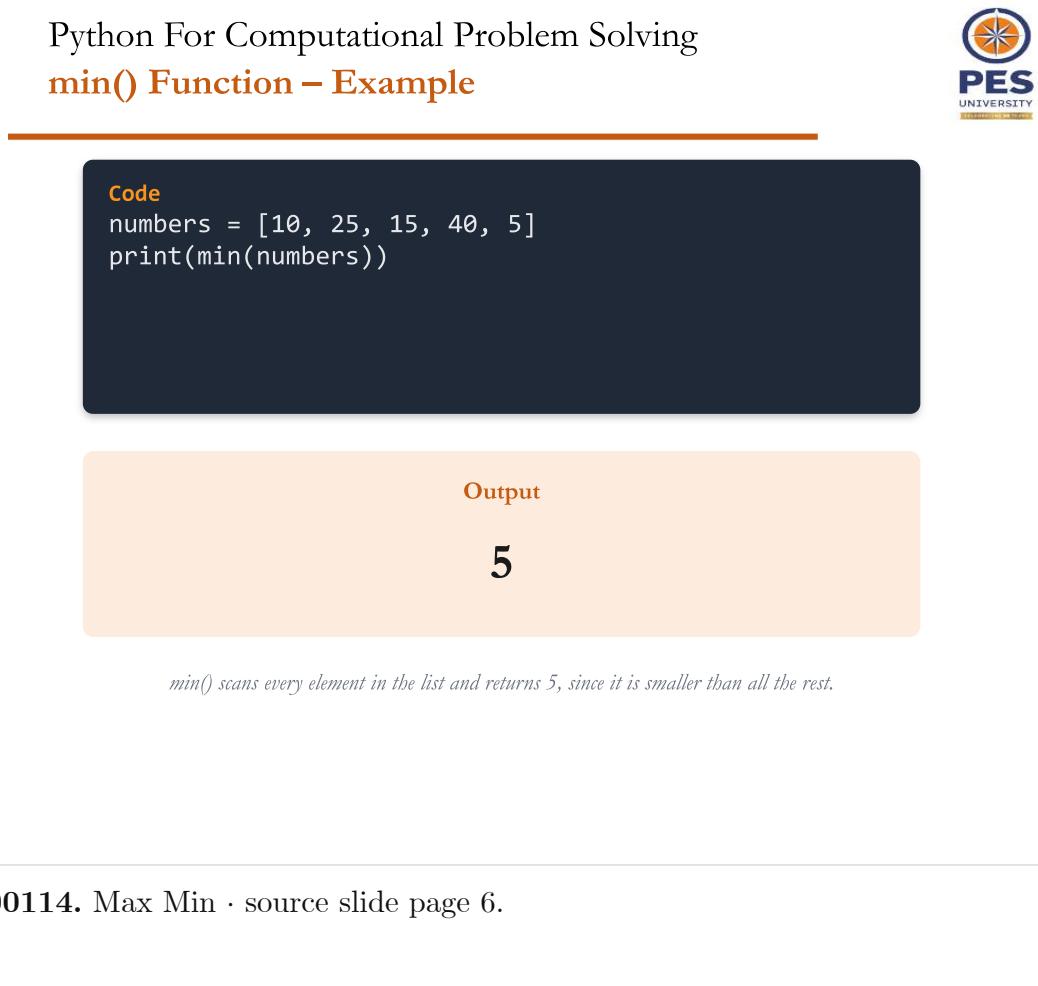

Figure G000114. Max Min · source slide page 6.

EV0000458

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

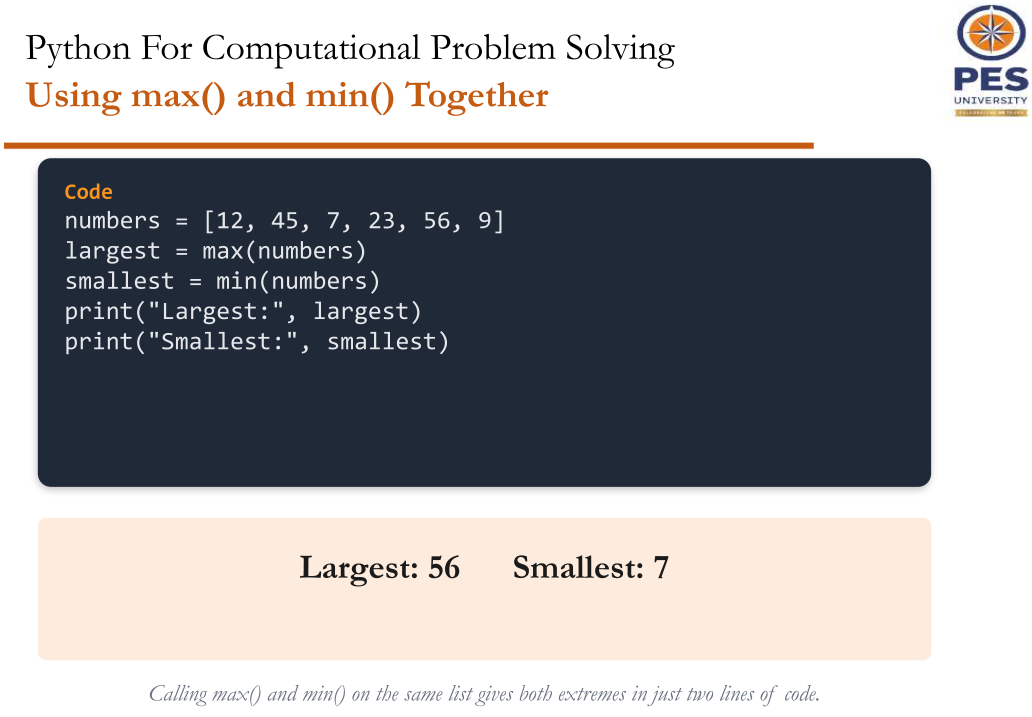

Figure G000115. Max Min · source slide page 7.

EV0000461

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

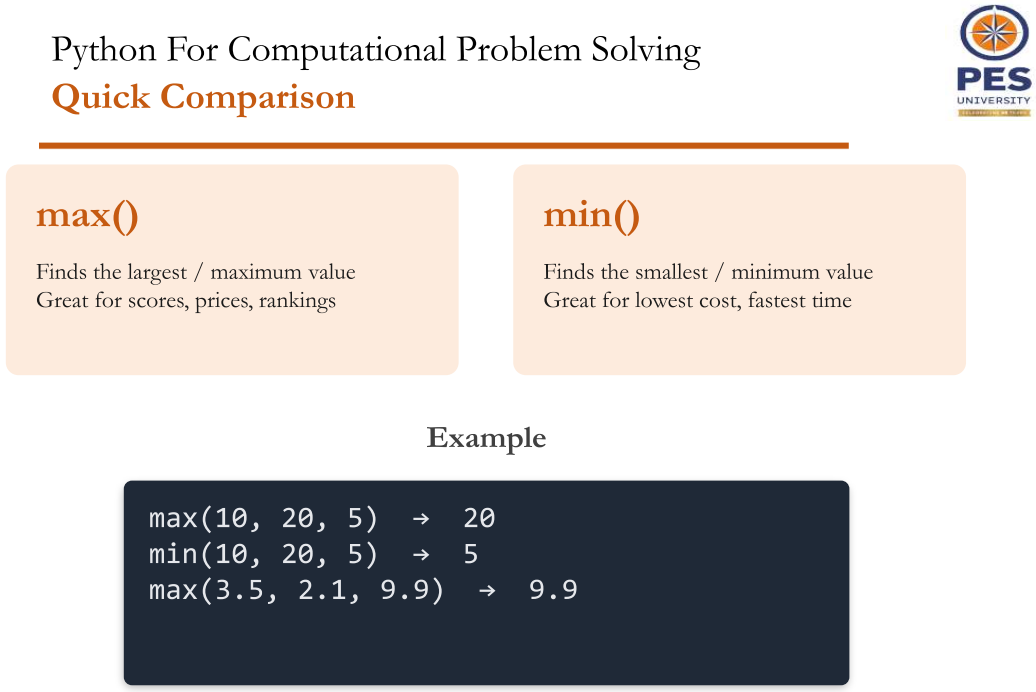

Figure G000116. Max Min · source slide page 8.

EV0000464

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

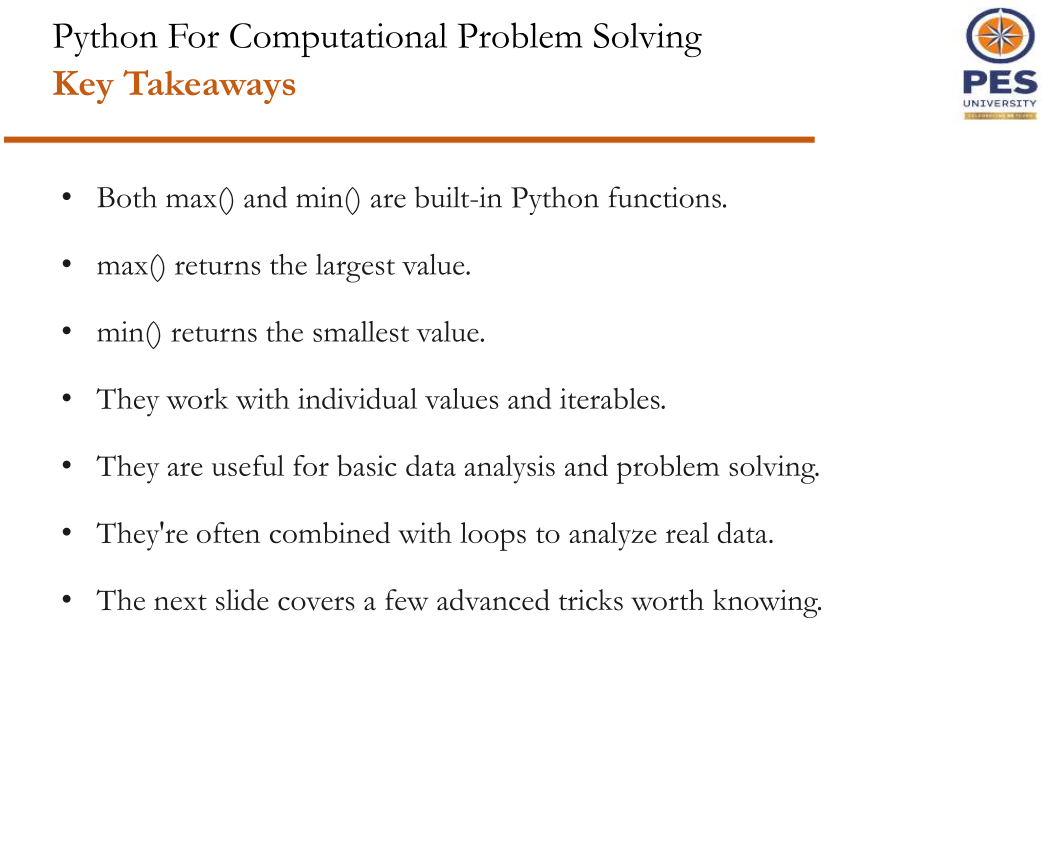

Figure G000117. Max Min · source slide page 9.

EV0000467

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

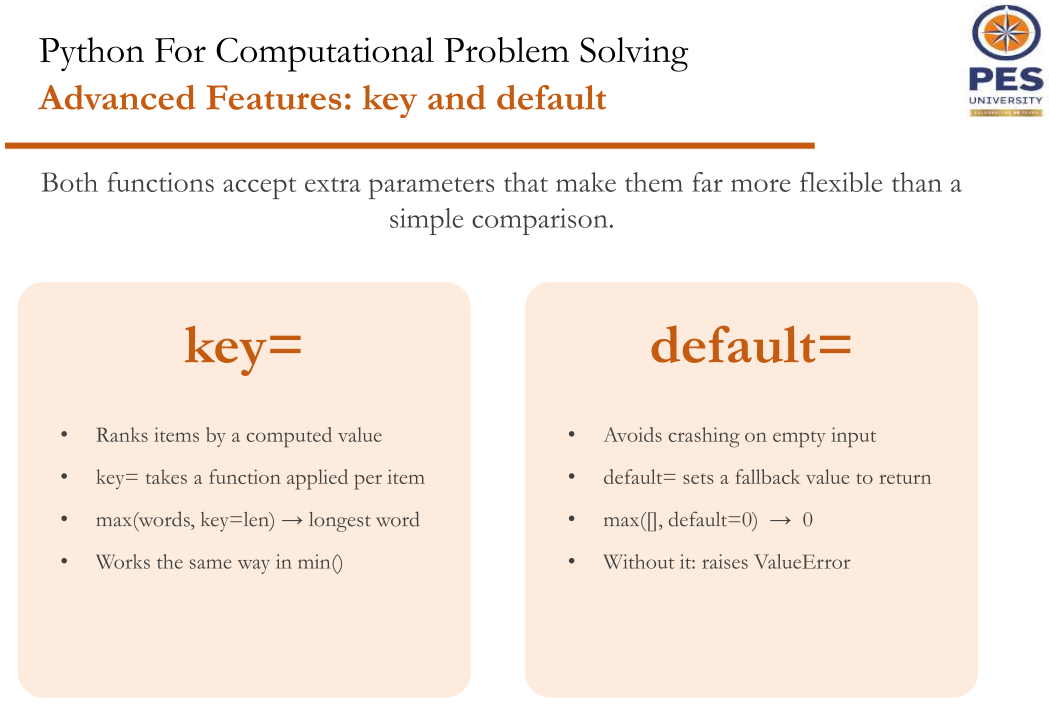

Figure G000118. Max Min · source slide page 10.

EV0000470

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Python For Computational Problem Solving

Practice Questions

1. What does max(4, 9, 2, 7) return?
2. Write one line of code to find the smallest number in marks = [67, 82, 45, 90, 71].
3. What will max("apple", "fig", "kiwi") return, and why?
4. Write a program that reads 5 numbers and prints both the largest and smallest.

Figure G000119. Max Min · source slide page 11.

EV0000473

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

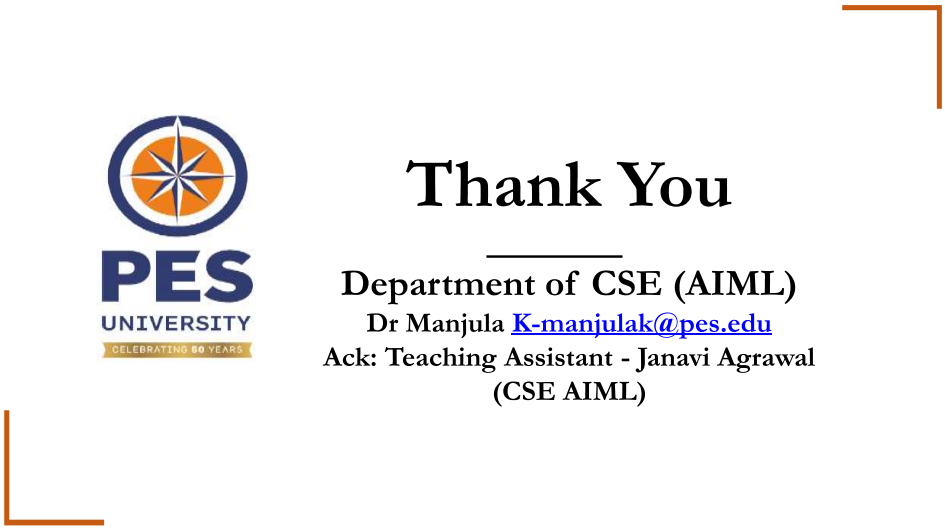

Figure G000120. Max Min · source slide page 12.

EV0000476

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): Python For Computational Problem Solving- UE26CS151A Dr Manjula K Professor, PCPS Anchor 2026 Department of CSE(AIML) EV0000442 · EV0000443 · EV0000444
2. Source slide page 2 (extracted text): Python For Computational Problem Solving max() and min() Functions in Python Python provides built-in functions to find the largest and smallest values among a set of numbers — no loops or manual comparisons needed. max() min() • Returns the largest value • Returns the smallest value • Works with lists, tuples, sets • Works with lists, tuples, sets • Takes multiple values directly • Takes multiple values directly • Great for finding the top score or • Great for finding the lowest score or price price EV0000445 · EV0000446 · EV0000447
3. Source slide page 3 (extracted text): Python For Computational Problem Solving max() Function – Introduction • max() is a built-in Python function. • It returns the largest value among the Syntax given values. max(iterable) • It can also be used with collections max(value1, value2, such as lists, tuples and sets. value3, ...) • Works with numbers and strings (alphabetical order). • On a tie, it returns the first maximum found. EV0000448 · EV0000449 · EV0000450

4. Source slide page 4 (extracted text): Python For Computational Problem Solving max() Function – Example Code numbers = [10, 25, 15, 40, 5] print(max(numbers)) Output 40 max() scans every element in the list and returns 40, since it is greater than all the rest. EV0000451 · EV0000452 · EV0000453
5. Source slide page 5 (extracted text): Python For Computational Problem Solving min() Function – Introduction • min() is a built-in Python function. • It returns the smallest value among the Syntax given values. min(iterable) • It can be used with lists, tuples, sets min(value1, value2, and other iterables. value3, ...) • Works with numbers and strings (alphabetical order). • On a tie, it returns the first minimum found. EV0000454 · EV0000455 · EV0000456
6. Source slide page 6 (extracted text): Python For Computational Problem Solving min() Function – Example Code numbers = [10, 25, 15, 40, 5] print(min(numbers)) Output 5 min() scans every element in the list and returns 5, since it is smaller than all the rest. EV0000457 · EV0000458 · EV0000459
7. Source slide page 7 (extracted text): Python For Computational Problem Solving Using max() and min() Together Code numbers = [12, 45, 7, 23, 56, 9] largest = max(numbers) smallest = min(numbers) print("Largest:", largest) print("Smallest:", smallest) Largest: 56 Smallest: 7 Calling max() and min() on the same list gives both extremes in just two lines of code. EV0000460 · EV0000461 · EV0000462
8. Source slide page 8 (extracted text): Python For Computational Problem Solving Quick Comparison max() min() Finds the largest / maximum value Finds the smallest / minimum value Great for scores, prices, rankings Great for lowest cost, fastest time Example max(10, 20, 5) → 20 min(10, 20, 5) → 5 max(3.5, 2.1, 9.9) → 9.9 EV0000463 · EV0000464 · EV0000465
9. Source slide page 9 (extracted text): Python For Computational Problem Solving Key Takeaways • Both max() and min() are built-in Python functions. • max() returns the largest value. • min() returns the smallest value. • They work with individual values and iterables. • They are useful for basic data analysis and problem solving. • They're often combined with loops to analyze real data. • The next slide covers a few advanced tricks worth knowing. EV0000466 · EV0000467 · EV0000468
10. Source slide page 10 (extracted text): Python For Computational Problem Solving Advanced Features: key and default Both functions accept extra parameters that make them far more flexible than a simple comparison. key= default= • Ranks items by a computed value • Avoids crashing on empty input • key= takes a function applied per item • default= sets a fallback value to return • max(words, key=len) → longest word • max([], default=0) → 0 • Works the same way in min() • Without it: raises ValueError EV0000469 · EV0000470 · EV0000471
11. Source slide page 11 (extracted text): Python For Computational Problem Solving Practice Questions 1. What does max(4, 9, 2, 7) return? 2. Write one line of code to find the smallest number in marks = [67, 82, 45, 90, 71]. 3. What will max("apple", "fig", "kiwi") return, and why? 4. Write a program that reads 5 numbers and prints both the largest and smallest. EV0000472 · EV0000473 · EV0000474
12. Source slide page 12 (extracted text): Thank You Department of CSE (AIML) Dr Manjula K-manjulak@pes.edu Ack: Teaching Assistant - Janavi Agrawal (CSE AIML) EV0000475 · EV0000476 · EV0000477

SHORT-NOTE RECAP

Source deck preserved: Max Min. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000442

18 Source slide deck: Intro to Modules

This entry preserves the complete slide deck titled Intro to Modules as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Python for computational Problem Solving, Unit_03, to avoid an unsupported match. EV0000601 · EV0000602

Graphs and diagrams

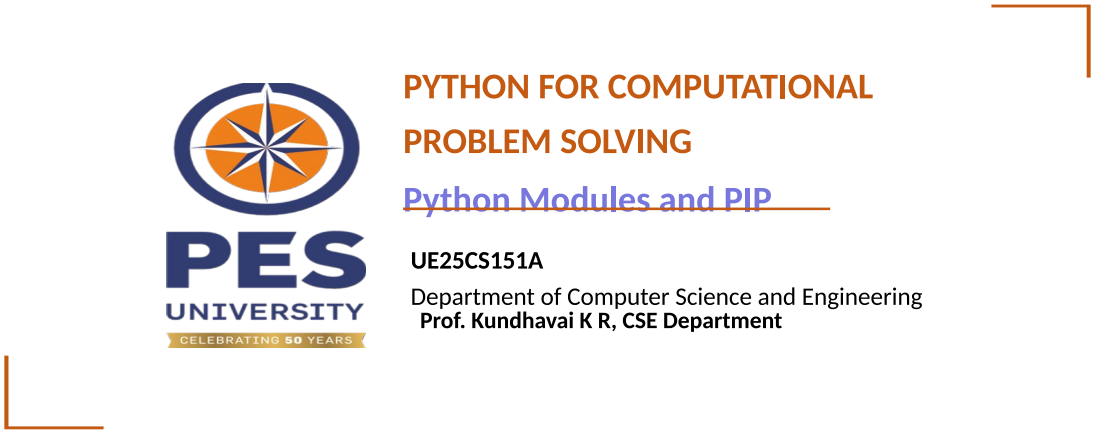

Figure G000162. Intro to Modules · source slide page 1.

EV0000602

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

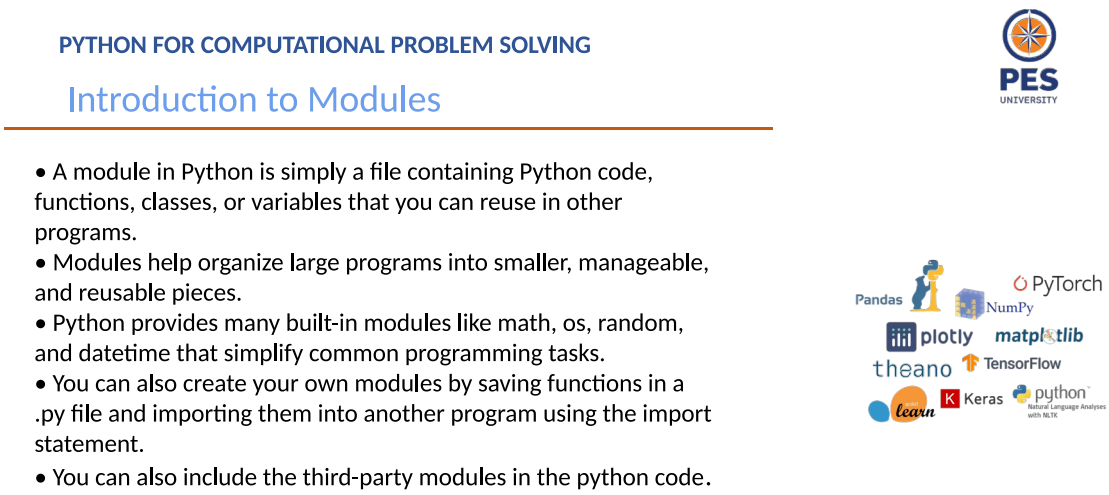

Figure G000163. Intro to Modules · source slide page 2.

EV0000605

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

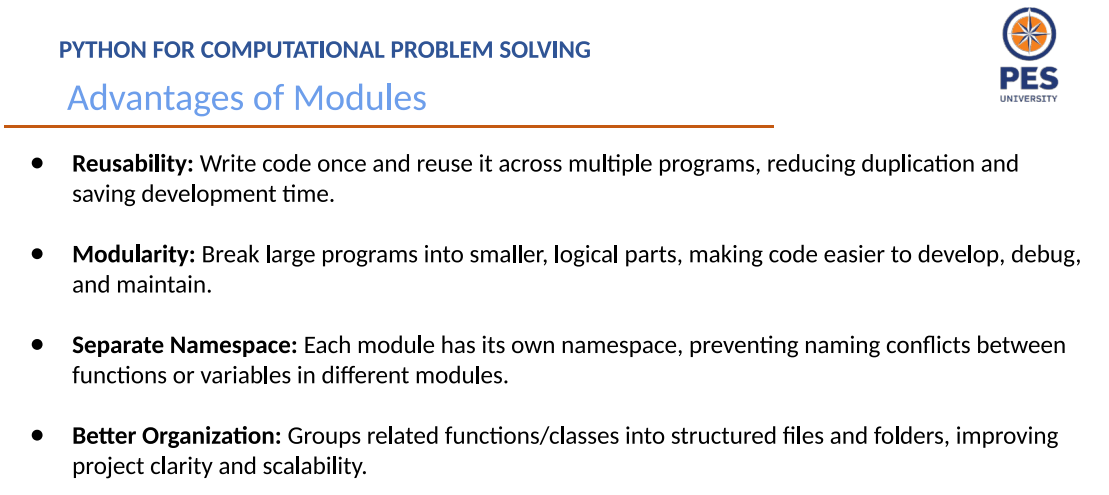

Figure G000164. Intro to Modules · source slide page 3.

EV0000608

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000165. Intro to Modules · source slide page 4.

EV0000611

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

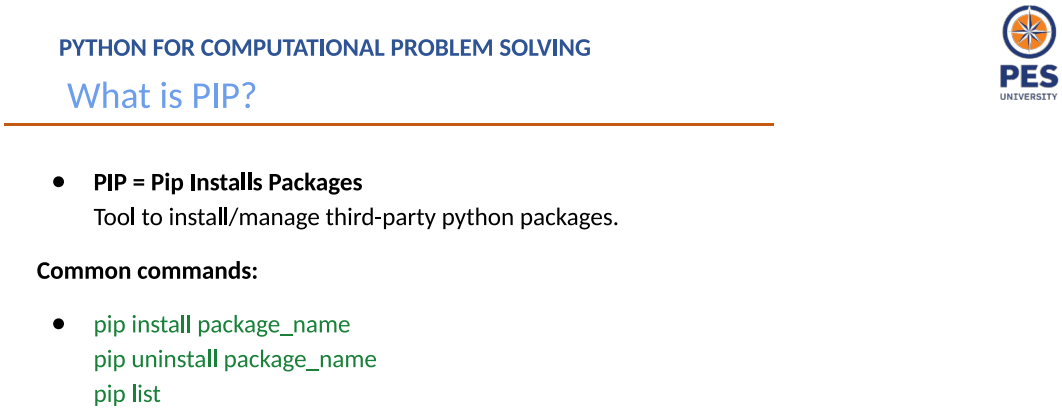

Figure G000166. Intro to Modules · source slide page 5.

EV0000614

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

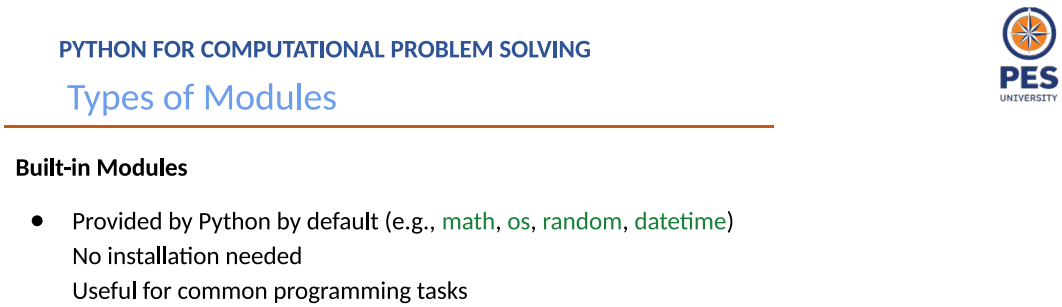

Figure G000167. Intro to Modules · source slide page 6.

EV0000617

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

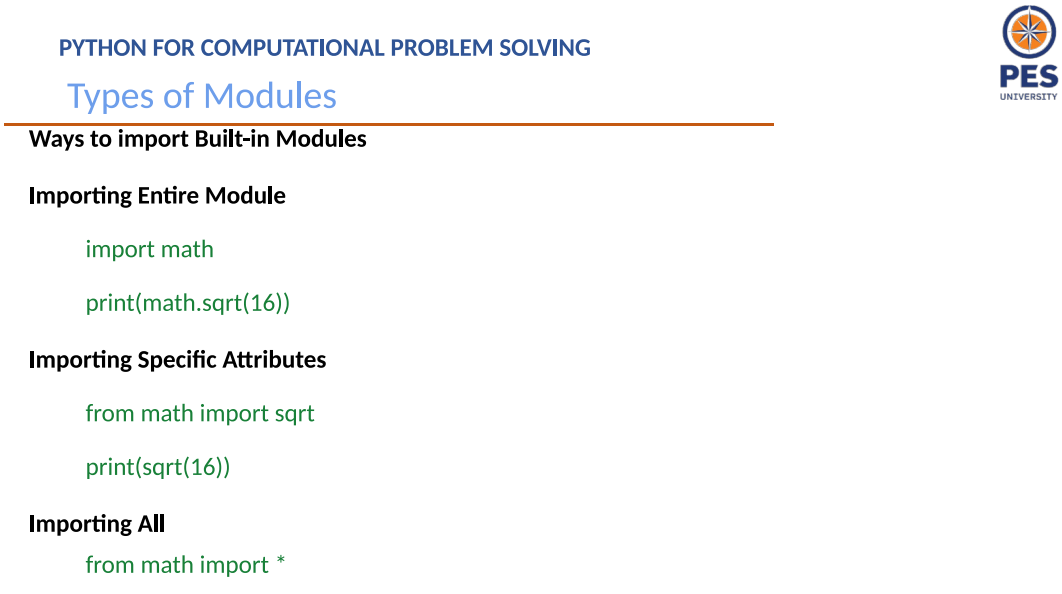

Figure G000168. Intro to Modules · source slide page 7.

EV0000620

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

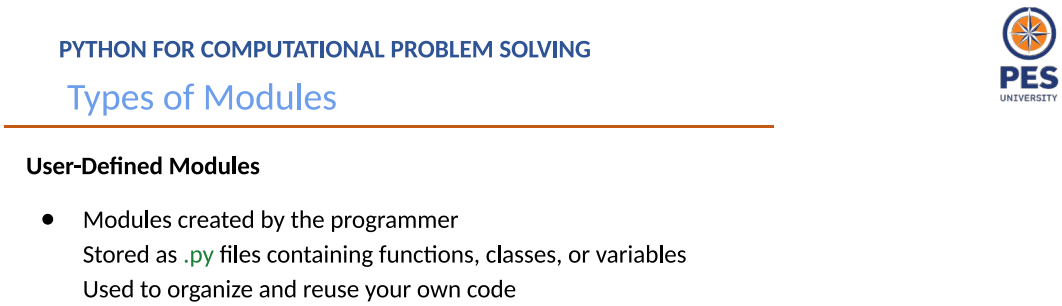

Figure G000169. Intro to Modules · source slide page 8.

EV0000623

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

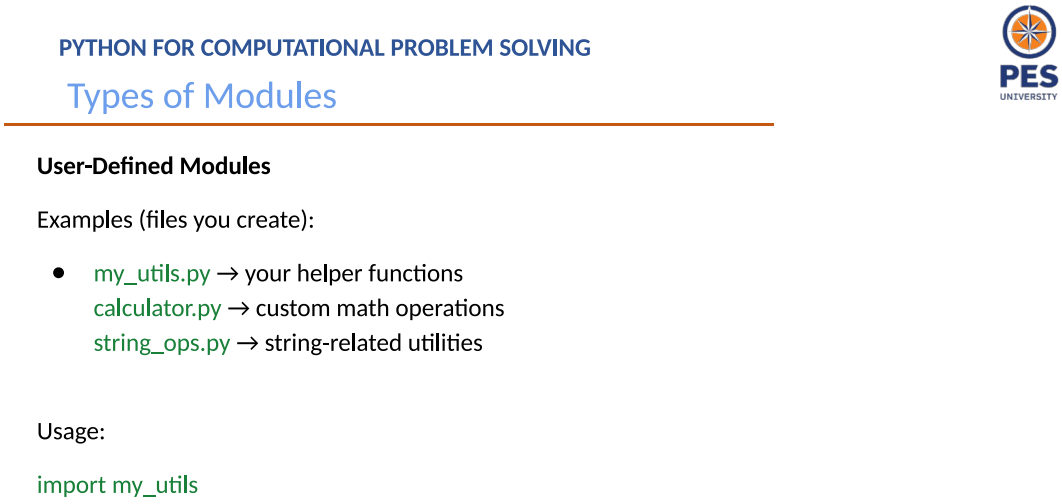

Figure G000170. Intro to Modules · source slide page 9.

EV0000626

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

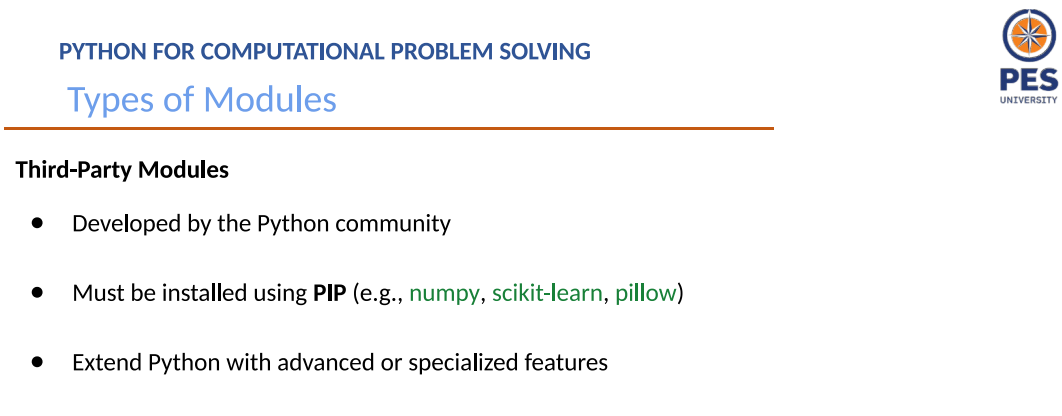

Figure G000171. Intro to Modules · source slide page 10.

EV0000629

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

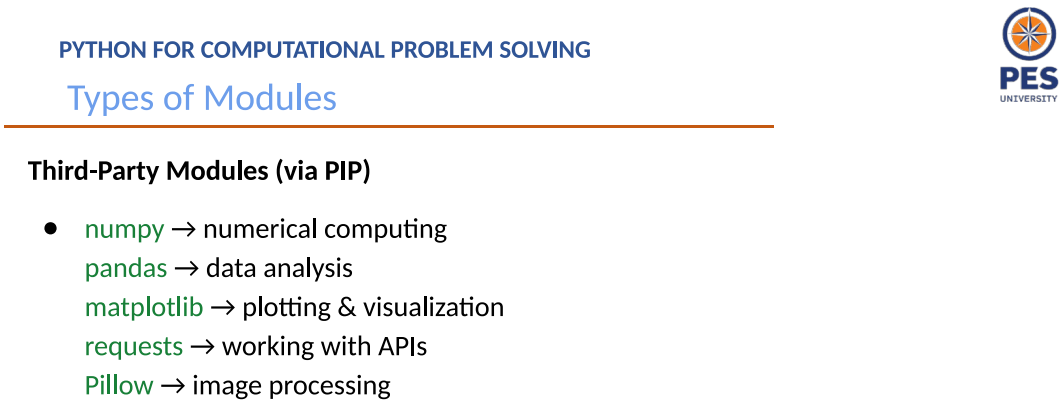

Figure G000172. Intro to Modules · source slide page 11.

EV0000632

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

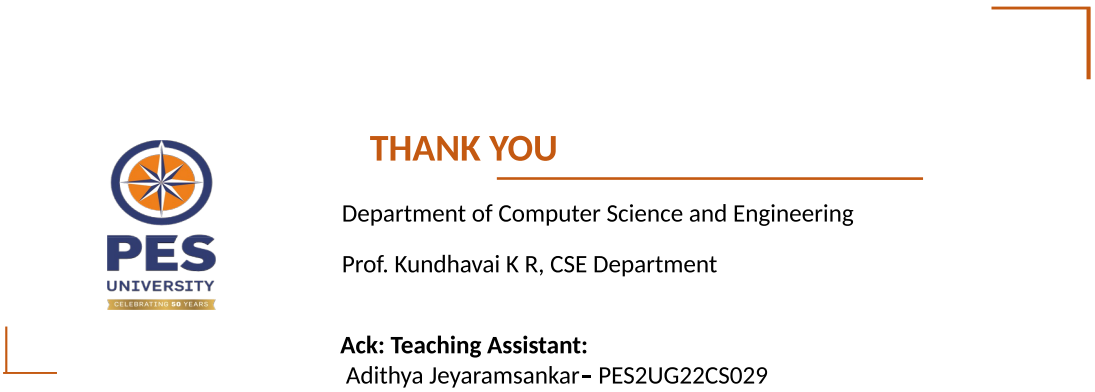

Figure G000173. Intro to Modules · source slide page 12.

EV0000635

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Python Modules and PIP UE25CS151A Department of Computer Science and Engineering Prof. Kundhavai K R, CSE Department EV0000601 · EV0000602 · EV0000603
2. Source slide page 2 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Introduction to Modules • A module in Python is simply a file containing Python code, functions, classes, or variables that you can reuse in other programs. • Modules help organize large programs into smaller, manageable, and reusable pieces. • Python provides many built-in modules like math, os, random, and datetime that simplify common programming tasks. • You can also create your own modules by saving functions in a .py file and importing them into another program using the import statement. • You can also include the third-party modules in the python code. EV0000604 · EV0000605 · EV0000606
3. Source slide page 3 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Advantages of Modules Reusability: Write code once and reuse it across multiple programs, reducing duplication and saving development time. Modularity: Break large programs into smaller, logical parts, making code easier to develop, debug, and maintain. Separate Namespace: Each module has its own namespace, preventing naming conflicts between functions or variables in different modules. Better Organization: Groups related functions/classes into structured files and folders, improving project clarity and scalability. EV0000607 · EV0000608 · EV0000609

4. Source slide page 4 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Types of Modules Built-in Modules. User-Defined Modules (Creating your own modules). PIP - Third Party Modules. EV0000610 · EV0000611 · EV0000612
5. Source slide page 5 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING What is PIP? PIP = Pip Installs Packages Tool to install/manage third-party python packages. Common commands: pip install package_name pip uninstall package_name pip list EV0000613 · EV0000614 · EV0000615
6. Source slide page 6 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Types of Modules Built-in Modules Provided by Python by default (e.g., math, os, random, datetime) No installation needed Useful for common programming tasks EV0000616 · EV0000617 · EV0000618
7. Source slide page 7 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Types of Modules Ways to import Built-in Modules Importing Entire Module import math print(math.sqrt(16)) Importing Specific Attributes from math import sqrt print(sqrt(16)) Importing All from math import * EV0000619 · EV0000620 · EV0000621
8. Source slide page 8 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Types of Modules User-Defined Modules Modules created by the programmer Stored as .py files containing functions, classes, or variables Used to organize and reuse your own code EV0000622 · EV0000623 · EV0000624
9. Source slide page 9 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Types of Modules User-Defined Modules Examples (files you create): my_utils.py → your helper functions calculator.py → custom math operations string_ops.py → string-related utilities Usage: import my_utils EV0000625 · EV0000626 · EV0000627
10. Source slide page 10 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Types of Modules Third-Party Modules Developed by the Python community Must be installed using PIP (e.g., numpy, scikit-learn, pillow) Extend Python with advanced or specialized features EV0000628 · EV0000629 · EV0000630
11. Source slide page 11 (extracted text): PYTHON FOR COMPUTATIONAL PROBLEM SOLVING Types of Modules Third-Party Modules (via PIP) numpy → numerical computing pandas → data analysis matplotlib → plotting & visualization requests → working with APIs Pillow → image processing EV0000631 · EV0000632 · EV0000633
12. Source slide page 12 (extracted text): THANK YOU Department of Computer Science and Engineering Prof. Kundhavai K R, CSE Department Ack: Teaching Assistant: Adithya Jeyaramsankar– PES2UG22CS029 EV0000634 · EV0000635 · EV0000636

SHORT-NOTE RECAP

Source deck preserved: Intro to Modules. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000601

19 Source slide deck: Matplotlib

This entry preserves the complete slide deck titled Matplotlib as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Python for computational Problem Solving, Unit_03, to avoid an unsupported match. EV0000676 · EV0000677

Graphs and diagrams

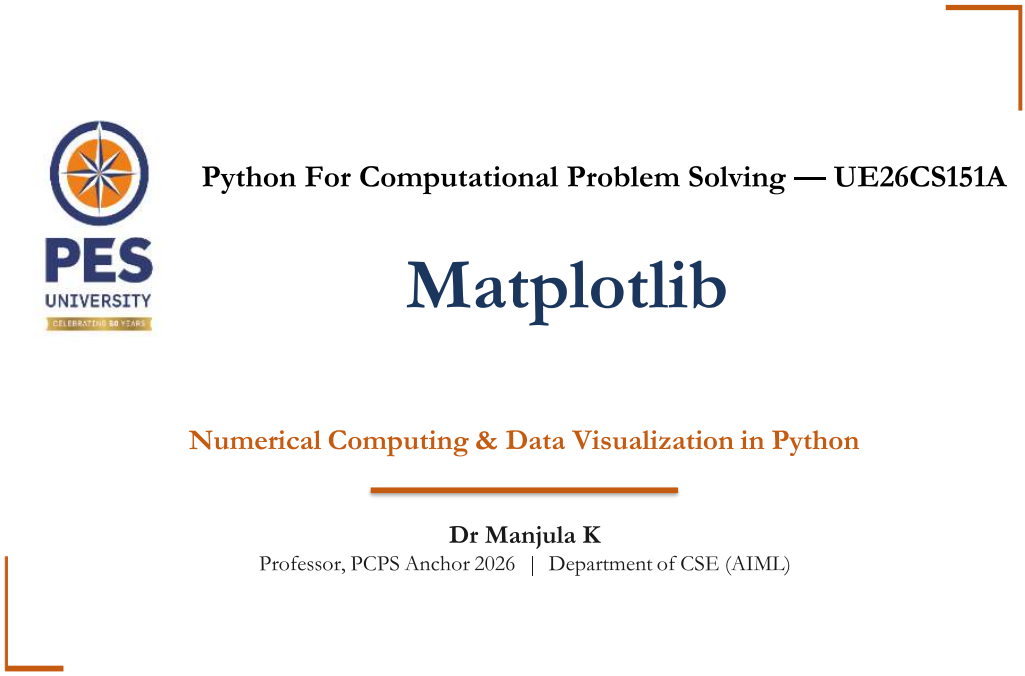

Figure G000187. Matplotlib · source slide page 1.

EV0000677

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000188. Matplotlib · source slide page 2.

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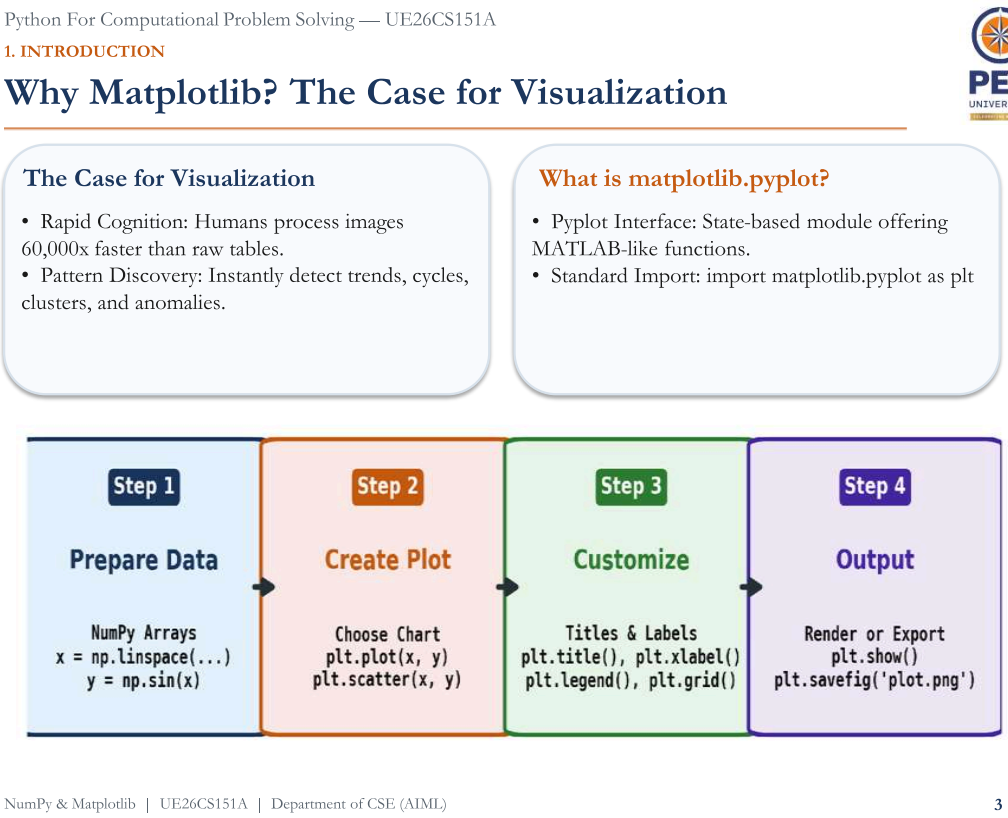

Figure G000189. Matplotlib · source slide page 3.

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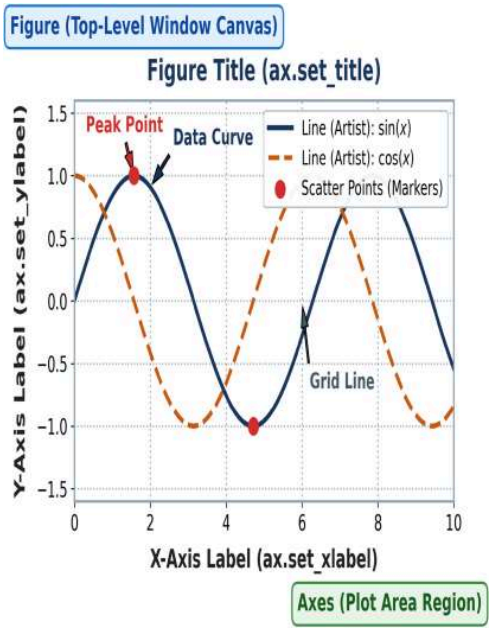
Python For Computational Problem Solving — UE26CS151A

1. INTRODUCTION

Anatomy of a Matplotlib Figure

Core Figure Hierarchy

- Figure: The top-level canvas window holding all subplots, titles, and legends.
- Axes: The actual plot area containing the coordinate space where data is drawn.
- Axis: Individual X and Y lines determining scale, limits, and tick locations.
- Ticks: Numbered markers denoting specific values.
- Spines: Bounding lines enclosing the plot area.
- Artists: Every visible element (lines, text, dots) is an Artist.



NumPy & Matplotlib | UE26CS151A | Department of CSE (AIML) 4

Figure G000190. Matplotlib · source slide page 4.

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Python For Computational Problem Solving — UE26CS151A

2. BASIC PLOTTING

Basic Plotting: Line Plots (plt.plot)

CODE: Line Plot Implementation

```
import matplotlib.pyplot as plt
import numpy as np

# 1. Prepare continuous data
x = np.linspace(0, 10, 15)
y_ideal = 2 * x + 3
y_sample = y_ideal + np.random.normal(0, 1.2, 15)

# 2. Draw line plots with formatting
plt.plot(x, y_ideal, color='#1B365D', lw=3.0,
         label='Ideal Model: y = 2x + 3')

plt.plot(x, y_sample, color='#C55A11', lw=2.2,
         linestyle='--', marker='o',
         markersize=8, label='Sampled Readings')

# 3. Label & Display
plt.title('Sensor Velocity Over Time')
plt.xlabel('Time (s)'); plt.ylabel('Velocity
(m/s)')
plt.legend(); plt.grid(True)
plt.show()
```

Key Formatting Parameters

- color: 'b', 'r', or hex codes like '#1B365D'
- linestyle: '-' (solid), '--' (dashed), ':' (dotted)
- marker: 'o' (circle), 's' (square), '^' (triangle)
- linewidth (lw): Line thickness in points

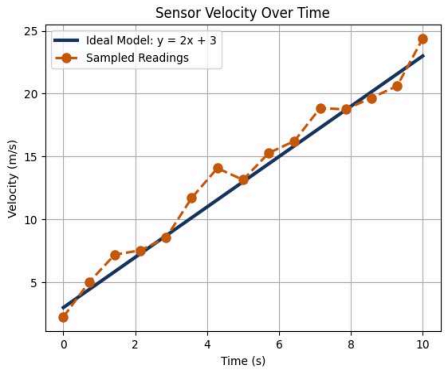

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Figure G000191. Matplotlib · source slide page 5.

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Python For Computational Problem Solving — UE26CS151A

2. BASIC PLOTTING

Basic Plotting: Scatter Plots (plt.scatter)

CODE: Scatter Plot Implementation

```
import matplotlib.pyplot as plt

# 1. Simple data points
x = [1, 2, 3, 4, 5, 6, 7, 8]
y = [5, 8, 10, 20, 15, 16, 32, 28]

# 2. Scatter plot
plt.scatter(x, y, color='red', marker='o',
s=50)

# 3. Labels
plt.title('Simple Scatter Plot')
plt.xlabel('X Values')
plt.ylabel('Y Values')
plt.grid(True)

plt.show()
```

When to Use Scatter Plots?

- Correlation: Inspect relationships between 2+ continuous variables.
- Multidimensional: Encode 4 variables at once (X, Y, Size, Color).
- Discrete Data: Distinct points without artificial connecting lines.

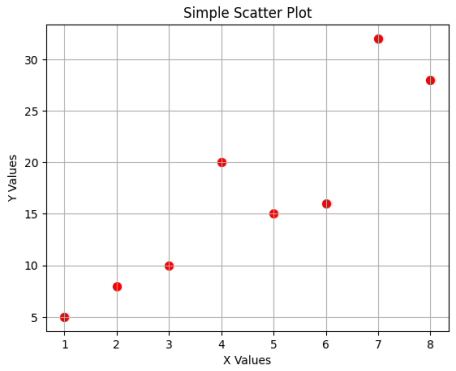

Figure G000192. Matplotlib · source slide page 6.

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Python For Computational Problem Solving — UE26CS151A

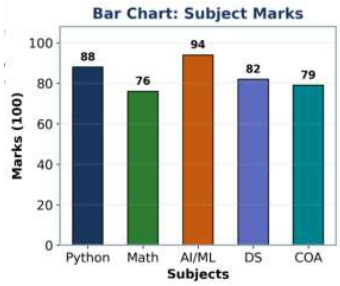
2. BASIC PLOTTING

Essential Plots: Bar Charts & Histograms

CODE: Bar Chart & Histogram Syntax

```
# 1. Bar Chart: Comparing Discrete Categories
subjects = ["Python", "Math", "AI/ML", "DS", "COA"]
marks = [88, 76, 94, 82, 79]
plt.bar(subjects, marks, color='#1B365D', width=0.55)
plt.title("Subject Marks"); plt.ylabel("Score")

# 2. Histogram: Continuous Data Distribution & Spread
scores = np.random.normal(loc=74, scale=9, size=120)
plt.hist(scores, bins=7, color='#D9531E',
edgecolor='white')
plt.title("Score Spread"); plt.xlabel("Marks Range");
plt.ylabel("Count")
```



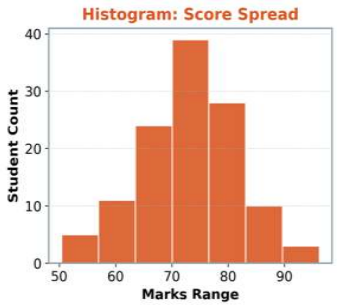


Figure G000193. Matplotlib · source slide page 7.

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3. CUSTOMIZATION

Customizing Plots: Labels, Legends & Grids

CODE: Multi-Series Customization

```
import matplotlib.pyplot as plt
import numpy as np

x = np.linspace(0, 2 * np.pi, 100)
# Plot multiple curves on same axes
plt.plot(x, np.sin(x), color='#1B365D',
         lw=3.0, label='sin(x)')

plt.plot(x, np.cos(x), color='#C55A11',
         lw=3.0, linestyle='--', label='cos(x)')

plt.plot(x, np.sin(x)*np.cos(x), color='#2E7D32',
         lw=2.6, linestyle='-.',
label='sin(x)cos(x)')

# Customize Titles, Labels, Grid & Legend
plt.title("Trigonometric Series with legend",
fontsize=16, fontweight='bold')
plt.xlabel("Angle θ (radians)");
plt.ylabel("Amplitude")
plt.grid(True, linestyle=':', alpha=0.7)
plt.legend(loc='lower left', frameon=True)
plt.show()
```

Essential Customization API

- plt.title(), plt.xlabel(), plt.ylabel(): Clear titles & units.
- plt.legend(loc='best'): Identifies multiple datasets.
- plt.grid(True, linestyle=':'): Adds reference gridlines.

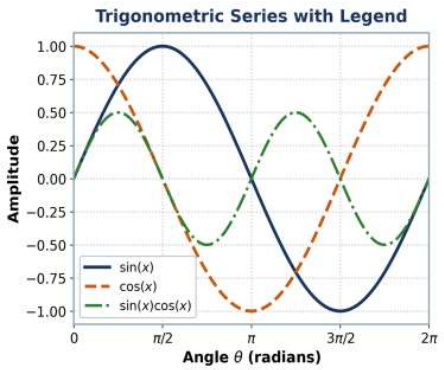

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Figure G000194. Matplotlib · source slide page 8.

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3. CUSTOMIZATION

Subplots: Multi-Panel Grids & Exporting

CODE: Subplots & savefig() Syntax

```
import matplotlib.pyplot as plt
import numpy as np

x = np.linspace(-3, 3, 100)
# Create 1 row, 2 columns of subplots
fig, (ax1, ax2) = plt.subplots(1, 2,
figsize=(8, 3.5))

# Subplot 1 (Left)
ax1.plot(x, x**2, color='#1B365D', lw=2.8)
ax1.set_title("y = x^2 (Parabola)")
ax1.grid(True, linestyle=':')

# Subplot 2 (Right)
ax2.plot(x, x**3, color='#C55A11', lw=2.8)
ax2.set_title("y = x^3 (Cubic)")
ax2.grid(True, linestyle=':')

plt.tight_layout()  # Fixes overlapping text
# Export high-resolution image to disk
plt.savefig("subplots.png", dpi=300,
bbox_inches='tight')
plt.show()
```

Key Subplot Rules

- fig, axes = plt.subplots(r, c): Multi-axis layout.
- plt.tight_layout(): Auto-adjusts padding between subplots.
- plt.savefig('file.png', dpi=300): Publication-ready export.

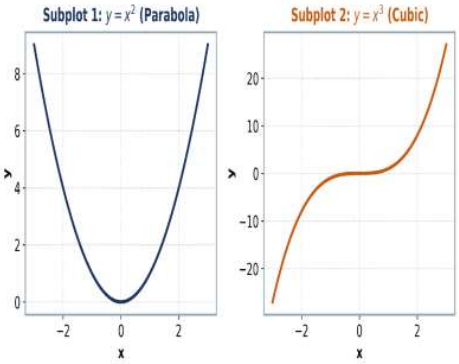

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Figure G000195. Matplotlib · source slide page 9.

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4. NUMPY + MATPLOTLIB

Putting It Together: NumPy + Matplotlib

CODE: NumPy Array Plotting (Sine & Cosine Example)

```
import matplotlib.pyplot as plt
import numpy as np

# 1. Generate 100 sample points with NumPy
x = np.linspace(0, 2 * np.pi, 100)

# 2. Compute mathematical arrays (vectorized)
y_sin = np.sin(x)
y_cos = np.cos(x)

# 3. Pass NumPy arrays directly to Matplotlib
plt.plot(x, y_sin, color="#1B365D", lw=3.0,
label="y = sin(x)")
plt.plot(
    x, y_cos, color="#C55A11", lw=3.0,
linestyle="--", label="y = cos(x)"
)

# 4. Customize & Display
plt.title("Sine and Cosine Waves")
plt.xlabel("Angle (radians)")
plt.ylabel("Value")
plt.legend()
plt.grid(True)
plt.show()
```

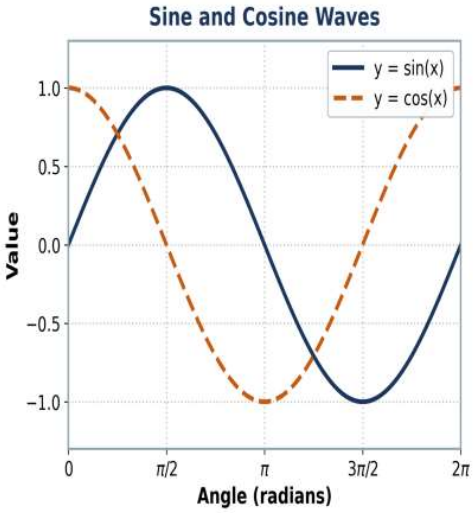

Figure G000196. Matplotlib · source slide page 10.

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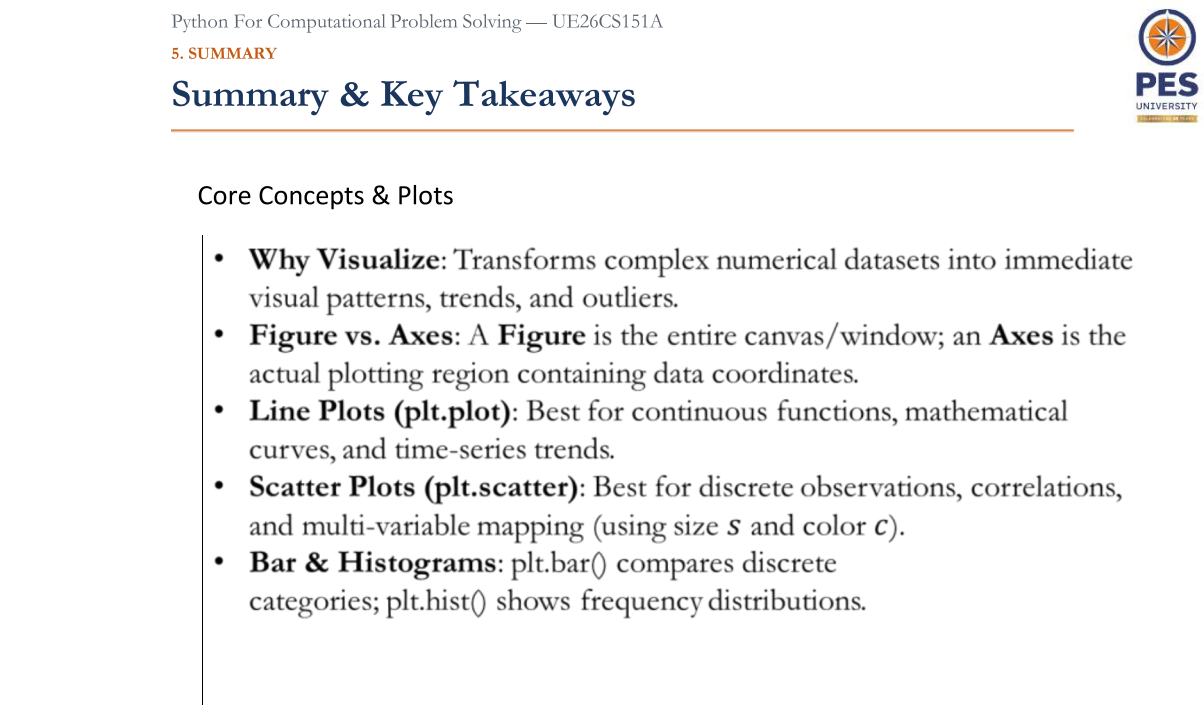

Figure G000197. Matplotlib · source slide page 11.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

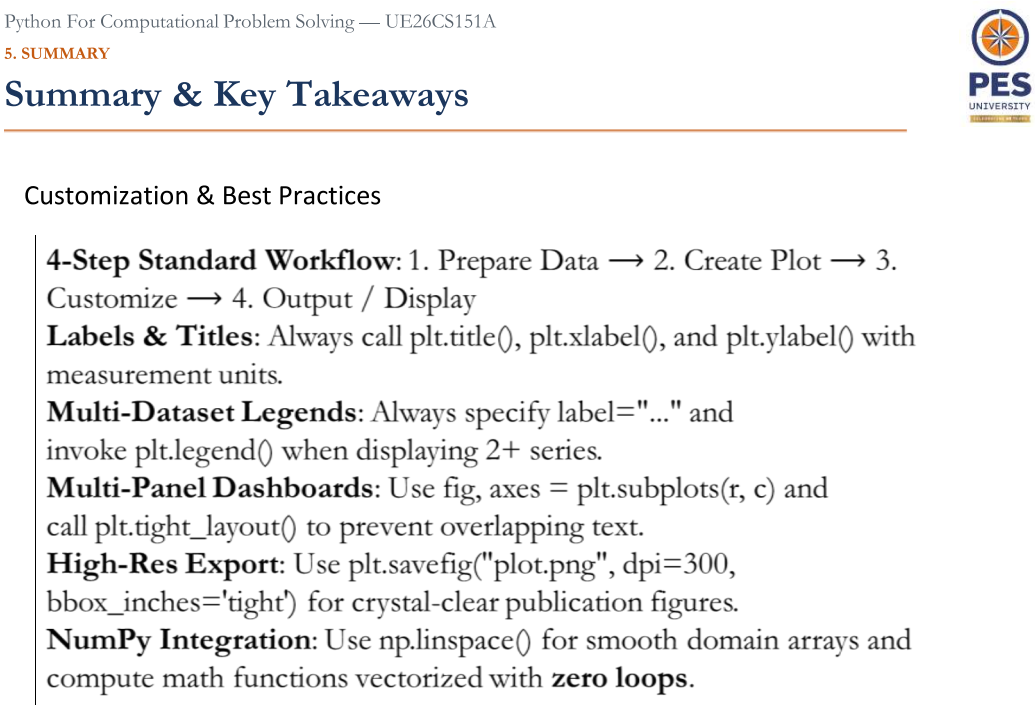

Figure G000198. Matplotlib · source slide page 12.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

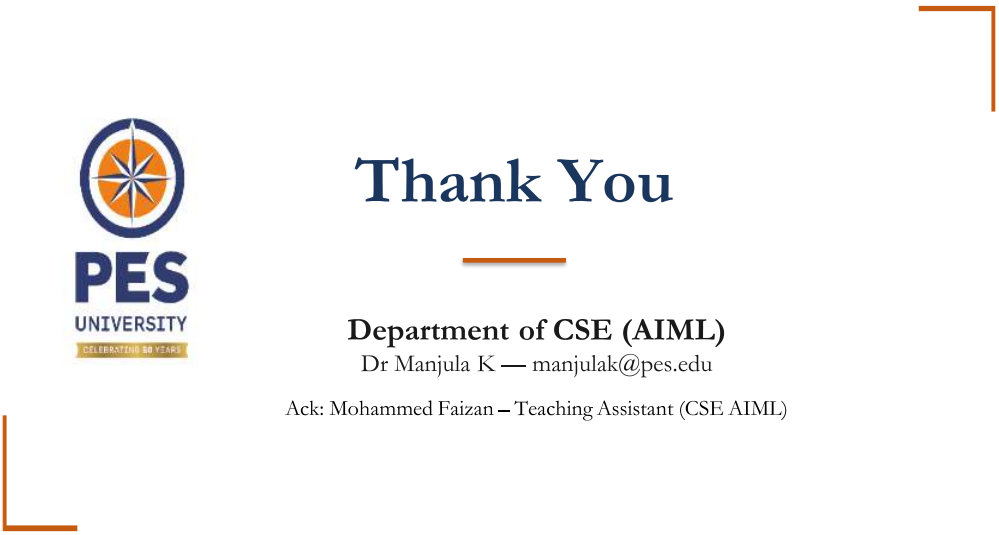

Figure G000199. Matplotlib · source slide page 13.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): Python For Computational Problem Solving — UE26CS151A Matplotlib Numerical Computing & Data Visualization in Python Dr Manjula K Professor, PCPS Anchor 2026 | Department of CSE (AIML) EV0000676 · EV0000677 · EV0000678
2. Source slide page 2 (extracted text): Python For Computational Problem Solving — UE26CS151A COURSE OVERVIEW Agenda & Lecture Roadmap Why Matplotlib? — The Case for Visualization 1 Visual cognition, Pyplot interface, and Figure vs Axes architecture. Basic Plotting: Line and Scatter Plots 2 Continuous trends with plt.plot(), discrete points with plt.scatter(), and bar/histograms. Customizing Plots: Labels, Legends & Subplots 3 Adding titles, units, grids, legends, and multi-panel figures with plt.subplots(). Putting It Together: NumPy + Matplotlib 4 Generating domain data with np.linspace(). Summary & Key Takeaways 5 "Comprehensive recap of the Session". NumPy & Matplotlib | UE26CS151A | Department of CSE (AIML) 2 EV0000679 · EV0000680 · EV0000681

3. Source slide page 3 (extracted text): Python For Computational Problem Solving — UE26CS151A 1. INTRODUCTION Why Matplotlib? The Case for Visualization The Case for Visualization What is matplotlib.pyplot? • Rapid Cognition: Humans process images • Pyplot Interface: State-based module offering 60,000x faster than raw tables. MATLAB-like functions. • Pattern Discovery: Instantly detect trends, cycles, • Standard Import: import matplotlib.pyplot as plt clusters, and anomalies. NumPy & Matplotlib | UE26CS151A | Department of CSE (AIML) 3 EV0000682 · EV0000683 · EV0000684

4. Source slide page 4 (extracted text): Python For Computational Problem Solving — UE26CS151A 1. INTRODUCTION Anatomy of a Matplotlib Figure Core Figure Hierarchy • Figure: The top-level canvas window holding all subplots, titles, and legends. • Axes: The actual plot area containing the coordinate space where data is drawn. • Axis: Individual X and Y lines determining scale, limits, and tick locations. • Ticks: Numbered markers denoting specific values. • Spines: Bounding lines enclosing the plot area. • Artists: Every visible element (lines, text, dots) is an Artist. NumPy & Matplotlib | UE26CS151A | Department of CSE (AIML) 4 EV0000685 · EV0000686 · EV0000687

5. Source slide page 5 (extracted text): Python For Computational Problem Solving — UE26CS151A 2. BASIC PLOTTING Basic Plotting: Line Plots (plt.plot) CODE: Line Plot Implementation Key Formatting Parameters import matplotlib.pyplot as plt • color: 'b', 'r', or hex codes like '#1B365D' import numpy as np • linestyle: '-' (solid), '--' (dashed), ':' (dotted) • marker: 'o' (circle), 's' (square), '^' (triangle) # 1. Prepare continuous data x = np.linspace(0, 10, 15) • linewidth (lw): Line thickness in points y_ideal = 2 * x + 3 y_sample = y_ideal + np.random.normal(0, 1.2, 15) # 2. Draw line plots with formatting plt.plot(x, y_ideal, color='#1B365D', lw=3.0, label='Ideal Model: y = 2x + 3') plt.plot(x, y_sample, color='#C55A11', lw=2.2, linestyle='--', marker='o', markersize=8, label=‘Sampled Readings') # 3. Label & Display plt.title('Sensor Velocity Over Time') plt.xlabel('Time (s)'); plt.ylabel('Velocity (m/s)') plt.legend(); plt.grid(True) plt.show() NumPy & Matplotlib | UE26CS151A | Department of CSE (AIML) 5 EV0000688 · EV0000689 · EV0000690

6. Source slide page 6 (extracted text): Python For Computational Problem Solving — UE26CS151A 2. BASIC PLOTTING Basic Plotting: Scatter Plots (plt.scatter) CODE: Scatter Plot Implementation When to Use Scatter Plots? import matplotlib.pyplot as plt • Correlation: Inspect relationships between 2+ # 1. Simple data points continuous variables. x = [1, 2, 3, 4, 5, 6, 7, 8] y = [5, 8, 10, 20, 15, 16, 32, 28] • Multidimensional: Encode 4 variables at once (X, Y, Size, Color). • Discrete Data: Distinct points without artificial # 2. Scatter plot plt.scatter(x, y, color='red', marker='o', connecting lines. s=50) # 3. Labels plt.title('Simple Scatter Plot') plt.xlabel('X Values') plt.ylabel('Y Values') plt.grid(True) plt.show() NumPy & Matplotlib | UE26CS151A | Department of CSE (AIML) 6 EV0000691 · EV0000692 · EV0000693

7. Source slide page 7 (extracted text): Python For Computational Problem Solving — UE26CS151A 2. BASIC PLOTTING Essential Plots: Bar Charts & Histograms CODE: Bar Chart & Histogram Syntax # 1. Bar Chart: Comparing Discrete Categories subjects = ["Python", "Math", "AI/ML", "DS", "COA"] marks = [88, 76, 94, 82, 79] plt.bar(subjects, marks, color='#1B365D', width=0.55) plt.title("Subject Marks"); plt.ylabel("Score") # 2. Histogram: Continuous Data Distribution & Spread scores = np.random.normal(loc=74, scale=9, size=120) plt.hist(scores, bins=7, color='#D9531E', edgecolor='white') plt.title("Score Spread"); plt.xlabel("Marks Range"); plt.ylabel("Count") NumPy & Matplotlib | UE26CS151A | Department of CSE (AIML) 7 EV0000694 · EV0000695 · EV0000696

8. Source slide page 8 (extracted text): Python For Computational Problem Solving — UE26CS151A 3. CUSTOMIZATION Customizing Plots: Labels, Legends & Grids CODE: Multi-Series Customization Essential Customization API import matplotlib.pyplot as plt import numpy as np • plt.title(), plt.xlabel(), plt.ylabel(): Clear titles & units. x = np.linspace(0, 2 * np.pi, 100) • plt.legend(loc='best'): Identifies multiple datasets. # Plot multiple curves on same axes plt.plot(x, np.sin(x), color='#1B365D', • plt.grid(True, linestyle=':'): Adds reference lw=3.0, label='sin(x)') gridlines. plt.plot(x, np.cos(x), color='#C55A11', lw=3.0, linestyle='--', label='cos(x)') plt.plot(x, np.sin(x)*np.cos(x), color='#2E7D32', lw=2.6, linestyle='-.', label='sin(x)cos(x)') # Customize Titles, Labels, Grid & Legend plt.title("Trigonometric Series with legend", fontsize=16, fontweight='bold') plt.xlabel("Angle ϑ (radians)"); plt.ylabel("Amplitude") plt.grid(True, linestyle=':', alpha=0.7) plt.legend(loc='lower left', frameon=True) plt.show() NumPy & Matplotlib | UE26CS151A | Department of CSE (AIML) 8 EV0000697 · EV0000698 · EV0000699
9. Source slide page 9 (extracted text): Python For Computational Problem Solving — UE26CS151A 3. CUSTOMIZATION Subplots: Multi-Panel Grids & Exporting CODE: Subplots & savefig() Syntax Key Subplot Rules import matplotlib.pyplot as plt import numpy as np • fig, axes = plt.subplots(r, c): Multi-axis layout. • plt.tight_layout(): Auto-adjusts padding between subplots. x = np.linspace(-3, 3, 100) # Create 1 row, 2 columns of subplots • plt.savefig('file.png', dpi=300): Publication-ready fig, (ax1, ax2) = plt.subplots(1, 2, export. figsize=(8, 3.5)) # Subplot 1 (Left) ax1.plot(x, x**2, color='#1B365D', lw=2.8) ax1.set_title("y = x^2 (Parabola)") ax1.grid(True, linestyle=':') # Subplot 2 (Right) ax2.plot(x, x**3, color='#C55A11', lw=2.8) ax2.set_title("y = x^3 (Cubic)") ax2.grid(True, linestyle=':') plt.tight_layout() # Fixes overlapping text # Export high-resolution image to disk plt.savefig("subplots.png", dpi=300, bbox_inches='tight') plt.show() NumPy & Matplotlib | UE26CS151A | Department of CSE (AIML) 9 EV0000700 · EV0000701 · EV0000702
10. Source slide page 10 (extracted text): Python For Computational Problem Solving — UE26CS151A 4. NUMPY + MATPLOTLIB Putting It Together: NumPy + Matplotlib CODE: NumPy Array Plotting (Sine & Cosine Example) import matplotlib.pyplot as plt import numpy as np # 1. Generate 100 sample points with NumPy x = np.linspace(0, 2 * np.pi, 100) # 2. Compute mathematical arrays (vectorized) y_sin = np.sin(x) y_cos = np.cos(x) # 3. Pass NumPy arrays directly to Matplotlib plt.plot(x, y_sin, color="#1B365D", lw=3.0, label="y = sin(x)") plt.plot(x, y_cos, color="#C55A11", lw=3.0, linestyle="--", label="y = cos(x)") # 4. Customize & Display plt.title("Sine and Cosine Waves") plt.xlabel("Angle (radians)") plt.ylabel("Value") plt.legend() plt.grid(True) plt.show() NumPy & Matplotlib | UE26CS151A | Department of CSE (AIML) 10 EV0000703 · EV0000704 · EV0000705
11. Source slide page 11 (extracted text): Python For Computational Problem Solving — UE26CS151A 5. SUMMARY Summary & Key Takeaways Core Concepts & Plots NumPy & Matplotlib | UE26CS151A | Department of CSE (AIML) 11 EV0000706 · EV0000707 · EV0000708
12. Source slide page 12 (extracted text): Python For Computational Problem Solving — UE26CS151A 5. SUMMARY Summary & Key Takeaways Customization & Best Practices NumPy & Matplotlib | UE26CS151A | Department of CSE (AIML) 1 2 EV0000709 · EV0000710 · EV0000711
13. Source slide page 13 (extracted text): Thank You Department of CSE (AIML) Dr Manjula K — manjulak@pes.edu Ack: Mohammed Faizan – Teaching Assistant (CSE AIML) EV0000712 · EV0000713 · EV0000714

SHORT-NOTE RECAP

Source deck preserved: Matplotlib. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000676

20 Source slide deck: argparse

This entry preserves the complete slide deck titled argparse as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Python for computational Problem Solving, Unit_03, to avoid an unsupported match. EV0000781 · EV0000782

Graphs and diagrams

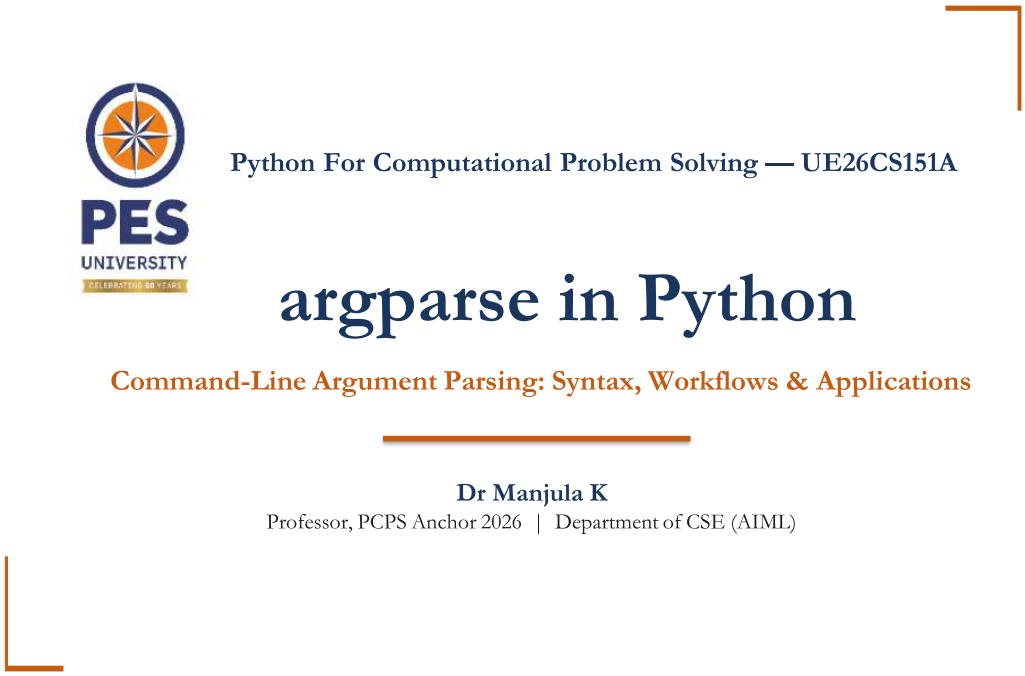

Figure G000222. argparse · source slide page 1.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

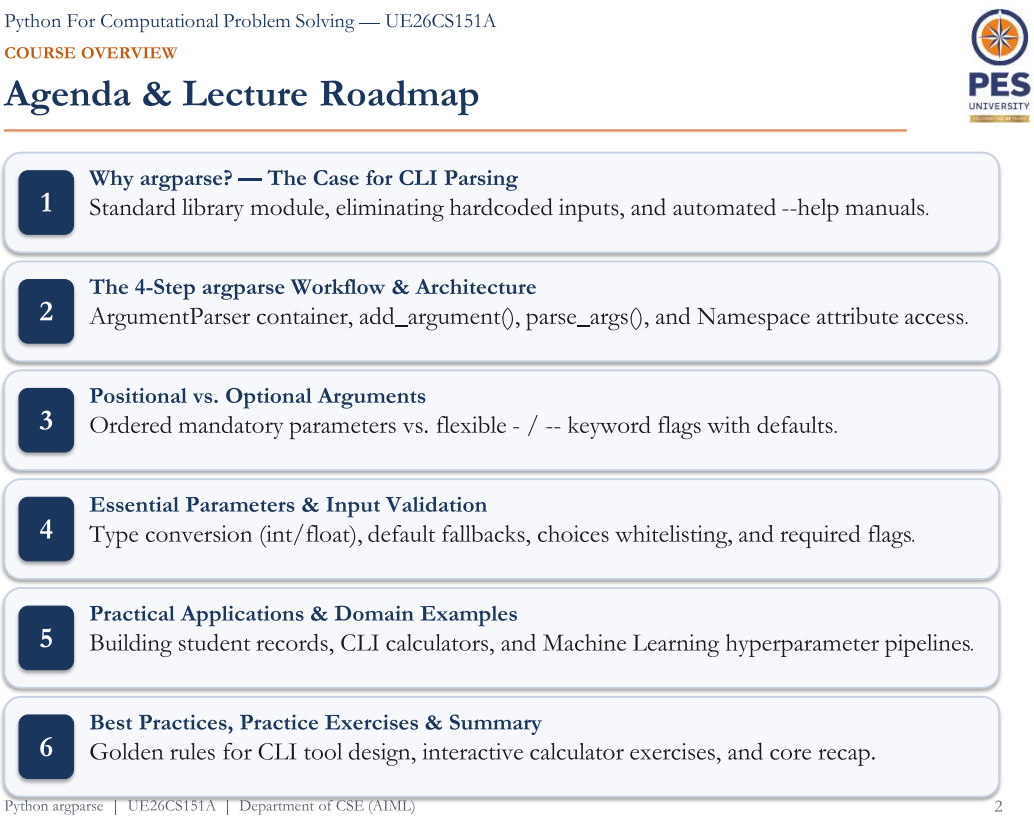

Figure G000223. argparse · source slide page 2.

EV0000785

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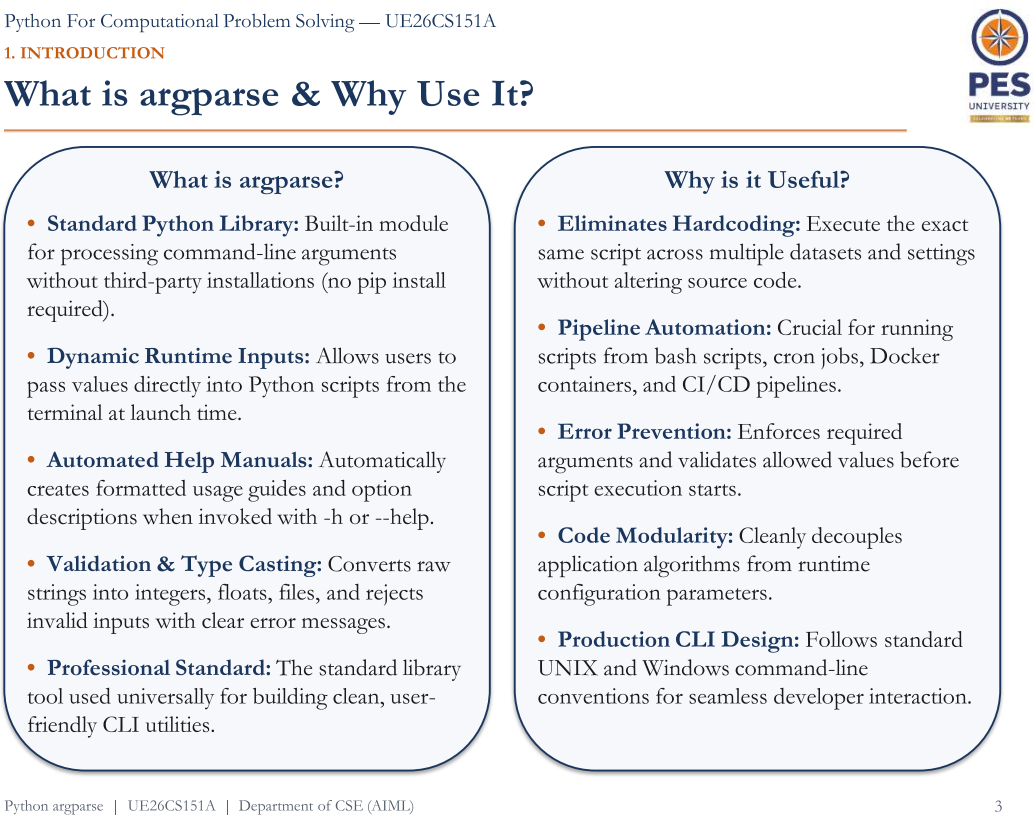

Figure G000224. argparse · source slide page 3.

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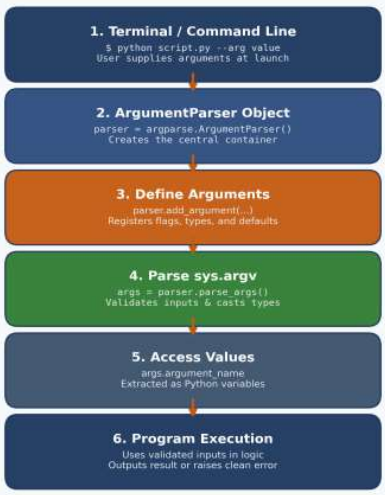
Python For Computational Problem Solving — UE26CS151A

2. CORE WORKFLOW

How argparse Works: Execution Pipeline

Execution Pipeline Flowchart



Step-by-Step Dataflow

- **1. Terminal Invocation:** User executes $ python app.py --flag value. The operating system shell passes arguments to Python via sys.argv.
- **2. ArgumentParser Object:** Creates the central container: parser = argparse.ArgumentParser(description=...). Holds all argument metadata.
- **3. Argument Registration:** parser.add_argument() defines expected flag names, data types, default fallbacks, choices, and help strings.
- **4. Parse Command Line:** args = parser.parse_args() inspects sys.argv, converts input types, and validates constraints.
- **5. Namespace Attribute Access:** Values are accessed as clean Python attributes via args.<name> (e.g. args.number, args.epochs).
- **6. Program Logic:** Application logic executes using verified input values. If input is invalid, a formatted error is displayed immediately.

Figure G000225. argparse · source slide page 4.

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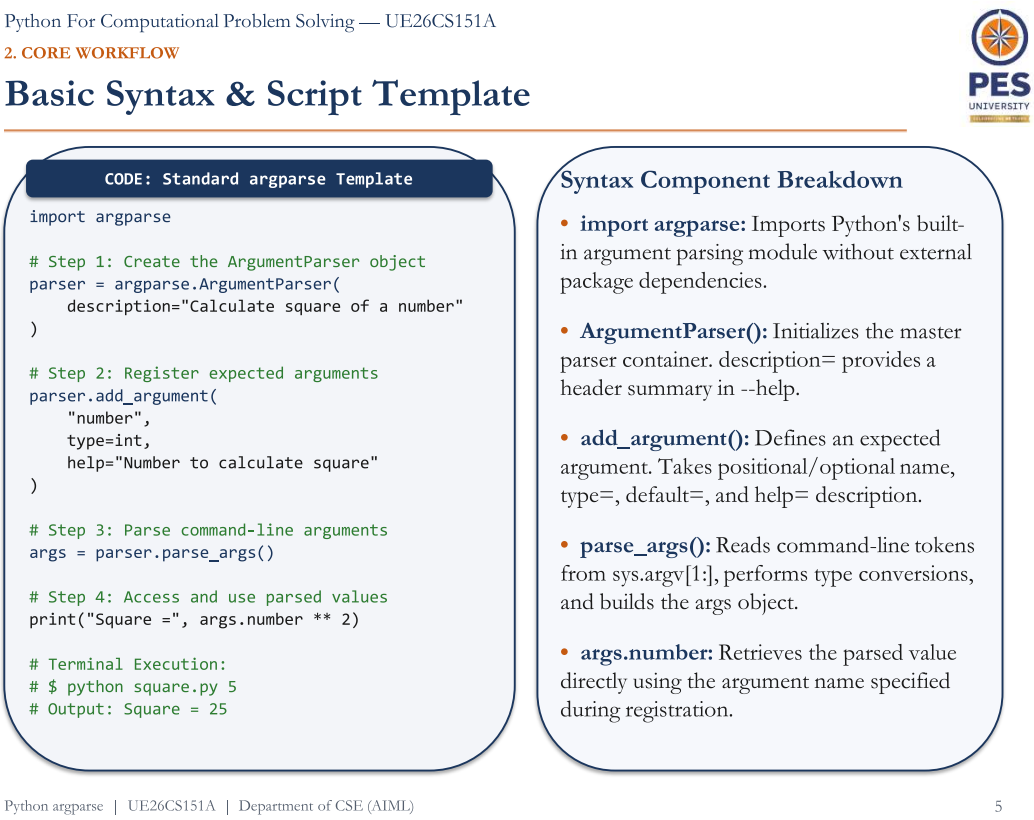

Figure G000226. argparse · source slide page 5.

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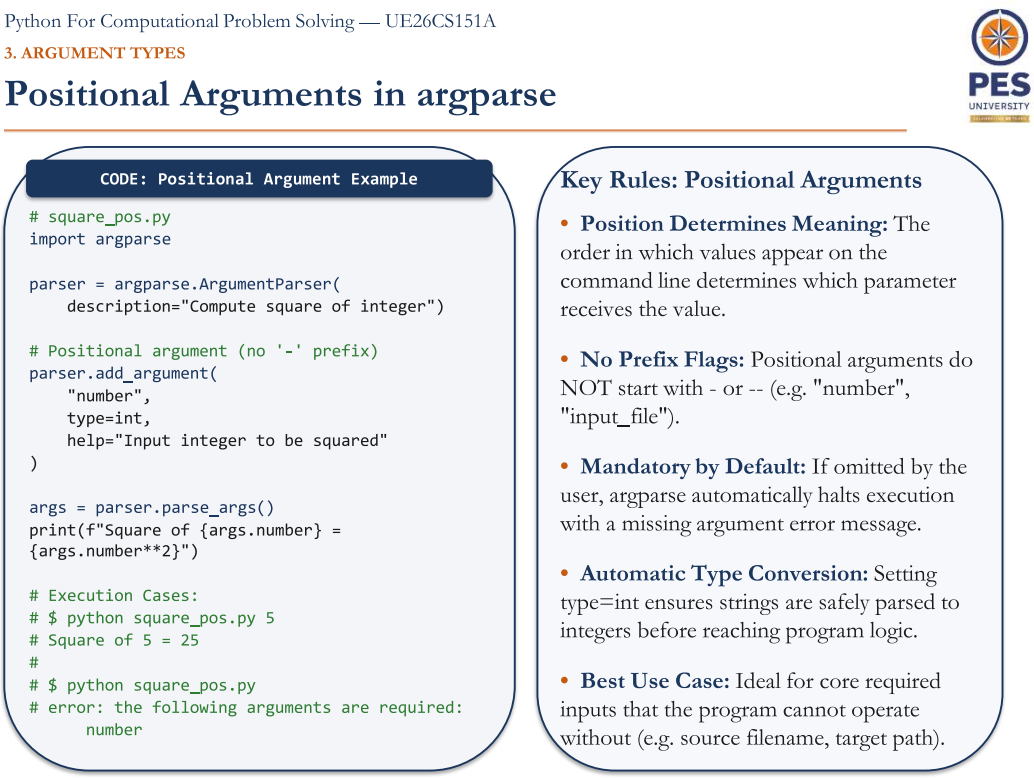

Figure G000227. argparse · source slide page 6.

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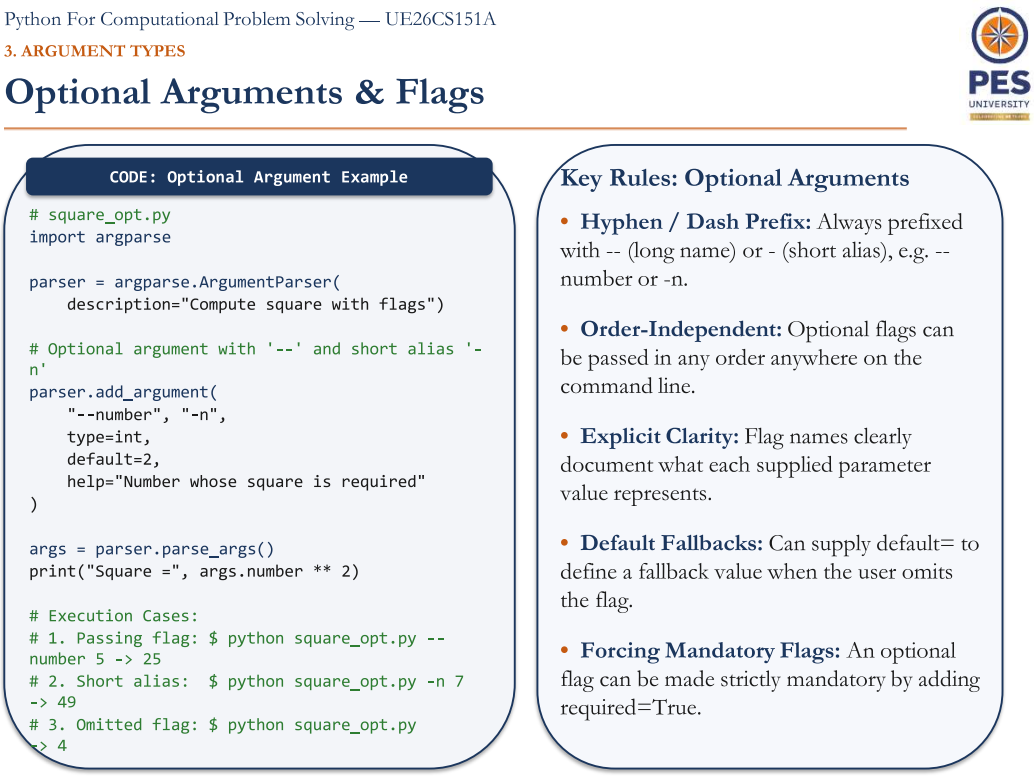

Figure G000228. argparse · source slide page 7.

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Python For Computational Problem Solving — UE26CS151A

4. PARAMETERS & VALIDATION

Essential add_argument() Parameters

Parameter	Purpose	Example Syntax	Description & Behavior
`type`	Data type conversion	`type=int, type=float`	Converts raw string input into specified Python data type automatically.
`default`	Fallback value	`default=10, default="app.log"`	Supplies default value if argument is omitted from command line.
`required`	Mandatory flag	`required=True`	Forces an optional flag (--name) to be mandatory at runtime.
`choices`	Restrict input set	`choices=["add", "sub", "mul"]`	Restricts valid input to a specified list; rejects other values.
`help`	Help documentation	`help="Path to input dataset"`	Text displayed next to argument in auto-generated --help manual.
`action`	Boolean flag control	`action="store_true"`	Sets attribute to True when flag is present, False when absent.

Python argparse | UE26CS151A | Department of CSE (AIML) 8

Figure G000229. argparse · source slide page 8.

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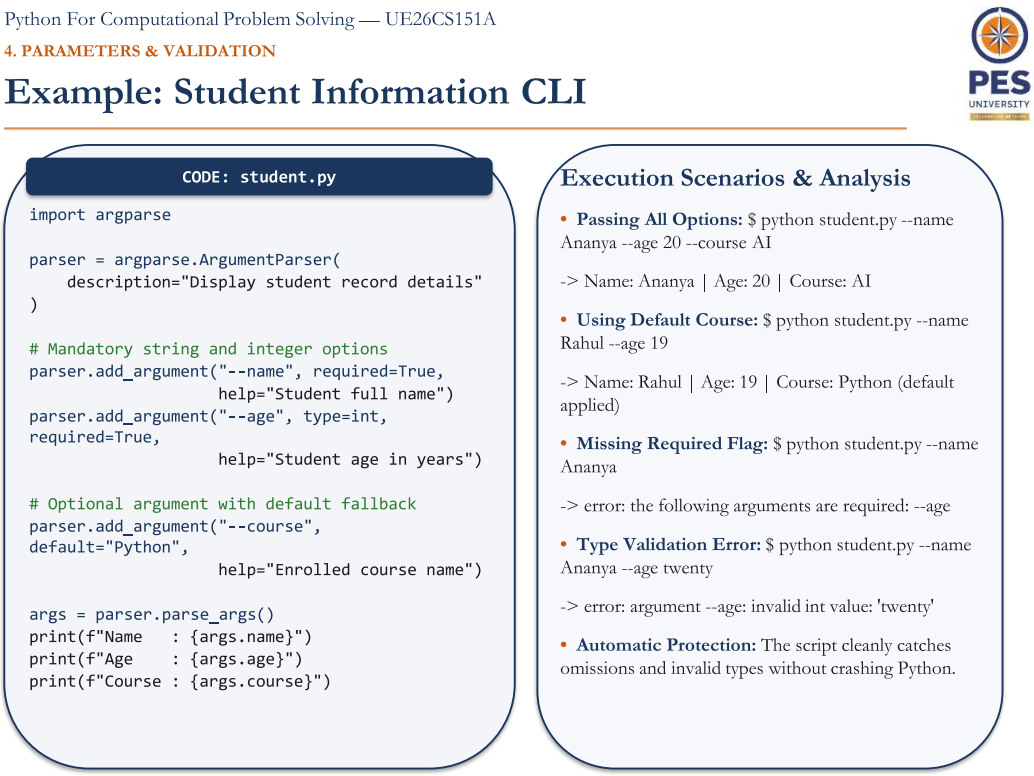

Figure G000230. argparse · source slide page 9.

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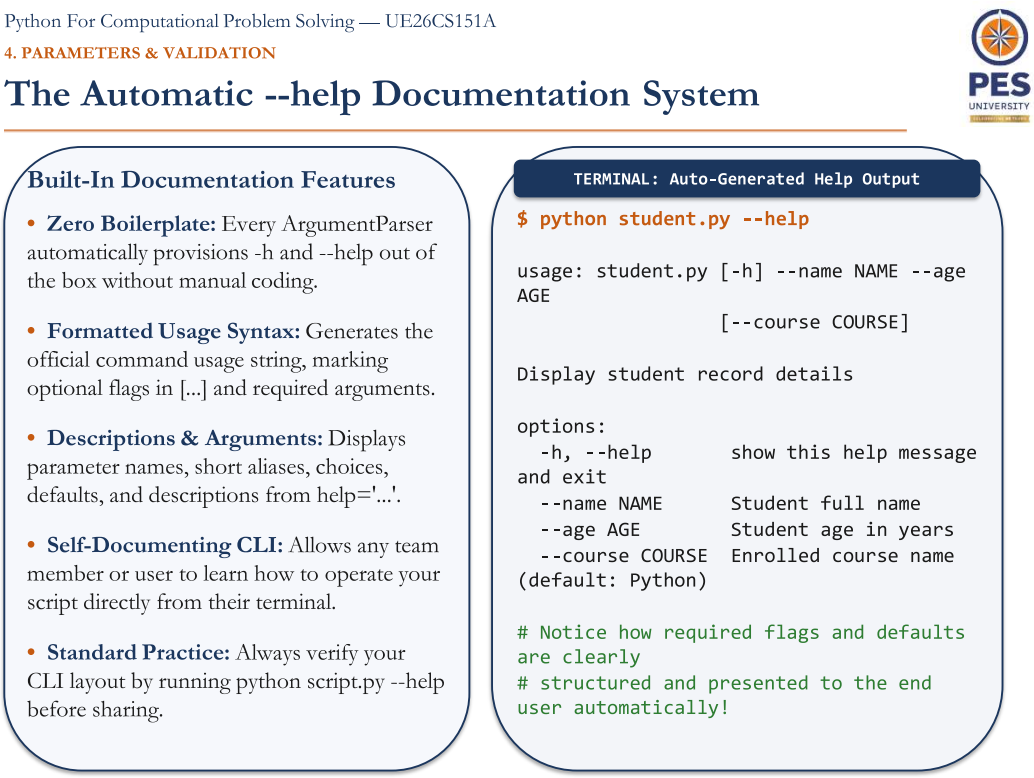

Figure G000231. argparse · source slide page 10.

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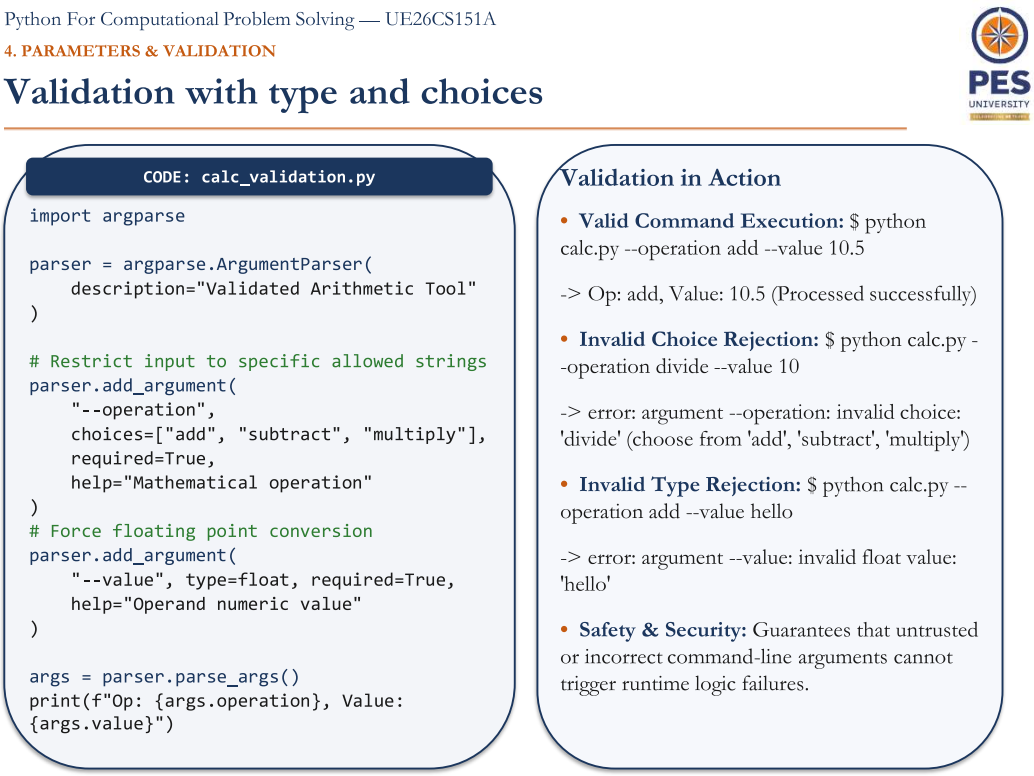

Figure G000232. argparse · source slide page 11.

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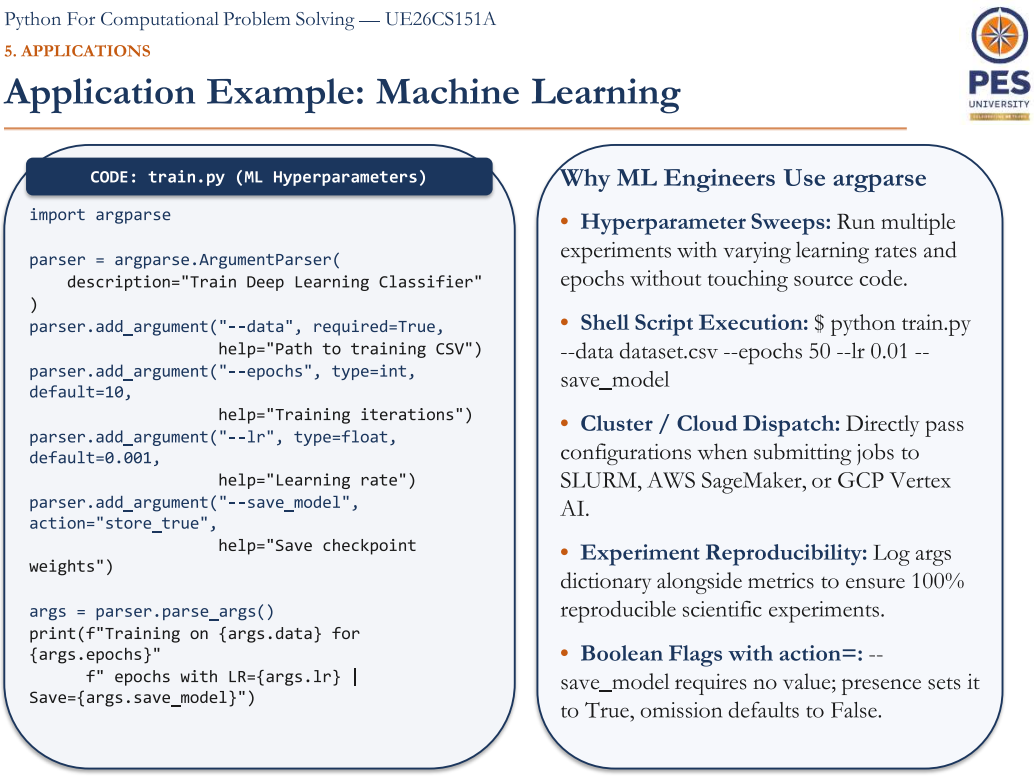

Figure G000233. argparse · source slide page 12.

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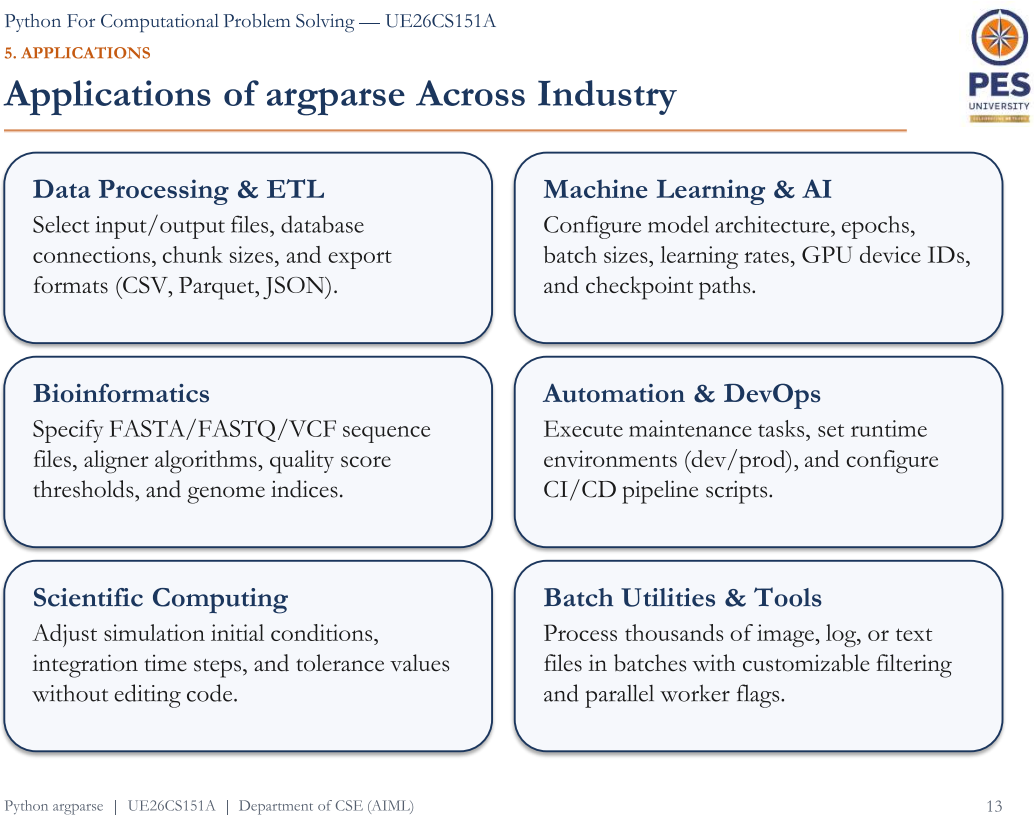

Figure G000234. argparse · source slide page 13.

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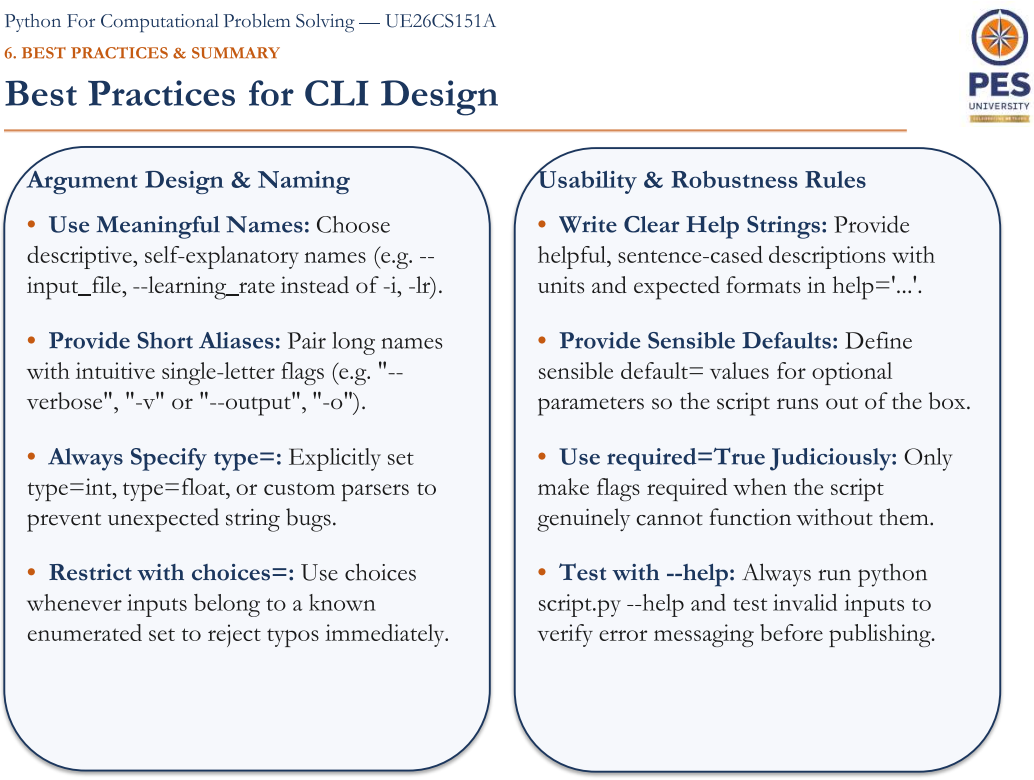

Figure G000235. argparse · source slide page 14.

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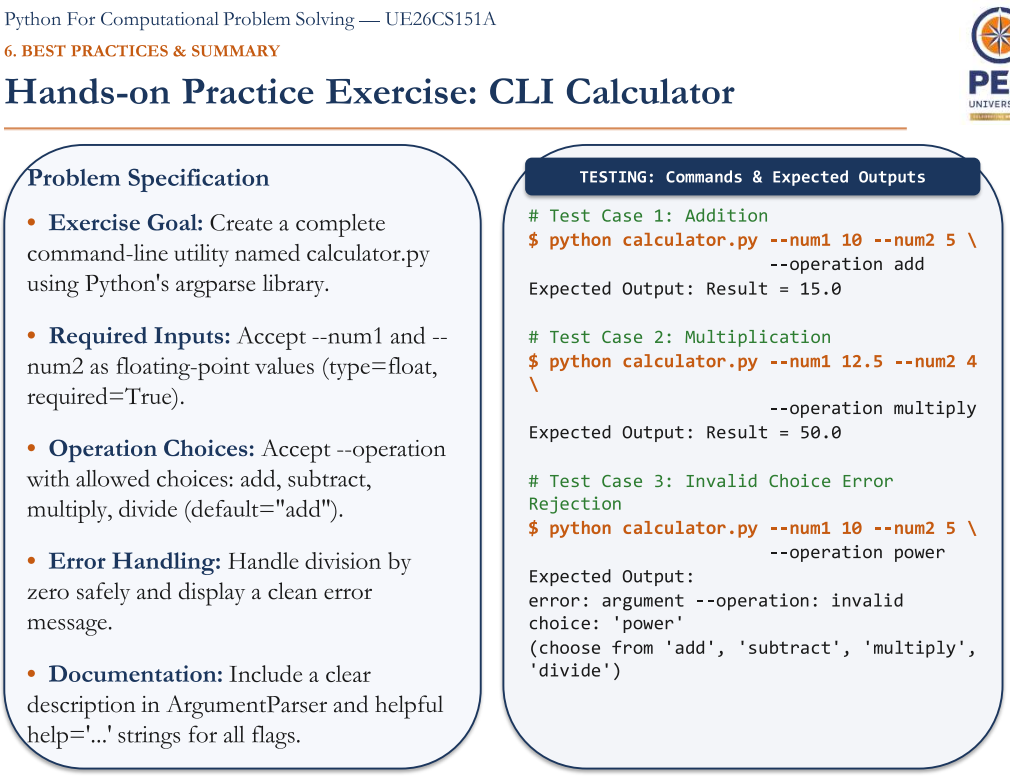

Figure G000236. argparse · source slide page 15.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

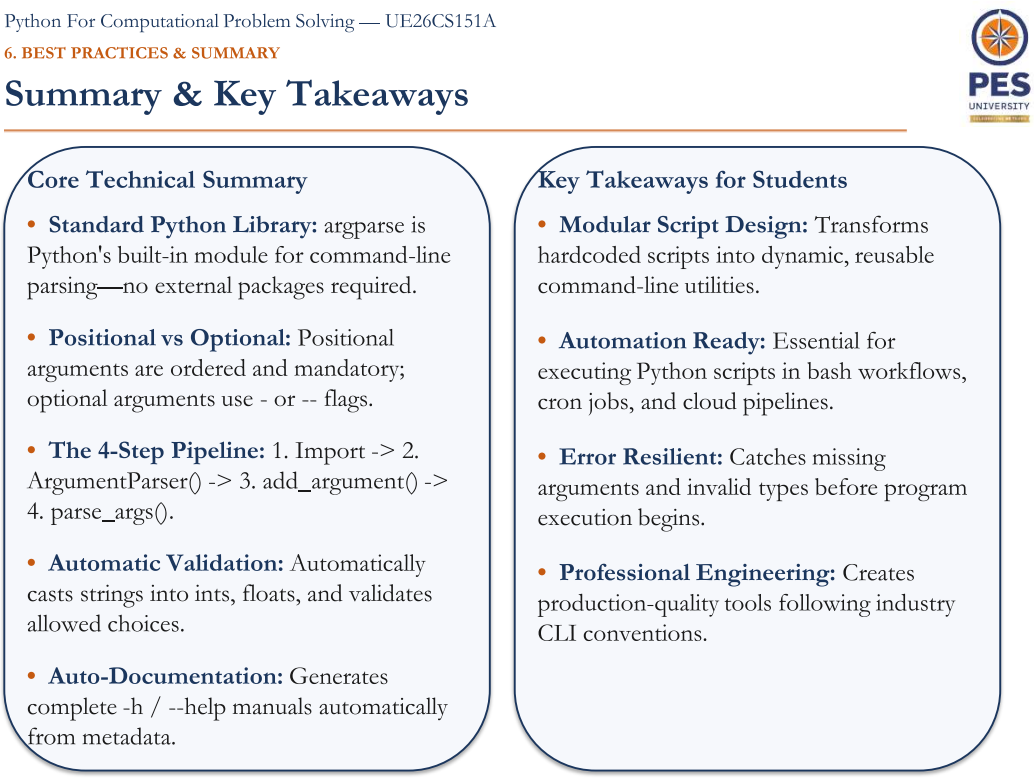

Figure G000237. argparse · source slide page 16.

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Figure G000238. argparse · source slide page 17.

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Important points and why

1. Source slide page 1 (extracted text): Python For Computational Problem Solving — UE26CS151A argparse in Python Command-Line Argument Parsing: Syntax, Workflows & Applications Dr Manjula K Professor, PCPS Anchor 2026 | Department of CSE (AIML) EV0000781 · EV0000782 · EV0000783
2. Source slide page 2 (extracted text): Python For Computational Problem Solving — UE26CS151A COURSE OVERVIEW Agenda & Lecture Roadmap Why argparse? — The Case for CLI Parsing 1 Standard library module, eliminating hardcoded inputs, and automated --help manuals. The 4-Step argparse Workflow & Architecture 2 ArgumentParser container, add_argument(), parse_args(), and Namespace attribute access. Positional vs. Optional Arguments 3 Ordered mandatory parameters vs. flexible - / -- keyword flags with defaults. Essential Parameters & Input Validation 4 Type conversion (int/float), default fallbacks, choices whitelisting, and required flags. Practical Applications & Domain Examples 5 Building student records, CLI calculators, and Machine Learning hyperparameter pipelines. Best Practices, Practice Exercises & Summary 6 Golden rules for CLI tool design, interactive calculator exercises, and core recap. Python argparse | UE26CS151A | Department of CSE (AIML) 2 EV0000784 · EV0000785 · EV0000786

3. Source slide page 3 (extracted text): Python For Computational Problem Solving — UE26CS151A 1. INTRODUCTION What is argparse & Why Use It? What is argparse? Why is it Useful? • Standard Python Library: Built-in module • Eliminates Hardcoding: Execute the exact for processing command-line arguments same script across multiple datasets and settings without third-party installations (no pip install without altering source code. required). • Pipeline Automation: Crucial for running • Dynamic Runtime Inputs: Allows users to scripts from bash scripts, cron jobs, Docker pass values directly into Python scripts from the containers, and CI/CD pipelines. terminal at launch time. • Error Prevention: Enforces required • Automated Help Manuals: Automatically arguments and validates allowed values before creates formatted usage guides and option script execution starts. descriptions when invoked with -h or --help. • Code Modularity: Cleanly decouples • Validation & Type Casting: Converts raw application algorithms from runtime strings into integers, floats, files, and rejects configuration parameters. invalid inputs with clear error messages. • Production CLI Design: Follows standard • Professional Standard: The standard library UNIX and Windows command-line tool used universally for building clean, user- conventions for seamless developer interaction. friendly CLI utilities. Python argparse | UE26CS151A | Department of CSE (AIML) 3 EV0000787 · EV0000788 · EV0000789
4. Source slide page 4 (extracted text): Python For Computational Problem Solving — UE26CS151A 2. CORE WORKFLOW How argparse Works: Execution Pipeline Execution Pipeline Flowchart Step-by-Step Dataflow • 1. Terminal Invocation: User executes $ python app.py --flag value. The operating system shell passes arguments to Python via sys.argv. • 2. ArgumentParser Object: Creates the central container: parser = argparse.ArgumentParser(description=...). Holds all argument metadata. • 3. Argument Registration: parser.add_argument() defines expected flag names, data types, default fallbacks, choices, and help strings. • 4. Parse Command Line: args = parser.parse_args() inspects sys.argv, converts input types, and validates constraints. • 5. Namespace Attribute Access: Values are accessed as clean Python attributes via args.<name> (e.g. args.number, args.epochs). • 6. Program Logic: Application logic executes using verified input values. If input is invalid, a formatted error is displayed immediately. Python argparse | UE26CS151A | Department of CSE (AIML) 4 EV0000790 · EV0000791 · EV0000792
5. Source slide page 5 (extracted text): Python For Computational Problem Solving — UE26CS151A 2. CORE WORKFLOW Basic Syntax & Script Template CODE: Standard argparse Template Syntax Component Breakdown import argparse • import argparse: Imports Python's built- in argument parsing module without external # Step 1: Create the ArgumentParser object parser = argparse.ArgumentParser(package dependencies. description="Calculate square of a number") • ArgumentParser(): Initializes the master parser container. description= provides a # Step 2: Register expected arguments parser.add_argument(header summary in --help. "number", type=int, • add_argument(): Defines an expected help="Number to calculate square" argument. Takes positional/optional name,) type=, default=, and help= description. # Step 3: Parse command-line arguments args = parser.parse_args() • parse_args(): Reads command-line tokens from sys.argv[1:], performs type conversions, # Step 4: Access and use parsed values and builds the args object. print("Square =", args.number ** 2) • args.number: Retrieves the parsed value # Terminal Execution: # $ python square.py 5 directly using the argument name specified # Output: Square = 25 during registration. Python argparse | UE26CS151A | Department of CSE (AIML) 5 EV0000793 · EV0000794 · EV0000795

6. Source slide page 6 (extracted text): Python For Computational Problem Solving — UE26CS151A 3. ARGUMENT TYPES Positional Arguments in argparse CODE: Positional Argument Example Key Rules: Positional Arguments # square_pos.py • Position Determines Meaning: The import argparse order in which values appear on the parser = argparse.ArgumentParser(command line determines which parameter description="Compute square of integer") receives the value. # Positional argument (no '-' prefix) • No Prefix Flags: Positional arguments do parser.add_argument(not start with - or -- (e.g. "number", type=int, "input_file"). help="Input integer to be squared") • Mandatory by Default: If omitted by the user, argparse automatically halts execution args = parser.parse_args() print(f"Square of {args.number} = with a missing argument error message. {args.number**2}") • Automatic Type Conversion: Setting # Execution Cases: type=int ensures strings are safely parsed to # $ python square_pos.py 5 integers before reaching program logic. # Square of 5 = 25 # # $ python square_pos.py • Best Use Case: Ideal for core required # error: the following arguments are required: inputs that the program cannot operate number without (e.g. source filename, target path). Python argparse | UE26CS151A | Department of CSE (AIML) 6 EV0000796 · EV0000797 · EV0000798
7. Source slide page 7 (extracted text): Python For Computational Problem Solving — UE26CS151A 3. ARGUMENT TYPES Optional Arguments & Flags CODE: Optional Argument Example Key Rules: Optional Arguments # square_opt.py • Hyphen / Dash Prefix: Always prefixed import argparse with -- (long name) or - (short alias), e.g. -- parser = argparse.ArgumentParser(number or -n. description="Compute square with flags") • Order-Independent: Optional flags can # Optional argument with '--' and short alias '- be passed in any order anywhere on the n' parser.add_argument(command line. "--number", "-n", type=int, • Explicit Clarity: Flag names clearly default=2, document what each supplied parameter help="Number whose square is required" value represents.) args = parser.parse_args() • Default Fallbacks: Can supply default= to print("Square =", args.number ** 2) define a fallback value when the user omits the flag. # Execution Cases: # 1. Passing flag: $ python square_opt.py -- • Forcing Mandatory Flags: An optional number 5 -> 25 # 2. Short alias: $ python square_opt.py -n 7 flag can be made strictly mandatory by adding -> 49 required=True. # 3. Omitted flag: $ python square_opt.py -> 4 Python argparse | UE26CS151A | Department of CSE (AIML) 7 EV0000799 · EV0000800 · EV0000801
8. Source slide page 8 (extracted text): Python For Computational Problem Solving — UE26CS151A 4. PARAMETERS & VALIDATION Essential add_argument() Parameters Parameter Purpose Example Syntax Description & Behavior type Data type conversion type=int, type=float Converts raw string input into specified Python data type automatically. default Fallback value default=10, Supplies default value if argument is omitted default="app.log" from command line. required Mandatory flag required=True Forces an optional flag (--name) to be mandatory at runtime. choices Restrict input set choices=["add", "sub", Restricts valid input to a specified list; rejects "mul"] other values. help Help documentation help="Path to input Text displayed next to argument in auto-dataset" generated --help manual. action Boolean flag control action="store_true" Sets attribute to True when flag is present, False when absent. Python argparse | UE26CS151A | Department of CSE (AIML) 8 EV0000802 · EV0000803 · EV0000804

9. Source slide page 9 (extracted text): Python For Computational Problem Solving — UE26CS151A 4. PARAMETERS & VALIDATION Example: Student Information CLI CODE: student.py Execution Scenarios & Analysis import argparse • Passing All Options: $ python student.py --name Ananya --age 20 --course AI parser = argparse.ArgumentParser(description="Display student record details" -> Name: Ananya | Age: 20 | Course: AI) • Using Default Course: $ python student.py --name Rahul --age 19 # Mandatory string and integer options parser.add_argument("--name", required=True, -> Name: Rahul | Age: 19 | Course: Python (default help="Student full name") applied) parser.add_argument("--age", type=int, required=True, • Missing Required Flag: $ python student.py --name help="Student age in years") Ananya # Optional argument with default fallback -> error: the following arguments are required: --age parser.add_argument("--course", default="Python", • Type Validation Error: $ python student.py --name help="Enrolled course name") Ananya --age twenty -> error: argument --age: invalid int value: 'twenty' args = parser.parse_args() print(f"Name : {args.name}") • Automatic Protection: The script cleanly catches print(f"Age : {args.age}") omissions and invalid types without crashing Python. print(f"Course : {args.course}") Python argparse | UE26CS151A | Department of CSE (AIML) 9 EV0000805 · EV0000806 · EV0000807
10. Source slide page 10 (extracted text): Python For Computational Problem Solving — UE26CS151A 4. PARAMETERS & VALIDATION The Automatic --help Documentation System TERMINAL: Auto-Generated Help Output Built-In Documentation Features • Zero Boilerplate: Every ArgumentParser $ python student.py --help automatically provisions -h and --help out of usage: student.py [-h] --name NAME --age the box without manual coding. AGE • Formatted Usage Syntax: Generates the [--course COURSE] official command usage string, marking Display student record details optional flags in [...] and required arguments. options: • Descriptions & Arguments: Displays -h, --help show this help message parameter names, short aliases, choices, and exit defaults, and descriptions from help='...'. --name NAME Student full name --age AGE Student age in years • Self-Documenting CLI: Allows any team --course COURSE Enrolled course name member or user to learn how to operate your (default: Python) script directly from their terminal. # Notice how required flags and defaults • Standard Practice: Always verify your are clearly CLI layout by running python script.py --help # structured and presented to the end before sharing. user automatically! Python argparse | UE26CS151A | Department of CSE (AIML) 10 EV0000808 · EV0000809 · EV0000810
11. Source slide page 11 (extracted text): Python For Computational Problem Solving — UE26CS151A 4. PARAMETERS & VALIDATION Validation with type and choices CODE: calc_validation.py Validation in Action import argparse • Valid Command Execution: $ python calc.py --operation add --value 10.5 parser = argparse.ArgumentParser(description="Validated Arithmetic Tool" -> Op: add, Value: 10.5 (Processed successfully)) • Invalid Choice Rejection: $ python calc.py - # Restrict input to specific allowed strings -operation divide --value 10 parser.add_argument("--operation", -> error: argument --operation: invalid choice: choices=["add", "subtract", "multiply"], 'divide' (choose from 'add', 'subtract', 'multiply') required=True, help="Mathematical operation" • Invalid Type Rejection: $ python calc.py --) operation add --value hello # Force floating point conversion parser.add_argument(-> error: argument --value: invalid float value: "--value", type=float, required=True, 'hello' help="Operand numeric value") • Safety & Security: Guarantees that untrusted or incorrect command-line arguments cannot args = parser.parse_args() trigger runtime logic failures. print(f"Op: {args.operation}, Value: {args.value}") Python argparse | UE26CS151A | Department of CSE (AIML) 11 EV0000811 · EV0000812 · EV0000813

12. Source slide page 12 (extracted text): Python For Computational Problem Solving — UE26CS151A 5. APPLICATIONS Application Example: Machine Learning CODE: train.py (ML Hyperparameters) Why ML Engineers Use argparse import argparse • Hyperparameter Sweeps: Run multiple experiments with varying learning rates and parser = argparse.ArgumentParser(description="Train Deep Learning Classifier" epochs without touching source code.) parser.add_argument("--data", required=True, • Shell Script Execution: $ python train.py help="Path to training CSV") --data dataset.csv --epochs 50 --lr 0.01 -- parser.add_argument("--epochs", type=int, save_model default=10, help="Training iterations") • Cluster / Cloud Dispatch: Directly pass parser.add_argument("--lr", type=float, default=0.001, configurations when submitting jobs to help="Learning rate") SLURM, AWS SageMaker, or GCP Vertex parser.add_argument("--save_model", AI. action="store_true", help="Save checkpoint • Experiment Reproducibility: Log args weights") dictionary alongside metrics to ensure 100% args = parser.parse_args() reproducible scientific experiments. print(f"Training on {args.data} for {args.epochs}" • Boolean Flags with action=: -- f" epochs with LR={args.lr} | save_model requires no value; presence sets it Save={args.save_model}") to True, omission defaults to False. Python argparse | UE26CS151A | Department of CSE (AIML) 12 EV0000814 · EV0000815 · EV0000816
13. Source slide page 13 (extracted text): Python For Computational Problem Solving — UE26CS151A 5. APPLICATIONS Applications of argparse Across Industry Data Processing & ETL Machine Learning & AI Select input/output files, database Configure model architecture, epochs, connections, chunk sizes, and export batch sizes, learning rates, GPU device IDs, formats (CSV, Parquet, JSON). and checkpoint paths. Bioinformatics Automation & DevOps Specify FASTA/FASTQ/VCF sequence Execute maintenance tasks, set runtime files, aligner algorithms, quality score environments (dev/prod), and configure thresholds, and genome indices. CI/CD pipeline scripts. Scientific Computing Batch Utilities & Tools Adjust simulation initial conditions, Process thousands of image, log, or text integration time steps, and tolerance values files in batches with customizable filtering without editing code. and parallel worker flags. Python argparse | UE26CS151A | Department of CSE (AIML) 13 EV0000817 · EV0000818 · EV0000819
14. Source slide page 14 (extracted text): Python For Computational Problem Solving — UE26CS151A 6. BEST PRACTICES & SUMMARY Best Practices for CLI Design Argument Design & Naming Usability & Robustness Rules • Use Meaningful Names: Choose • Write Clear Help Strings: Provide descriptive, self-explanatory names (e.g. -- helpful, sentence-cased descriptions with input_file, --learning_rate instead of -i, -lr). units and expected formats in help='...'. • Provide Short Aliases: Pair long names • Provide Sensible Defaults: Define with intuitive single-letter flags (e.g. "-- sensible default= values for optional verbose", "-v" or "--output", "-o"). parameters so the script runs out of the box. • Always Specify type=: Explicitly set • Use required=True Judiciously: Only type=int, type=float, or custom parsers to make flags required when the script prevent unexpected string bugs. genuinely cannot function without them. • Restrict with choices=: Use choices • Test with --help: Always run python whenever inputs belong to a known script.py --help and test invalid inputs to enumerated set to reject typos immediately. verify error messaging before publishing. Python argparse | UE26CS151A | Department of CSE (AIML) 14 EV0000820 · EV0000821 · EV0000822

15. Source slide page 15 (extracted text): Python For Computational Problem Solving — UE26CS151A 6. BEST PRACTICES & SUMMARY Hands-on Practice Exercise: CLI Calculator TESTING: Commands & Expected Outputs Problem Specification • Exercise Goal: Create a complete # Test Case 1: Addition $ python calculator.py --num1 10 --num2 5 \ command-line utility named calculator.py --operation add using Python's argparse library. Expected Output: Result = 15.0 • Required Inputs: Accept --num1 and -- # Test Case 2: Multiplication num2 as floating-point values (type=float, $ python calculator.py --num1 12.5 --num2 4 \ required=True). --operation multiply Expected Output: Result = 50.0 • Operation Choices: Accept --operation with allowed choices: add, subtract, # Test Case 3: Invalid Choice Error multiply, divide (default="add"). Rejection $ python calculator.py --num1 10 --num2 5 \ • Error Handling: Handle division by --operation power Expected Output: zero safely and display a clean error error: argument --operation: invalid message. choice: 'power' (choose from 'add', 'subtract', 'multiply', • Documentation: Include a clear 'divide') description in ArgumentParser and helpful help='...' strings for all flags. Python argparse | UE26CS151A | Department of CSE (AIML) 15 EV0000823 · EV0000824 · EV0000825
16. Source slide page 16 (extracted text): Python For Computational Problem Solving — UE26CS151A 6. BEST PRACTICES & SUMMARY Summary & Key Takeaways Core Technical Summary Key Takeaways for Students • Standard Python Library: argparse is • Modular Script Design: Transforms Python's built-in module for command-line hardcoded scripts into dynamic, reusable parsing—no external packages required. command-line utilities. • Positional vs Optional: Positional • Automation Ready: Essential for arguments are ordered and mandatory; executing Python scripts in bash workflows, optional arguments use - or -- flags. cron jobs, and cloud pipelines. • The 4-Step Pipeline: 1. Import -> 2. • Error Resilient: Catches missing ArgumentParser() -> 3. add_argument() -> arguments and invalid types before program 4. parse_args(). execution begins. • Automatic Validation: Automatically • Professional Engineering: Creates casts strings into ints, floats, and validates production-quality tools following industry allowed choices. CLI conventions. • Auto-Documentation: Generates complete -h / --help manuals automatically from metadata. Python argparse | UE26CS151A | Department of CSE (AIML) 16 EV0000826 · EV0000827 · EV0000828
17. Source slide page 17 (extracted text): Thank You Department of CSE (AIML) Dr Manjula K — manjulak@pes.edu Ack: Mohammed Faizan – Teaching Assistant (CSE AIML) EV0000829 · EV0000830 · EV0000831

SHORT-NOTE RECAP

Source deck preserved: argparse. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000781

21 Historical question-source pages · unit mapping candidates

This section preserves historical paper/question-bank pages whose extracted terms matched Python for computational Problem Solving, Unit_03 (Unit 3: Functions, Functional Programming and Modules) above the dataset's candidate threshold. The association has not been manually confirmed, source text/OCR may be incomplete, and the pages are not rewritten or presented as solved questions. EV0000832 · EV0000833

Important points and why

1. Archive page 5 from July__17cJZWr8dZ.pdf; unit match candidate score 0.75; source text/OCR may be partial. (4.0 Marks) (6.0 Marks) (5.0 Marks) (6.0 Marks) 4.b. Given the below mentioned list. Write a python code using lambda list1 = ['Java', 'Python', 'Csharp', 'Javascript', 'Go', 'Swift'] i) To convert the strings to lowercase if the length of it less than 4 and display it. ii) Display all the strings in the list which has the letter 'o' in it. 4.c. Explain the following module attributes. 1. __name__ 2. __doc__ 3. __dict__ 4.d. Write a function to mimic filter() - called myfilter(). Test this with the following. i)Given a list of strings, display all words whose length exceeds 6. 5.a. Explain the following with an example code. 1. Constructor 2. Inheritance 1 name of the module 2 has docstring 3 has all attributes(var,functions,modules) of class,object 1 used to initialize the attributes of a class when an object is created __init__ 2 repeat EV0000832
2. Archive page 4 from May__1SdbvafL_8.pdf; unit match candidate score 0.4167; source text/OCR may be partial. (4.0 Marks) (5.0 Marks) (6.0 Marks) (5.0 Marks) 3.d. Explain the following functions with respect to the tkinter module in python with an example code. Tk() mainloop() 4.a. Mention any two ways of importing functions from a module. Explain one way with python code. 4.b. 1) Given a list, create a new list by removing negative numbers.
numbers = [1, -2, 3, -4, 5, -6] 2) Given a list having numbers in terms of strings, nums_strs = ["4", "8", "6", "5", "3", "2", "8", "9", "2", "5"], create a new list having the integers. 4.c. Given list1 = [101,3,3234,2331,31670,10], create a list of tuples using list comprehension where each tuple having an element from the given list and its length. initialise the main window of tkinter mainloop keeps the onscreen window to keep displaying in a constant loop unless closed by user import module module.function() from module import func1, fun2 func1() EV0000833
3. Archive page 5 from July__1e0jrB6Yh8.pdf; unit match candidate score 0.4167; source text/OCR may be partial. (4.0 Marks) (6.0 Marks) (5.0 Marks) (6.0 Marks) 4.b. Given the below mentioned list. Write a python code using lambda list1 = ['Java', 'Python', 'Csharp', 'Javascript', 'Go', 'Swift'] i) To convert the strings to lowercase if the length of it less than 4 and display it. ii) Display all the strings in the list which has the letter 'o' in it. 4.c. Explain the following module attributes. 1. __name__ 2. __doc__ 3. __dict__ 4.d. Write a function to mimic filter() - called myfilter(). Test this with the following. i)Given a list of strings, display all words whose length exceeds 6. 5.a. Explain the following with an example code. 1. Constructor 2. Inheritance EV0000834

SHORT-NOTE RECAP

3 historical archive pages are included as question-source references. Inspect the original paper before relying on OCR or unit placement; no answers are newly authored here.

EV0000832

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
Recursion transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/001_Recursion__57028129e466db02.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Recursion transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/001_Recursion__57028129e466db02.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Recursion transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/001_Recursion__57028129e466db02.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9997; only the JSON segmentation supplies the displayed timestamps.
Recursion transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/001_Recursion__57028129e466db02.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Callback transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/003_Callback__da5d1859294b2983.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Callback transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/003_Callback__da5d1859294b2983.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Callback transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/003_Callback__da5d1859294b2983.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9997; only the JSON segmentation supplies the displayed timestamps.
Callback transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/003_Callback__da5d1859294b2983.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Functional and object oriented programming transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/007_Functional_and_object_oriented_programming__8545ef40cb22fe08.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Functional and object oriented programming transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/007_Functional_and_object_oriented_programming__8545ef40cb22fe08.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Functional and object oriented programming transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/007_Functional_and_object_oriented_programming__8545ef40cb22fe08.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9986; only the JSON segmentation supplies the displayed timestamps.
Functional and object oriented programming transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/007_Functional_and_object_oriented_programming__8545ef40cb22fe08.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Lambda Function transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/008_Lambda_Function__4c5d893e819615b4.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Lambda Function transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/008_Lambda_Function__4c5d893e819615b4.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Lambda Function transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/008_Lambda_Function__4c5d893e819615b4.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9992; only the JSON segmentation supplies the displayed timestamps.
Lambda Function transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/008_Lambda_Function__4c5d893e819615b4.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
List Comprehension transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/009_List_Comprehension__27fdb808f3d32a2b.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
List Comprehension transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/009_List_Comprehension__27fdb808f3d32a2b.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
List Comprehension transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/009_List_Comprehension__27fdb808f3d32a2b.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
List Comprehension transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/009_List_Comprehension__27fdb808f3d32a2b.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Closure transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/010_Closure__193f0d4e9567775c.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Closure transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/010_Closure__193f0d4e9567775c.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Closure transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/010_Closure__193f0d4e9567775c.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9996; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Closure transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/010_Closure__193f0d4e9567775c.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Decorators transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/011_Decorators__6b4052527d795156.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Decorators transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/011_Decorators__6b4052527d795156.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Decorators transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/011_Decorators__6b4052527d795156.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9998; only the JSON segmentation supplies the displayed timestamps.
Decorators transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/011_Decorators__6b4052527d795156.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Generators transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/012_Generators__c5a09f63b1677476.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Generators transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/012_Generators__c5a09f63b1677476.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Generators transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/012_Generators__c5a09f63b1677476.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9996; only the JSON segmentation supplies the displayed timestamps.
Generators transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/012_Generators__c5a09f63b1677476.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
module transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/013_module__3e602706ded0701e.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
module transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/013_module__3e602706ded0701e.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
module transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/013_module__3e602706ded0701e.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
module transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/013_module__3e602706ded0701e.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Numpy transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/014_Numpy__7d37477e1600226d.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Numpy transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/014_Numpy__7d37477e1600226d.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Numpy transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/014_Numpy__7d37477e1600226d.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
Numpy transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/014_Numpy__7d37477e1600226d.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
PIL transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/015_PIL__b612657fcad391f3.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
PIL transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/015_PIL__b612657fcad391f3.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
PIL transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/015_PIL__b612657fcad391f3.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
PIL transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/AV_Summaries/015_PIL__b612657fcad391f3.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Slide page text: UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE24CS151A PCPS Unit 3 Sindhu Class 1 2 Recursion (converted PDF; original hash 9420a0698c7b988bfbe4b209380d4b71f1ee5c8c566a505ecba3f200bc31627c)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE24CS151A PCPS Unit 3 Sindhu Class 3 4 Callback	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/003_UE24CS151A_PCPS_Unit_3_Sindhu_Class__3__4_Callback__afde03d096f1327d.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
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Page-render source for UE24CS151A PCPS Unit 3 Sindhu Class 3 4 Callback (converted PDF; original hash d3ab1b6e2604ab559bbb7b935b2c18db5640c671de427ee1efc8a4d1c4d4459b)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/003_UE24CS151A_PCPS_Unit_3_Sindhu_Class__3__4_Callback__afde03d096f1327d.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Slide page text: UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE24CS151A PCPS Unit 4 SindhuRPai Class 2 Lambda (converted PDF; original hash c69fa1d290757f15ef100e1f1b00046aafaf411696deac13bc756b8db6abc0bf)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE24CS151A PCPS Unit 4 Sindhu Class 1 4 functional programming map lambda	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
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Page-render source for UE24CS151A PCPS Unit 4 Sindhu Class 1 4 functional programming map lambda (converted PDF; original hash c69fa1d290757f15ef100e1f1b00046aafaf411696deac13bc756b8db6abc0bf)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE24CS151A PCPS Unit 4 Sindhu Class 2 Functional programming filter	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/008_UE24CS151A_PCPS_Unit_4_Sindhu_Class__2_Functional_programming_filter__128520dfe54dc9f3.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
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Page-render source for UE24CS151A PCPS Unit 4 Sindhu Class 2 Functional programming filter (converted PDF; original hash b2b970153fac1bdc58b45e8e8c15d4d698bf046bc67cb73bf7cd52312c40f600)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/008_UE24CS151A_PCPS_Unit_4_Sindhu_Class__2_Functional_programming_filter__128520dfe54dc9f3.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE24CS151A PCPS Unit 4 SindhuRPai Class 5 reduce	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__5_reduce__da5942d6e378890e.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

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Slide page text: UE24CS151A PCPS Unit 4 SindhuRPai class 6 list comprehension	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
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Slide page text: UE24CS151A PCPS Unit 4 SindhuRPai class 6 zip	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

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Alternate extracted slide page text: UE24CS151A PCPS Unit 4 SindhuRPai class 6 zip	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
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Slide page text: Max Min	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Max Min	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
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Slide page text: UE24CS151A PCPS Unit 3 Sindhu Class 5 Closures	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

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Slide page text: UE24CS151A PCPS Unit 3 Sindhu Class 6 Decorators	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
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Slide page text: UE24CS151A PCPS Unit 3 Sindhu Class 7 Generators	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

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Slide page text: Intro to Modules	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
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Page-render source for Intro to Modules (converted PDF; original hash 3ee2df63ab4e17c6d560369f5eb890fe24dcf446b09e940e4d4e0bb7a64a938c)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Numpy	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
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Page-render source for Numpy (converted PDF; original hash 1f45831a80858ac542eab36ae1f6314923a768837afc74abdb2920f0862c95c5)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Matplotlib	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Matplotlib	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Matplotlib (converted PDF; original hash 5cd46f020e8794e90586ff1945ceeb71d0a2877bac205bf8f679cf08d833925c)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: PIL	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_PIL__5a5c2ba9aaa68100.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: PIL	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_PIL__5a5c2ba9aaa68100.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for PIL (converted PDF; original hash 9745d2e78ab3b998e455db5c82600e448708bce7c19b6d765270dbc9663fc345)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_PIL__5a5c2ba9aaa68100.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Re Module	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Re Module	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Re Module (converted PDF; original hash ebeace81bbef2fa2539f3644e690a4ca793e495a51782decd5bd38e1b76a8b2f)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: argparse	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: argparse	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for argparse (converted PDF; original hash 4bbb04f9d373166155a288e6defac4b1bb748b0d64c1ca351c00b27f9790911e)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Extracted historical pages: July__17cJZWr8dZ.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Python/Papers/ESA Papers/UE22CS151A/Answers/July__17cJZWr8dZ.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: May__1SdbvafL_8.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Python/Papers/ESA Papers/UE22CS151A/Answers/May__1SdbvafL_8.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: July__1e0jrB6Yh8.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Python/Papers/ESA Papers/UE22CS151A/July__1e0jrB6Yh8.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
2. 4 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
3. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
4. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
5. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

Coverage

Every inventoried source unit has an explicit coverage decision. Covered means reviewed and mapped to content; exclusions and gaps remain visible.

Evidence	Source unit	Status	Decision
EV0000001	SRC00001/catalogue	covered	Unit title and index copied from the official units.json catalogue; human content review is pending.
EV0000002	SRC00002/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000003	SRC00003/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000004	SRC00004/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare at similarity 0.9997; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000005	SRC00005/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000006	SRC00002/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0024 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000007	SRC00003/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0024 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000008	SRC00004/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0024 to s0041. Sidecar variants compare at similarity 0.9997; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000009	SRC00005/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0024 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000010	SRC00002/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0050. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000011	SRC00003/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0050. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000012	SRC00004/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0050. Sidecar variants compare at similarity 0.9997; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000013	SRC00005/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0050. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000014	SRC00006/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000015	SRC00007/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000016	SRC00008/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9997; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000017	SRC00009/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000018	SRC00006/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000019	SRC00007/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000020	SRC00008/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 0.9997; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000021	SRC00009/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000022	SRC00006/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0049. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000023	SRC00007/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0049. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000024	SRC00008/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0049. Sidecar variants compare at similarity 0.9997; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000025	SRC00009/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0049. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000026	SRC00010/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000027	SRC00011/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000028	SRC00012/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000029	SRC00013/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000030	SRC00010/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000031	SRC00011/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000032	SRC00012/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000033	SRC00013/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000034	SRC00010/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000035	SRC00011/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000036	SRC00012/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0053. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000037	SRC00013/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000038	SRC00014/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000039	SRC00015/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000040	SRC00016/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000041	SRC00017/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000042	SRC00014/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000043	SRC00015/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000044	SRC00016/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000045	SRC00017/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000046	SRC00018/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000047	SRC00019/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000048	SRC00020/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000049	SRC00021/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000050	SRC00018/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000051	SRC00019/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000052	SRC00020/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000053	SRC00021/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000054	SRC00018/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0049. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000055	SRC00019/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0049. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000056	SRC00020/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0049. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000057	SRC00021/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0049. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000058	SRC00022/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000059	SRC00023/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000060	SRC00024/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000061	SRC00025/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000062	SRC00022/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000063	SRC00023/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000064	SRC00024/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0038. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000065	SRC00025/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000066	SRC00026/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000067	SRC00027/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000068	SRC00028/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 0.9998; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000069	SRC00029/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000070	SRC00026/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000071	SRC00027/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000072	SRC00028/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 0.9998; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000073	SRC00029/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000074	SRC00026/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0067. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000075	SRC00027/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0067. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000076	SRC00028/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0067. Sidecar variants compare at similarity 0.9998; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000077	SRC00029/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0067. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000078	SRC00026/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0068 to s0088. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000079	SRC00027/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0068 to s0088. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000080	SRC00028/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0068 to s0088. Sidecar variants compare at similarity 0.9998; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000081	SRC00029/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0068 to s0088. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000082	SRC00026/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0089 to s0091. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000083	SRC00027/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0089 to s0091. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000084	SRC00028/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0089 to s0091. Sidecar variants compare at similarity 0.9998; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000085	SRC00029/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0089 to s0091. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000086	SRC00030/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000087	SRC00031/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000088	SRC00032/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000089	SRC00033/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000090	SRC00030/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000091	SRC00031/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000092	SRC00032/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000093	SRC00033/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000094	SRC00034/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000095	SRC00035/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000096	SRC00036/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000097	SRC00037/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000098	SRC00034/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0029. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000099	SRC00035/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0029. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000100	SRC00036/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0029. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000101	SRC00037/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0029. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000102	SRC00038/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000103	SRC00039/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000104	SRC00040/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000105	SRC00041/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000106	SRC00038/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000107	SRC00039/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000108	SRC00040/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0037. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000109	SRC00041/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000110	SRC00042/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000111	SRC00043/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000112	SRC00044/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000113	SRC00045/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000114	SRC00042/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0025. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000115	SRC00043/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0025. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000116	SRC00044/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0025. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000117	SRC00045/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0025. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000118	SRC00046/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000119	SRC00048/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000120	SRC00047/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000121	SRC00046/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000122	SRC00048/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000123	SRC00047/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000124	SRC00046/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000125	SRC00048/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000126	SRC00047/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000127	SRC00046/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000128	SRC00048/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000129	SRC00047/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000130	SRC00046/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000131	SRC00048/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000132	SRC00047/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000133	SRC00046/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000134	SRC00048/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000135	SRC00047/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000136	SRC00046/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000137	SRC00048/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000138	SRC00047/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000139	SRC00046/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000140	SRC00048/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000141	SRC00047/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000142	SRC00046/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000143	SRC00048/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000144	SRC00047/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000145	SRC00046/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000146	SRC00048/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000147	SRC00047/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000148	SRC00046/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000149	SRC00048/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000150	SRC00047/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000151	SRC00046/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000152	SRC00048/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000153	SRC00047/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000154	SRC00046/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000155	SRC00048/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000156	SRC00047/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000157	SRC00046/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000158	SRC00048/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000159	SRC00047/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000160	SRC00046/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000161	SRC00048/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000162	SRC00047/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000163	SRC00046/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000164	SRC00048/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000165	SRC00047/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV0000166	SRC00046/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000167	SRC00048/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000168	SRC00047/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000169	SRC00046/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000170	SRC00048/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000171	SRC00047/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV0000172	SRC00046/p19	covered	Extracted slide page 19 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000173	SRC00048/p19	covered	Original slide page 19 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000174	SRC00047/p19	covered	Alternate extracted slide text page 19; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000175	SRC00046/p20	covered	Extracted slide page 20 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000176	SRC00048/p20	covered	Original slide page 20 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000177	SRC00047/p20	covered	Alternate extracted slide text page 20; normalized text matches the .slides.md page. Human comparison is pending.
EV0000178	SRC00046/p21	covered	Extracted slide page 21 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000179	SRC00048/p21	covered	Original slide page 21 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000180	SRC00047/p21	covered	Alternate extracted slide text page 21; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000181	SRC00046/p22	covered	Extracted slide page 22 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000182	SRC00048/p22	covered	Original slide page 22 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/001_UE24CS151A_PCPS_Unit_3_Sindhu_Class__1__2_Recursion__163e7093cd196fcc.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000183	SRC00047/p22	covered	Alternate extracted slide text page 22; normalized text matches the .slides.md page. Human comparison is pending.
EV0000184	SRC00049/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/003_UE24CS151A_PCPS_Unit_3_Sindhu_Class__3__4_Callback__afde03d096f1327d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000185	SRC00051/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/003_UE24CS151A_PCPS_Unit_3_Sindhu_Class__3__4_Callback__afde03d096f1327d.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000186	SRC00050/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000187	SRC00049/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/003_UE24CS151A_PCPS_Unit_3_Sindhu_Class__3__4_Callback__afde03d096f1327d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000188	SRC00051/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/003_UE24CS151A_PCPS_Unit_3_Sindhu_Class__3__4_Callback__afde03d096f1327d.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000189	SRC00050/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000190	SRC00049/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/003_UE24CS151A_PCPS_Unit_3_Sindhu_Class__3__4_Callback__afde03d096f1327d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000191	SRC00051/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/003_UE24CS151A_PCPS_Unit_3_Sindhu_Class__3__4_Callback__afde03d096f1327d.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000192	SRC00050/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000193	SRC00049/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/003_UE24CS151A_PCPS_Unit_3_Sindhu_Class__3__4_Callback__afde03d096f1327d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000194	SRC00051/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/003_UE24CS151A_PCPS_Unit_3_Sindhu_Class__3__4_Callback__afde03d096f1327d.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000195	SRC00050/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000196	SRC00049/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/003_UE24CS151A_PCPS_Unit_3_Sindhu_Class__3__4_Callback__afde03d096f1327d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000197	SRC00051/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/003_UE24CS151A_PCPS_Unit_3_Sindhu_Class__3__4_Callback__afde03d096f1327d.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000198	SRC00050/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000199	SRC00049/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/003_UE24CS151A_PCPS_Unit_3_Sindhu_Class__3__4_Callback__afde03d096f1327d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000200	SRC00051/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/003_UE24CS151A_PCPS_Unit_3_Sindhu_Class__3__4_Callback__afde03d096f1327d.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000201	SRC00050/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000202	SRC00049/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/003_UE24CS151A_PCPS_Unit_3_Sindhu_Class__3__4_Callback__afde03d096f1327d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000203	SRC00051/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/003_UE24CS151A_PCPS_Unit_3_Sindhu_Class__3__4_Callback__afde03d096f1327d.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000204	SRC00050/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000205	SRC00049/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/003_UE24CS151A_PCPS_Unit_3_Sindhu_Class__3__4_Callback__afde03d096f1327d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000206	SRC00051/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/003_UE24CS151A_PCPS_Unit_3_Sindhu_Class__3__4_Callback__afde03d096f1327d.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000207	SRC00050/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000208	SRC00052/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000209	SRC00054/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000210	SRC00053/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000211	SRC00052/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000212	SRC00054/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000213	SRC00053/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000214	SRC00052/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000215	SRC00054/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000216	SRC00053/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000217	SRC00052/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000218	SRC00054/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000219	SRC00053/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000220	SRC00052/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000221	SRC00054/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000222	SRC00053/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000223	SRC00052/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000224	SRC00054/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000225	SRC00053/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000226	SRC00052/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000227	SRC00054/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000228	SRC00053/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000229	SRC00052/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000230	SRC00054/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000231	SRC00053/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000232	SRC00052/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000233	SRC00054/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000234	SRC00053/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000235	SRC00052/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000236	SRC00054/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000237	SRC00053/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000238	SRC00052/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000239	SRC00054/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000240	SRC00053/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000241	SRC00052/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000242	SRC00054/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000243	SRC00053/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000244	SRC00052/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000245	SRC00054/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000246	SRC00053/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000247	SRC00052/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000248	SRC00054/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000249	SRC00053/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000250	SRC00052/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000251	SRC00054/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000252	SRC00053/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000253	SRC00052/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000254	SRC00054/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000255	SRC00053/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV0000256	SRC00052/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000257	SRC00054/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000258	SRC00053/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000259	SRC00052/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000260	SRC00054/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/005_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__2_Lambda__caf6259b1397b6cd.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000261	SRC00053/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV0000262	SRC00055/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000263	SRC00057/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000264	SRC00056/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000265	SRC00055/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000266	SRC00057/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000267	SRC00056/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000268	SRC00055/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000269	SRC00057/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000270	SRC00056/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002 71	SRC00055/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 72	SRC00057/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 73	SRC00056/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 74	SRC00055/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 75	SRC00057/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 76	SRC00056/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000277	SRC00055/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000278	SRC00057/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000279	SRC00056/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000280	SRC00055/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000281	SRC00057/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000282	SRC00056/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000283	SRC00055/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000284	SRC00057/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000285	SRC00056/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000286	SRC00055/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000287	SRC00057/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000288	SRC00056/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000289	SRC00055/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000290	SRC00057/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000291	SRC00056/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000292	SRC00055/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000293	SRC00057/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000294	SRC00056/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000295	SRC00055/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000296	SRC00057/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000297	SRC00056/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000298	SRC00055/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000299	SRC00057/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000300	SRC00056/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000301	SRC00055/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000302	SRC00057/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000303	SRC00056/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000304	SRC00055/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000305	SRC00057/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000306	SRC00056/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000307	SRC00055/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000308	SRC00057/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000309	SRC00056/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV0000310	SRC00055/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000311	SRC00057/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000312	SRC00056/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000313	SRC00055/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000314	SRC00057/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/007_UE24CS151A_PCPS_Unit_4_Sindhu_Class__1__4_functional_programming_map_lambda__d6cfcde0f8bc4626.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000315	SRC00056/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV0000316	SRC00058/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/008_UE24CS151A_PCPS_Unit_4_Sindhu_Class__2_Functional_programming_filter__128520dfe54dc9f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000317	SRC00060/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/008_UE24CS151A_PCPS_Unit_4_Sindhu_Class__2_Functional_programming_filter__128520dfe54dc9f3.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000318	SRC00059/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003319	SRC00058/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/008_UE24CS151A_PCPS_Unit_4_Sindhu_Class__2_Functional_programming_filter__128520dfe54dc9f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003320	SRC00060/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/008_UE24CS151A_PCPS_Unit_4_Sindhu_Class__2_Functional_programming_filter__128520dfe54dc9f3.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003321	SRC00059/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00003322	SRC00058/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/008_UE24CS151A_PCPS_Unit_4_Sindhu_Class__2_Functional_programming_filter__128520dfe54dc9f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003323	SRC00060/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/008_UE24CS151A_PCPS_Unit_4_Sindhu_Class__2_Functional_programming_filter__128520dfe54dc9f3.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003324	SRC00059/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003325	SRC00058/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/008_UE24CS151A_PCPS_Unit_4_Sindhu_Class__2_Functional_programming_filter__128520dfe54dc9f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003326	SRC00060/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/008_UE24CS151A_PCPS_Unit_4_Sindhu_Class__2_Functional_programming_filter__128520dfe54dc9f3.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003327	SRC00059/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00003328	SRC00058/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/008_UE24CS151A_PCPS_Unit_4_Sindhu_Class__2_Functional_programming_filter__128520dfe54dc9f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003329	SRC00060/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/008_UE24CS151A_PCPS_Unit_4_Sindhu_Class__2_Functional_programming_filter__128520dfe54dc9f3.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003330	SRC00059/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000331	SRC00058/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/008_UE24CS151A_PCPS_Unit_4_Sindhu_Class__2_Functional_programming_filter__128520dfe54dc9f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000332	SRC00060/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/008_UE24CS151A_PCPS_Unit_4_Sindhu_Class__2_Functional_programming_filter__128520dfe54dc9f3.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000333	SRC00059/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000334	SRC00058/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/008_UE24CS151A_PCPS_Unit_4_Sindhu_Class__2_Functional_programming_filter__128520dfe54dc9f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000335	SRC00060/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/008_UE24CS151A_PCPS_Unit_4_Sindhu_Class__2_Functional_programming_filter__128520dfe54dc9f3.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000336	SRC00059/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000337	SRC00058/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/008_UE24CS151A_PCPS_Unit_4_Sindhu_Class__2_Functional_programming_filter__128520dfe54dc9f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000338	SRC00060/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/008_UE24CS151A_PCPS_Unit_4_Sindhu_Class__2_Functional_programming_filter__128520dfe54dc9f3.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000339	SRC00059/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000340	SRC00058/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/008_UE24CS151A_PCPS_Unit_4_Sindhu_Class__2_Functional_programming_filter__128520dfe54dc9f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000341	SRC00060/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/008_UE24CS151A_PCPS_Unit_4_Sindhu_Class__2_Functional_programming_filter__128520dfe54dc9f3.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000342	SRC00059/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000343	SRC00061/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__5_reduce__da5942d6e378890e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000344	SRC00063/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__5_reduce__da5942d6e378890e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000345	SRC00062/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000346	SRC00061/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__5_reduce__da5942d6e378890e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000347	SRC00063/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__5_reduce__da5942d6e378890e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000348	SRC00062/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000349	SRC00061/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__5_reduce__da5942d6e378890e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000350	SRC00063/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__5_reduce__da5942d6e378890e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000351	SRC00062/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000352	SRC00061/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__5_reduce__da5942d6e378890e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000353	SRC00063/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__5_reduce__da5942d6e378890e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000354	SRC00062/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000355	SRC00061/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__5_reduce__da5942d6e378890e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000356	SRC00063/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__5_reduce__da5942d6e378890e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000357	SRC00062/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000358	SRC00061/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__5_reduce__da5942d6e378890e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000359	SRC00063/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__5_reduce__da5942d6e378890e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000360	SRC00062/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000361	SRC00061/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__5_reduce__da5942d6e378890e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000362	SRC00063/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_Class__5_reduce__da5942d6e378890e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000363	SRC00062/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000364	SRC00064/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000365	SRC00066/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000366	SRC00065/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000367	SRC00064/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000368	SRC00066/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000369	SRC00065/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000370	SRC00064/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000371	SRC00066/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000372	SRC00065/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000373	SRC00064/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000374	SRC00066/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000375	SRC00065/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000376	SRC00064/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000377	SRC00066/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000378	SRC00065/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003379	SRC00064/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003380	SRC00066/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003381	SRC00065/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00003382	SRC00064/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003383	SRC00066/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003384	SRC00065/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000385	SRC00064/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000386	SRC00066/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000387	SRC00065/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000388	SRC00064/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000389	SRC00066/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000390	SRC00065/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000391	SRC00064/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000392	SRC00066/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000393	SRC00065/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000394	SRC00064/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000395	SRC00066/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000396	SRC00065/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000397	SRC00064/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000398	SRC00066/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000399	SRC00065/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000400	SRC00064/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000401	SRC00066/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000402	SRC00065/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000403	SRC00064/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000404	SRC00066/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000405	SRC00065/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000406	SRC00064/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000407	SRC00066/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_list_comprehension__a50c71c4eae54791.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000408	SRC00065/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000409	SRC00067/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000410	SRC00069/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000411	SRC00068/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000412	SRC00067/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000413	SRC00069/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000414	SRC00068/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000415	SRC00067/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000416	SRC00069/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000417	SRC00068/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000418	SRC00067/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000419	SRC00069/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000420	SRC00068/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000421	SRC00067/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000422	SRC00069/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000423	SRC00068/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000424	SRC00067/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000425	SRC00069/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000426	SRC00068/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000427	SRC00067/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000428	SRC00069/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000429	SRC00068/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000430	SRC00067/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000431	SRC00069/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000432	SRC00068/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000433	SRC00067/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000434	SRC00069/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000435	SRC00068/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000436	SRC00067/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000437	SRC00069/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000438	SRC00068/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000439	SRC00067/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000440	SRC00069/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/009_UE24CS151A_PCPS_Unit_4_SindhuRPai_class__6_zip__33aedb5fc4db15f6.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000441	SRC00068/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000442	SRC00070/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000443	SRC00072/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000444	SRC00071/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000445	SRC00070/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000446	SRC00072/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000447	SRC00071/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000448	SRC00070/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000449	SRC00072/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000450	SRC00071/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000451	SRC00070/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000452	SRC00072/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000453	SRC00071/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000454	SRC00070/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000455	SRC00072/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000456	SRC00071/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000457	SRC00070/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000458	SRC00072/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000459	SRC00071/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000460	SRC00070/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000461	SRC00072/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000462	SRC00071/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000463	SRC00070/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000464	SRC00072/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000465	SRC00071/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000466	SRC00070/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000467	SRC00072/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000468	SRC00071/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000469	SRC00070/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000470	SRC00072/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000471	SRC00071/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000472	SRC00070/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000473	SRC00072/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000474	SRC00071/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000475	SRC00070/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000476	SRC00072/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_Max_Min__dc1351408347fdb8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000477	SRC00071/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000478	SRC00073/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000479	SRC00075/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000480	SRC00074/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000481	SRC00073/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000482	SRC00075/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000483	SRC00074/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000484	SRC00073/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000485	SRC00075/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000486	SRC00074/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000487	SRC00073/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000488	SRC00075/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000489	SRC00074/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000490	SRC00073/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000491	SRC00075/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000492	SRC00074/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000493	SRC00073/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000494	SRC00075/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000495	SRC00074/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000496	SRC00073/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000497	SRC00075/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000498	SRC00074/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000499	SRC00073/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000500	SRC00075/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000501	SRC00074/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000502	SRC00073/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000503	SRC00075/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000504	SRC00074/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000505	SRC00073/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000506	SRC00075/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000507	SRC00074/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000508	SRC00073/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000509	SRC00075/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000510	SRC00074/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000511	SRC00073/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000512	SRC00075/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000513	SRC00074/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000514	SRC00073/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000515	SRC00075/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/010_UE24CS151A_PCPS_Unit_3_Sindhu_Class__5_Closures__8889b68d5d038029.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000516	SRC00074/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000517	SRC00076/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000518	SRC00078/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000519	SRC00077/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000520	SRC00076/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000521	SRC00078/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000522	SRC00077/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000523	SRC00076/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000524	SRC00078/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000525	SRC00077/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000526	SRC00076/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000527	SRC00078/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000528	SRC00077/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000529	SRC00076/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000530	SRC00078/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000531	SRC00077/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000532	SRC00076/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000533	SRC00078/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000534	SRC00077/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000535	SRC00076/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000536	SRC00078/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000537	SRC00077/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000538	SRC00076/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000539	SRC00078/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000540	SRC00077/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000541	SRC00076/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000542	SRC00078/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000543	SRC00077/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000544	SRC00076/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000545	SRC00078/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000546	SRC00077/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000547	SRC00076/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000548	SRC00078/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000549	SRC00077/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000550	SRC00076/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000551	SRC00078/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000552	SRC00077/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000553	SRC00076/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000554	SRC00078/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000555	SRC00077/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000556	SRC00076/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000557	SRC00078/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000558	SRC00077/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000559	SRC00076/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000560	SRC00078/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000561	SRC00077/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000562	SRC00076/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000563	SRC00078/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/011_UE24CS151A_PCPS_Unit_3_Sindhu_Class__6_Decorators__abed17696290de23.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000564	SRC00077/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000565	SRC00079/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000566	SRC00081/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000567	SRC00080/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000568	SRC00079/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000569	SRC00081/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000570	SRC00080/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000571	SRC00079/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000572	SRC00081/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000573	SRC00080/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000574	SRC00079/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000575	SRC00081/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000576	SRC00080/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000577	SRC00079/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000578	SRC00081/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000579	SRC00080/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000580	SRC00079/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000581	SRC00081/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000582	SRC00080/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000583	SRC00079/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000584	SRC00081/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000585	SRC00080/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000586	SRC00079/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000587	SRC00081/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000588	SRC00080/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000589	SRC00079/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000590	SRC00081/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000591	SRC00080/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000592	SRC00079/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000593	SRC00081/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000594	SRC00080/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000595	SRC00079/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000596	SRC00081/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000597	SRC00080/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000598	SRC00079/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000599	SRC00081/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/012_UE24CS151A_PCPS_Unit_3_Sindhu_Class__7_Generators__6cae1ddd3709b089.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000600	SRC00080/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000601	SRC00082/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000602	SRC00084/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000603	SRC00083/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000604	SRC00082/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000605	SRC00084/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000606	SRC00083/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000607	SRC00082/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000608	SRC00084/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000609	SRC00083/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000610	SRC00082/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000611	SRC00084/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000612	SRC00083/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000613	SRC00082/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000614	SRC00084/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000615	SRC00083/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000616	SRC00082/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000617	SRC00084/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000618	SRC00083/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000619	SRC00082/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000620	SRC00084/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000621	SRC00083/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000622	SRC00082/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000623	SRC00084/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000624	SRC00083/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000625	SRC00082/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000626	SRC00084/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000627	SRC00083/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000628	SRC00082/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000629	SRC00084/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000630	SRC00083/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000631	SRC00082/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000632	SRC00084/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000633	SRC00083/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000634	SRC00082/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000635	SRC00084/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/013_Intro_to_Modules__9bad64887fc1f1ee.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000636	SRC00083/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000637	SRC00085/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000638	SRC00087/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000639	SRC00086/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000640	SRC00085/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000641	SRC00087/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000642	SRC00086/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000643	SRC00085/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000644	SRC00087/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000645	SRC00086/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000646	SRC00085/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000647	SRC00087/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000648	SRC00086/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000649	SRC00085/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000650	SRC00087/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000651	SRC00086/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000652	SRC00085/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000653	SRC00087/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000654	SRC00086/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000655	SRC00085/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000656	SRC00087/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000657	SRC00086/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000658	SRC00085/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000659	SRC00087/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000660	SRC00086/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000661	SRC00085/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000662	SRC00087/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000663	SRC00086/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000664	SRC00085/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000665	SRC00087/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000666	SRC00086/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000667	SRC00085/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000668	SRC00087/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000669	SRC00086/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000670	SRC00085/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000671	SRC00087/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000672	SRC00086/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000673	SRC00085/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000674	SRC00087/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/014_Numpy__084bdca3b5d7a7d1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000675	SRC00086/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000676	SRC00088/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000677	SRC00090/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000678	SRC00089/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000679	SRC00088/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000680	SRC00090/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000681	SRC00089/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000682	SRC00088/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000683	SRC00090/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000684	SRC00089/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000685	SRC00088/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000686	SRC00090/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000687	SRC00089/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000688	SRC00088/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000689	SRC00090/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000690	SRC00089/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000691	SRC00088/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000692	SRC00090/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000693	SRC00089/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000694	SRC00088/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000695	SRC00090/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000696	SRC00089/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000697	SRC00088/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000698	SRC00090/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000699	SRC00089/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000700	SRC00088/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000701	SRC00090/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000702	SRC00089/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000703	SRC00088/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000704	SRC00090/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000705	SRC00089/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000706	SRC00088/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000707	SRC00090/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000708	SRC00089/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000709	SRC00088/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000710	SRC00090/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000711	SRC00089/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000712	SRC00088/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000713	SRC00090/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_Matplotlib__26db950762ee81e0.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000714	SRC00089/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000715	SRC00091/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_PIL__5a5c2ba9aaa68100.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000716	SRC00093/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_PIL__5a5c2ba9aaa68100.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000717	SRC00092/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000718	SRC00091/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_PIL__5a5c2ba9aaa68100.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000719	SRC00093/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_PIL__5a5c2ba9aaa68100.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000720	SRC00092/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000721	SRC00091/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_PIL__5a5c2ba9aaa68100.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000722	SRC00093/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_PIL__5a5c2ba9aaa68100.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000723	SRC00092/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000724	SRC00091/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_PIL__5a5c2ba9aaa68100.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000725	SRC00093/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_PIL__5a5c2ba9aaa68100.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000726	SRC00092/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000727	SRC00091/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_PIL__5a5c2ba9aaa68100.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000728	SRC00093/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_PIL__5a5c2ba9aaa68100.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000729	SRC00092/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000730	SRC00091/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_PIL__5a5c2ba9aaa68100.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000731	SRC00093/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_PIL__5a5c2ba9aaa68100.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000732	SRC00092/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000733	SRC00091/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_PIL__5a5c2ba9aaa68100.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000734	SRC00093/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_PIL__5a5c2ba9aaa68100.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000735	SRC00092/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000736	SRC00091/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_PIL__5a5c2ba9aaa68100.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000737	SRC00093/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_PIL__5a5c2ba9aaa68100.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000738	SRC00092/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000739	SRC00091/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_PIL__5a5c2ba9aaa68100.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000740	SRC00093/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/015_PIL__5a5c2ba9aaa68100.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000741	SRC00092/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000742	SRC00094/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000743	SRC00096/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000744	SRC00095/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000745	SRC00094/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000746	SRC00096/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000747	SRC00095/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000748	SRC00094/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000749	SRC00096/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000750	SRC00095/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000751	SRC00094/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000752	SRC00096/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000753	SRC00095/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000754	SRC00094/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000755	SRC00096/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000756	SRC00095/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000757	SRC00094/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000758	SRC00096/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000759	SRC00095/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000760	SRC00094/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000761	SRC00096/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000762	SRC00095/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000763	SRC00094/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000764	SRC00096/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000765	SRC00095/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000766	SRC00094/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000767	SRC00096/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000768	SRC00095/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000769	SRC00094/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00007 70	SRC00096/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00007 71	SRC00095/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV00007 72	SRC00094/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00007 73	SRC00096/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00007 74	SRC00095/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV00007 75	SRC00094/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00007 76	SRC00096/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000777	SRC00095/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000778	SRC00094/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000779	SRC00096/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_Re_Module__9097da6befb51bf8.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000780	SRC00095/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000781	SRC00097/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000782	SRC00099/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000783	SRC00098/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000784	SRC00097/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000785	SRC00099/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000786	SRC00098/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000787	SRC00097/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000788	SRC00099/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000789	SRC00098/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000790	SRC00097/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000791	SRC00099/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000792	SRC00098/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000793	SRC00097/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000794	SRC00099/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000795	SRC00098/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000796	SRC00097/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000797	SRC00099/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00007 98	SRC00098/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00007 99	SRC00097/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008 00	SRC00099/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00008 01	SRC00098/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00008 02	SRC00097/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008 03	SRC00099/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00008 04	SRC00098/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000805	SRC00097/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000806	SRC00099/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000807	SRC00098/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000808	SRC00097/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000809	SRC00099/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000810	SRC00098/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000811	SRC00097/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000812	SRC00099/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000813	SRC00098/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000814	SRC00097/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000815	SRC00099/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000816	SRC00098/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000817	SRC00097/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000818	SRC00099/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000819	SRC00098/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000820	SRC00097/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000821	SRC00099/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000822	SRC00098/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000823	SRC00097/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000824	SRC00099/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000825	SRC00098/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00008026	SRC00097/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008027	SRC00099/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00008028	SRC00098/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV00008029	SRC00097/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008030	SRC00099/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_03/Slides/016_argparse__7901fba9314fe002.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00008031	SRC00098/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV00008032	SRC00100/q5	covered	Original archive PDF page 5; candidate unit association is text-based and unconfirmed.
EV00008033	SRC00101/q4	covered	Original archive PDF page 4; candidate unit association is text-based and unconfirmed.
EV00008034	SRC00102/q5	covered	Original archive PDF page 5; candidate unit association is text-based and unconfirmed.

C Revision and retrieval practice